NATIONAL MARINE ELECTRONICS ASSOCIATION



NMEA 2000 ®

STANDARD FOR SERIAL-DATA NETWORKING OF MARINE ELECTRONIC DEVICES

Appendix C

Test Criteria

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C.1 Introduction

This document is intended to provide the information necessary to ensure that devices meet the minimum requirements of the NMEA 2000 Standard For Serial-Data Networking Of Marine Electronic Devices Specification.

C.1.1 Scope

This document is not intended to encompass all aspects of designing, building, and testing NMEA 2000 devices. Rather, the goal is to identify specific requirements within the NMEA 2000 Specification, which are critical to achieving consistent and proper network behavior from any device from any manufacturer. In addition, the methods to validate specific requirements and the expected validation results are provided. The section numbers in square brackets,[ ],references these requirements to the NMEA 2000 standard.

NMEA 2000 Devices are certified to a Level of Compliance to the Standard. The level of certification is determined by the functionality of the device to the network, and by the information the device must provide or receive from the network. There are three levels of certification for each Class: Level A, Level B, and Listen Only. To be certified, the device must meet all mandatory test conditions as defined in the table below for the desired certification level.

Device Class	Certification Level	Test Sections of This Document
Class 1	Level A	C.2 and C.3
Class 1	Level B	C.2 and C.4
Class 1	Listen Only	C.2 and C.5
Class 2	Level A	C.2 and C.6
Class 2	Level B	C.2 and C.7
Class 2	Listen Only	C.2 and C.8

For instance, if a device needs to send or receive a PGN with a data size greater than the fast packet message size of 223 bytes, then it would require the Transport Protocol. This protocol is part of the Level A Certification. Another device such as a speed transducer may only need to transmit the PGN for Speed, which is a single frame message. This type of device would not need the functionality or capabilities required for Level A certification, and could be certified to Level B compliance. This allows the manufacturer to scale their product to the desired NMEA 2000 Certification Level.

A manufacturer’s product is not limited to only Level A or Level B. A product certified to Level B may offer one or more of the Level A capabilities if desired.

Devices that properly meet the criteria defined in the NMEA 2000 Standard For Serial-Data Networking Of Marine Electronic Devices Specification, and this document, are eligible for Certification by the National Marine Electronics Association, and may thus be granted permission to use and display the "Certified for NMEA 2000" Logo.

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C.1.2 References

C.1.3 Equipment

C.1.3.1 Class 1 Device Test Equipment

Device to be tested.
Computer
CAN Interface:
- One bus interface
- A message timestamp resolution of 1 millisecond or better

Certification software
Network Cables
- Two network backbone cable segments
- Two network terminators
- Two branch junctions (T-connectors or equivalent)
- One power junction
- Two branch cables
- One power branch cable

One network power supply
Multi-meter
Diskette (for sending test results to NMEA)

C.1.3.2 Class 2 Device Test Equipment

Device to be tested
Computer
CAN Interface:
- Two CAN bus interfaces
- A message timestamp resolution of 1 millisecond or better
- Message timestamps provided on the two channels must be synchronized to each other

Certification software
Network Cables:
- Four network backbone cable segments
- Four network terminators
- Four branch junctions (T-connectors)
- Two power junctions
- Four branch cables
- Two power branch cables

Two network power supplies
Multi-meter
Diskette

C.1.4 Definitions

DUT - Device Under Test
PGN – Parameter Group Number

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C.2 Certification Criteria

This section contains the requirements that cannot be certified through automatic means. The majority of the requirements identified in this section are mandatory. Devices that do not meet the mandatory minimum requirements cannot be certified as NMEA 2000 compliant. Some of the requirements in this section are not mandatory, but a recommended as good practice, and identified as non-mandatory. Devices that do not meet the non-mandatory requirements may still be certified as long as all of the mandatory minimum requirements are satisfied.

C.2.1 Durability and Resistance to Environmental Conditions [2.1.1.1] – Non Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.2 Extended storage (non-operating) temperature [2.1.1.2] – Non Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.3 Unwanted Electromagnetic Emissions [2.1.2.1] – Non Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.4 Immunity to Electromagnetic Environment [2.1.2.2] – Non Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.5 *Controller Area Network (CAN) [2.2.1] – Mandatory*

Requirement fulfilled: Yes_____ No_____

C.2.6 Network Signaling Rate [2.3] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.7 Shield Connections [2.4.1] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.8 Shield Isolation [2.4.1.1] – Mandatory

Requirement fulfilled: Yes_____ No_____

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C.2.9 DC Isolation [2.4.2.1] - Mandatory

Provide measured resistance values for each interface on the DUT in units of K Ohms

From:	Net-H	Net-L	Net-S	Net-C	Shield
To product power positive terminal					
To product power negative terminal					
To product AC terminal 1 (if applicable)					
To product AC terminal 2 (if applicable)					
To product ground terminal					
to					
to					
to					
to					

C.2.10 AC Isolation [2.4.2.2] - Mandatory

Provide maximum capacitance measurements (mandatory), the maximum value is a recommended value.

From:	Net-H	Net-L	Net-S	Net-C	Shield
To product power positive terminal					
To product power negative terminal					
To product AC terminal 1 (if applicable)					
To product AC terminal 2 (if applicable)					
To product ground terminal					
to					
to					
to					
to					

Capacitance meter used: _____ (Yes / No)
Other means used (describe): _____

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C.2.11 NET-H and NET-L signal lines [2.4.3.2] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.12 Timing Limit [2.4.4.1] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.13 Input timing delays [2.4.4.2] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.14 Output timing delays [2.4.4.3] - Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.15 Voltage level protection [2.4.6.1] - Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.16 Wiring protection [2.4.6.2] - Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.17 Voltage range [2.4.7.1] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.18 Maximum node current [2.4.7.2.1] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.19 Dedicated node power cable isolation [2.4.7.2.2] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.20 Dedicated node power cable labeling [2.4.7.2.3] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.21 Node power and common isolation [2.4.7.4] – Mandatory

Requirement fulfilled: Yes_____ No_____

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C.2.22 Node power and common currents [2.4.7.5] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.23 Load equivalency number [2.4.7.6] – Mandatory

Load Equivalency number specified?: Yes_____ No_____
Value of Load Equivalency Number: _____

C.2.24 Node noise, ripple, and transients [2.4.7.7.] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.25 Conducted emissions [2.4.7.7.1] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.26 Power disturbances [2.4.7.8] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.27 DC Ground isolation [2.5.1.1] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.28 Output Voltage Range [2.5.1.2] - Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.29 AC input line variation [2.5.1.3] - Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.30 DC input line variation [2.5.1.4] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.31 Current capacity shall be specified [2.5.1.5] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.32 Output current surge capacity [2.5.1.6] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.33 Output ripple voltage [2.5.1.7] – Mandatory

Requirement fulfilled: Yes_____ No_____

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C.2.34 Output over voltage protection [2.5.1.8] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.35 Output over current protection [2.5.1.9] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.36 Power supply sink current [2.5.1.10] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.37 Power supply shield connection [2.5.1.11] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.38 Selectable isolated supply [2.5.1.12] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.39 Excessive noise, ripple, or transients [2.5.1.13] - Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.40 Maximum conducted emissions [2.5.1.13.1] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.41 Termination resistors [2.6.2] – Mandatory

Requirement fulfilled: Yes_____ No_____

C.2.42 Number of NMEA 2000 Connections [2.7.1] – Mandatory

Number of NMEA 2000 protocol connections on the device: _____

If more than 1 an explanation is required:

_____

_____

_____

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C.2.43 Field Programmability of the three Instance Fields [8.3.1] – Mandatory

System Instance Requirement fulfilled: Yes_____ No_____

Device Instance Upper Requirement fulfilled: Yes_____ No_____

Device Instance Lower Requirement fulfilled: Yes_____ No_____

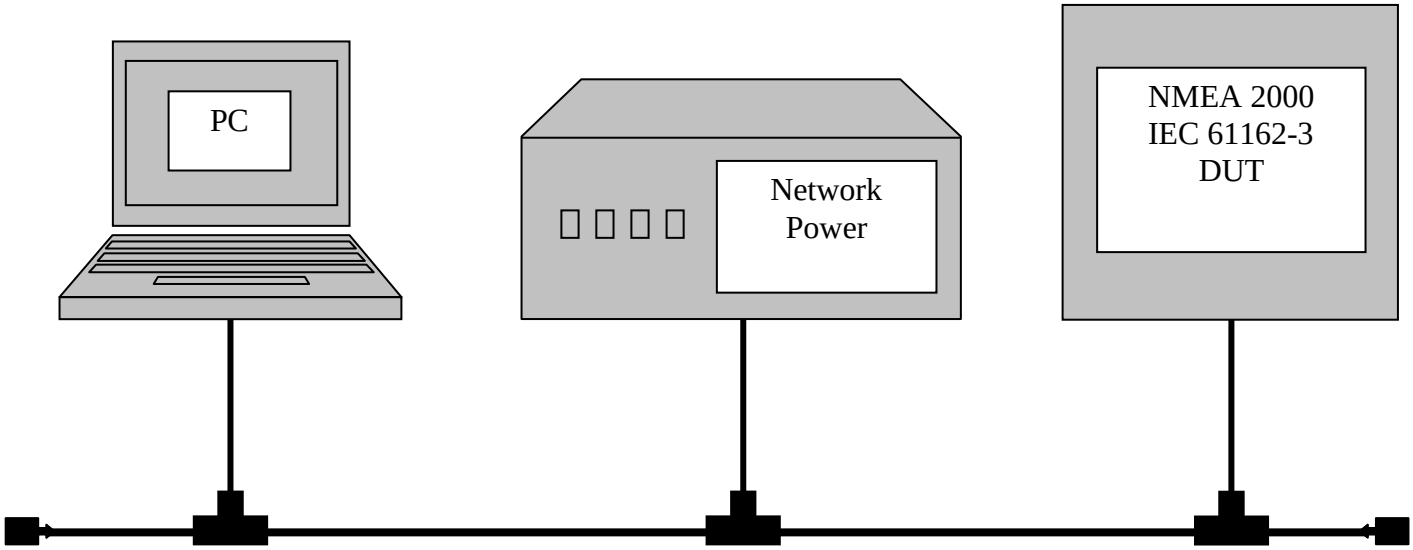
Explain how this requirement has been satisfied:

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C.3 Class 1 Level A - Automated Test Methods

This section contains the requirements and methods of measurement that can be performed automatically via a suite of software and hardware designed to certify compliance with the Level A minimum requirements of the NMEA 2000 standard for a Class 1 device.

To perform the automated tests, you must assemble the network as shown below:



The first part of performing the automated test methods is to determine what PGNs are provided by the DUT. This determination includes both NMEA 2000 defined PGNs and Manufacturers Proprietary PGNs. The Tool will do the following:

1. It should prompt the user to identify ALL (including proprietary) the PGNs (enter the PGNs) that are generated by the DUT.
2. It can query the DUT for each and every NMEA 2000 defined PGN and Proprietary PGN (including all PGN numbers slated for use by NMEA 2000 but not yet assigned) with the ISO request.

The responses to this extensive interrogation provide several important things. This adds some extensibility to the Certification Tool as the PGN data base changes or grows, in between updates to the Tool, the Tool can now create a list of all PGNS supported by the device, the default priority of provided PGNs can be checked, and the consistency between user's notification of what PGNs are supported versus actual responses to ISO Request can be made. Further, any fast packet PGNs that are identified as being provided by the DUT, other than Product Information, can be used to conduct additional tests. The results of this automated PGN determination shall be provided in a table or tabular form with the certification test results.

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C.3.1 Message Concurrency [3.1.2] – Mandatory

This section certifies the requirement that "*The minimum NMEA 2000 requirement is to support the concurrent reception of the following messages:*
One single frame message
One fast-packet message
One multi-packet Broadcast message
One multi-packet RTS/CTS session"
This requires four messages to be sent to the DUT concurrently, meaning that the individual message frames from each message will be interleaved in some manner with those of the other three messages. The DUT will need to handle (receive, process, and respond if required) all four messages properly. The four methods that follow must all be synchronized so that the four messages actually transit on the NMEA 2000 network at the same time.

C.3.1.1 Method of measurement (multi-packet RTS/CTS session):

This method involves the first concurrent message to be sent to the DUT. This message should be the longest message with respect to time, and should be complete after the other three concurrent messages. This message is sent using the ISO 11783 multi-packet RTS/CTS session. In order for this message to take the longest time, and to require many frames, the Request Group Function, requesting the NAME (PGN 60928), is chosen as the first message.

Send an ISO Request PGN 059904 requesting the ISO Address Claim PGN 060928. Record the values of the fields returned in PGN 060928 that, according to the PGN 060928 definition, can be requested. Next, generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 60928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 7
Field 6 - Field number of first parameter = 2 (field 2 from 060928)
Field 7 - Value of first parameter = value of field 2, Manufacturer code
Field 8 - Field number of second parameter = 3 (field 3 from 060928)
Field 9 - Value of second parameter = value of field 3, ECU Instance
Field 10 - Field number of third parameter = 4 (field 4 from 060928)
Field 11 - Value of third parameter = value of field 4, Function Instance
Field 12 - Field number of fourth parameter = 5 (field 5 from 060928)
Field 13 - Value of fourth parameter = value of field 5, Function
Field 14 - Field number of fifth parameter = 7 (field 7 from 060928)
Field 15 - Value of fifth parameter = value of field 7, Device Class
Field 16 - Field number of sixth parameter = 8 (field 8 from 060928)
Field 17 -Value of sixth parameter = value of field 8, Device Class Instance
Field 18 - Field number of seventh parameter = 9 (field 9 from 060928)
Field 19 - Value of seventh parameter = value of field 9, Industry Group

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Initiate a connection mode transfer session with the DUT with the Request Group Function PGN created above. Begin transferring this PGN data, and proceed to 3.1.2, Method of Measurement (multi-packet Broadcast Message).

C.3.1.1.1 Required results:

A data transfer connection should be established with the DUT. Data should be transferred in accordance with the multi-packet RTS/CTS session protocol. The DUT should receive and accept this message. This transfer session should begin before the other three concurrent messages are started as defined in Methods 3.1.2, 3.1.3, and 3.1.4. This transfer session should finish after the other three concurrent messages have been completed. The DUT should receive and accept this message. The DUT should respond with the ISO Address Claim PGN 060928.

C.3.1.2 Method of measurement (multi-packet Broadcast Message):

This method involves the second concurrent message to be sent to the DUT. This message should be the second longest message with respect to time, and should be complete before the first concurrent message, and after the third and fourth concurrent messages. This message is sent using the ISO 11783 multi-packet Broadcast Message protocol, requiring the use of the Broadcast Announce Message (BAM). This method will send a Request Group Function, PGN 126208, requesting the Transmitted PGN List Group Function, PGN 126464. Generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 0 (Transmit PGN List Message)

Begin transferring this PGN data, and proceed to 3.1.3, Method of Measurement (fast-packet).

C.3.1.2.1 Required results

Data should be transferred in accordance with the multi-packet Broadcast transfer protocol. This message transfer sequence should start after the first concurrent message has started (3.1.2), and should finish after the last two concurrent messages (3.1.3 and 3.1.4) have been completed. The DUT should receive and accept this message. The DUT should respond with the Transmitted List Group Function PGN 126464 using the ISO 11783 multi-packet Broadcast Message protocol.

C.3.1.3 Method of measurement (fast-packet):

This method involves the third concurrent message to be sent to the DUT. This message should be the third longest message with respect to time, and should start before and finish after the fourth concurrent message. This message is sent using the NMEA 2000 Fast-Packet protocol, and should be globally addressed. This method will send the Request Group Function PGN 126208, requesting the Received PGN List Group Function, PGN 126464. Generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)

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Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 1 (Receive PGN List Message)

Begin transferring this PGN data, and proceed to 3.1.4, Method of Measurement (single frame).

C.3.1.3.1 Required results

Data should be transferred in accordance with the NMEA 2000 Fast Packet protocol. This message transfer sequence should start after the second concurrent message has started (3.1.2), and should finish after the last concurrent message (3.1.4) has been completed. The DUT should receive and accept this message. The DUT should respond with the Received List Group Function PGN 126464.

C.3.1.4 Method of measurement (Single Frame):

This method involves the fourth concurrent message to be sent to the DUT. This message should be the shortest message with respect to time, and should start after the previous three concurrent messages have started. This message is sent using the ISO Single Frame Transmission protocol, as an addressed message. This method will send an ISO Request PGN 059904, requesting the Product Information PGN 126996. Send this PGN data.

C.3.1.4.1 Required results

This should be the last concurrent PGN to be sent, and should be the first to finish. The other concurrent PGNS should come to completion in the reverse order they were started, i.e. method 3.1.3 followed by method 3.1.2, followed by the completion of method 3.1.1.
The DUT should receive and accept this concurrent addressed single frame message. The DUT should respond with the Product Information PGN 126996.

C.3.1.5 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.1 passed or failed. Provide the Certification Test major section number (such as 3.1), and each Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement, and the results obtained by each method. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.1.5.1 Example text:

The following text shows x different results. The first example shows the result when the test, consisting of four methods, is passed. The second example depicts the results showing that the DUT could not handle concurrent message reception. Many other types of results may be possible and should be similarly documented.

Test 3.1 Message Concurrency [3.1.2] - PASSED
Method 3.1.1: PGN 059904 sent
PGN 060928 received
PGN 126208 sent
PGN 060928 received
Method 3.1.2: PGN 126208 sent
PGN 126464 received

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Method 3.1.3: PGN 126208 sent
PGN 126464 received
Method 3.1.4: PGN 059904 sent
PGN 126996 received

Or

Test 3.1 Message Concurrency [3.1.2] - FAILED*
Method 3.1.1: PGN 059904 sent
PGN 060928 received
PGN 126208 sent
PGN 060928 received
Method 3.1.2: PGN 126208 sent
PGN 126464 NOT received*
Method 3.1.3: PGN 126208 sent
PGN 126464 NOT received*
Method 3.1.4: PGN 059904 sent
PGN 126996 NOT received*

C.3.2 Fast Packet Sequence Counter [3.1.6.3] – Mandatory

This section certifies the requirement that "*The modulo 8 sequence counter shall increment each time the same PGN is transmitted from the same source address*." This is accomplished with two specific tests. The purpose of the first test (3.2.1) is to verify proper sequence number behavior with a mandatory PGN. The second test (3.2.2) examines the same requirement with all non-mandatory PGNs.

C.3.2.1 Method of measurement (Mandatory PGN):

Send an ISO request for Product Information, PGN 126996. Record the sequence number value. Send eight more requests for PGN 126996, and verify that the sequence number is incremented by one for each successive PGN and that the sequence number properly wraps around from 7 to 0.

C.3.2.1.1 Required results:

The sequence counter should be seen to start at some value between 0 and 7, and should increment up to 7 and then should start back at 0.

C.3.2.2 Method of measurement (Non-Mandatory PGNs):

Perform the same test described in 3.2.1 above for every non-mandatory PGN identified during the automated PGN determination process. This will verify that other PGNs, not just the mandatory PGNs, are implemented properly, with respect to the sequence number. If the Device Under Test (DUT) does not transmit any Fast Packet PGNs, then perform this test by requesting successive Transmitted/Received PGN List Group Function, PGN 126464, Received Group Function.

C.3.2.2.1 Required results:

For every non-mandatory PGN supported by the DUT, the sequence counter should be seen to start at some value between 0 and 7, and should increment up to 7 and then should start back at 0.

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C.3.2.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.2 passed or failed. Provide the Certification Test major section number (such as 3.2), and each Method of measurement section number, with the PGNs that were utilized in each Method of measurement, and the values of the sequence numbers recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.2.3.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of two methods, is passed. The second example depicts the results showing that the non-mandatory PGN measurements have failed. Notice that the sequence numbers were not incremented, but fixed at zero. Other results may be possible and should be similarly documented.

Test 3.2 Fast Packet Sequence Counter [3.1.6.3] - PASSED
Method 3.2.1: PGN 126996 2,3,4,5,6,7,0,1,2
Method 3.2.2: PGN 127493 0,1,2,3,4,5,6,7,0
PGN126998 5,6,7,0,1,2,3,4,5

Or

Test 3.2 Fast Packet Sequence Counter [3.1.6.3] - FAILED*
Method 3.2.1: PGN 126996 2,3,4,5,6,7,0,1,2
Method 3.2.2: PGN 127493 0,0,0,0,0,0,0,0,0*
PGN 126998 5,6,7,0,1,2,3,4,5

C.3.3 Unique Fast Packet Sequence Counter [3.1.6.4] – Mandatory

This section certifies the requirement that "*A separate sequence counter shall be utilized for each different PGN transmitted from the same source address.*" The purpose of this test is to ensure there is no linkage of the sequence number between Fast Packet PGNs.

C.3.3.1 Method of measurement (Multiple PGNs):

This test is performed with every non-mandatory PGN identified during the automated PGN determination process. If no non-mandatory PGNs are provided by the DUT, use PGN 126464 Transmit List Group Function as the “non-mandatory” PGN. Send one or more requests for PGN 126996, and one or more requests for each non-mandatory PGN, until both have different sequence numbers. Then, send alternate requests, one for PGN 126996 and one for the non-mandatory PGN, eight times. This will verify that the behavior is proper for all supported PGNs given the interleaved requests.

C.3.3.1.1 Required results:

The sequence number behavior should be independent of PGNs, that is, the series of sequence numbers from one PGN should have no relationship to those of another PGN.

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C.3.3.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.3 passed or failed. Provide the Certification Test major section number (such as 3.3), and each Method of measurement section number, with the PGNs that were utilized in each Method of measurement, and the values of the sequence numbers recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.3.2.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the results showing that the Multiple PGN measurements have failed. Notice that the sequence numbers were not incremented, but fixed at zero. The third example shows that the sequence numbers were not unique across multiple PGNs. Other results may be possible and should be similarly documented.

Test 3.3 Unique Fast Packet Sequence Counter [3.1.6.4] - PASSED
Method 3.3.1: PGN 126996 3,4,5,6,7,0,1,2,3
PGN 126464 1,2,3,4,5,6,7,0,1

Or

Test 3.3 Unique Fast Packet Sequence Counter [3.1.6.4] - FAILED*
Method 3.3.1: PGN 126996 3,4,5,6,7,0,1,2,3
PGN 126464 0,0,0,0,0,0,0,0,0*

Or

Test 3.3 Unique Fast Packet Sequence Counter [3.1.6.4] - FAILED*
Method 3.3.1: PGN 126996 4,6,0,2,4,6,0,2,4*
PGN 126464 5,7,1,3,5,7,1,3,5*

C.3.4 Successive Fast Packet Message Delay [3.1.6.7] – Mandatory

This section is intentionally blank due to requirement removal.

C.3.5 Fast Packet Message Time Out [3.1.6.8] - Mandatory

This section certifies the requirement that "*A fast-packet message shall not be considered as incomplete until a time-out of 750 ms from the last received frame occurs*." The purpose of this test is to verify that the DUT will function properly on a heavily loaded network, i.e. that it does not drop messages prematurely, when dealing with significant yet acceptable delays between a given message's frames. Though this test is conducted with only a single fast packet message, the reality is that this situation will occur in cases where devices may need to handle multiple concurrent fast packet transmissions on the network.

C.3.5.1 Method of measurement (Fast Packet Message Time Out):

This test requires the ability of the test hardware and software to be able to delay or not complete sending all the frames of a message for specified period of time. To conduct this test, a Request Group Function PGN 126208, requesting the Received PGN List Group Function PGN 126464, should be generated. Some portion

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of the request PGN should be sent, and then a delay of 745ms should occur, followed by the remaining frames of the request PGN.

C.3.5.1.1 Required results:

The DUT, upon receiving the completed request, should provide the Received PGN List Group Function PGN 126464.

C.3.5.2 Reporting requirements:

Provide detailed information as to what part (Method of measurement) of test 3.5 passed or failed. Provide the Certification Test major section number (such as 3.5), and the Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.5.2.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the results showing that the delay between the Transmit List PGN and the Receive List PGN were less than 10 frames. Other results may be possible and should be similarly documented.

Test 3.5 Fast Packet Message Time Out [3.1.6.8] - Mandatory - PASSED
Method 3.5.1: PGN 126208 requesting PGN 126464 sent
PGN 126464 Received PGN List received

Or

Test 3.5 Fast Packet Message Time Out [3.1.6.8] - Mandatory - FAILED*
Method 3.5.1: PGN 126208 requesting PGN 126464 sent
PGN 126464 Received PGN List NOT received*

C.3.6 Complex Request Message Support [3.2.1.1] – Mandatory

This section certifies the requirement that " *Support for the Complex Request Group Function message shall be implemented based on device certification level, in accordance with the minimum message implementation defined in Section 9.0.*". The purpose of this test is to verify that the DUT supports the Complex Request Message, PGN 126208. Though this group function also provides expanded acknowledgment and command capability, the scope of this particular test is limited to the Request function.

C.3.6.1 Method of measurement (Complex Request Message Support):

To conduct this test, a Request Group Function PGN 126208, requesting the Received PGN List Group Function PGN 126464, should be generated and sent to the DUT.

C.3.6.1.1 Required results:

The DUT should accept and process this request, and provide the Received PGN List Group Function PGN 126464. The PGNs listed in the response should include PGN 126208 as one of the PGNs that the DUT is capable of receiving.

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C.3.6.2 Reporting requirements:

Provide detailed information as to what part (Method of measurement) of test 3.6 passed or failed. Provide the Certification Test major section number (such as 3.6), and the Method of measurement section number, with the PGN number(s) that were utilized. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.6.2.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the case where the Transmit Group Function was provided versus the Received Group Function. The third example shows that though the Received Group Function PGN was provided, it did not list the Received Group Function PGN 126208 as a supported PGN in the PGN list. Other results may be possible and should be similarly documented.

Test 3.6 Complex Request Message Support [3.2.1.1] - Mandatory - PASSED
Method 3.6.1: PGN 126208 requesting PGN 126464 sent
PGN 126464 Received PGN List received
PGN 126464 contains the 126208 PG Number.

Or

Test 3.6 Complex Request Message Support [3.2.1.1] - Mandatory - FAILED*
Method 3.6.1: PGN 126208 requesting PGN 126464 sent
PGN 126464 Transmitted PGN List received*
PGN 126464 contains the 126208 PG Number.

Or

Test 3.6 Complex Request Message Support [3.2.1.1] - Mandatory - FAILED*
Method 3.6.1: PGN 126208 requesting PGN 126464 sent
PGN 126464 Received PGN List received
PGN 126464 missing the 126208 PG Number*

C.3.7 Acknowledgement of Complex Request Message [3.2.1.4] - Mandatory

This section certifies the requirement that "*All devices shall be capable of acknowledging a Complex Request message addressed to it. The acknowledgment is the data requested or the Expanded Acknowledgment message containing the appropriate error codes.*" The purpose of this test is to verify that the DUT provides the correct response to receiving a Complex Request PGN 126208. This is accomplished with five specific tests. The purpose of the first test (3.7.1) is to verify proper response when a mandatory PGN is requested. The second test (3.7.2) is designed to verify that a request for a mandatory PGN that cannot be met due to non-matching values of request fields, is properly acknowledged. The third (3.7.3) and fourth (3.7.4) tests are conducted in the same manner as the first two tests, but with all supported non-mandatory PGNs. The fifth test (3.7.5) observes the result when a request is made for a PGN not supported by the DUT.

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C.3.7.1 Method of measurement (Mandatory PGN Request):

To conduct this test, a Request Group Function PGN 126208, requesting the NAME contained in PGN 060928, should be generated and sent as an addressed PGN to the DUT. First, send an ISO Request PGN 059904 requesting the ISO Address Claim PGN 060928. Record the value of field two (Manufacturer code) returned in PGN 060928 that, according to the PGN 060928 definition, can be requested. Next, generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)

Field 2 - Requested PGN = 60928

Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)

Field 4 - Transmission interval offset = 0 (0000_{16})

Field 5 - Number of Pairs of Request Parameters to follow = 1

Field 6 - Field number of first parameter = 2 (field 2 from 060928)

Field 7 - Value of first parameter = value of field 2, Manufacturer code

C.3.7.1.1 Required results:

The DUT should respond to this test with two transmissions of the ISO Address Claim PGN 060928. The first response is to an addressed ISO request. The second response is to an addressed request containing a specific Manufacturer code parameter.

C.3.7.2 Method of measurement (Mandatory PGN Acknowledgment):

To conduct this test, a Request Group Function PGN 126208, requesting the NAME contained in PGN 060928, should be generated and sent as an addressed PGN to the DUT. First, send an ISO Request PGN 059904 requesting the ISO Address Claim PGN 060928. Record the value of field two (manufacturer code) returned in PGN 060928 that, according to the PGN 060928 definition, can be requested. Next, generate a Request Group Function PGN 126208, with the destination field set to the network address of the DUT, and with the data fields set as follows. Note that the value of the manufacturer code returned in PGN 060928 is different than what is being requested by PGN 126208 below.

Field 1 - Complex Request Group Function Code = 0 (Request group function)

Field 2 - Requested PGN = 60928

Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)

Field 4 - Transmission interval offset = 0 (0000_{16})

Field 5 - Number of Pairs of Request Parameters to follow = 1

Field 6 - Field number of first parameter = 2 (field 2 from 060928)

Field 7 - Value of first parameter = value of field 2 (Manufacturer code) minus 1

C.3.7.2.1 Required results:

The DUT should respond to this test with the ISO Address Claim PGN 060928, followed by the Acknowledgment Group Function PGN 126208, with the field 6 (First parameter error code) set to a value of 3 meaning "Request or command parameter field out-of-range". This indicates that the request group function was properly processed.

C.3.7.3 Method of measurement (Non-Mandatory PGN Request):

To conduct this test, a Request Group Function PGN 126208 requesting each non-mandatory PGN supported by the DUT should be generated with the destination field set to the network address of the DUT, and as shown below.

Field 1 - Complex Request Group Function Code = 0 (Request group function)

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Field 2 - Requested PGN = selected non-mandatory PG Number
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.3.7.3.1 Required results:

The required response to this test is either the non-mandatory PGN requested or the Acknowledgment PGN 126208 with the PGN error code value of 2 (PGN temporarily not available), or 3 (Access denied), or 4 (Request or Command not supported) meaning requests for this PGN are not accepted.

C.3.7.4 Method of measurement (Non-Mandatory PGN Acknowledgment):

This test is conducted using the same PGN as in test 3.7.3 above, but in a different manner. A transmission interval will be requested that will likely not be supported by the DUT. Generate a Request Group Function PGN 126208 with the destination field set to the network address of the DUT, and requesting the chosen non-mandatory PGN from test 3.7.3 but with the fields set to the following values.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = non-mandatory PGN from test 3.7.3
Field 3 - Transmission Interval = 1 (This is a value of .001 seconds or 1000Hz)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.3.7.4.1 Required results:

The DUT should respond with an acknowledgment group function PGN 126208 , with a PGN error code of 0 and a Transmission interval error code of 1, 2, or 3, or with a PGN error code of 2, 3, or 4.

A PGN error code value of 0 indicates no error with the PGN number requested. A value of 3 indicates Access denied, meaning the PGN cannot be requested.

A Transmission interval error code value of 1 indicates that the specific transmit interval value requested is not supported. A value of 2 indicates that the specific transmit interval requested is less than the measurement/calculation interval of the data (PGN) requested. A value of 3 indicates that the requested PGN does not support any transmit interval assignments or changes over the network.

C.3.7.5 Method of measurement (Unknown PGN Requested):

To conduct this test, a Request Group Function PGN 126208 requesting a non-supported PGN should be generated with the destination field set to the network address of the DUT, and sent to the DUT. Perform this test by selecting a PGN value of 130537.

C.3.7.5.1 Required results:

The DUT should respond with an acknowledgment group function PGN 126208, with a PGN error code value of 1 indicating that the PGN requested is not supported.

C.3.7.6 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.7 passed or failed. Provide the Certification Test major section number (such as 3.7). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure,

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such as the various error code values returned in the acknowledgment PGN 126208. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.7.6.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of five methods, is passed. The second example depicts the case where methods 3.7.1, 3.7.2, 3.7.3, and 3.7.5 were not completely successful. Method 3.7.1 shows that the DUT did not recognize or utilize the parameter provided with the request. Method 3.7.2 should have resulted in an acknowledgment with an error code, but instead the DUT provided the PGN. The DUT in method 3.7.3 incorrectly indicated that the PGN requested is not supported, and in method 3.7.5, the DUT also provided an incorrect error code. Other results may be possible and should be similarly documented.

Test 3.7 Acknowledgement of Complex Request Message [3.2.1.4] - Mandatory - PASSED
Method 3.7.1: PGN 059904 requesting PGN 060928 sent
PGN 060928 received
PGN 126208 request sent with Manufacturer code ###
PGN 060928 received
Method 3.7.2: PGN 059904 requesting PGN 060928 sent
PGN 060928 received
PGN 126208 request sent with Manufacturer code ###
PGN 126208 acknowledgment received
First parameter error code = 3, Request Parameter out of range
Method 3.7.3: PGN 126208 requesting PGN ###### sent
PGN ###### received
Method 3.7.4: PGN 126208 requesting PGN ###### with a transmission
interval of 1 (.001 seconds or 1000 hz) sent
PGN 126206 acknowledgment received
PGN error code = 0 No error/acknowledge
Transmission interval error code = 2 Transmit interval
Less that the measurement/calculation interval
Method 3.7.5: PGN 126208 requesting PGN 130537 sent
PGN 126208 acknowledgment received
PGN error code = 1, PGN not supported

Or

Test 3.7 Acknowledgement of Complex Request Message [3.2.1.4] - Mandatory - FAILED*
Method 3.7.1: PGN 059904 requesting PGN 060928 sent
PGN 060928 received
PGN 126208 request sent with Manufacturer code ###
PGN 126208 acknowledgment received*
First parameter error code = 1, Invalid request parameter field*
Method 3.7.2: PGN 059904 requesting PGN 060928 sent
PGN 060928 received
PGN 126208 request sent with Manufacturer code ###

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PGN 060928 received*

Method 3.7.3: PGN 126208 requesting the Transmit List PGN 126464 sent
PGN 126464 Transmit List received
PGN 126208 requesting PGN ###### sent
PGN 126208 acknowledgment received
PGN error code = 1, PGN not supported*

Method 3.7.4: PGN 126208 requesting PGN ###### with a transmission interval of 1 (.001 seconds or 1000 hz) sent
PGN 126206 acknowledgment received
PGN error code = 0 No error/acknowledge
Transmission interval error code = 2 Transmit interval Less that the measurement/calculation interval

Method 3.7.5: PGN 126208 requesting PGN 130537 sent
PGN 126208 acknowledgment received*
PGN error code = 2, PGN temporarily not available*

C.3.8 Request Parameter Field Support [3.2.1.5] - Mandatory

This sections certifies the requirement that "*When a device provides mandatory or optional query support for a PGN containing "yes" for the Request Parameter fields (see Appendix A Section A-1.) it shall at a minimum support requests specified by those fields.*" The purpose of this test is to verify that the DUT responds correctly to requests for PGN's with mandatory query support and requestable parameters.

C.3.8.1 Method of measurement (Mandatory Query Support):

To conduct this test, a NMEA 2000 Request Group Function PGN 126208 requesting the Receive/Transmit PGN Group Function PGN 126464 is sent to the DUT. The request group function should contain the following information:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 of 126464)
Field 7 - Value of first parameter = 1 (Receive PGN list message)

C.3.8.1.1 Required Results:

The required result of this test is that the DUT responds with the Receive PGN Group Function containing the Receive PGN list message.

C.3.8.2 Reporting Requirements:

Provide detailed information as to what part (Methods of measurement) of test 3.8 passed or failed. Provide the Certification Test major section number (such as 3.8). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as the value of field 1 returned in PGN 126464. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

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C.3.8.2.1 Example text:

The following text shows two different results. The first example shows the result when the test is passed. The second example depicts the case where the test failed. The failed example shows that the transmit list was returned rather than the requested receive list. Other results may be possible and should be similarly documented.

Test 3.8 Request Parameter Field Support [3.2.1.5] - Mandatory - PASSED
Method 3.8.1: PGN 126208 sent requesting PGN 126464 with receive list value
PGN 126464 receive PGN list message received
PGN 126464 field 1 = 1

Or

Test 3.8 Request Parameter Field Support [3.2.1.5] - Mandatory - FAILED*
Method 3.8.1: PGN 126208 sent requesting PGN 126464 with receive list value
PGN 126464 transmit PGN List message received*
PGN 126464 field 1 = 0*

C.3.9 Request for Address Claim PGN 60928 [3.2.1.7] – Mandatory

This sections certifies the requirement that "*NMEA 2000 devices shall support a Global Complex Data Request message requesting the Address Claim PGN 60928 and reply with the Address Claim parameter group when the Parameter Request fields contain no information (null) or when the contents of these fields match those of the device for any of the following fields:*.

Industry Group
Device Class field
System Instance field
Function field
Device Instance (upper) field
Device Instance (lower) field
Manufacturer's Code field"

The purpose of this test is to verify that the DUT responds correctly to requests for PGN 060928 with any possible combination of requested parameters. This is accomplished with 5 specific tests. The purpose of the first test (3.9.1) is to verify proper responses when a global and destination specific ISO Request is made. The second test (3.9.2) verifies that a globally addressed and destination specific NMEA 2000 Complex Request with no parameters is properly handled. The third test (3.9.3) consists 63 Complex requests with ordered request parameters sent both globally and addressed. The fourth test (3.9.4) checks the DUTs ability to detect and respond properly to a mixture of valid and invalid request parameters. This test also contains 63 complex requests sent both globally and addressed. The fifth and final test confirms that the DUT can properly handle request parameters in any order. This test consists of 53 complex requests.

C.3.9.1 Method of measurement (ISO Requests):

To conduct this test, two different ISO Requests, PGN 059904, requesting the ISO Address Claim PGN 060928, are sent to the DUT. The first ISO Request is sent globally while the second is sent with the DUT's specific address. Save the NAME fields of the returned ISO Address Claim PGN for use in the remaining methods of measurement in section 3.9.

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C.3.9.1.1 Required results:

The required results of this test are the receipt of the Address Claim PGN 060928 for each of two ISO Requests.

C.3.9.2 Method of measurement (Complex Requests - no parameters):

To conduct this test two different NMEA 2000 Complex Requests, PGN 126208, containing no request parameters, are sent to the DUT. Each requests the DUT to return the ISO Address Claim PGN 060928. The first Complex request is sent globally while the second is sent with the DUT's specific address.

C.3.9.2.1 Required results:

The required results of this test are the receipt of the Address Claim PGN 060928 for each of two NMEA 2000 Complex Requests.

C.3.9.3 Method of measurement (Requests by ordered NAME fields):

This test is conducted with PGN 126208 complex request, by requesting the ISO Address Claim PGN 060928 with specific sets of request parameters. The test utilizes 63 valid combinations of request parameters. They will first be tested with the request being sent globally, and then with the request being sent to the DUT's specific address. The request group function shall contain the following information as specified below and will be sent globally addressed. After all sixty-three combinations have been tested globally; they will be repeated with the specific address of the DUT. The first four fields of every request shall contain the following values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 060928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})

The remaining fields shall be set according to the following table. Each row in the table identifies a specific combination of request parameters. Each row in the table as such, represents a single Request PGN. The first column contains the number of request parameters in a given request PGN. The "#" column contains the field number of the request parameter as ordered in PGN 060928. The "VALUE" column contains the value of the request parameter, as recorded from the DUT in section 3.9.1 above. Note that some padding is necessary after most parameters, as they do not end on byte boundaries. The "#" and "VALUE" columns repeat for each request parameter. Empty cells in the table indicate that that field is not used in the request PGN.

Shorthand notation is used to save space, rather than write out the name of each parameter in each cell of the table. The following single characters in the table are representative of the following fields:
A = Manufacturer's Code - (field #2 of 060928) - Requires a 5 bit pad field
B = Device Instance (lower) - (field #3 of 060928) - Requires a 5 bit pad field
C = Device Instance (upper) - (field #4 of 060928) - Requires a 3 bit pad field
D = Device Function - (field #5 of 060928) – Requires no pad field
E = Device Class - (field #7 of 060928) – Requires a 1 bit pad field
F = System Instance - (field #8 of 060928) – Requires a 4 bit pad field
G = Industry Group - (field #9 of 060928) – Requires a 5 bit pad field

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Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
1	1	2	A												
2	2	2	A	3	B										
3	3	2	A	3	B	4	C								
4	4	2	A	3	B	4	C	5	D						
5	5	2	A	3	B	4	C	5	D	7	E				
6	6	2	A	3	B	4	C	5	D	7	E	8	F		
7	7	2	A	3	B	4	C	5	D	7	E	8	F	9	G
8	2	2	A	4	C										
9	3	2	A	4	C	5	D								
10	4	2	A	4	C	5	D	7	E						
11	5	2	A	4	C	5	D	7	E	8	F				
12	6	2	A	4	C	5	D	7	E	8	F	9	G		
13	2	2	A	5	D										
14	3	2	A	5	D	7	E								
15	4	2	A	5	D	7	E	8	F						
16	5	2	A	5	D	7	E	8	F	9	G				
17	2	2	A	7	E										
18	3	2	A	7	E	8	F								
19	4	2	A	7	E	8	F	9	G						
20	2	2	A	8	F										
21	3	2	A	8	F	9	G								
22	2	2	A	9	G										
23	1	3	B												
24	2	3	B	4	C										
25	3	3	B	4	C	5	D								
26	4	3	B	4	C	5	D	7	E						
27	5	3	B	4	C	5	D	7	E	8	F				
28	6	3	B	4	C	5	D	7	E	8	F	9	G		
29	2	3	B	5	D										
30	3	3	B	5	D	7	E								
31	4	3	B	5	D	7	E	8	F						
32	5	3	B	5	D	7	E	8	F	9	G				
33	2	3	B	7	E										
34	3	3	B	7	E	8	F								
35	4	3	B	7	E	8	F	9	G						
36	2	3	B	8	F										
37	3	3	B	8	F	9	G								
38	2	3	B	9	G										
39	1	4	C												
40	2	4	C	5	D										

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Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
41	3	4	C	5	D	7	E								
42	4	4	C	5	D	7	E	8	F						
43	5	4	C	5	D	7	E	8	F	9	G				
44	2	4	C	7	E										
45	3	4	C	7	E	8	F								
46	4	4	C	7	E	8	F	9	G						
47	2	4	C	8	F										
48	3	4	C	8	F	9	G								
49	2	4	C	9	G										
50	1	5	D												
51	2	5	D	7	E										
52	3	5	D	7	E	8	F								
53	4	5	D	7	E	8	F	9	G						
54	2	5	D	8	F										
55	3	5	D	8	F	9	G								
56	2	5	D	9	G										
57	1	7	E												
58	2	7	E	8	F										
59	3	7	E	8	F	9	G								
60	2	7	E	9	G										
61	1	8	F												
62	2	8	F	9	G										
63	1	9	G												

C.3.9.3.1 Required results:

The required results of this test are the receipt of 126 Address Claim PGN 060928 for the 63 global requests and for the 63 address specific requests.

C.3.9.4 Method of measurement (Requests with invalid values):

This test is conducted by requesting the ISO Address Claim PGN 060928 with specific sets of request parameters. The test utilizes 63 valid combinations of request parameters. They will first be tested with the request being sent globally, and then with the request being sent to the DUT's specific address. The request group function shall contain the following information as specified below and will be sent globally addressed. After all sixty-three combinations have been tested globally; they will be repeated with the specific address of the DUT. The first four fields of every request shall contain the following values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)

Field 2 - Requested PGN = 060928

Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)

Field 4 - Transmission interval offset = 0 (0000_{16})

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The remaining fields shall be set according to the following table. Each row in the table identifies a specific combination of request parameters. Each row in the table as such, represents a single Request PGN. The first column contains the number of request parameters in a given request PGN. The "#" column contains the field number of the request parameter as ordered in PGN 060928. The "VALUE" column contains the value of the request parameter, as recorded from the DUT in section 3.9.1 above, with a slight modification as shown below. Note that some padding is necessary after most parameters, as they do not end on byte boundaries. The "#" and "VALUE" columns repeat for each request parameter. Empty cells in the table indicate that that field is not used in the request PGN.

Shorthand notation is used to save space, rather than write out the name of each parameter in each cell of the table. The following single characters in the table are representative of the following fields:
A = Manufacturer's Code - (field #2 of 060928) - Requires a 5 bit pad field
B = Device Instance (lower) - (field #3 of 060928) - Requires a 5 bit pad field
C = Device Instance (upper) - (field #4 of 060928) - Requires a 3 bit pad field
D = Device Function - (field #5 of 060928) – Requires no pad field
E = Device Class - (field #7 of 060928) – Requires a 1 bit pad field
F = System Instance - (field #8 of 060928) – Requires a 4 bit pad field
G = Industry Group - (field #9 of 060928) – Requires a 5 bit pad field

Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
1	1	2	A+1												
2	2	2	A+1	3	B										
3	3	2	A	3	B+1	4	C								
4	4	2	A	3	B	4	C+1	5	D						
5	5	2	A	3	B	4	C	5	D+1	7	E				
6	6	2	A	3	B	4	C	5	D	7	E+1	8	F		
7	7	2	A	3	B	4	C	5	D	7	E	8	F+1	9	G
8	2	2	A+1	4	C										
9	3	2	A+1	4	C	5	D								
10	4	2	A	4	C	5	D+1	7	E						
11	5	2	A	4	C	5	D	7	E+1	8	F				
12	6	2	A	4	C	5	D	7	E	8	F+1	9	G		
13	2	2	A+1	5	D										
14	3	2	A	5	D	7	E+1								
15	4	2	A	5	D	7	E	8	F+1						
16	5	2	A	5	D	7	E	8	F	9	G+1				
17	2	2	A+1	7	E										
18	3	2	A	7	E	8	F+1								
19	4	2	A	7	E	8	F	9	G+1						
20	2	2	A+1	8	F										
21	3	2	A	8	F	9	G+1								

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Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
22	2	2	A+1	9	G										
23	1	3	B+1												
24	2	3	B+1	4	C										
25	3	3	B	4	C+1	5	D								
26	4	3	B	4	C	5	D+1	7	E						
27	5	3	B	4	C	5	D	7	E+1	8	F				
28	6	3	B	4	C	5	D	7	E	8	F+1	9	G		
29	2	3	B+1	5	D										
30	3	3	B	5	D+1	7	E								
31	4	3	B	5	D	7	E+1	8	F						
32	5	3	B	5	D	7	E	8	F+1	9	G				
33	2	3	B+1	7	E										
34	3	3	B	7	E+1	8	F								
35	4	3	B	7	E	8	F+1	9	G						
36	2	3	B+1	8	F										
37	3	3	B	8	F+1	9	G								
38	2	3	B+1	9	G										
39	1	4	C+1												
40	2	4	C+1	5	D										
41	3	4	C	5	D+1	7	E								
42	4	4	C	5	D	7	E+1	8	F						
43	5	4	C	5	D	7	E	8	F+1	9	G				
44	2	4	C+1	7	E										
45	3	4	C	7	E	8	F+1								
46	4	4	C	7	E	8	F	9	G+1						
47	2	4	C+1	8	F										
48	3	4	C	8	F+1	9	G								
49	2	4	C+1	9	G										
50	1	5	D+1												
51	2	5	D	7	E+1										
52	3	5	D	7	E	8	F+1								
53	4	5	D	7	E	8	F	9	G+1						
54	2	5	D+1	8	F										
55	3	5	D	8	F+1	9	G								
56	2	5	D	9	G+1										
57	1	7	E+1												
58	2	7	E+1	8	F										
59	3	7	E	8	F+1	9	G								
60	2	7	E+1	9	G										
61	1	8	F+1												

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Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
62	2	8	F	9	G+1										
63	1	9	G+1												

C.3.9.4.1 Required results:

The required results of this test are the receipt of 63 Acknowledge Group Functions, PGN 126208, in response to the destination-addressed requests. There should be no response to the globally addressed requests.

C.3.9.5 Method of measurement (Requests with random ordered fields):

This test is conducted by requesting the ISO Address Claim PGN 060928 with specific sets of request parameters in random order. The test utilizes 56 randomized combinations of request parameters. They will be tested with the request being sent to the DUT's specific address. The request group function shall contain the following information as specified below. The first four fields of every request shall contain the following values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 060928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})

The remaining fields shall consist of those identified in the table of section 3.9.3 that contain two or more request fields. Each row from the table shall have the order of the fields randomized. For instance, test 3 in the table may be ordered as B, A, C or C, B, A, or A, C, B. Any random order is valid as long as the result is properly measured against the random order used in the request. The first column contains the number of request parameters in a given request PGN. The "#" column contains the field number of the request parameter as ordered in PGN 060928. The "VALUE" column contains the value of the request parameter, as recorded from the DUT in section 3.9.1 above. Note that some padding is necessary after most parameters, as they do not end on byte boundaries. The "#" and "VALUE" columns repeat for each request parameter. Empty cells in the table indicate that that field is not used in the request PGN.

C.3.9.5.1 Required results:

The required results to this test are the reception of 56 Address Claim PGN 060928.

C.3.9.6 Reporting Requirements:

Provide detailed information as to what part (Methods of measurement) of test 3.9 passed or failed. Provide the Certification Test major section number (such as 3.9). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

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C.3.9.6.1 Example text:

The following text shows two different results. The first example shows the result when the test is passed. The second example depicts the case where the test failed. The failed example shows that the DUT improperly rejected complex requests containing request parameters that were not in the same order as defined in the ISO Address Claim PGN 060928. Though it is not shown here, the specific details of each complex request that failed shall be provided in the text file for use by the manufacturer and also in the Certification Results file for NMEA verification. Other results may be possible and should be similarly documented.

Test 3.9 Request for Address Claim PGN 60928 [3.2.1.7] - Mandatory - PASSED
Method 3.9.1: PGN 059904 sent requesting PGN 060928 Globally
PGN 060928 received
PGN 059904 sent requesting PGN 060928 Addressed
PGN 060928 received
Method 3.9.2: PGN 126208 sent requesting PGN 060928 Globally
PGN 060928 received
PGN 126208 sent requesting PGN 060928 Addressed
PGN 060928 received
Method 3.9.3: 63 variations of PGN 126208 sent globally requesting PGN 060928
63 PGN 060928 received
63 variations of PGN 126208 sent addressed requesting PGN 060928
63 PGN 060928 received
Method 3.9.4: 63 variations of PGN 126208 sent globally requesting PGN 060928
No response to global requests
63 variations of PGN 126208 sent addressed requesting PGN 060928
63 PGN 126208 Acknowledge Group functions returned indicating a parameter error code of 0x3
Method 3.9.5: 56 variations of PGN 126208 sent addressed requesting PGN 060928
56 PGN 060928 received

Or

Test 3.9 Request for Address Claim PGN 60928 [3.2.1.7] - Mandatory - FAILED*
Method 3.9.1: PGN 059904 sent requesting PGN 060928 Globally
PGN 060928 received
PGN 059904 sent requesting PGN 060928 Addressed
PGN 060928 received
Method 3.9.2: PGN 126208 sent requesting PGN 060928 Globally
PGN 060928 received
PGN 126208 sent requesting PGN 060928 Addressed
PGN 060928 received
Method 3.9.3: 63 variations of PGN 126208 sent globally requesting PGN 060928
63 PGN 060928 received
63 variations of PGN 126208 sent addressed requesting PGN 060928
63 PGN 060928 received
Method 3.9.4: 63 variations of PGN 126208 sent globally requesting PGN 060928
No response to global requests

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63 variations of PGN 126208 sent addressed requesting PGN 060928
63 PGN 126208 Acknowledge Group functions returned indicating a parameter error code of 0x3

Method 3.9.5: 56 variations of PGN 126208 sent addressed requesting PGN 060928
56 PGN 126208 Acknowledge Group functions returned indicating a parameter error code of 0x1*

C.3.10 Expanded Acknowledgment Group Function Command Response [3.2.2.4] - Mandatory

This test is designed to ensure the requirement that "*The Expanded Acknowledgment Group Function message shall be transmitted in response to each Command Message indicating acknowledgment or containing the appropriate error code.*" is met by the DUT. The purpose of this test is to verify that the DUT provides the correct response to receiving a Command Group Function PGN 126208. This is accomplished with five specific tests. The purpose of the first test (3.10.1) is to verify proper response when a mandatory PGN is commanded. The second test (3.10.2) is designed to verify that a command for a mandatory PGN that cannot be met is properly acknowledged. The third (3.10.3) test is conducted by commanding a mandatory PGN with an invalid value. The fourth test commands all supported non-mandatory PGNs to change their priority. The fifth test (3.10.5) observes the result when a command is made for a PGN not supported by the DUT. It is important to understand that the DUT does not have to accept these commands, but does have to respond with the acknowledgement PGN containing appropriate responses to the commands given.

C.3.10.1 Method of measurement (Mandatory PGN Commanded with no changes):

To conduct this test, a Command Group Function PGN 126208, commanding no fields of the NAME contained in PGN 060928, should be generated and sent as an addressed PGN to the DUT. Generate a Command Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 060928
Field 3 – Priority Setting = 8 (do not change priority)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 0

C.3.10.1.1 Required results:

The DUT should respond to this test with one transmission of the Acknowledgment Group Function PGN 126208. Valid PGN error codes are 0, 3, and 4. Valid Priority error codes are 0 and 3. Any other returned values to this test, or no response, would constitute a failure.

C.3.10.2 Method of measurement (Mandatory PGN Commanded with value):

To conduct this test, a Command Group Function PGN 126208, commanding the system instance field of the NAME contained in PGN 060928, should be generated and sent as an addressed PGN to the DUT. Generate a Command Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 060928
Field 3 – Priority Setting = 8 (do not change priority)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 1

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Field 6 - Field number of first parameter = 8 (field 8 from 060928)
Field 7 - Value of first command parameter = 1

If the Acknowledgement response indicates acceptance of the command a Request Group Function PGN 126208 requesting the Address Claim PGN 060928 should be made to verify the NAME with the modified System Instance field.

C.3.10.2.1 Required results:

The DUT should respond to this test with one transmission of the Acknowledgment Group Function PGN 126208. The commanded value may or may not be accepted, with the acknowledgment reporting the appropriate response. If accepted, the DUT will have a new NAME and must enter the Address Claim Process. Valid PGN error codes are 0, 3, and 4. Valid Parameter error codes are 0, 4, and 5. Any other returned values to this test, or no response, would constitute a failure.

C.3.10.3 Method of measurement (Mandatory PGN Commanded with bad value):

To conduct this test, a Command Group Function PGN 126208, commanding the reserved (ISO self-configuring) bit field of the NAME contained in PGN 060928, should be generated and sent as an addressed PGN to the DUT. Generate a Command Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 060928
Field 3 – Priority Setting = 8 (do not change priority)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 1
Field 6 - Field number of first parameter = 10 (field 10 from 060928)
Field 7 - Value of first command parameter = 0 (This value is not allowed)

C.3.10.3.1 Required results:

The DUT should respond to this test with one transmission of the Acknowledgment Group Function PGN with field 6 (First parameter error code) set to a value of 1, 4, or 5. This indicates that the Command group function was properly processed and a proper acknowledge was generated. A returned first parameter error code of 0 would not necessarily mean failure. This would indicate that the device is intended to work on other networks, and may be appropriate. A returned value of 0 requires manual validation as to the capabilities and planned markets for this device. The Acknowledge Group function's abbreviated version using a PGN error code of 4 is also an acceptable response.

C.3.10.4 Method of measurement (Non-Mandatory PGN Commanded):

To conduct this test, generate a Command Group Function PGN 126208 commanding each non-mandatory PGN as shown below.

Field 1 - Complex Request Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = PGN chosen from PGN 126464 response
Field 3 – Priority Setting = 7 (Set priority to 7)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 0.

If the value of the PGN and Priority error codes in the Acknowledgement response are both 0, than a request for this non-mandatory PGN should be made and the priority bits in the CAN frame header must be examined to verify the priority change.

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C.3.10.4.1 Required results:

The required response to this test is the Acknowledgment PGN 126208. The Acknowledgment should contain a PGN error code value of 0 (command for this PGN are accepted), 2 (temporarily not available), 3 (access denied), or 4 (request or command not supported) meaning no commands for this PGN are accepted. The Acknowledgment should contain a Priority error code value of 0 if the priority change is accepted, or a value of 1 or 3 if not accepted.

C.3.10.5 Method of measurement (Unknown PGN Commanded):

To conduct this test, a Command Group Function PGN 126208 requesting a non-supported PGN should be generated and sent addressed to the DUT. Perform this test by selecting a PGN value of 130537, and commanding the priority to “return to default”.

C.3.10.5.1 Required results:

The DUT should respond with an acknowledgment group function PGN 126208, with a PGN error code value of 1 indicating that the PGN requested is not supported.

C.3.10.6 Method of measurement (Global command to Unknown PGN Commanded):

To conduct this test, a Command Group Function PGN 126208 requesting a non-supported PGN should be generated and sent globally to the DUT. Perform this test by selecting a PGN value of 130537, and commanding the priority to “return to default”.

C.3.10.6.1 Required results:

The DUT should not respond since Command Group Functions are only to be sent addressed. The DUT should ignore this command.

C.3.10.7 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.10 passed or failed. Provide the Certification Test major section number (such as 3.10). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as the various error code values returned in the acknowledgment PGN 126208. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.10.7.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of six methods, is passed. The second example depicts the case where 3.10.5 is not successful. In method 3.10.5 no Acknowledgment is received within the 1-second time out period. Other results may be possible and should be similarly documented.

Test 3.10 Expanded Acknowledgment Group Function Command Response [3.2.2.4] Mandatory - PASSED
Method 3.10.1: PGN 126208 Commanding PGN 060928 sent
PGN 126208 Acknowledgement received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 3 (Value from field 4 of response)
Method 3.10.2: PGN 126208 Commanding PGN 060928 sent with System Instance value

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PGN 126208 acknowledgment received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Parameter error code = 4 (Value from field 6 of response)

Method 3.10.3: PGN 126208 Commanding PGN 060928 sent with bad self-configuring value
PGN 126208 acknowledgment received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Parameter error code = 5 (Value from field 6 of response)

Method 3.10.4:
PGN 126208 commanding priority of PGN ###### sent
PGN 126208 acknowledgment received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = (Value of field 3 of response)
Priority error code = 3 (Value from field 4 of response)

Method 3.10.5: PGN 126208 commanding priority of PGN 130537 sent
PGN 126208 acknowledgment received
PGN error code = 1, PGN not supported
PGN of message being acknowledged = 130537
Priority error code = 1 (Value from field 4 of response)

Method 3.10.6: PGN 126208 commanding priority of PGN 130537 sent
No PGN 126208 acknowledgment received

Or

Test 3.10 Expanded Acknowledgment Group Function Command Response [3.2.2.4] Mandatory – FAILED*

Method 3.10.1: PGN 126208 Commanding PGN 060928 sent
PGN 126208 Acknowledgement received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 3 (Value from field 4 of response)

Method 3.10.2: PGN 126208 Commanding PGN 060928 sent with System Instance value
PGN 126208 acknowledgment received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Parameter error code = 4 (Value from field 6 of response)

Method 3.10.3: PGN 126208 Commanding PGN 060928 sent with bad self-configuring value
PGN 126208 acknowledgment received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Parameter error code = 5 (Value from field 6 of response)

Method 3.10.4:

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PGN 126208 commanding priority of PGN ###### sent
PGN 126208 acknowledgment received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = (Value of field 3 of response)
Priority error code = 3 (Value from field 4 of response)

Method 3.10.5: PGN 126208 commanding priority of PGN 130537 sent
PGN 126208 acknowledgment not received*

Method 3.10.6: PGN 126208 commanding priority of PGN 130537 sent
No PGN 126208 acknowledgment received

C.3.11 Expanded Acknowledgment Group Function [3.2.3.1] - Mandatory

This validates the requirement that "*All NMEA 2000 devices shall support the Expanded Acknowledgment Group Function message*."

C.3.11.1 Method of measurement (Previous Tests):

This is accomplished by examining the results of two previous tests, test 3.7 and test 3.10.

C.3.11.1.1 Required results:

If both tests 3.7 and 3.10 are passed then this requirement is satisfied. If either tests 3.7 or 3.10 fail, then this test is also failed.

C.3.11.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.11 passed or failed. Provide the Certification Test major section number (such as 3.11). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.11.2.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of one method based upon previous tests, is passed. The second example depicts the case where one of the previous tests is not successful. Other results may be possible and should be similarly documented.

Test 3.11 Expanded Acknowledgment Group Function [3.2.3.1] – Mandatory – PASSED
Method 3.11.1 Test 3.7 Passed
Test 3.10 Passed

Or

Test 3.11 Expanded Acknowledgment Group Function [3.2.3.1] – Mandatory – FAILED*
Method 3.11.1 Test 3.7 Passed
Test 3.10 Failed*

C.3.12 Global Use Of Expanded Acknowledgment Message [3.2.3.3] - Mandatory

This section certifies the requirement that " *The Expanded Acknowledgment message shall not be used in response to Global requests.*" The purpose of this test is to verify that the DUT does not respond to global requests with the Acknowledgement Group Function PGN 126208. This is accomplished with three specific tests. The purpose of the first test (3.12.1) is to check behavior when a global request is made for a mandatory

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PGN that cannot be accommodated. The second test (3.12.2) examines the same requirement with all supported non-mandatory PGNs. The third test (3.12.3) requests a PGN not supported by the DUT.

C.3.12.1 Method of measurement (Mandatory PGN with invalid field request):

This method tests the ability of the DUT to properly discriminate between an addressed request that would require a response instead of the data requested, and global request of the same. A globally addressed Request Group Function PGN 126208 is generated and sent to the DUT. The fields of the PGN are as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 3 (Out of range value)

C.3.12.1.1 Required results:

The DUT should not be able to accommodate this request. The value of the parameter supplied does not match the range of field 1 in PGN 126464. Instead of sending the PGN requested, the DUT should ignore the request and send nothing in response. If no response is received from the DUT for a period of 2 seconds (1 second is the maximum timeout period), then the DUT has passed the test.

C.3.12.2 Method of measurement (Non-mandatory PGN with invalid transmission interval):

This method tests the ability of the DUT to properly discriminate between an addressed request that would require a response instead of the data requested, and global request of the same. To conduct this test, generate a globally addressed Request Group Function, PGN 126208, requesting each non-mandatory PGN as shown below.

The fields of the PGN are as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = selected non-mandatory PG Number
Field 3 - Transmission Interval = 4294967294 (According to the uint32 Type Definition, this value indicates "Out of range" and should not be used.)
Field 4 - Transmission interval offset = 0 (0000_{16} or no offset so transmit immediately)
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.3.12.2.1 Required results:

The DUT should not be able to accommodate this request. The value of the transmission interval is so large that it should not be accepted. Instead of sending the PGN requested, the DUT should ignore the request and send nothing in response. If no response is received from the DUT for a period of 2 seconds (1 second is the maximum timeout period), then the DUT has passed the test.

C.3.12.3 Method of measurement (Non-supported PGN):

This method tests the ability of the DUT to properly discriminate between an addressed request that would require a response instead of the data requested, and global request of the same. To conduct this test, generate a globally addressed Request Group Function, PGN 126208, requesting the non-supported PGN 130537 as shown below.

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The fields of the PGN are as follows:
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 130537
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.3.12.3.1 Required results:

The DUT should not be able to accommodate this request. The request for a non-supported PGN should not be accepted. Instead of sending the PGN requested, or acknowledging the request, the DUT should ignore the request and send nothing in response. If no response is received from the DUT for a period of 2 seconds (1 second is the maximum timeout period), then the DUT has passed the test.

C.3.12.4 Reporting Requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.12 passed or failed. Provide the Certification Test major section number (such as 3.12). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of an acknowledge Group function or some or the PGN requested. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.12.4.1 Example Text:

The following text shows two different results. The first example shows the result when the test, consisting of three methods is passed. The second example depicts the case where the second test (3.12.2) results in the DUT responding with an Acknowledgement Group Function containing an error code for requested transmission interval. Other results may be possible and should be similarly documented.

Test 3.12 Global Use Of Expanded Acknowledgment Message [3.2.3.3] - Mandatory – PASSED
Method 3.12.1: PGN 126208 Request Group Function requesting PGN 126464 with out of range request parameter sent globally addressed.
No response from DUT for a 2 second period.
Method 3.12.2: PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN with extreme transmission interval sent globally addressed.
No response from DUT for a 2 second period.
Method 3.12.3: PGN 126208 Request Group Function requesting PGN 130537 sent globally addressed.
No response from DUT for a 2 second period.

Or

Test 3.12 Global Use Of Expanded Acknowledgment Message [3.2.3.3] - Mandatory – FAILED*
Method 3.12.1: PGN 126208 Request Group Function requesting PGN 126464 with out of range request parameter sent globally addressed.
No response from DUT for a 2 second period.
Method 3.12.2: PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN with extreme transmission interval sent globally addressed.
PGN 126208 Acknowledgement received*

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PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = ######
Transmission Interval code = 1 (Value from field 4 of response)

Method 3.12.3: PGN 126208 Request Group Function requesting PGN 130537 sent globally addressed. No response from DUT for a 2 second period.

C.3.13 Expanded Acknowledgment Message Timing [3.2.3.4] - Mandatory

This section certifies the requirement that "*An Expanded Acknowledgment message, when required, shall be transmitted within 1 second of receiving the request or command.* " The purpose of this test is to measure the response time of the DUT when provided with destination addressed requests and commands, requiring an Acknowledgement Group Function PGN 126208 response. This is accomplished with four specific tests. The purpose of the first test (3.13.1) is to check behavior when a destination specific addressed request is made for a mandatory PGN that cannot be accommodated. The second test (3.13.2) examines the same requirement with every non-mandatory PGN supported by the DUT. The third test (3.13.3) requests a PGN not supported by the DUT. The fourth test (3.13.4) commands a mandatory PGN 060928 with an out of range parameter.

C.3.13.1 Method of measurement (Mandatory PGN with invalid field request):

This method tests the ability of the DUT to respond with in the maximum allowed time of 1 second to a request that requires a response instead of the data requested. A destination specific addressed Request Group Function PGN 126208 is generated and sent to the DUT. The fields of the PGN are as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 3 (Out of range value)

C.3.13.1.1 Required results:

The DUT should not be able to accommodate this request. The value of the parameter supplied does not match the range of field 1 in PGN 126464. Instead of sending the PGN requested the DUT should send an acknowledgement Group Function with a PGN Error code of 0 (No error/Acknowledge), Transmission Interval/Priority error code of 0 (No error/Acknowledge), and First Parameter error code of 3 (Request or Command parameter out-of-range). If a response is received from the DUT within a period of 1 second (1 second is the maximum timeout period) with the appropriate error codes, then the DUT has passed the test. An elapsed time from request to response greater than 1 second constitutes failure.

C.3.13.2 Method of measurement (Non-mandatory PGN with invalid transmission interval):

This method tests the ability of the DUT to respond within the maximum allowed time of 1 second to a request that requires a response instead of the data requested. To conduct this test, a destination specific addressed Request Group Function, PGN 126208 is generated, requesting each supported non-mandatory PGN as shown below.

The fields of the PGN are as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = selected non-mandatory PG Number

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Field 3 - Transmission Interval = 4294967294 ($FFFFFFFE_{16}$) (According to the uint32 Type Definition, this value indicates “Out of range” and should not be used.)
Field 4 - Transmission interval offset = 0 (0000_{16} or no offset so transmit immediately)
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.3.13.2.1 Required results:

The DUT should not be able to accommodate this request. The value of the transmission interval is so large that it should not be accepted. Instead of sending the PGN requested, the DUT should send an acknowledgement Group Function with the appropriate error codes. Acceptable error code combinations are PGN error code of 0 (No error), 2 (Temporarily not available) or 3 (Access denied) with a Transmission Interval/Priority error code of 1 (Transmit interval not supported), 2 (Transmit interval less than measurement/calculation interval) or 3 (Access denied). A PGN error code of 4 (Request not supported) with a Transmission Interval/Priority error code of 4 (Request not supported) is also acceptable. The Number of requested parameters in the response must be set to 0xFF (indicating no fields to follow). If a response is received from the DUT within a period of 1 second (1 second is the maximum timeout period), then the DUT has passed the test. An elapsed time from request to response greater than 1 second constitutes failure.

C.3.13.3 Method of measurement (Non-supported PGN):

This method tests the ability of the DUT to respond with in the maximum allowed time of 1 second to a request that requires a response instead of the data requested. To conduct this test, generate a destination specific addressed Request Group Function, PGN 126208, requesting the non-supported PGN 130537 as shown below.
The fields of the PGN are as follows:
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 130537
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.3.13.3.1 Required results:

The DUT should not be able to accommodate this request. The request for a non-supported PGN should not be accepted. Instead of sending the PGN requested, the DUT should send an acknowledgement Group Function with the PGN Error code of 1 (PGN not supported); Transmission Interval/Priority error code of 0 (No error), 1 (Transmit Interval not supported) or 3 (Access denied); and the Number of requested parameters set to 0xFF (indicating no fields to follow). If a response is received from the DUT within a period of 1 second (1 second is the maximum timeout period), then the DUT has passed the test. An elapsed time from request to response greater than 1 second constitutes failure.

C.3.13.4 Method of measurement (Command Mandatory PGN with out of range parameter):

This method tests the ability of the DUT to respond with in the maximum allowed time of 1 second to a command that requires a response instead of the data requested. To conduct this test, generate a destination specific addressed Command Group Function, PGN 126208, commanding the Reserved (Self-configuring bit) bit field in the NAME to 0 as shown below.
The fields of the PGN are as follows:
Field 1 - Complex Command Group Function Code = 1 (Command group function)

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Field 2 - Commanded PGN = 060928
Field 3 – Priority Setting = 8 (8_{16} meaning do not change)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 1
Field 6 – Field Number of First Parameter = 10 (Reserved – ISO Self-Configurable)
Field 7 – Value of First Parameter = 0 (out of range value).

C.3.13.4.1 Required results:

The DUT should not be able to accommodate this command. The command to change the reserved bit field of the NAME entity should not be accepted. The DUT should send an acknowledgement Group Function with the appropriate error codes. Acceptable values are a PGN Error code of 0 (No error) or 3 (Access denied), the Priority Error code set to 0 (No error) or 3 (Access denied), the Number of parameters set to 1, and the First Parameter Error code set to 1 (Invalid commanded parameter field), 4 (Access denied) or 5 (Command Group Function not supported). A PGN Error of 4 (Command not supported), a Priority Error of 4 (Command not supported) and the Number of Parameters set to 0xFF is also an acceptable response. If a response is received from the DUT within a period of 1 second (1 second is the maximum timeout period), then the DUT has passed the test. An elapsed time from request to response greater than 1 second constitutes failure.

C.3.13.5 Reporting Requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.13 passed or failed. Provide the Certification Test major section number (such as 3.13). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of an acknowledge Group function or some or the PGN requested. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.13.5.1 Example Text:

The following text shows two different results. The first example shows the result when the test, consisting of four methods is passed. The second example depicts the case where the second test (3.13.2) results in the DUT responding with an Acknowledgement Group Function containing an error code for requested transmission interval. Other results may be possible and should be similarly documented.

Test 3.13 Expanded Acknowledgment Message Timing [3.2.3.4] - Mandatory – PASSED
Method 3.13.1: PGN 126208 Request Group Function requesting PGN 126464 with out of range request parameter sent with the DUT's destination address.
PGN 126208 Acknowledgement received within 1 second.
PGN of message being acknowledged = 126464
PGN error code = 0 (Value from field 3 of response)
Transmission Interval/Priority error code = 0 (Value from field 4 of response)
Number of requested parameters = 1 (Value from field 5 of response)
Parameter error code = 3 (Value from field 6 of response)
Method 3.13.2: PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN with extreme transmission interval sent with the DUT's destination address.
PGN 126208 Acknowledgement received within 1 second.
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = ######

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Transmission Interval code = 1 (Value from field 4 of response)
Number of requested parameters = 0xFF (Value from field 5 of response)

Method 3.13.3: PGN 126208 Request Group Function requesting PGN 126464 Transmitted PGN List Group Function.
PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN with extreme transmission interval sent with the DUT's destination address.
PGN 126208 Acknowledgement received within 1 second.
PGN error code = 1 (Value from field 2 of response)
PGN of message being acknowledged = ######
Transmission Interval code = 1 (Value from field 4 of response)
Number of requested parameters = 0xFF (Value from field 5 of response)

Method 3.13.4: PGN 126208 Command Group Function commanding PGN 060928 sent.
PGN 126208 acknowledgment received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Number of commanded parameters = 1 (Value from field 5 of response)
Parameter error code = 1 (Value from field 6 of response)

Or

Test 3.13 Expanded Acknowledgment Message Timing [3.2.3.4] - Mandatory – FAILED*

Method 3.13.1: PGN 126208 Request Group Function requesting PGN 126464 with out of range request parameter sent with the DUT's destination address.
No response from DUT for a 2 second period.*

Method 3.13.2: PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN with extreme transmission interval sent with the DUT's destination address.
No response from DUT for a 2 second period.*

Method 3.13.3: PGN 126208 Request Group Function requesting PGN 130537 sent with the DUT's destination address.
No response from DUT for a 2 second period.*

Method 3.13.4: PGN 126208 Command Group Function commanding PGN 060928 sent.
PGN 126208 acknowledgment received within 1 second.
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Number of commanded parameters = 1 (Value from field 5 of response)
Parameter error code = 3 (Value from field 6 of response)*

C.3.14 ISO 11783-3 3.2.2 Reserved Bit ® [Table 4] – Mandatory

This test is intended to confirm that the DUT implements the reserved bit in bit-position five of the can frame. The requirement is this "*ISO 11783-3 Section 3.2.2 Reserved Bit, specifies that all messages shall set the reserved bit (bit number 5) in the CAN ID field to zero on transmit*." This will be accomplished with two tests. The first test (3.14.1) will check the 5th bit of the CAN frame from the transmitted PGN List function, and then from some non-mandatory PGN.

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C.3.14.1 Method of measurement (Reserved bit in Transmitted List PGN):

The purpose of this method is to verify that the reserved bit is properly set in CAN frames of mandatory messages. To conduct this test a Request PGN 126208 will be used to request PGN 126464, the Transmit list group function. The reserved bit, bit number 5 in the CAN frame will be examined, and the list of PGNS contained in the response will be used in method 3.14.2. The fields of the request PGN should be filled out as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 0 (Transmitted list))

C.3.14.1.1 Required results:

The DUT should respond to the request with PGN 126464 Transmit list group function. Bit 5 of the CAN frame must be set to zero. If bit 5 is not zero, the DUT fails this test.

C.3.14.2 Method of measurement (Reserved bit in non-mandatory PGN):

The purpose of this method is to verify that the reserved bit is properly set in CAN frames of all non-mandatory PGNs supported by the DUT. To conduct this test, generate a Request PGN 126208 asking for the immediate transmission of each supported non-mandatory PGN. The reserved bit, bit number 5 in the CAN frame will be examined. The fields of the request PGN should be filled out as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = ###### (random non-mandatory PGN)
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})

C.3.14.2.1 Required results:

The DUT should respond to the request with PGN ###### requested or with NACK. In either case, bit 5 of the CAN frame must be set to zero. If bit 5 is not zero, the DUT fails this test.

C.3.14.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.14 passed or failed. Provide the Certification Test major section number (such as 3.14). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs and the value of the reserved bit from bit position 5 in the CAN frame. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.14.3.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of two methods is passed. The second example depicts the case where the second test (3.14.2) fails because the value of the reserved bit is a one instead of a zero. Other results may be possible and should be similarly documented.

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Test 3.14 ISO 11783-3 3.2.2 Reserved Bit ® [Table 3-1] - Mandatory – PASSED
Method 3.14.1: PGN 126208 Request Group Function requesting PGN 126464 sent.
PGN 126464 Transmit PGN List received.
Reserved bit from bit position 5 of CAN frame = 0
Method 3.14.2: PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN sent
PGN ###### received.
Reserved bit from bit position 5 of CAN frame = 0

Or

Test 3.14 ISO 11783-3 3.2.2 Reserved Bit ® [Table 3-1] - Mandatory – FAILED*
Method 3.14.1: PGN 126208 Request Group Function requesting PGN 126464 sent.
PGN 126464 Transmit PGN List received.
Reserved bit from bit position 5 of CAN frame = 0
Method 3.14.2: PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN sent
PGN ###### received.
Reserved bit from bit position 5 of CAN frame = 1*

C.3.15 ISO 11783-3 Data field reserved bytes [Table 4] - Mandatory

This test is to certify that the "fill-bits" placed in a multi-packet protocol message are all binary ones. The requirement is as follows: "*ISO 11783-3 Section 3.10.4 Transport Protocol - Data Transfer Messages (PGN 60160) requires that unused Data field bytes in that last packet of the multi-packet message shall be filled with* $FF_{16.}$" This test is conducted by sending a ISO multi-packet formatted Request PGN 126208 to the DUT requesting PGN 126464.

C.3.15.1 Method of measurement (ISO Multi-Packet request):

This method requires that PGN 126208 is sent using the ISO 11783 multi-packet Broadcast Message protocol, requiring the use of the Broadcast Announce Message (BAM). This method will send an ISO 060160 PGN containing the Request Group Function, PGN 126208, requesting the Transmitted PGN List Group Function, PGN 126464. Generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 0 (Transmit PGN List Message)

C.3.15.1.1 Required results

The DUT should respond with PGN 126464 contained within ISO 060160 in accordance with the multi-packet Broadcast transfer protocol. The last packet needs to be examined, specifically, any non-data "fill-bits". The fill bits must be all binary ones. If the ISO 060160 PGN has no fill bits in the last packet, then this method should be repeated with a request for the receive list function instead.

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C.3.15.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.15 passed or failed. Provide the Certification Test major section number (such as 3.15). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs and the value of the "fill-bits" from the last packet of the ISO 060160 Multi-packet PGN. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.15.2.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of one method is passed. The second example depicts the case where the test fails because the values of the "fill-bits" are binary zeros instead of binary ones. Other results may be possible and should be similarly documented.

Test 3.15 ISO 11783-3 Data field reserved bytes [Table 3-1] - Mandatory – PASSED
Method 3.15.1: PGN 126208 Request Group Function requesting PGN 126464 sent utilizing ISO 060160 Multi-packet BAM protocol.
ISO Multi-packet PGN 060160 received containing PGN 126464 Transmit PGN List. Fill-bits in last multi-packet frame = all binary ones

Or

Test 3.15 ISO 11783-3 Data field reserved bytes [Table 3-1] - Mandatory – FAILED*
Method 3.15.1: PGN 126208 Request Group Function requesting PGN 126464 sent utilizing ISO 060160 Multi-packet BAM protocol.
ISO Multi-packet PGN 060160 received containing PGN 126464 Transmit PGN List. Fill-bits in last multi-packet frame = all binary zeros*

C.3.16 ISO 11783-3 Annex C, NACK [Table 3-1] – Mandatory

This test certifies that the ISO NACK is used as required. The requirement being tested is as follows: "*ISO 11783-3 Annex C, NACK - If the Parameter Group Number in a Command or Request is not recognized by the destination, it shall reply with the NACK message*." This is achieved with one test for proper NACK response to an ISO request.

C.3.16.1 Method of measurement (NACK - ISO Request):

This method is accomplished by sending a destination specific addressed ISO PGN 059904 Request message to the DUT. The ISO request will contain an invalid NMEA 2000 PGN number of 130537.

C.3.16.1.1 Required results:

The DUT, upon reception of the ISO Request PGN 059904 should send back an ISO 059392 Acknowledgement indicating a negative acknowledgement. Anything else or no response constitutes failure of this test.

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C.3.16.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.16 passed or failed. Provide the Certification Test major section number (such as 3.16). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs and the field values from the ISO Acknowledgment PGN. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.16.2.1 Example text:

The following text shows to different results. The first example shows what the text might look like if the test is passed. The second example illustrates one possibility of a failed test, where the acknowledgement contains the PGN of the ISO Request message instead of the requested PGN. Other results may be possible and should be similarly documented.

Test 3.16 ISO 11783-3 Annex C, NACK [Table 3-1] - Mandatory – PASSED
Method 3.16.1: PGN 059904 ISO Request requesting PGN 130537 sent.
PGN 059392 received.
Control Byte = 1
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537

Or

Test 3.16 ISO 11783-3 Annex C, NACK [Table 3-1] - Mandatory – FAILED*
Method 3.16.1: PGN 059904 ISO Request requesting PGN 130537 sent.
PGN 059392 received.
Control Byte = 1
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 059904*

C.3.17 NAME Support [8.3] – Mandatory

This test is designed to verify the following requirement: "*NMEA 2000 devices shall implement the NAME field, and field dependencies of the Address Claim message as defined by PGN 60928 in Appendix B database file "2000vx.x.mdb"*." Each field of the NAME entity will be examined to determine if they contain appropriate values. One test is used to conduct examination of the NAME fields.

C.3.17.1 Method of measurement (Examine NAME fields):

This method requests the address claim PGN 060928 containing the NAME fields from the DUT. Fields 1, 2, 5, and 7 of PGN 060928 are not tested but simply reported. These fields shall be reviewed and verified by NMEA prior to granting NMEA 2000 certification. The remaining fields (fields 3, 4, 6, 8, 9, 10) are examined and compared to the requirements to ensure proper values.

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C.3.17.1.1 Required results:

Upon receipt of PGN 060928, the NAME fields will be examined as follows. Field 6 should always be a zero, field 9 should be set to four, and field 10 should always be set to one. Other values will constitute failure.

C.3.17.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.17 passed or failed. Provide the Certification Test major section number (such as 3.17). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs and all the field values from the PGN 060928. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.17.2.1 Example text:

The text in this section shows two of many possible outcomes. The first example text shows what a valid set of parameters may look like. The second example shows that the DUT failed the test due to an incorrect Industry Group (field 9) value. Other results may be possible and should be similarly documented.

Test 3.17 NAME Support [8.3] - Mandatory – PASSED
Method 3.17.1: PGN 059904 ISO Request requesting PGN 060928 sent.
PGN 060928 received.
Field 1 (Unique number) = 5570 => Requires Verification
Field 2 (Manufacturer Code) = 1850 => Requires Verification
Field 3 (Device Instance Lower) = 0
Field 4 (Device Instance Upper) = 0
Field 5 (Device Function) = 150 => Requires Verification
Field 6 (Reserved) = 0
Field 7 (Device Class) = 60 => Requires Verification
Field 8 (System Instance) = 0
Field 9 (Industry Group) = 4
Field 10 (Reserved) = 1

Or

Test 3.17 NAME Support [8.3] - Mandatory – FAILED*
Method 3.17.1: PGN 059904 ISO Request requesting PGN 060928 sent.
PGN 060928 received.
Field 1 (Unique number) = 63953 => Requires Verification
Field 2 (Manufacturer Code) = 1851 => Requires Verification
Field 3 (Device Instance Lower) = 0 => Requires Verification
Field 4 (Device Instance Upper) = 0 => Requires Verification
Field 5 (Device Function) = 140 => Requires Verification
Field 6 (Reserved) = 0
Field 7 (Device Class) = 50 => Requires Verification
Field 8 (System Instance) = 0 => Requires Verification
Field 9 (Industry Group) = 1*
Field 10 (Reserved) = 1

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C.3.18 Product Information [8.3.2.1] – Mandatory

This test certifies the following requirement. "*All NMEA 2000 devices (except the Listen Only Device) shall support the Product Information message containing at a minimum the "NMEA 2000 Version Supported", "Manufacturer's Product Code", "NMEA 2000 Certification Level". and "Load Equivalency Number".*"

C.3.18.1 Method of measurement (Version field request parameter):

This method validates that the DUT can distinguish between valid and invalid requests for the Product Information PGN 126996 based upon the NMEA 2000 Version Supported request parameter. To conduct this test, an ISO request is sent requesting PGN 126996. The contents of the NMEA 2000 Database Version Number are checked and saved. Additionally, the Manufacturers Product Code, NMEA 2000 Certification Level, and Load Equivalency fields are recorded and checked. A Request PGN 126208 requesting PGN 126996 with an invalid Version number is then sent addressed to the DUT. An acknowledgement should be received from the DUT. Then a Request PGN requesting PGN 126996 with the NMEA version supported by the DUT is sent.

C.3.18.1.1 Required results:

The DUT should, upon reception of the ISO request, provide PGN 126996. Upon receipt of the request PGN 126208, the DUT should send back an acknowledgement indicating invalid request parameter. The DUT, upon reception of a second valid request for PGN 126996 should respond by providing PGN 126996.

C.3.18.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.18 passed or failed. Provide the Certification Test major section number (such as 3.18). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs and applicable field values from the PGNs 126208 and 126996. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.18.2.1 Example text:

The text in this section shows two of many possible outcomes. The first example text shows the results when the DUT passes the test. The second example shows that the DUT failed the test by not properly detecting a non-matching NMEA version request parameter value. Other results may be possible and should be similarly documented.

Test 3.18 Product Information [8.3.2.1] - Mandatory – PASSED
Method 3.18.1: PGN 059904 ISO Request requesting PGN 126996 sent.
PGN 126996 received.
Field 1 (NMEA 2000 Database Version) = 1000 => Requires Verification
Field 2 (NMEA Manufacturer's Prodcut Code = 127
Field 7 (Certification Level) = 0
Field 8 (Load Equivalency) = 1 => Requires Verification
PGN 126208 Request destination addressed containing the following parameters sent
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)

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Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = #### (NMEA 2000 Version + 1 from 126996 returned above)
PGN 126208 acknowledgment received
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 126996
Priority error code = 0 (Value from field 4 of response)
Parameter error code = 3 (Value from field 6 of response)
PGN 126208 Request destination addressed containing the following parameters sent
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = #### (NMEA 2000 Version from 126996 returned above)
PGN 126996 received.

Or

Test 3.18 Product Information [8.3.2.1] - Mandatory – FAILED*
Method 3.18.1: PGN 059904 ISO Request requesting PGN 126996 sent.
PGN 126996 received.
Field 1 (NMEA 2000 Database Version) = 1000 => Requires Verification
Field 2 (NMEA Manufacturer's Prodcut Code = 127
Field 7 (Certification Level) = 0
Field 8 (Load Equivalency) = 1 => Requires Verification
PGN 126208 Request destination addressed containing the following parameters sent
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = #### (NMEA 2000 Version + 1 from 126996 returned above)
PGN 126996 received*
PGN 126208 Request destination addressed containing the following parameters sent
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = #### (NMEA 2000 Version from 126996 returned above)

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PGN 126996 received.

C.3.19 Transmitted PGN List [8.4.1] - Mandatory

This section certifies the following requirement. "*All NMEA 2000 devices shall support the Transmitted PGN List message containing all transmitted messages supported by the device, including Data Link and Network Management messages. The Transmitted PGN List message shall be implemented as described in Appendix B.*"

C.3.19.1 Method of measurement (Transmitted PGN List):

This method requires that the DUT provide the Transmitted PGN list. The List is then examined to ensure that all the mandatory PGNs are included. Non-mandatory PGNs require manual verification. To conduct this test, a destination specific addressed request group function PGN 126208 is sent to the DUT asking for PGN 126464 Transmit PGN List.

C.3.19.1.1 Required results:

The DUT must respond to the request with the transmitted PGN list containing all the required PGNs. This includes those that are mandatory and all other PGNs that are produced by the DUT. The entire list of PGNs in Table 9-1 of the Standard, with the exception of PGN 059904, is required to be in this list.

C.3.19.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.19 passed or failed. Provide the Certification Test major section number (such as 3.19). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs and all the field values (PGNS) from PGN 126464. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.19.2.1 Example text:

The example text shows two possible outcomes. The first example shows results that pass the test. The second example shows a deficiency in the mandatory PGNs listed. Note that PGN 126464 is missing. Other results may be possible and should be similarly documented.

Test 3.19 Transmitted PGN List [8.3.1] – Mandatory – PASSED
Method 3.19.1: PGN 126208 Request for Transmitted PGN List PGN 126464 sent.
PGN 126464 Transmitted PGN List received.
Mandatory PGNS Listed: 126208, 059392, 060160, 060416, 060928, 129996, 126464
Non-Mandatory PGNs Listed: 126998, 129025, 129026, 129029, 129044, 129045, 129284 => Require Verification

Or

Test 3.19 Transmitted PGN List [8.3.1] – Mandatory – FAILED*
Method 3.19.1: PGN 126208 Request for Transmitted PGN List PGN 126464 sent.
PGN 126464 Transmitted PGN List received.
Mandatory PGNS Listed: 126208, 059392, 060160, 060416, 060928, 129996*

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Non-Mandatory PGNs Listed: 126998, 129025, 129026, 129029, 129044, 129045, 129284 => Require Verification

C.3.20 Received PGN List [8.4.2] - Mandatory

This section certifies the following requirement, "*All NMEA 2000 devices shall support the Received PGN List message containing all received messages supported by the device, including Data Link and Network Management messages.*
The Received PGN List message shall be implemented as described in Appendix B."

C.3.20.1 Method of measurement (Transmitted PGN List):

This method requires that the DUT provide the Received PGN list. The List is then examined to ensure that all the mandatory PGNs are included. Non-mandatory PGNs require manual verification. To conduct this test, a destination specific addressed request group function PGN 126208 is sent to the DUT asking for PGN 126464 Received PGN List.

C.3.20.1.1 Required results:

The DUT must respond to the request with the received PGN list containing all the required PGNs. This includes those that are mandatory and all other PGNs that are accepted and processed by the DUT. The entire list of PGNs in Table 9-1 of the Standard is required to be in this list, with the exception of PGNs 126464 and 126996.

C.3.20.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.20 passed or failed. Provide the Certification Test major section number (such as 3.20). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs and all the field values (PGNS) from PGN 126464. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.20.2.1 Example text:

The example text shows two possible outcomes. The first example shows results that pass the test. The second example shows a deficiency in the mandatory PGNs listed. Note that PGN 060160 and 060416 are missing. Other results may be possible and should be similarly documented.

Test 3.20 Received PGN List [8.3.2] - Mandatory – PASSED
Method 3.20.1: PGN 126208 Request for Received PGN List PGN 126464 sent.
PGN 126464 Received PGN List received.
Mandatory PGNS Listed: 126208, 059392, 059904, 060160, 060416, 060928
Non-Mandatory PGNs Listed: 129538 => Requires Verification

Or

Test 3.20 Received PGN List [8.3.2] - Mandatory – FAILED*
Method 3.20.1: PGN 126208 Request for Received PGN List PGN 126464 sent.
PGN 126464 Received PGN List received.
Mandatory PGNS Listed: 126208, 059392, 059904, 060928*

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Non-Mandatory PGNs Listed: 129538 => Requires Verification

C.3.21 Address Claim Test – Mandatory

This section certifies one of the most basic and important characteristics of the NMEA 2000 standard, Automatic self-configuring Address Claiming, at power up. There are several very important aspects to dynamic address claiming and assignment that must be adhered to in order to ensure interoperability with other NMEA 2000 devices. This requires multiple tests. Each of the following tests has one specific goal or criteria to be satisfied.

C.3.21.1 Method of measurement (Transmit Address Claim on Startup):

The specific item of interest is that the DUT, upon power up and normal (self test or initialization) delays, sends out an address claim message, PGN 060928. In order to conduct this test, the DUT will be powered up after the test equipment is ready to listen. The address claim PGN will be recorded, and the requested address will be documented.

C.3.21.1.1 Required results:

The DUT must provide the address claim PGN 060928 shortly after power up. The Address must be a valid NMEA 2000 address in the range of 0 to 251. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 constitutes failure of this test.

C.3.21.2 Method of measurement (Response to lower priority address contention):

This test verifies that the DUT properly handles address contention with devices that have a lower priority (higher value NAME means lower priority). After the DUT is powered up, and issues an address claim message, an address claim message is sent (broadcast by the test equipment) with a higher value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done within the 250ms period after the DUT sends its address claim message, so that contention is achieved before the DUT actually claims an address.

C.3.21.2.1 Required results:

The DUT should receive and process the lower priority address claim message. The DUT should respond to the contention by retransmitting the same address claim message again and then begin the 250ms waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT changes to some other address as a result of the contention, or does not respond to the contention, then it has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 constitutes failure of this test.

C.3.21.3 Method of measurement (Response to higher priority Address contention):

This test verifies that the DUT properly handles address contention with devices that have a higher priority (lower value NAME means higher priority). After the DUT is powered up, and issues an address claim message, an address claim message is sent (broadcast by the test equipment) with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done within the 250ms period after the DUT sends its address claim message, so that contention is achieved before the DUT actually claims an address.

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C.3.21.3.1 Required results:

The DUT should receive and process the higher priority address claim message. The DUT should respond to the contention by choosing a different address and transmitting a new address claim message, and then begin the 250ms waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT keeps the original contested address in its second address claim message, or does not respond to the contention, then it has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 also constitutes failure of this test.

C.3.21.4 Method of measurement (Response to no addresses available):

This test verifies two items. The first item is that the DUT attempts to claim a valid address across the full range of valid address space, from 0 to 251. Second, that the DUT indicates when it is unable to claim an address. The sequences of events to conduct this test are as follows. After the DUT is powered up, and issues an address claim message, an address claim message is sent (broadcast by the test equipment) with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done within the 250ms period after the DUT sends its address claim message, so that contention is achieved before the DUT actually claims an address. This should cause the DUT to try for another address. The test equipment should respond with another address claim with the same address as desired by the DUT, and with a lower value NAME than the DUT. This should cause the DUT to try for another address. This cycle should be repeated until the DUT sends a can not claim address message, or begins repeating address claim attempts with a previously attempted address.

C.3.21.4.1 Required results:

The DUT should, over the course of this test, send 252 address claim messages followed by a can not claim address message, which is an address claim message with the source address set to 254. The DUT will fail this test if it does not try all 252 addresses (order does not matter), begins repeating claims for the same address, or does not to send a can not claim address message.

C.3.21.5 Method of measurement (Address Claim Timeout):

This is intended to determine if the DUT waits for 250ms from sending it's own address claim message before assuming that it has the address. The DUT, upon sending its address claim message, should not respond to anything other than another address claim for the same address during the 250ms period. The test equipment should, upon receipt of the DUTs initial address claim, wait for 100ms and then send a request for address claim PGN globally. The test equipment should not see any response from the DUT until at least an additional 150ms has elapsed.

C.3.21.5.1 Required results:

The DUT should not transmit any PGNs for a period of 250ms after transmitting the address claim PGN. Any transmission from the DUT during the 250ms constitutes failure.

C.3.21.6 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.21 passed or failed. Provide the Certification Test major section number (such as 3.21). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as Source Address and NAME values. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

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C.3.21.6.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes all five methods. The second example shows a variety of problems. The first problem is that the address used in 3.21.1 is not valid. The problem in 3.21.2 is that the DUT changed addresses in response to a device with a lower priority, when it should not have. The problem in 3.21.3 is that the DUT did not change its address in response to a device with a higher priority. In 3.21.4 the DUT did not attempt to claim all possible addresses before sending the can not claim message. The last problem, shown in 3.21.5, is that the DUT responded to a "request for address claim message" before the 250ms period was complete. Other results may be possible and should be similarly documented.

Test 3.21 Address Claim Test – Mandatory – PASSED
Method 3.21.1: PGN 060928 received
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
Method 3.21.2: PGN 060928 received
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 sent
Address = 12
NAME = 1383963202651702067 6 ($C0104000AAAAA004_{16}$)
PGN 060928 received
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
Method 3.21.3: PGN 060928 received
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 sent
Address = 12
NAME = 1383963202651702067 4 ($C0104000AAAAA002_{16}$)
PGN 060928 received
Address = 13
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
Method 3.21.4: PGN 060928 received
252 Addresses tried.
PGN 060928 received with Address = 254
Method 3.21.5: PGN 060928 received
100 ms elapsed
PGN 059904 Requesting 060928 sent Globally
150 ms elapsed

Or

Test 3.21 Address Claim Test – Mandatory – FAILED*
Method 3.21.1: PGN 060928 received
Address = 252*

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NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
Method 3.21.2: PGN 060928 received
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
Address = 12
NAME = 1383963202651702067 6 (C0104000AAAAA004$_{16}$)
PGN 060928 received
Address = 13*
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
Method 3.21.3: PGN 060928 received
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
Address = 12
NAME = 1383963202651702067 4 (C0104000AAAAA002$_{16}$)
PGN 060928 received
Address = 12*
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
Method 3.21.4: PGN 060928 received
Addresses missed, 0,1,2,3,4,5,6,7,8,9,10,11*
PGN 060928 received with Address = 254
Method 3.21.5: PGN 060928 received
100 ms elapsed
PGN 059904 Requesting 060928 sent Globally
PGN 060928 received with Address = 254*

C.3.22 Minimum Message Implementation [Table 9-1] – Mandatory

This section is intended to certify, "*The messages listed in Table 9-1 are all required to be implemented*." Many previous test sections utilize these messages, both to and from the DUT, to conduct the certification. The verification of this requirement, when applicable, is based upon those other tests as described in the methods below.

C.3.22.1 Method of measurement (PGN 126208):

This method is dependant upon tests 3.6 and 3.7. If test 3.6 is passed, the DUT has implemented the Request Group Function PGN 126208. If test 3.7 is passed the DUT has implemented the Acknowledge Group Function PGN 126208. Implementation of the Command Group Function PGN 126208 is optional.

C.3.22.1.1 Required Results:

Check that tests 3.6 and 3.7 have been passed. Failure of either test constitutes failure of this method.

C.3.22.2 Method of measurement (PGN 059392):

This method is dependant upon test 3.16. If test 3.16 is passed, the DUT has implemented the ISO Acknowledgement PGN 059392.

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C.3.22.2.1 Required Results:

Check that test 3.16 has passed. Failure of test 3.16 constitutes failure of this method.

C.3.22.3 Method of measurement (PGN 059904):

There are no previous tests that rely solely upon this PGN, so an additional test is needed. The test equipment will send a PGN 059904 to the DUT, both destination and globally addressed, requesting the Address Claim PGN 060928. A proper response to this test will verify that the DUT has implemented the ISO Request.

C.3.22.3.1 Required Results:

The DUT should respond with two PGN 060928 messages, one for the destination specific, and one for the global ISO request 059904.

C.3.22.4 Method of measurement (BAM - PGNs 060416 and 060160):

This test verifies both reception and transmission of the ISO transport protocol using the broadcast announcement message and multi-packet data transfer. If successfully passed, it shows that the DUT can receive, process, and transmit data using PGNs 060416 and 060160 globally addressed. The test is conducted in the following manner. First, send a PGN 060416 BAM with the Request PGN 126208 to the DUT. Then send PGN 060160 containing the contents of PGN 126208, requesting the PGN 126464 Transmit PGN List. All communications shall be in accordance with the timing requirements of the transport protocol.

C.3.22.4.1 Required Results:

The DUT should receive and decode the 060416. It should then expect to receive the Request PGN 126208 via the Transport Protocol Data Transfer PGN 060160. The DUT should then respond with a Transport Protocol Broadcast Announcement Message, PGN 060416, indicating that PGN 126464 is to follow. The DUT should then send the content of PGN 126464, within the Transport Protocol Data Transfer PGN 060160. Failure to provide the data using the transport protocol or exceeding any of the timing requirements constitutes failure for this method.

C.3.22.5 Method of measurement (RTS/CTS - PGNs 060416 and 060160):

This test verifies both reception and transmission of the ISO transport protocol using the destination addressed RTS/CTS connection management and multi-packet data transfer capabilities. If successfully passed, it shows that the DUT can receive, process, and transmit data using PGNs 060416 and 060160 via RTS/CTS connection management with destination specific addresses. The test is conducted in the following manner. First, send a PGN 060416 RTS with the Request PGN 126208 to the DUT. Wait for the DUT to respond with the CTS message. Then begin sending the contents of PGN 126208 requesting PGN 126464 Received PGN List. The data transfer session should take place in accordance with ISO 11783-3 Annex B, Transport Protocol Transport Sequences. The DUT should process and respond to the request by providing PGN 126464 Received PGN List via the transport protocol using the RTS/CTS connection management. All communications shall be in accordance with the timing requirements of the transport protocol.

C.3.22.5.1 Required Results:

The DUT should accept and process the PGN 126208 request for PGN 126464 Received PGN List via the ISO Transport Protocol using RTS/CTS connection management. The DUT should then Initiate a RTS/CTS manages session with the test equipment, and transfer the contents of PGN 126464 in the established transport protocol session.

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Failure to provide the data using the transport protocol or exceeding any of the timing requirements constitutes failure for this method.

C.3.22.6 Method of measurement (PGN 060928):

This method is dependant upon tests 3.9 and 3.17. If tests 3.9 and 3.17 are passed, the DUT has implemented the ISO Address Claim PGN 060928 correctly.

C.3.22.6.1 Required Results:

Check that tests 3.9 and 3.17 have been passed. Failure of test 3.9 or 3.17 constitutes failure of this method.

C.3.22.7 Method of measurement (PGN 126996):

This method is dependant upon test 3.18. If test 3.18 is passed, the DUT has implemented the Product Information PGN 126996 with the NMEA 2000 Version request parameter correctly.

C.3.22.7.1 Required Results:

Check that test 3.18 has been passed. Failure of test 3.18 constitutes failure of this method.

C.3.22.8 Method of measurement (PGN 126464):

This method is dependant upon tests 3.19 and 3.20. If tests 3.19 and 3.20 are passed, the DUT has implemented the Transmit/Receive PGN List function, PGN 126464, correctly.

C.3.22.8.1 Required Results:

Check that tests 3.19 and 3.20 have been passed. Failure of test 3.19 or 3.20 constitutes failure of this method.

C.3.22.9 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.22 passed or failed. Provide the Certification Test major section number (such as 3.22). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. Contents of the connection management PGNs 060416 should be reported, along with all PGNs involved in the transport protocol sessions. The report is intended to provide a complete record of the data transfer sequences that take place during this test. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.22.9.1 Example text:

The text in this section is merely an outline of two of many possible outcomes. The first example text outlines the results when the DUT passes the test. The second example illustrates that the DUT failed the test by not properly responding to the Transport Protocol Broadcast Announce Message in test 3.22.4. Other results may be possible and should be similarly documented.

Test 3.22 Minimum Message Implementation [Table 9-1] – Mandatory – PASSED
Method 3.22.1: Method 3.6 Passed
Method 3.7 Passed
Method 3.22.2: Method 3.16 Passed
Method 3.22.3: PGN 059904 ISO Request requesting PGN 060928 sent destination specific.
PGN 060928 received.
PGN 059904 ISO Request requesting PGN 060928 sent globally.

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PGN 060928 received.
Method 3.22.4: PGN 060416 BAM sent
List contents of 060416….
PGNs 060160 sent….containing PGN 126208 request for PGN 126464
PGN 060416 received from DUT.
List and validate contents….
PGNs 060160 received…..containing PGN 126464 Transmit PGN List.
Method 3.22.5: PGN 060416 RTS sent … list contents
List PGNs sent and received.
List contents of PGNs 060416 sent
List PGNs received
List contents of PGNs 060416 received
PGNs 060160 received …. Containing PGN 126464 Received PGN List
Method 3.22.6: Method 3.9 Passed
Method 3.17 Passed
Method 3.22.7: Method 3.18 Passed
Method 3.22.8: Method 3.19 Passed
Method 3.20 Passed

Or

Test 3.22 Minimum Message Implementation [Table 9-1] – Mandatory – FAILED*
Method 3.22.1: Method 3.6 Passed
Method 3.7 Passed
Method 3.22.2: Method 3.16 Passed
Method 3.22.3: PGN 059904 ISO Request requesting PGN 060928 sent destination specific.
PGN 060928 received.
PGN 059904 ISO Request requesting PGN 060928 sent globally.
PGN 060928 received.
Method 3.22.4: PGN 060416 BAM sent
List contents of 060416….
PGNs 060160 sent….containing PGN 126208 request for PGN 126464
No response from DUT*
Method 3.22.5: PGN 060416 RTS sent … list contents
List PGNs sent and received.
List contents of PGNs 060416 sent
List PGNs received
List contents of PGNs 060416 received
PGNs 060160 received …. Containing PGN 126464 Received PGN List
Method 3.22.6: Method 3.9 Passed
Method 3.17 Passed
Method 3.22.7: Method 3.18 Passed
Method 3.22.8: Method 3.19 Passed
Method 3.20 Passed

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C.3.23 Address Bumping – Mandatory

This test is intended to verify that once a DUT has claimed an address, and begun normal operation, that it properly handles having it's address bumped (taken) by a higher priority device entering the network. Tests with lower priority claims will be made as well as testing the case when no addresses are available. Though these tests are similar to those for power-up address claim, these tests help verify that the DUT properly executes the address claim functions after having been operating in a normal state.

C.3.23.1 Method of measurement (Response to lower priority address contention):

This test verifies that the DUT properly handles address contention with devices that have a lower priority (higher value NAME means lower priority). This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) with a higher value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done after it has been confirmed that the DUT is in normal operation with a claimed address.

C.3.23.1.1 Required results:

The DUT should receive and process the lower priority address claim message. The DUT should respond to the contention by retransmitting the same address claim message again. All PGNs are documented. If the DUT changes to some other address as a result of the contention, or does not respond to the contention, then it has failed this test.

C.3.23.2 Method of measurement (Response to higher priority Address contention):

This test verifies that the DUT properly handles address contention with devices that have a higher priority (lower value NAME means higher priority). This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done after it has been confirmed that the DUT is in normal operation with a claimed address.

C.3.23.2.1 Required results:

The DUT should receive and process the higher priority address claim message. The DUT should respond to the contention by choosing a different address and transmitting a new address claim message, and then begin the 250ms-waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT keeps the original contested address in its second address claim message, or does not respond to the contention, then it has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 also constitutes failure of this test.

C.3.23.3 Method of measurement (Response to no addresses available):

This test verifies two items. The first item is that the DUT attempts to claim a valid address across the full range of valid address space, from 0 to 251. Second, that the DUT indicates that it is unable to claim an address. The sequences of events to conduct this test are as follows. This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done after it has been confirmed that the DUT is in normal operation with a claimed address. This should cause the DUT to try for another address. The test equipment

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should respond with another address claim with the same address as desired by the DUT, and with a lower value NAME than the DUT. This should cause the DUT to try for another address. This cycle should be repeated until the DUT sends a can not claim address message, or begins repeating address claim attempts with a previously attempted address.

C.3.23.3.1 Required results:

The DUT should, over the course of this test, send 252 address claim messages followed by a can not claim address message, which is an address claim message with the source address set to 254. The DUT will fail this test if it does not try all 252 addresses (order does not matter), begins repeating claims for the same address, or does not to send a can not claim address message.

C.3.23.4 Method of measurement (reduced communication during address bumping):

This test verifies that when the DUT is bumped by a higher priority device, it ceases communication until it's new address has been successfully claimed. This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done after it has been confirmed that the DUT is in normal operation with a claimed address. Once the DUT has issued a new address claim for a new address, two requests will be sent before the 250ms time period expires. An ISO request, PGN 059904 is sent to the DUT's previous address, requesting the Product Information PGN 126996, and an ISO request, PGN 059904 is sent to the DUT's new attempted but not yet claimed address, requesting the Product Information PGN 126996.

C.3.23.4.1 Required results:

The DUT should receive and process the higher priority address claim message. The DUT should respond to the contention by choosing a different address and transmitting a new address claim message, and then begin the 250ms-waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT keeps the original contested address in its second address claim message, or does not respond to the contention, then it has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 also constitutes failure of this test.

C.3.23.5 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.23 passed or failed. Provide the Certification Test major section number (such as 3.23). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as Source Address and NAME values. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.23.5.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes all four methods. The second example shows a variety of problems. The problem in 3.23.1 is that the DUT changed addresses in response to a device with a lower priority, when it should not have. The problem in 3.23.2 is that the DUT did not change its address in response to a device with a higher priority. In 3.23.3 the DUT did not attempt to claim all possible addresses before sending the can not claim message. The last problem, shown in 3.23.4, is that the DUT responded to requests for PGN 126996 while it was in transition between addresses and before the 250ms period was complete. Other results may be possible and should be similarly documented.

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Test 3.23 Address Claim Test – Mandatory – PASSED
Method 3.23.1: DUT Address = 12
DUT NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
Address = 12
NAME = 13839632026517020676 (C0104000AAAAA004$_{16}$)
PGN 060928 received
Address = 12
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
Method 3.23.2: DUT Address = 12
DUT NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
Address = 12
NAME = 13839632026517020674 (C0104000AAAAA002$_{16}$)
PGN 060928 received
Address = 13
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
Method 3.23.3: DUT Address = 12
DUT NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
252 Addresses tried.
PGN 060928 received with Address = 254
Method 3.23.4: DUT Address = 12
DUT NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
Address = 12
NAME = 13839632026517020674 (C0104000AAAAA002$_{16}$)
PGN 060928 received
Address = 13
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
ISO Request PGN 050904 sent to Address 12 requesting PGN 126996
ISO Request PGN 050904 sent to Address 13 requesting PGN 126996
No response from DUT during the 250ms timeout period.

Or

Test 3.23 Address Claim Test – Mandatory – FAILED*
Method 3.23.1: DUT Address = 12
DUT NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
Address = 12
NAME = 13839632026517020676 (C0104000AAAAA004$_{16}$)
PGN 060928 received
Address = 105*
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)

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Method 3.23.2: DUT Address = 12
DUT NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
Address = 12
NAME = 13839632026517020674 (C0104000AAAAA002$_{16}$)
PGN 060928 received
Address = 12*
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)

Method 3.23.3: DUT Address = 12
DUT NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
Addresses missed, 10,13,21,34,44,51,62,71,80,90,101,113*
PGN 060928 received with Address = 254

Method 3.23.4: DUT Address = 12
DUT NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 sent
Address = 12
NAME = 13839632026517020674 (C0104000AAAAA002$_{16}$)
PGN 060928 received
Address = 13
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
ISO Request PGN 050904 sent to Address 12 requesting PGN 126996
PGN 126996 received from DUT before 250 ms timeout from old address*
ISO Request PGN 050904 sent to Address 13 requesting PGN 126996
PGN 126996 received from DUT before 250 ms timeout from new address*

C.3.24 Default Reporting Intervals – Mandatory

The purpose of this is to verify that the DUT, operating with factory default settings, adheres the default reporting intervals for the NMEA 2000 published PGNs. Reporting intervals of Proprietary PGNs will be observed if possible.

C.3.24.1 Method of measurement (Default reporting intervals):

This test requires the DUT to have claimed an address and be operating in normal mode. The test consists of listening to the normal PGN outputs of the DUT and recording the time between like PGN transmissions. A Complex Request PGN 126208 is sent to the DUT requesting PGN 126464 Transmit PGN List. This forms the basis of what PGN outputs are expected from the DUT. Any anomalies, i.e. PGNs in the list but not transmitted by the DUT or PGNs not in the list but transmitted by the DUT should be noted.

C.3.24.1.1 Required results:

The PGNs provided by the DUT under normal operation should be transmitted at the default rates listed in the NMEA 2000 PGN descriptions. Reporting intervals or update rates that are smaller, i.e. less time between PGN reports that the default constitutes failure. The results will require manual verification by NMEA

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C.3.24.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.24 passed or failed. Provide the Certification Test major section number (such as 3.24). Provide the Method of measurement section numbers, with the PGN number(s) that were received. Also include information that determines pass or failure, such as the time between like PGN transmissions. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.24.2.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes. The second example shows a PGN being transmitted at much faster intervals on the network than the required defaults. Other results may be possible and should be similarly documented.

Test 3.24 Default Reporting Intervals – Mandatory – PASSED
Method 3.24.1: PGN 129025 received
100 ms
PGN 129025 received
120 ms
PGN 129025 received

Or

Test 3.24 Default Reporting Intervals – Mandatory – FAILED*
Method 3.24.1: PGN 129025 received
50 ms* => Requires Validation
PGN 129025 received
55 ms* => Requires Validation
PGN 129025 received

C.3.25 Default Priority – Mandatory

The purpose of this is to verify that the DUT, operating with factory default settings, adheres the default priority settings for the NMEA 2000 published PGNs. Priority values of Proprietary PGNs will be observed if possible.

C.3.25.1 Method of measurement (Default reporting intervals):

This test requires the DUT to have claimed an address and be operating in normal mode. The test consists of listening to the normal PGN outputs of the DUT and recording the priority value of each PGN sent by the DUT. A Complex Request PGN 126208 is sent to the DUT requesting PGN 126464 Transmit PGN List. This forms the basis of what PGN outputs are expected from the DUT. Any anomalies, i.e. PGNs in the list but not transmitted by the DUT or PGNs not in the list but transmitted by the DUT should be noted.

C.3.25.1.1 Required results:

The PGNs provided by the DUT under normal operation should be transmitted with the default priority listed in the NMEA 2000 PGN descriptions. Higher Priority (lower values) than the default constitutes failure. The results will require manual verification by NMEA

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C.3.25.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 3.25 passed or failed. Provide the Certification Test major section number (such as 3.25). Provide the Method of measurement section numbers, with the PGN number(s) that were received. Also include information that determines pass or failure, such as the priority value of PGN transmissions. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.3.25.2.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes. The second example shows a PGN being transmitted with a higher priority (lower value) than the required defaults. Other results may be possible and should be similarly documented.

Test 3.24 Default Reporting Intervals – Mandatory – PASSED
Method 3.24.1: PGN 129025 received
Priority 2
PGN 129029 received
Priority 3
PGN 129539 received
Priority 7

Or

Test 3.24 Default Reporting Intervals – Mandatory – FAILED*
Method 3.24.1: PGN 129025 received
Priority 2
PGN 129029 received
Priority 2* => Requires Validation
PGN 129539 received
Priority 6

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C.4 Class 1 Level B - Automated Test Methods

This section contains the requirements and methods of measurement that can be performed automatically via a suite of software and hardware designed to certify compliance with the Level B minimum requirements of the NMEA 2000 standard for a Class 1 device. If the DUT contains capabilities beyond the Level B minimum requirements, such as the Request/Command/Acknowledge Group Function PGN 126208, then the corresponding Level A certification test will be conducted to verify that implementation.

To perform these tests, you must assemble the network as shown in section 3.0.

The first part of performing a Level B certification test is to determine what PGNs are provided by the Level B DUT. This determination includes both NMEA 2000 defined PGNs and Manufacturers Proprietary PGNs. Since the TX/RX List PGN 126464 is not required, other methods of finding out which data PGNs are provided are implemented in the Certification Tool for Level B. The Tool will do the following:

1. It should prompt the user to identify ALL (including proprietary) the PGNs (enter the PGNs) that are generated by the DUT.
2. It can query the DUT for each and every NMEA 2000 defined PGN and Proprietary PGN (including all PGN numbers slated for use by NMEA 2000 but not yet assigned) with the ISO request.

The responses to this extensive interrogation provide several important things. This adds some extensibility to the Certification Tool as the PGN data base changes or grows, in between updates to the Tool, the Tool can now create a list of all PGNS supported by the device, the default priority of provided PGNs can be checked, and the consistency between user's notification of what PGNS are supported versus actual responses to ISO Request can be made. Further, any fast packet PGNS that are identified as being provided by the DUT, other than Product Information, can be used to conduct additional tests. The results of this PGN determination shall be provided in a table or tabular form with the certification test results.

C.4.1 Fast Packet Sequence Counter [3.1.3.3] – Mandatory

This section certifies the requirement that "*The modulo 8 sequence counter shall increment each time the same PGN is transmitted from the same source address*." This is accomplished with two specific tests.

The purpose of the first test (4.1.1) is to verify proper sequence number behavior with a mandatory PGN.

The second test (4.1.2) examines the same requirement with a non-mandatory Fast Packet PGN. This second test can only be performed if the DUT transmits a non-mandatory Fast Packet PGN. Determination of the actual PGN to be tested is dependent upon either a user entered Fast Packet PGN number(s) or a Fast Packet PGN as determined by the certification tool. If the DUT does not transmit any non-mandatory PGN, then the second test is not necessary, no failure will be indicated for test 4.1.2.

C.4.1.1 Method of measurement (Mandatory PGN):

Using the ISO Request PGN 059904, send a request for Product Information, PGN 126996. Record the sequence number value. Send eight more requests for PGN 126996, and verify that the sequence number is incremented by one for each successive PGN and that the sequence number properly wraps around from 7 to 0.

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C.4.1.1.1 Required results:

The sequence counter should be seen to start at some value between 0 and 7, and should increment up to 7 and then should start back at 0.

C.4.1.2 Method of measurement (Non-Mandatory PGN):

This test is only performed if the DUT transmits one or more non-mandatory Fast Packet PGNs. This is determined by user entry or automatic tool determination efforts. Perform the same test procedure described in 4.1.1 above. This will verify that all supported PGNs, not just the mandatory PGNs, are implemented properly, with respect to the sequence number. If the Device Under Test (DUT) does not transmit any Fast Packet PGNs, then this test is not performed.

C.4.1.2.1 Required results:

If the DUT does not transmit any non-mandatory Fast Packet PGNs, there are no required results.
If the DUT does transmit any non-mandatory Fast Packet PGNs, then see 4.1.1.1

C.4.1.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 4.1 passed or failed. Provide the Certification Test major section number (such as 4.1), and each Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement, and the values of the sequence numbers recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.4.1.3.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of two methods, is passed. The second example depicts the results showing that the non-mandatory Fast Packet PGN measurements have failed. Notice that the sequence numbers were not incremented, but fixed at zero. The third example illustrates the results with a Level B device that does not transmit any non-mandatory Fast Packet PGNs. Other results may be possible and should be similarly documented.

Test 4.1 Fast Packet Sequence Counter [3.1.3.3] - PASSED
Method 4.1.1: PGN 126996 2,3,4,5,6,7,0,1,2
Method 4.1.2: PGN 126998 0,1,2,3,4,5,6,7,0

Or

Test 4.1 Fast Packet Sequence Counter [3.1.3.3] - FAILED*
Method 4.1.1: PGN 126996 2,3,4,5,6,7,0,1,2
Method 4.1.2: PGN 126998 0,0,0,0,0,0,0,0,0*

Or

Test 4.1 Fast Packet Sequence Counter [3.1.3.3] - PASSED
Method 4.1.1: PGN 126996 2,3,4,5,6,7,0,1,2
Method 4.1.2: Not Applicable

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C.4.2 Unique Fast Packet Sequence Counter [3.1.3.4] – Mandatory

This section certifies the requirement that "*A separate sequence counter shall be utilized for each different PGN transmitted from the same source address.*" The purpose of this test is to ensure there is no linkage of the sequence number between Fast Packet PGNs. This test is only performed on devices that transmit non-mandatory Fast Packet PGNs in addition to the mandatory Product Information PGN 126996. Determination of the actual non-mandatory Fast Packet PGN to be tested is dependent upon either user entered Fast Packet PGN number(s) or a Fast Packet PGN as determined by the certification tool. If the DUT does not transmit any non-mandatory PGN, then this test cannot be conducted, and no failure will be indicated for test 4.2.1.

C.4.2.1 Method of measurement (Multiple PGNs):

Using the ISO Request PGN 059904, send one or more requests for the Product Information PGN 126996, and one or more requests for a supported non-mandatory Fast Packet PGN, until both have different sequence numbers. Then, send alternate requests, one for PGN 126996 and one for the selected non-mandatory Fast Packet PGN, eight times. This will verify that the behavior is proper given interleaved requests. Repeat this procedure for every supported non-mandatory Fast Packet PGN.

C.4.2.1.1 Required results:

The sequence number behavior should be independent of PGNs, that is, the series of sequence numbers from one PGN should have no relationship to those of another PGN. If the DUT does not transmit any non-mandatory Fast Packet PGNs, there are no required results.

C.4.2.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 4.2 passed or failed. Provide the Certification Test major section number (such as 4.2), and each Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement, and the values of the sequence numbers recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.4.2.2.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the results showing that the Multiple PGN measurements have failed. Notice that the sequence numbers were not incremented, but fixed at zero. The third example shows that the sequence numbers were not unique across multiple PGNs. The fourth example illustrates the documented result when the test can not be run due to no non-mandatory Fast Packet PGNs supported by the DUT. Other results may be possible and should be similarly documented.

Test 4.2 Unique Fast Packet Sequence Counter [3.1.3.4] - PASSED
Method 4.2.1: PGN 126996 3,4,5,6,7,0,1,2,3
PGN 126998 1,2,3,4,5,6,7,0,1

Or

Test 4.2 Unique Fast Packet Sequence Counter [3.1.3.4] - FAILED*
Method 4.2.1: PGN 126996 3,4,5,6,7,0,1,2,3

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PGN 126998 0,0,0,0,0,0,0,0,0*

Or

Test 4.2 Unique Fast Packet Sequence Counter [3.1.3.4] - FAILED*
Method 4.2.1: PGN 126996 4,6,0,2,4,6,0,2,4*
PGN 126998 5,7,1,3,5,7,1,3,5*

Or

Test 4.2 Unique Fast Packet Sequence Counter [3.1.3.4] – Not Applicable

C.4.3 Fast Packet Message Time Out [3.1.4.8] – Mandatory

This section certifies the requirement that "*A fast-packet message shall not be considered as incomplete until a time-out of 750 ms from the last received frame occurs*." The purpose of this test is to verify that the DUT will function properly on a heavily loaded network, i.e. that it does not drop messages prematurely, when dealing with significant yet acceptable delays between a given message's frames. Though this test is conducted with only a single fast packet message, this situation will occur in cases where devices may need to handle multiple concurrent fast packet transmissions on the network. This test is intended to verify that the DUT adheres to the timeout period on reception of fast packet PGNs, however, Level B devices are not required to receive any Fast Packet PGNs, but if they do they must do so in accordance with this standard. This test and verification that the DUT adheres to the specification can only be accomplished if the Fast Packet Request Group Function PGN 126208 has been implemented.

C.4.3.1 Method of measurement (Fast Packet Message Time Out):

This test requires the ability of the test hardware and software to be able to delay or not complete sending all the frames of a message for specified period of time. To conduct this test, a Request Group Function PGN 126208, requesting the Product Information PGN 126996, should be generated. Some portion of the request PGN should be sent, and then a delay of 745ms should occur, followed by the remaining frames of the request PGN.

C.4.3.1.1 Required results:

If the DUT supports the Request Group Function PGN, then upon receiving the completed request, it should provide the Product Information PGN 126996. If the DUT does not support the Request Group Function PGN, than adherence to this requirement cannot be validated, this test is not run, and no failure is indicated.

C.4.3.2 Reporting requirements:

Provide detailed information as to what part (Method of measurement) of test 4.3 passed or failed. Provide the Certification Test major section number (such as 4.3), and the Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.4.3.2.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the results showing that the delay between the Transmit

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List PGN and the Receive List PGN were less than 10 frames. The third example illustrates the situation when the DUT does not support the PGNs necessary to conduct this test. Other results may be possible and should be similarly documented.

Test 4.3 Fast Packet Message Time Out [3.1.4.8] – Mandatory - PASSED
Method 4.3.1: PGN 126208 requesting PGN 126996 sent
PGN 126996 Product Information PGN received

Or

Test 4.3 Fast Packet Message Time Out [3.1.4.8] - Mandatory - FAILED*
Method 4.3.1: PGN 126208 requesting PGN 126996 sent
PGN 126996 Product Information PGN NOT received*

Or

Test 4.3 Fast Packet Message Time Out [3.1.4.8] – Not Applicable

C.4.4 ISO 11783-3 3.2.2 Reserved Bit ® [Table 3-1] – Mandatory

This test is intended to confirm that the DUT implements the reserved bit in bit-position five of the can frame. The requirement is this "*ISO 11783-3 Section 3.2.2 Reserved Bit, specifies that all messages shall set the reserved bit (bit number 5) in the CAN ID field to zero on transmit*." This will be accomplished with two tests. The first test (4.4.1) will check the 5th bit of the CAN frame from the Product Information PGN 126996, and then from some non-mandatory PGN.

C.4.4.1 Method of measurement (Reserved bit in Product Information PGN):

The purpose of this method is to verify that the reserved bit is properly set in CAN frames of mandatory messages. To conduct this test an addressed ISO Request PGN 059904 will be used to request the Product Information PGN 126996. The reserved bit, bit number 5 in the CAN frame will be examined.

C.4.4.1.1 Required results:

The DUT should respond to the ISO Request with PGN 126996 Product Information. Bit 5 of the CAN frame must be set to zero. If bit 5 is not zero, the DUT fails this test.

C.4.4.2 Method of measurement (Reserved bit in non-mandatory PGN):

The purpose of this method is to verify that the reserved bit is properly set in CAN frames of all supported non-mandatory messages. To conduct this test, generate and send an ISO Request PGN 059904 asking for the immediate transmission of eachsupported non-mandatory PGN. The reserved bit, bit number 5 in the CAN frame will be examined.

C.4.4.2.1 Required results:

The DUT should respond to the request with PGN ###### requested. Bit 5 of the CAN frame must be set to zero. If bit 5 is not zero, the DUT fails this test. No response also constitutes failure.

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C.4.4.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 4.4 passed or failed. Provide the Certification Test major section number (such as 4.4). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs and the value of the reserved bit from bit position 5 in the CAN frame. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.4.4.3.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of two methods is passed. The second example depicts the case where the second test (4.4.2) fails because the value of the reserved bit is a one instead of a zero. Other results may be possible and should be similarly documented.

Test 4.4 ISO 11783-3 3.2.2 Reserved Bit [Table 3-1] - Mandatory – PASSED
Method 4.4.1: PGN 059904 ISO Request requesting PGN 126996 sent.
PGN 126996 Product Information PGN received.
Reserved bit from bit position 5 of CAN frame = 0
Method 4.4.2: PGN 059904 ISO Request requesting PGN ###### non-mandatory PGN sent
PGN ###### received.
Reserved bit from bit position 5 of CAN frame = 0

Or

Test 4.4 ISO 11783-3 3.2.2 Reserved Bit [Table 3-1] - Mandatory – FAILED*
Method 4.4.1: PGN 059904 ISO Request requesting PGN 126996 sent.
PGN 126996 Product Information PGN received.
Reserved bit from bit position 5 of CAN frame = 0
Method 4.4.2: PGN 059904 ISO Request requesting PGN ###### non-mandatory PGN sent
PGN ###### received.
Reserved bit from bit position 5 of CAN frame = 1*

C.4.5 ISO 11783-3 Annex C, NACK [Table 3-1] – Mandatory – Refer to C.3.16

This test certifies that the ISO NACK is used as required. The requirement being tested is as follows: "*ISO 11783-3 Annex C, NACK - If the Parameter Group Number in a Command or Request is not recognized by the destination, it shall reply with the NACK message*." This is achieved with one test for proper NACK response to an ISO request. See section C.3.16 for specific test methods, results, and reporting requirements.

C.4.6 NAME Support [8.3] – Mandatory - Refer to C.3.17

This test is designed to verify the following requirement: "*NMEA 2000 devices shall implement the NAME field, and field dependencies of the Address Claim message as defined by PGN 60928 in Appendix B database file "2000vx.x.mdb"*." Each field of the NAME entity will be examined to determine if they contain appropriate values. One test is used to conduct examination of the NAME fields. See section C.3.17 for specific test methods, results, and reporting requirements.

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C.4.7 Product Information [8.3.2] – Mandatory

This test certifies the following requirement. “All NMEA 2000 devices shall support the Product Information message containing at a minimum the “NMEA 2000 Version Supported” data field. The Product Information message shall be implemented as described in Appendix B.” If the Level B DUT supports the Request Group Function PGN 126208 see section C.3.18 for specific test methods, results, and reporting requirements. If the DUT does not support the Request Group Function PGN 126208, the ISO Request portion of test C.3.18 will be utilized to satisfy then this test.

C.4.8 Address Claim Test – Mandatory – Refer to C.3.21

This section certifies one of the most basic and important characteristics of the NMEA 2000 standard, Automatic self-configuring Address Claiming, at power up. There are several very important aspects to dynamic address claiming and assignment that must be adhered to in order to ensure interoperability with other NMEA 2000 devices. This requires multiple tests. Each of the following tests has one specific goal or criteria to be satisfied. See section C.3.21 for specific test methods, results, and reporting requirements.

C.4.9 Minimum Message Implementation [Table 9-2] – Mandatory

This section is intended to certify, “*The messages listed in Table 9-2 are all required to be implemented*.” Many previous test sections utilize these messages, both to and from the DUT, to conduct the certification. The verification of this requirement, when applicable, is based upon those other tests as described in the methods below.

C.4.9.1 Method of measurement (PGN 059392):

This method is dependant upon test 4.5. If test 4.5 is passed, the DUT has implemented the ISO Acknowledgement PGN 059392.

C.4.9.1.1 Required Results:

Check that test 4.5 has passed. Failure of test 4.5 constitutes failure of this method.

C.4.9.2 Method of measurement (PGN 059904):

There are no previous tests that rely solely upon this PGN, so an additional test is needed. The test equipment will send a PGN 059904 to the DUT, both destination and globally addressed, requesting the Address Claim PGN 060928. A proper response to this test will verify that the DUT has implemented the ISO Request.

C.4.9.2.1 Required Results:

The DUT should respond with two PGN 060928 messages, one for the destination specific, and one for the global ISO request 059904.

C.4.9.3 Method of measurement (PGN 060928):

This method is dependant upon tests 4.6 and 4.8. If tests 4.6 and 4.8 are passed, the DUT has implemented the ISO Address Claim PGN 060928 correctly.

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C.4.9.3.1 Required Results:

Check that tests 4.6 and 4.8 have been passed. Failure of test 4.6 or 4.8 constitutes failure of this method.

C.4.9.4 Method of measurement (PGN 126996):

This method is dependant upon test 4.7. If test 4.7 is passed, the DUT has implemented the Product Information PGN 126996.

C.4.9.4.1 Required Results:

Check that test 4.7 has been passed. Failure of test 4.7 constitutes failure of this test.

C.4.9.5 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 4.9 passed or failed. Provide the Certification Test major section number (such as 4.9). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.4.9.5.1 Example text:

The text in this section is merely an outline of two of many possible outcomes. The first example text outlines the results when the DUT passes the test. The second example illustrates that the DUT failed test 4.5. Other results may be possible and should be similarly documented.

Test 4.9 Minimum Message Implementation [Table 9-2] – Mandatory – PASSED
Method 4.9.1: Method 3.16 Passed
Method 4.9.2: PGN 059904 ISO Request requesting PGN 060928 sent destination specific.
PGN 060928 received.
PGN 059904 ISO Request requesting PGN 060928 sent globally.
PGN 060928 received.

Or

Method 4.9.1: Method 3.16 Failed*
Method 4.9.2: PGN 059904 ISO Request requesting PGN 060928 sent destination specific.
PGN 060928 received.
PGN 059904 ISO Request requesting PGN 060928 sent globally.
PGN 060928 received.

C.4.10 Address Bumping – Mandatory – Refer to C.3.23

This test is intended to verify that once a DUT has claimed an address, and begun normal operation, that it properly handles having it's address bumped (taken) by a higher priority device entering the network. Tests with lower priority claims will be made as well as testing the case when no addresses are available. Though these tests are similar to those for power-up address claim, these tests help verify that the DUT properly executes the address claim functions after having been operating in a normal state. See section C.3.23 for specific test methods, results, and reporting requirements.

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C.4.11 Default Reporting Intervals – Mandatory

The purpose of this is to verify that the DUT, operating with factory default settings, adheres the default reporting intervals for the NMEA 2000 published PGNs. Reporting intervals of Proprietary PGNs will be observed if possible.

C.4.11.1 Method of measurement (Default reporting intervals):

This test requires the DUT to have claimed an address and be operating in normal mode. The test consists of listening to the normal PGN outputs of the DUT and recording the time between like PGN transmissions. Determination of the actual PGNs expected is dependent upon the user entered PGN number(s) or the PGNs as determined by the certification tool. This forms the basis of what PGN outputs are expected from the DUT. Any anomalies, i.e. PGNs in the list but not transmitted by the DUT or PGNs not in the list but transmitted by the DUT should be noted. See test 3.24 for results and reporting requirements.

C.4.12 Default Priority – Mandatory

The purpose of this is to verify that the DUT, operating with factory default settings, adheres the default priority settings for the NMEA 2000 published PGNs. Priority values of Proprietary PGNs will be observed if possible.

C.4.12.1 Method of measurement (Default reporting intervals):

This test requires the DUT to have claimed an address and be operating in normal mode. The test consists of listening to the normal PGN outputs of the DUT and recording the priority value of each PGN sent by the DUT. Determination of the actual PGNs expected is dependent upon the user entered PGN number(s) or the PGNs as determined by the certification tool. This forms the basis of what PGN outputs are expected from the DUT. Any anomalies, i.e. PGNs in the list but not transmitted by the DUT or PGNs not in the list but transmitted by the DUT should be noted. See test 3.25 for results and reporting requirements.

C.4.13 Verification of unsupported Level A capabilities – Mandatory

The purpose of this test is verify that level B devices respond consistently and properly to any requests, commands, or attempts to access capabilities, that may be supported by the device.

If the device implements and supports any capability of the NMEA 2000 Request and Command Group Function PGN 126208, then it must respond properly to any requests or commands received. If the device does not support any form of PGN 126208, which is optional for Level B, then the device will be checked.

If the device implements and supports the Transmit/Received PGN List, PGN 126464, then the DUT must respond properly to requests received for PGN 126464. If the device does not support PGN 126464, which is optional for Level B, then the device will be checked. If the device supports PGN 126464, but does not support PGN 126208, a test will be conducted to confirm this. If the device supports PGN 126464 and PGN 126208, then the Level A tests of 3.19 and 3.20 shall be conducted.

The following test methods are designed to confirm proper behavior by the device under these circumstances.

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C.4.13.1 Method of measurement (Acknowledgement of Request Group Function):

The test is conducted by sending a request for a Level B mandatory PGN using the Request Group Function, PGN 126208. The Request Group Function PGN 126208 requesting the Product Information PGN 126996 will be sent to the DUT. Though the Product Information PGN 126996 is a mandatory PGN for all NMEA 2000 devices, Level B devices are only required to provide this PG from requests made using the ISO Request PGN 059904. Generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)

C.4.13.1.1 Required Results:

There are three acceptable responses, depending upon the support for PGN 126208.

No PGN 126208 Support:

If the DUT does not support any PGN 126208 capability, then the required response is no response. Any Acknowledgment under this circumstance is considered a failure.

Any PGN 126208 Support:

If the DUT supports any PGN 126208 Request or Command capabilities, then it must fully support PGN 126208 Acknowledgements to ALL Requests and Commands. There are two types or forms of acknowledgement responses. The first form is the full response with individual error codes for each item requested or commanded. The second form is an abbreviated acknowledgment with no individual field error codes for each item requested or commanded. See the Example Text.

C.4.13.2 Method of measurement (Acknowledgement of Command Group Function):

The test is conducted by sending a command to change the priority and manufacturers product code value of a Level B mandatory PGN using the Command Group Function, PGN 126208. The Command Group Function PGN 126208 attempting to modify the priority and manufacturers product code of the Product Information PGN 126996 will be sent to the DUT. Generate a Command Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)

C.4.13.2.1 Required Results:

There are three acceptable responses, depending upon the support for PGN 126208.

No PGN 126208 Support:

If the DUT does not support any PGN 126208 capability, then the required response is no response. Any Acknowledgment under this circumstance is considered a failure.

Any PGN 126208 Support:

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If the DUT supports any PGN 126208 Request or Command capabilities, then it must fully support PGN 126208 Acknowledgements to ALL Requests and Commands. There are two types or forms of acknowledgement responses. The first form is the full response with individual error codes for each item requested or commanded. The second form is an abbreviated acknowledgment with no individual field error codes for each item requested or commanded. See the Example Text.

C.4.13.3 Method of measurement (PGN 126464 Transmit/Received PGN List):

This method validates the DUT's implementation of the Transmit/Received PGN list. The contents of the PGN if supported are examined to ensure that all the mandatory PGNs are included. Non-mandatory PGNs require manual verification. To conduct this test, a destination specific addressed ISO request PGN 059904 is sent to the DUT asking for PGN 126464 Transmit/Received PGN List.

C.4.13.3.1 Required Results:

There are three acceptable responses, depending upon the support for PGN 126464 and PGN 126208.

No PGN 126464 Support:

The required response is an ISO Acknowledgement PGN 059392 indicating a negative acknowledgement.

PGN 126464 Supported but no support for PGN 126208:

The required response is the return of PGN 126464 Transmit PGN List and PGN 126464 Received PGN list.

Both PGN 126464 and PGN 126208 supported:

In this case, tests 3.19 and 3.20 shall be conducted and the results of those tests shall determine pass or failure of this test.

C.4.13.4 Reporting Requirements:

Provide detailed information as to what parts (Methods of measurement) of test 4.13 passed or failed. Provide the Certification Test major section number (such as 4.13). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.4.13.4.1 Example Text:

The text in this section shows four of many possible outcomes. The first example text shows the results when the DUT does not support PGN 126208 or PGN 126464 and passes the tests. The second example shows the result when the DUT does not support PGN 126208, but does support PGN 126464. The third example shows the result when the DUT supports both PGN 126208 and PGN 126464, and passes all the tests. The fourth example shows that though the DUT supports PGN 126208 and PGN 126464, it failed the test by not responding properly to the Command Group Function PGN 126208 Command. Other results may be possible and should be similarly documented.

Example 1 – No support for PGN 126208 or PGN 126464
Test 4.13 Verification of unsupported Level A capabilities - Mandatory – PASSED
Method 4.13.1: PGN 126208 Request destination addressed containing the following parameters sent.
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)

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Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT

Method 4.13.2: PGN 126208 Command destination addressed containing the following parameters sent.
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT

Method 4.13.3: PGN 059904 ISO Request requesting PGN 126464 sent.
PGN 059392 received from DUT.
Field 1 - Control Byte = 1
Field 2 - Group Function Value = 255 (FF_{16})
Field 3 - Reserved bits = 16777215 ($FFFFFF_{16}$)
Field 4 - PGN of requested information = 126464

Example 2 – No support for PGN 126208, but PGN 126464 supported.
Test 4.13 Verification of unsupported Level A capabilities - Mandatory – PASSED

Method 4.13.1: PGN 126208 Request destination addressed containing the following parameters sent.
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT

Method 4.13.2: PGN 126208 Command destination addressed containing the following parameters sent.
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT

Method 4.13.3: PGN 059904 ISO Request requesting PGN 126464 sent.
PGN 126464 Transmit PGN List received.
Mandatory PGNS Listed: 059392, 060928, 129996

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Non-Mandatory PGNs Listed: 126464, 126998, 129025, 129026, 129029, 129044, 129045, 129284 => Require Verification
PGN 126464 Received PGN List received.
Mandatory PGNS Listed: 059392, 059904, 060928
Non-Mandatory PGNs Listed: 129538 => Requires Verification

Example 3 – Support for PGN 126208 and PGN 126464
Test 4.13 Verification of unsupported Level A capabilities - Mandatory – PASSED
Method 4.13.1: PGN 126208 Request destination addressed containing the following parameters sent.
Field 1 - Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Acknowledgment received (contents below)
Acknowledgment Group Function Code = 2
PGN of message being acknowledged = 126996
PGN error code = 0
Transmission Interval code = 0
Number of Requested or Commanded Parameters = 1
Parameter error code = 3
Method 4.13.2: PGN 126208 Command destination addressed containing the following parameters sent.
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Acknowledgment received (contents below)
Acknowledgment Group Function Code = 2
PGN of message being acknowledged = 126996
PGN error code = 4
Priority error code = 4
Number of Requested or Commanded Parameters = FF
Method 4.13.3: Method 3.19 Passed (Full results of 3.19 to be listed)
Method 3.20 Passed (Full results of 3.20 to be listed)

Example 4 – Support for 126208
Test 4.13 Verification of unsupported Level A capabilities - Mandatory – FAILED*
Method 4.13.1: PGN 126208 Request destination addressed containing the following parameters sent.
Field 1 - Request Group Function Code = 0 (Request group function)

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Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Acknowledgment received (contents below)
Acknowledgment Group Function Code = 2
PGN of message being acknowledged = 126996
PGN error code = 0
Transmission Interval code = 0
Number of Requested or Commanded Parameters = 1
Parameter error code = 3

Method 4.13.2: PGN 126208 Command destination addressed containing the following parameters sent.
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Requested PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT*

Method 4.13.3: Method 3.19 Passed (Full results of 3.19 to be listed)
Method 3.20 Passed (Full results of 3.20 to be listed)

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C.5 Class 1 Listen Only - Automated Test Methods

This section contains the requirements and methods of measurement that can be performed automatically via a suite of software and hardware designed to certify compliance with the minimum requirements of the NMEA 2000 standard for Listen Only Devices.

To perform this test, you must assemble the network as shown in section 3.0.

C.5.1 No Response Verification – Mandatory

The purpose of this test is to verify that the Listen Only device does not transmit any acknowledge or error frames on the CAN bus.

C.5.1.1 Method of measurement (Repsonse to Address Claim):

This method validates that the DUT does not respond with either at all when an address claim PGN is placed on the bus. To conduct this test there should be only one other device on the bus, the certification tool. The tool will generate an address claim PGN on the bus, and look for any response from the DUT.

C.5.1.1.1 Required Results:

The only acceptable response from the DUT is no response at all and the address claim PGN transmission on the bus from the tool should fail. The DUT will fail this test if the address claim PGN transmission succeeds, because this can only happen if the DUT provides an acknowledgment frame on the bus.

C.5.1.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 5.1 passed or failed. Provide the Certification Test major section number (such as 5.1). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.5.1.2.1 Example Text:

The text is this section shows two possible results. The first result illustrates th DUT passing the test while the second result shows the DUT failing the test.

Example 1 – DUT does not respond to address claim message
Test 5.1 No Response Verification – Mandatory - PASSED
Method 5.1.1: PGN 060928 Address Claim PGN transmitted by tool
Transmission of PGN 060928 failed due to no response from DUT

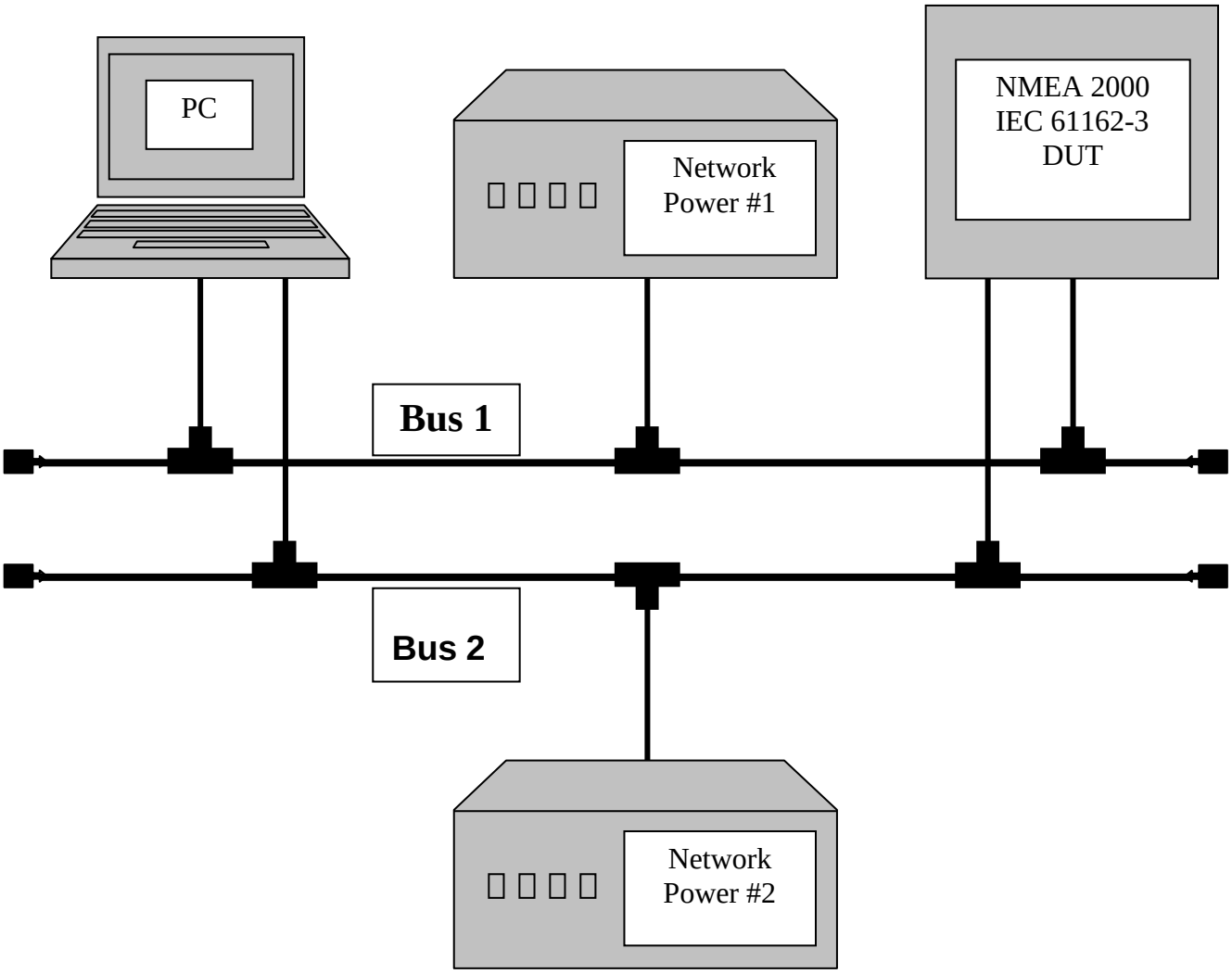
Example 2 – DUT responds to address claim message
Test 5.1 No Response Verification – Mandatory – FAILED*
Method 5.1.1: PGN 060928 Address Claim PGN transmitted by tool
Transmission of PGN 060928 successful due to acknowledgement frame response from DUT*

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C.6 Class 2 Level A - Automated Test Methods

This section contains the requirements and measurement methods to be performed automatically via a software/hardware suite designed to certify compliance with the NMEA 2000 standard's Level A minimum requirements.

To perform the automated tests, you must assemble the network as shown below:



From here on, the terms "bus 1" and "bus 2" will be used to distinguish between the redundant networks as shown in the picture. Realize that as far as the tests are concerned, the networks could have been labeled with Bus 2 on top and Bus 1 on the bottom in the picture. Likewise, the network power supplies are located in the middle of their respective buses for convenience in the drawing.

The first part of performing the automated test methods is to determine what PGNs are provided by the DUT. This determination includes both NMEA 2000 defined PGNs and Manufacturers Proprietary PGNs. The Tool will do the following:

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1. It should prompt the user to identify ALL (including proprietary) PGNs (enter the PGNs) that are generated by the DUT.
2. It can query the DUT for each and every NMEA 2000 defined PGN and Proprietary PGN (including all PGN numbers slated for use by NMEA 2000 but not yet assigned) with the ISO request.

When performing the Class 2 automated tests, the test tool is responsible for correctly representing itself on the buses as a Class 1 or Class 2 device as each test method requires. This is accomplished by claiming an network address with appropriate NAME values. Class 2 representation by the test tool should have the same NAME value for each node except for the System Instance field. When representing a Class 1 device, the NAME values should also differentiate in some other field of the NAME (Manufacturer's code is recommended).

C.6.1 Message Concurrency [3.1.2] – Mandatory

This section certifies the requirement that "*The minimum NMEA 2000 requirement is to support the concurrent reception of the following messages:*

One single frame message
One fast-packet message
One multi-packet Broadcast message
One multi-packet RTS/CTS session"

This requires four messages to be sent to the DUT concurrently, meaning that the individual message frames from each message will be interleaved in some manner with those of the other three messages. The DUT will need to handle (receive, process, and respond if required) all four messages properly. The four methods that follow must all be synchronized so that the four messages actually transit on the NMEA 2000 network at the same time.

C.6.1.1 Method of measurement (multi-packet RTS/CTS session):

This method involves the first concurrent message to be sent to the DUT. This message should be the longest message with respect to time, and should be complete after the other three concurrent messages. This message is sent using the ISO 11783 multi-packet RTS/CTS session. In order for this message to take the longest time, and to require many frames, the Request Group Function, requesting the NAME (PGN 60928), is chosen as the first message.

Send an ISO Request PGN 059904 requesting the ISO Address Claim PGN 060928 on both buses. Record the values of the fields returned in PGN 060928 that, according to the PGN 060928 definition, can be requested. Next, generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 60928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 7
Field 6 - Field number of first parameter = 2 (field 2 from 060928)
Field 7 - Value of first parameter = value of field 2, Manufacturer code
Field 8 - Field number of second parameter = 3 (field 3 from 060928)
Field 9 - Value of second parameter = value of field 3, ECU Instance
Field 10 - Field number of third parameter = 4 (field 4 from 060928)
Field 11 - Value of third parameter = value of field 4, Function Instance

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Field 12 - Field number of fourth parameter = 5 (field 5 from 060928)
Field 13 - Value of fourth parameter = value of field 5, Function
Field 14 - Field number of fifth parameter = 7 (field 7 from 060928)
Field 15 - Value of fifth parameter = value of field 7, Device Class
Field 16 - Field number of sixth parameter = 8 (field 8 from 060928)
Field 17 -Value of sixth parameter = value of field 8, Device Class Instance
Field 18 - Field number of seventh parameter = 9 (field 9 from 060928)
Field 19 - Value of seventh parameter = value of field 9, Industry Group

Initiate a connection mode transfer session with the DUT with the Request Group Function PGN created above on both buses. The transfer sessions shall use the same adjusted DLC value to indicate the same Complex Request is being transmitted. Begin transferring this PGN data, and proceed to 6.1.2, Method of Measurement (multi-packet Broadcast Message).

C.6.1.1.1 Required results:

A data transfer connection should be established with the DUT on both buses. Data should be transferred in accordance with the multi-packet RTS/CTS session protocol on both buses. The DUT should receive and accept this message from both buses. The transfer sessions should begin before the other three concurrent messages are started as defined in Methods 6.1.2, 6.1.3, and 6.1.4. The transfer sessions should finish after the other three concurrent messages have been completed. The DUT should receive and accept the message on both buses. The DUT should respond with the ISO Address Claim PGN 060928 on both buses with the same adjusted DLC.

C.6.1.2 Method of measurement (multi-packet Broadcast Message):

This method involves the second concurrent message to be sent to the DUT. This message should be the second longest message with respect to time, and should be complete before the first concurrent message, and after the third and fourth concurrent messages. This message is sent on both buses using the ISO 11783 multi-packet Broadcast Message protocol, requiring the use of the Broadcast Announce Message (BAM). This method will send a Request Group Function, PGN 126208, requesting the Transmitted PGN List Group Function, PGN 126464. Generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 0 (Transmit PGN List Message)

The broadcast sessions will use the same DLC offset for both buses to indicate the same Complex Request is being transmitted. The DLC offset used will be the next in the DLC series from the offset used in 6.1.1. Begin transferring this PGN data, and proceed to 6.1.3, Method of Measurement (fast-packet).

C.6.1.2.1 Required results

Data should be transferred in accordance with the multi-packet Broadcast transfer protocol on both buses. The message transfer sequence should start after the first concurrent message has started (6.1.1), and should finish after the last two concurrent messages (6.1.3 and 6.1.4) have been completed. The DUT should receive and

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accept this message on both buses. The DUT should respond with the Transmitted List Group Function PGN 126464 using the ISO 11783 multi-packet Broadcast Message protocol on both buses with the same adjusted DLC.

C.6.1.3 Method of measurement (fast-packet):

This method involves the third concurrent message to be sent to the DUT. This message should be the third longest message with respect to time, and should start before and finish after the fourth concurrent message. This message is sent using the NMEA 2000 Fast-Packet protocol on both buses, and should be globally addressed. This method will send the Request Group Function PGN 126208, requesting the Received PGN List Group Function, PGN 126464. Generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 1 (Receive PGN List Message)

The same DLC offset will be used for both buses to indicate the same Complex Request is being transmitted. The DLC offset used will be the next in the DLC series from the offset used in 6.1.2. Begin transferring this PGN data, and proceed to 3.1.4, Method of Measurement (single frame).

C.6.1.3.1 Required results

Data should be transferred in accordance with the NMEA 2000 Fast Packet protocol on both buses. The message transfer sequence should start after the second concurrent message has started (6.1.2), and should finish after the last concurrent message (6.1.4) has been completed. The DUT should receive and accept this message on both buses. The DUT should respond with the Received List Group Function PGN 126464 on both buses with the same adjusted DLC. The adjusted DLC must be the next in the DLC series from the adjusted DLC in the received response in 6.1.2.

C.6.1.4 Method of measurement (Single Frame):

This method involves the fourth concurrent message to be sent to the DUT. This message should be the shortest message with respect to time, and should start after the previous three concurrent messages have started. This message is sent on both buses using the ISO Single Frame Transmission protocol, as an addressed message. This method will send an ISO Request PGN 059904, requesting the Product Information PGN 126996. Send this PGN data.

C.6.1.4.1 Required results

This should be the last concurrent PGN to be sent, and should be the first to finish. The other concurrent PGNS should come to completion in the reverse order they were started, i.e. method 6.1.3 followed by method 6.1.2, followed by the completion of method 6.1.1.

The DUT should receive and accept this concurrent addressed single frame message on both buses. The DUT should respond with the Product Information PGN 126996 on both buses with the same adjusted DLC.

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C.6.1.5 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.1 passed or failed. Provide the Certification Test major section number (such as 6.1), and each Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement, and the results obtained by each method. Provide the adjusted DLC received on each response message. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.1.5.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of four methods, is passed. The second example depicts the results showing that the DUT could not handle concurrent message reception. Many other types of results may be possible and should be similarly documented.

Test 6.1 Message Concurrency [3.1.2] - PASSED
Method 6.1.1: PGN 059904 sent bus 1
PGN 059904 sent bus 2
PGN 060928 received bus 1 with DLC 10
PGN 060928 received bus 2 with DLC 10
PGN 126208 sent bus 1
PGN 126208 sent bus 2
PGN 060928 received bus 1 with DLC 11
PGN 060928 received bus 2 with DLC 11
Method 6.1.2: PGN 126208 sent bus 1
PGN 126208 sent bus 2
PGN 126464 received bus 1 with DLC 10
PGN 126464 received bus 2 with DLC 10
Method 6.1.3: PGN 126208 sent bus 1
PGN 126208 sent bus 2
PGN 126464 received bus 1 with DLC 9
PGN 126464 received bus 2 with DLC 9
Method 6.1.4: PGN 059904 sent bus 1
PGN 059904 sent bus 2
PGN 126996 received bus 1 with DLC 13
PGN 126996 received bus 2 with DLC 13

Or

Test 6.1 Message Concurrency [3.1.2] - FAILED*
Method 6.1.1: PGN 059904 sent bus 1
PGN 059904 sent bus 2
PGN 060928 received bus 1 with DLC 10
PGN 060928 received bus 2 with DLC 10
PGN 126208 sent bus 1
PGN 126208 sent bus 2
PGN 060928 received bus 2 with DLC 11
PGN 060928 received bus 1 with DLC 11

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Method 6.1.2: PGN 126208 sent bus 1
PGN 126208 sent bus 2
PGN 126464 received bus 2 with DLC 10
PGN 126464 NOT received bus 1*
Method 6.1.3: PGN 126208 sent bus 2
PGN 126208 sent bus 1
PGN 126464 received bus 1 with DLC 9
PGN 126464 received bus 2 with incorrect DLC 11*
Method 6.1.4: PGN 059904 sent bus 1
PGN 059904 sent bus 2
PGN 126996 NOT received bus 1*
PGN 126996 NOT received bus 2*

C.6.2 Fast Packet Sequence Counter [3.1.4.3] – Mandatory

This section certifies the requirement that "*The modulo 8 sequence counter shall increment each time the same PGN is transmitted from the same source address*." This is accomplished with two specific tests. The purpose of the first test (6.2.1) is to verify proper sequence number behavior with a mandatory PGN. The second test (6.2.2) examines the same requirement with all non-mandatory PGNs.

C.6.2.1 Method of measurement (Mandatory PGN):

Send an ISO request for Product Information, PGN 126996, on both buses. Record the sequence number value and adjusted DLC. Send eight more requests for PGN 126996 on both buses, and verify that the sequence number and adjusted DLC is incremented by one for each successive PGN. The sequence number must properly wrap from 7 to 0. The adjusted DLC must wrap from 15 to 9.

C.6.2.1.1 Required results:

The sequence counter should be seen to start at some value between 0 and 7, and should increment up to 7 and then should start back at 0. The adjusted DLC must start at some value between 9 and 15, should increment up to 15, and should start back at 9. The adjust DLC should be the same on both buses for a given redundant transmission.

C.6.2.2 Method of measurement (Non-Mandatory PGNs):

Perform the same test described in 6.2.1 above for every non-mandatory PGN identified during the automated PGN determination process. This will verify that other PGNs, not just the mandatory PGNs, are implemented properly, with respect to the sequence number. If the DUT does not transmit any Fast Packet PGNs, then perform this test by requesting successive Transmitted/Received PGN List Group Function, PGN 126464, Received Group Function.

C.6.2.2.1 Required results:

For every non-mandatory PGN supported by the DUT, the sequence counter should be seen to start at some value between 0 and 7, and should increment up to 7 and then should start back at 0. The adjusted DLC must start at some value between 9 and 15, should increment up to 15, and should start back at 9. The adjust DLC should be the same on both buses for a given redundant transmission.

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C.6.2.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.2 passed or failed. Provide the Certification Test major section number (such as 6.2), and each Method of measurement section number, with the PGNs that were utilized in each Method of measurement, and the values of the sequence numbers and adjusted DLCs recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.2.3.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of two methods, is passed. The second example depicts the results showing that the non-mandatory PGN measurements have failed. Notice that the sequence numbers were not incremented, but fixed at zero. Other results may be possible and should be similarly documented.

Test 6.2 Fast Packet Sequence Counter [3.1.4.3] - PASSED
Method 6.2.1: PGN 126996 bus 1 DLC 10 sequence 2
PGN 126996 bus 2 DLC 10 sequence 2
PGN 126996 bus 1 DLC 11 sequence 3
PGN 126996 bus 2 DLC 11 sequence 3
PGN 126996 bus 2 DLC 12 sequence 4
PGN 126996 bus 1 DLC 12 sequence 4
PGN 126996 bus 1 DLC 13 sequence 5
PGN 126996 bus 2 DLC 13 sequence 5
PGN 126996 bus 1 DLC 14 sequence 6
PGN 126996 bus 2 DLC 14 sequence 6
PGN 126996 bus 2 DLC 15 sequence 7
PGN 126996 bus 1 DLC 15 sequence 7
PGN 126996 bus 1 DLC 9 sequence 0
PGN 126996 bus 2 DLC 9 sequence 0
PGN 126996 bus 1 DLC 10 sequence 1
PGN 126996 bus 2 DLC 10 sequence 1
PGN 126996 bus 1 DLC 11 sequence 2
PGN 126996 bus 2 DLC 11 sequence 2
Method 6.2.2: PGN 127493 bus 1 DLC 9 sequence 0
PGN 127493 bus 2 DLC 9 sequence 0
PGN 127493 bus 1 DLC 10 sequence 1
PGN 127493 bus 2 DLC 10 sequence 1
PGN 127493 bus 1 DLC 11 sequence 2
PGN 127493 bus 2 DLC 11 sequence 2
PGN 127493 bus 1 DLC 12 sequence 3
PGN 127493 bus 2 DLC 12 sequence 3
PGN 127493 bus 1 DLC 13 sequence 4
PGN 127493 bus 2 DLC 13 sequence 4
PGN 127493 bus 1 DLC 14 sequence 5
PGN 127493 bus 2 DLC 14 sequence 5
PGN 127493 bus 1 DLC 15 sequence 6
PGN 127493 bus 2 DLC 15 sequence 6

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PGN 127493 bus 1 DLC 9 sequence 7
PGN 127493 bus 2 DLC 9 sequence 7
PGN 127493 bus 1 DLC 10 sequence 0
PGN 127493 bus 2 DLC 10 sequence 0
PGN 126998 bus 2 DLC 15 sequence 5
PGN 126998 bus 1 DLC 15 sequence 5
PGN 126998 bus 2 DLC 9 sequence 6
PGN 126998 bus 1 DLC 9 sequence 6
PGN 126998 bus 2 DLC 10 sequence 7
PGN 126998 bus 1 DLC 10 sequence 7
PGN 126998 bus 2 DLC 11 sequence 0
PGN 126998 bus 1 DLC 11 sequence 0
PGN 126998 bus 2 DLC 12 sequence 1
PGN 126998 bus 1 DLC 12 sequence 1
PGN 126998 bus 2 DLC 13 sequence 2
PGN 126998 bus 1 DLC 13 sequence 2
PGN 126998 bus 2 DLC 14 sequence 3
PGN 126998 bus 1 DLC 14 sequence 3
PGN 126998 bus 2 DLC 15 sequence 4
PGN 126998 bus 1 DLC 15 sequence 4
PGN 126998 bus 2 DLC 9 sequence 5
PGN 126998 bus 1 DLC 9 sequence 5

Or

Test 6.2 Fast Packet Sequence Counter [3.1.4.3] - FAILED*
Method 6.2.1: PGN 126996 bus 1 DLC 10 sequence 2
PGN 126996 bus 2 DLC 10 sequence 2
PGN 126996 bus 1 DLC 11 sequence 3
PGN 126996 bus 2 DLC 11 sequence 3
PGN 126996 bus 2 DLC 12 sequence 4
PGN 126996 bus 1 DLC 12 sequence 4
PGN 126996 bus 1 DLC 13 sequence 5
PGN 126996 bus 2 DLC 13 sequence 5
PGN 126996 bus 1 DLC 14 sequence 6
PGN 126996 bus 2 DLC 14 sequence 6
PGN 126996 bus 2 DLC 15 sequence 7
PGN 126996 bus 1 DLC 15 sequence 7
PGN 126996 bus 1 DLC 9 sequence 0
PGN 126996 bus 2 DLC 9 sequence 0
PGN 126996 bus 1 DLC 10 sequence 1
PGN 126996 bus 2 DLC 10 sequence 1
PGN 126996 bus 1 DLC 11 sequence 2
PGN 126996 bus 2 DLC 11 sequence 2
Method 6.2.2: PGN 127493 bus 1 DLC 9 sequence 0
PGN 127493 bus 2 DLC 9 sequence 0

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PGN 127493 bus 1 DLC 10 sequence 0*
PGN 127493 bus 2 DLC 10 sequence 0*
PGN 127493 bus 1 DLC 11 sequence 0*
PGN 127493 bus 2 DLC 11 sequence 0*
PGN 127493 bus 1 DLC 12 sequence 0*
PGN 127493 bus 2 DLC 12 sequence 0*
PGN 127493 bus 1 DLC 13 sequence 0*
PGN 127493 bus 2 DLC 13 sequence 0*
PGN 127493 bus 1 DLC 14 sequence 0*
PGN 127493 bus 2 DLC 14 sequence 0*
PGN 127493 bus 1 DLC 15 sequence 0*
PGN 127493 bus 2 DLC 15 sequence 0*
PGN 127493 bus 1 DLC 9 sequence 0*
PGN 127493 bus 2 DLC 9 sequence 0*
PGN 127493 bus 1 DLC 10 sequence 0*
PGN 127493 bus 2 DLC 10 sequence 0*
PGN 126998 bus 2 DLC 15 sequence 5
PGN 126998 bus 1 DLC 15 sequence 5
PGN 126998 bus 2 DLC 8 sequence 6*
PGN 126998 bus 1 DLC 8 sequence 6*
PGN 126998 bus 2 DLC 9 sequence 7
PGN 126998 bus 1 DLC 9 sequence 7
PGN 126998 bus 2 DLC 10 sequence 0
PGN 126998 bus 1 DLC 10 sequence 0
PGN 126998 bus 2 DLC 11 sequence 1
PGN 126998 bus 1 DLC 11 sequence 1
PGN 126998 bus 2 DLC 12 sequence 2
PGN 126998 bus 1 DLC 12 sequence 2
PGN 126998 bus 2 DLC 13 sequence 3
PGN 126998 bus 1 DLC 13 sequence 3
PGN 126998 bus 2 DLC 14 sequence 4
PGN 126998 bus 1 DLC 14 sequence 4
PGN 126998 bus 2 DLC 15 sequence 5
PGN 126998 bus 1 DLC 15 sequence 5

C.6.3 Unique Fast Packet Sequence Counter [3.1.4.4] – Mandatory

This section certifies the requirement that "*A separate sequence counter shall be utilized for each different PGN transmitted from the same source address.*" The purpose of this test is to ensure there is no linkage of the sequence number between Fast Packet PGNs.

C.6.3.1 Method of measurement (Multiple PGNs):

This test is performed with every non-mandatory PGN identified during the automated PGN determination process. If no non-mandatory, fast packet PGNs are provided by the DUT, use PGN 126464 Transmit List Group Function as the “non-mandatory” PGN. Send one or more requests for PGN 126996 on both buses, and one or more requests for each non-mandatory PGN on both buses, until both have different sequence numbers.

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Then, send alternate requests, one for PGN 126996 and one for the non-mandatory PGN on both buses, eight times. This will verify that the behavior is proper for all supported PGNs given the interleaved requests.

C.6.3.1.1 Required results:

The sequence number behavior should be independent of PGNs, that is, the series of sequence numbers from one PGN should have no relationship to those of another PGN. The adjusted DLC on each non-mandatory, response message should be in the range of 9 to 15, should increment up to 15, should wrap from 15 to 9, and should be the same on both buses for a given redundant transmission.

C.6.3.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.3 passed or failed. Provide the Certification Test major section number (such as 6.3), and each Method of measurement section number, with the PGNs that were utilized in each Method of measurement, and the values of the sequence numbers and adjusted DLCs recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.3.2.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the results showing that the Multiple PGN measurements have failed. Notice that the sequence numbers were not incremented, but fixed at zero. The third example shows that the sequence numbers were not unique across multiple PGNs. Other results may be possible and should be similarly documented.

Test 6.3 Unique Fast Packet Sequence Counter [3.1.4.4] - PASSED
Method 6.3.1: PGN 126996 bus 1 DLC 11 sequence 3
PGN 126996 bus 2 DLC 11 sequence 3
PGN 126464 bus 1 DLC 14 sequence 1
PGN 126464 bus 2 DLC 14 sequence 1
PGN 126996 bus 1 DLC 12 sequence 4
PGN 126996 bus 2 DLC 12 sequence 4
PGN 126464 bus 1 DLC 15 sequence 2
PGN 126464 bus 2 DLC 15 sequence 2
PGN 126996 bus 1 DLC 13 sequence 5
PGN 126996 bus 2 DLC 13 sequence 5
PGN 126464 bus 1 DLC 9 sequence 3
PGN 126464 bus 2 DLC 9 sequence 3
PGN 126996 bus 1 DLC 14 sequence 6
PGN 126996 bus 2 DLC 14 sequence 6
PGN 126464 bus 1 DLC 10 sequence 4
PGN 126464 bus 2 DLC 10 sequence 4
PGN 126996 bus 1 DLC 15 sequence 7
PGN 126996 bus 2 DLC 15 sequence 7
PGN 126464 bus 1 DLC 11 sequence 5
PGN 126464 bus 2 DLC 11 sequence 5
PGN 126996 bus 1 DLC 9 sequence 0
PGN 126996 bus 2 DLC 9 sequence 0

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PGN 126464 bus 1 DLC 12 sequence 6
PGN 126464 bus 2 DLC 12 sequence 6
PGN 126996 bus 1 DLC 10 sequence 1
PGN 126996 bus 2 DLC 10 sequence 1
PGN 126464 bus 1 DLC 13 sequence 7
PGN 126464 bus 2 DLC 13 sequence 7
PGN 126996 bus 1 DLC 11 sequence 2
PGN 126996 bus 2 DLC 11 sequence 2
PGN 126464 bus 1 DLC 14 sequence 0
PGN 126464 bus 2 DLC 14 sequence 0
PGN 126996 bus 1 DLC 12 sequence 3
PGN 126996 bus 2 DLC 12 sequence 3
PGN 126464 bus 1 DLC 15 sequence 1
PGN 126464 bus 2 DLC 15 sequence 1

Or

Test 6.3 Unique Fast Packet Sequence Counter [3.1.4.4] - FAILED*
Method 6.3.1: PGN 126996 bus 1 DLC 11 sequence 3
PGN 126996 bus 2 DLC 11 sequence 3
PGN 126464 bus 1 DLC 14 sequence 0
PGN 126464 bus 2 DLC 14 sequence 0
PGN 126996 bus 1 DLC 12 sequence 4
PGN 126996 bus 2 DLC 12 sequence 4
PGN 126464 bus 1 DLC 15 sequence 0*
PGN 126464 bus 2 DLC 15 sequence 0*
PGN 126996 bus 1 DLC 13 sequence 5
PGN 126996 bus 2 DLC 13 sequence 5
PGN 126464 bus 1 DLC 9 sequence 0*
PGN 126464 bus 2 DLC 9 sequence 0*
PGN 126996 bus 1 DLC 14 sequence 6
PGN 126996 bus 2 DLC 14 sequence 6
PGN 126464 bus 1 DLC 10 sequence 0*
PGN 126464 bus 2 DLC 10 sequence 0*
PGN 126996 bus 1 DLC 15 sequence 7
PGN 126996 bus 2 DLC 15 sequence 7
PGN 126464 bus 1 DLC 11 sequence 0*
PGN 126464 bus 2 DLC 11 sequence 0*
PGN 126996 bus 1 DLC 9 sequence 0
PGN 126996 bus 2 DLC 9 sequence 0
PGN 126464 bus 1 DLC 12 sequence 0*
PGN 126464 bus 2 DLC 12 sequence 0*
PGN 126996 bus 1 DLC 10 sequence 1
PGN 126996 bus 2 DLC 10 sequence 1
PGN 126464 bus 1 DLC 13 sequence 0*
PGN 126464 bus 2 DLC 13 sequence 0*

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PGN 126996 bus 1 DLC 11 sequence 2
PGN 126996 bus 2 DLC 11 sequence 2
PGN 126464 bus 1 DLC 14 sequence 0*
PGN 126464 bus 2 DLC 14 sequence 0*
PGN 126996 bus 1 DLC 12 sequence 3
PGN 126996 bus 2 DLC 12 sequence 3
PGN 126464 bus 1 DLC 15 sequence 0*
PGN 126464 bus 2 DLC 15 sequence 0*

Or

Test 6.3 Unique Fast Packet Sequence Counter [3.1.4.4] - FAILED*
Method 6.3.1: PGN 126996 bus 1 DLC 11 sequence 4
PGN 126996 bus 2 DLC 11 sequence 4
PGN 126464 bus 1 DLC 14 sequence 5
PGN 126464 bus 2 DLC 14 sequence 5
PGN 126996 bus 1 DLC 12 sequence 6*
PGN 126996 bus 2 DLC 12 sequence 6*
PGN 126464 bus 1 DLC 15 sequence 7*
PGN 126464 bus 2 DLC 15 sequence 7*
PGN 126996 bus 1 DLC 13 sequence 0*
PGN 126996 bus 2 DLC 13 sequence 0*
PGN 126464 bus 1 DLC 9 sequence 1*
PGN 126464 bus 2 DLC 9 sequence 1*
PGN 126996 bus 1 DLC 14 sequence 2*
PGN 126996 bus 2 DLC 14 sequence 2*
PGN 126464 bus 1 DLC 10 sequence 3*
PGN 126464 bus 2 DLC 10 sequence 3*
PGN 126996 bus 1 DLC 15 sequence 4*
PGN 126996 bus 2 DLC 15 sequence 4*
PGN 126464 bus 1 DLC 11 sequence 5*
PGN 126464 bus 2 DLC 11 sequence 5*
PGN 126996 bus 1 DLC 9 sequence 6*
PGN 126996 bus 2 DLC 9 sequence 6*
PGN 126464 bus 1 DLC 12 sequence 7*
PGN 126464 bus 2 DLC 12 sequence 7*
PGN 126996 bus 1 DLC 10 sequence 0*
PGN 126996 bus 2 DLC 10 sequence 0*
PGN 126464 bus 1 DLC 13 sequence 1*
PGN 126464 bus 2 DLC 13 sequence 1*
PGN 126996 bus 1 DLC 11 sequence 2*
PGN 126996 bus 2 DLC 11 sequence 2*
PGN 126464 bus 1 DLC 14 sequence 3*
PGN 126464 bus 2 DLC 14 sequence 3*
PGN 126996 bus 1 DLC 12 sequence 4*
PGN 126996 bus 2 DLC 12 sequence 4*

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PGN 126464 bus 1 DLC 15 sequence 5*
PGN 126464 bus 2 DLC 15 sequence 5*

C.6.4 Fast Packet Message Time Out [3.1.4.8] - Mandatory

This section certifies the requirement that "*A fast-packet message shall not be considered as incomplete until a time-out of 750 ms from the last received frame occurs*." The purpose of this test is to verify that the DUT will function properly on a heavily loaded network, i.e. that it does not drop messages prematurely, when dealing with significant yet acceptable delays between a given message's frames. Though this test is conducted with only a single fast packet message, the reality is that this situation will occur in cases where devices may need to handle multiple concurrent fast packet transmissions on the network.

C.6.4.1 Method of measurement (Fast Packet Message Time Out):

This test requires the ability of the test hardware and software to be able to delay or not complete sending all the frames of a message for a specified period of time. To conduct this test, a Request Group Function PGN 126208, requesting the Received PGN List Group Function PGN 126464, should be generated on both buses. Some portion of the request PGN should be sent on both buses, and then a delay of 745ms should occur, followed by the remaining frames of the request PGN on both buses.

C.6.4.1.1 Required results:

The DUT, upon receiving the completed request, should provide the Received PGN List Group Function PGN 126464 on both buses. The adjusted DLC on the response message should be the same on both buses and in the range of 9 to 15.

C.6.4.2 Reporting requirements:

Provide detailed information as to what part (Method of measurement) of test 6.4 passed or failed. Provide the Certification Test major section number (such as 6.4), and the Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement, and the response message DLCs recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.4.2.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts a failure to handle mid-request delay appropriately. Other results may be possible and should be similarly documented.

Test 6.4 Fast Packet Message Time Out [3.1.4.8] - Mandatory - PASSED
Method 6.4.1: PGN 126208 requesting PGN 126464 sent on bus 1
PGN 126208 requesting PGN 126464 sent on bus 2
PGN 126464 Received PGN List received on bus 1 DLC 13
PGN 126464 Received PGN List received on bus 2 DLC 13

Or

Test 6.4 Fast Packet Message Time Out [3.1.4.8] - Mandatory - FAILED*
Method 6.4.1: PGN 126208 requesting PGN 126464 sent on bus 1

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PGN 126208 requesting PGN 126464 sent on bus 2
PGN 126464 Received PGN List NOT received on bus 1*
PGN 126464 Received PGN List NOT received on bus 2*

C.6.5 Complex Request Message Support [3.2.1.1] – Mandatory

This section certifies the requirement that " *Support for the Complex Request Group Function message shall be implemented based on device certification level, in accordance with the minimum message implementation defined in Section 9.0.*" The purpose of this test is to verify that the DUT supports the Complex Request Message, PGN 126208. Though this group function also provides expanded acknowledgment and command capability, the scope of this particular test is limited to the Request function.

C.6.5.1 Method of measurement (Complex Request Message Support):

To conduct this test, a Request Group Function PGN 126208, requesting the Received PGN List Group Function PGN 126464, should be generated and sent to the DUT on both buses.

C.6.5.1.1 Required results:

The DUT should accept and process this request, and provide the Received PGN List Group Function PGN 126464 on both buses with the same adjusted DLC. The PGNs listed in the response should include PGN 126208 as one of the PGNs that the DUT is capable of receiving.

C.6.5.2 Reporting requirements:

Provide detailed information as to what part (Method of measurement) of test 6.5 passed or failed. Provide the Certification Test major section number (such as 6.5), and the Method of measurement section number, with the PGN number(s) that were utilized and the response messages' DLCs recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.5.2.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the case where the Transmit Group Function was provided versus the Received Group Function. The third example shows that though the Received Group Function PGN was provided, it did not list the Received Group Function PGN 126208 as a supported PGN in the PGN list. Other results may be possible and should be similarly documented.

Test 6.5 Complex Request Message Support [3.2.1.1] - Mandatory - PASSED
Method 6.5.1: PGN 126208 requesting PGN 126464 sent bus 1
PGN 126208 requesting PGN 126464 sent bus 2
PGN 126464 Received PGN List received bus 1 DLC 14
PGN 126464 Received PGN List received bus 2 DLC 14
PGN 126464 contains the 126208 PG Number.

Or

Test 6.5 Complex Request Message Support [3.2.1.1] - Mandatory - FAILED*
Method 6.5.1: PGN 126208 requesting PGN 126464 sent bus 1
PGN 126208 requesting PGN 126464 sent bus 2

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PGN 126464 Transmitted PGN List received bus 1 DLC 14*
PGN 126464 Transmitted PGN List received bus 2 DLC 14*

Or

Test 6.5 Complex Request Message Support [3.2.1.1] - Mandatory - FAILED*
Method 6.5.1: PGN 126208 requesting PGN 126464 sent bus 1
PGN 126208 requesting PGN 126464 sent bus 2
PGN 126464 Received PGN List received bus 1 DLC 14
PGN 126464 Received PGN List received bus 2 DLC 14
PGN 126464 missing the 126208 PG Number*

C.6.6 Acknowledgement of Complex Request Message [3.2.1.4] - Mandatory

This section certifies the requirement that "*All devices shall be capable of acknowledging a Complex Request message addressed to it. The acknowledgment is the data requested or the Expanded Acknowledgment message containing the appropriate error codes.*" The purpose of this test is to verify that the DUT provides the correct response to receiving a Complex Request PGN 126208. This is accomplished with five specific tests. The purpose of the first test (6.6.1) is to verify proper response when a mandatory PGN is requested. The second test (6.6.2) is designed to verify that a request for a mandatory PGN that cannot be met due to non-matching values of request fields, is properly acknowledged. The third (6.6.3) and fourth (6.6.4) tests are conducted in the same manner as the first two tests, but with all supported non-mandatory PGNs. The fifth test (6.6.5) observes the result when a request is made for a PGN not supported by the DUT.

C.6.6.1 Method of measurement (Mandatory PGN Request):

To conduct this test, a Request Group Function PGN 126208, requesting the NAME contained in PGN 060928, should be generated and sent as an addressed PGN to the DUT on both buses. First, send an ISO Request PGN 059904 requesting the ISO Address Claim PGN 060928 on both buses. Record the value of field two (Manufacturer code) returned in PGN 060928 that, according to the PGN 060928 definition, can be requested. Next, generate on both buses a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 60928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field 2 from 060928)
Field 7 - Value of first parameter = value of field 2, Manufacturer code

C.6.6.3.1 Required results:

The DUT should respond to this test with two transmissions of the ISO Address Claim PGN 060928 on both buses. The first response is to an addressed ISO request. The second response is to an addressed request containing a specific Manufacturer code parameter. Both responses should be received on both buses with the redundant copies having the same adjusted DLC in the range of 9 to 15. The second response's adjusted DLC should be the next in the DLC series relative to the DLC value returned in the first response.

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C.6.6.4 Method of measurement (Mandatory PGN Acknowledgment):

To conduct this test, a Request Group Function PGN 126208, requesting the NAME contained in PGN 060928, should be generated and sent as an addressed PGN to the DUT on both buses. First, send an ISO Request PGN 059904 requesting the ISO Address Claim PGN 060928 on both buses. Record the value of field two (manufacturer code) returned in PGN 060928 that, according to the PGN 060928 definition, can be requested. Next, generate on both buses a Request Group Function PGN 126208, with the destination field set to the network address of the DUT, and with the data fields set as follows. Note that the value of the manufacturer code returned in PGN 060928 is different than what is being requested by PGN 126208 below.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 60928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field 2 from 060928)
Field 7 - Value of first parameter = value of field 2 (Manufacturer code) minus 1

C.6.6.4.1 Required results:

The DUT should respond to this test with the ISO Address Claim PGN 060928 on both buses, followed by the Acknowledgment Group Function PGN 126208 on both buses, with the field 6 (First parameter error code) set to a value of 3 meaning "Request or command parameter field out-of-range". This indicates that the request group function was properly processed. The adjusted DLC values in the redundant responses must be the same and in the range of 9 to 15.

C.6.6.5 Method of measurement (Non-Mandatory PGN Request):

To conduct this test, a Request Group Function PGN 126208 requesting each non-mandatory PGN supported by the DUT should be generated on both buses with the destination field set to the network address of the DUT, and as shown below.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = selected non-mandatory PGN Number
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.6.6.5.1 Required results:

The required response to this test is either the non-mandatory PGN requested or the Acknowledgment PGN 126208 with the PGN error code value of 2 (PGN temporarily not available), or 3 (Access denied), or 4 (Request or Command not supported) meaning requests for this PGN are not accepted. The response message must be received on both buses with the same adjusted DLC in the range of 9 to 15. As this test walks through all non-mandatory PGNs, if multiple PGNs cause an Acknowledgment PGN response, the adjusted DLC on the response PGN must walk through the DLC range appropriately.

C.6.6.6 Method of measurement (Non-Mandatory PGN Acknowledgment):

This test is conducted using the same PGN as in test 6.6.3 above, but in a different manner. A transmission interval will be requested that will likely not be supported by the DUT. Generate a Request Group Function PGN 126208 on both buses with the destination field set to the network address of the DUT, and requesting the chosen non-mandatory PGN from test 6.6.3 but with the fields set to the following values.

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Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = non-mandatory PGN from test 6.6.3
Field 3 - Transmission Interval = 1 (This is a value of .001 seconds or 1000Hz)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.6.6.6.1 Required results:

The DUT should respond with an acknowledgment group function PGN 126208 on both buses, with a PGN error code of 0 and a Transmission interval error code of 1, 2, or 3, or with a PGN error code of 2, 3, or 4.

A PGN error code value of 0 indicates no error with the PGN number requested. A value of 3 indicates Access denied, meaning the PGN cannot be requested.

A Transmission interval error code value of 1 indicates that the specific transmit interval value requested is not supported. A value of 2 indicates that the specific transmit interval requested is less than the measurement/calculation interval of the data (PGN) requested. A value of 3 indicates that the requested PGN does not support any transmit interval assignments or changes over the network.

The adjusted DLC in the response message must be the same on both buses and in the range of 9 to 15.

C.6.6.7 Method of measurement (Unknown PGN Requested):

To conduct this test, a Request Group Function PGN 126208 requesting a non-supported PGN should be generated with the destination field set to the network address of the DUT, and sent to the DUT on both buses. Perform this test by selecting a PGN value of 130537.

C.6.6.7.1 Required results:

The DUT should respond with an acknowledgment group function PGN 126208 on both buses, with a PGN error code value of 1 indicating that the PGN requested is not supported. The adjusted DLC in the response message must be the same on both buses and in the range of 9 to 15.

C.6.6.8 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.6 passed or failed. Provide the Certification Test major section number (such as 6.6). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as the various error code values returned in the acknowledgment PGN 126208. The adjusted DLC value in the response messages will also be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.6.8.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of five methods, is passed. The second example depicts the case where methods 6.6.1, 6.6.2, 6.6.3, 6.6.4 and 6.6.5 were not completely successful. Method 6.6.1 shows that the DUT did not recognize or utilize the parameter provided with the request. Method 6.6.2 should have resulted in an acknowledgment with an error code, but instead the DUT provided the PGN. The DUT in method 6.6.3 incorrectly indicated that the PGN requested is not supported. The adjusted DLC is not the same in the 6.6.4 response. In method 6.6.5, the DUT also provided an incorrect error code. Other results may be possible and should be similarly documented.

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Test 6.6 Acknowledgement of Complex Request Message [3.2.1.4] - Mandatory - PASSED
Method 6.6.1: PGN 059904 requesting PGN 060928 sent bus 1
PGN 059904 requesting PGN 060928 sent bus 2
PGN 060928 received bus 1 DLC 12
PGN 060928 received bus 2 DLC 12
PGN 126208 request sent bus 1 with Manufacturer code ###
PGN 126208 request sent bus 2 with Manufacturer code ###
PGN 060928 received bus 1 DLC 13
PGN 060928 received bus 2 DLC 13
Method 6.6.2: PGN 059904 requesting PGN 060928 sent bus 1
PGN 059904 requesting PGN 060928 sent bus 2
PGN 060928 received bus 1 DLC 14
PGN 060928 received bus 2 DLC 14
PGN 126208 request sent bus 1 with Manufacturer code ###
PGN 126208 request sent bus 2 with Manufacturer code ###
PGN 126208 acknowledgment received bus 1 DLC 9
PGN 126208 acknowledgment received bus 2 DLC 9
First parameter error code = 3, Request Parameter out of range
Method 6.6.3: PGN 126208 requesting PGN ###### sent bus 1
PGN 126208 requesting PGN ###### sent bus 2
PGN ###### received bus 1 DLC 14
PGN ###### received bus 2 DLC 14
Method 6.6.4: PGN 126208 requesting PGN ###### with a transmission
interval of 1 (.001 seconds or 1000 hz) sent bus 1
PGN 126208 requesting PGN ###### with a transmission
interval of 1 (.001 seconds or 1000 hz) sent bus 2
PGN 126208 acknowledgment received bus 1 DLC 10
PGN 126208 acknowledgment received bus 2 DLC 10
PGN error code = 0 No error/acknowledge
Transmission interval error code = 2 Transmit interval
Less that the measurement/calculation interval
Method 6.6.5: PGN 126208 requesting PGN 130537 sent bus 2
PGN 126208 requesting PGN 130537 sent bus 1
PGN 126208 acknowledgment received bus 2 DLC 11
PGN 126208 acknowledgment received bus 1 DLC 11
PGN error code = 1, PGN not supported

Or

Test 6.6 Acknowledgement of Complex Request Message [3.2.1.4] - Mandatory - FAILED*
Method 6.6.1: PGN 059904 requesting PGN 060928 sent bus 1
PGN 059904 requesting PGN 060928 sent bus 2
PGN 060928 received bus 1 DLC 15
PGN 060928 received bus 2 DLC 15
PGN 126208 request sent bus 1 with Manufacturer code ###

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PGN 126208 request sent bus 2 with Manufacturer code ###
PGN 126208 acknowledgment received bus 1 DLC 9*
PGN 126208 acknowledgment received bus 1 DLC 9*
First parameter error code = 1, Invalid request parameter field*

Method 6.6.2: PGN 059904 requesting PGN 060928 sent bus 2
PGN 059904 requesting PGN 060928 sent bus 1
PGN 060928 received bus 1 DLC 9
PGN 060928 received bus 2 DLC 9
PGN 126208 request sent bus 2 with Manufacturer code ###
PGN 126208 request sent bus 1 with Manufacturer code ###
PGN 060928 received bus 1 DLC 10*
PGN 060928 received bus 2 DLC 10*

Method 6.6.3: PGN 126208 requesting the Transmit List PGN 126464 sent bus 1
PGN 126208 requesting the Transmit List PGN 126464 sent bus 2
PGN 126464 Transmit List received bus 1 DLC 14
PGN 126464 Transmit List received bus 2 DLC 14
PGN 126208 requesting PGN ###### sent bus 2
PGN 126208 requesting PGN ###### sent bus 1
PGN 126208 acknowledgment received bus 2 DLC 10
PGN 126208 acknowledgment received bus 1 DLC 10
PGN error code = 1, PGN not supported*

Method 6.6.4: PGN 126208 requesting PGN ###### with a transmission interval of 1 (.001 seconds or 1000 hz) sent bus 1
PGN 126208 requesting PGN ###### with a transmission interval of 1 (.001 seconds or 1000 hz) sent bus 2
PGN 126208 acknowledgment received bus 2 DLC 10
PGN 126208 acknowledgment received bus 1 DLC 11*
PGN error code = 0 No error/acknowledge
Transmission interval error code = 2 Transmit interval Less that the measurement/calculation interval

Method 6.6.5: PGN 126208 requesting PGN 130537 sent bus 2
PGN 126208 requesting PGN 130537 sent bus 1
PGN 126208 acknowledgment received bus 2 DLC 12
PGN 126208 acknowledgment received bus 1 DLC 12
PGN error code = 2, PGN temporarily not available*

C.6.7 Request Parameter Field Support [3.2.1.5] - Mandatory

This section certifies the requirement that "*When a device provides mandatory or optional query support for a PGN containing "yes" for the Request Parameter fields (see Appendix A Section A-1.) it shall at a minimum support requests specified by those fields.*" The purpose of this test is to verify that the DUT responds correctly to requests for PGN's with mandatory query support and requestable parameters.

C.6.7.1 Method of measurement (Mandatory Query Support):

To conduct this test, a NMEA 2000 Request Group Function PGN 126208 requesting the Receive/Transmit PGN Group Function PGN 126464 is sent to the DUT on both buses. The request group function should contain the following information:

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Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 of 126464)
Field 7 - Value of first parameter = 1 (Receive PGN list message)

C.6.7.1.1 Required Results:

The required result of this test is that the DUT responds with the Receive PGN Group Function containing the Receive PGN list message on both buses. The response message must have the same adjusted DLC on both buses with a value in the range of 9 to 15.

C.6.7.2 Reporting Requirements:

Provide detailed information as to what part (Methods of measurement) of test 6.7 passed or failed. Provide the Certification Test major section number (such as 6.7). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as the value of field 1 returned in PGN 126464 and the recorded response message DLC values. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.7.2.1 Example text:

The following text shows two different results. The first example shows the result when the test is passed. The second example depicts the case where the test failed. The failed example shows that the transmit list was returned rather than the requested receive list. Other results may be possible and should be similarly documented.

Test 6.7 Request Parameter Field Support [3.2.1.5] - Mandatory - PASSED
Method 6.7.1: PGN 126208 sent bus 1 requesting PGN 126464 with receive list value
PGN 126208 sent bus 2 requesting PGN 126464 with receive list value
PGN 126464 receive PGN list message received bus 1 DLC 14
PGN 126464 receive PGN list message received bus 2 DLC 14
PGN 126464 field 1 = 1

Or

Test 6.7 Request Parameter Field Support [3.2.1.5] - Mandatory - FAILED*
Method 6.7.1: PGN 126208 sent bus 2 requesting PGN 126464 with receive list value
PGN 126208 sent bus 1 requesting PGN 126464 with receive list value
PGN 126464 transmit PGN List message received bus 2 DLC 14*
PGN 126464 transmit PGN List message received bus 1 DLC 14*
PGN 126464 field 1 = 0*

C.6.8 Request for Address Claim PGN 60928 [3.2.1.7] – Mandatory

This sections certifies the requirement that "*NMEA 2000 devices shall support a Global Complex Data Request message requesting the Address Claim PGN 60928 and reply with the Address Claim parameter*

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group when the Parameter Request fields contain no information (null) or when the contents of these fields match those of the device for any of the following fields:.

Industry Group
Device Class field
System Instance field
Function field
Device Instance (upper) field
Device Instance (lower) field
Manufacturer's Code field"

The purpose of this test is to verify that the DUT responds correctly to requests for PGN 060928 with any possible combination of requested parameters. This is accomplished with 5 specific tests. The purpose of the first test (6.8.1) is to verify proper responses when a global and destination specific ISO Request is made. The second test (6.8.2) verifies that a globally addressed and destination specific NMEA 2000 Complex Request with no parameters is properly handled. The third test (6.8.3) consists of 63 Complex requests with ordered request parameters sent both globally and addressed. The fourth test (6.8.4) checks the DUTs ability to detect and respond properly to a mixture of valid and invalid request parameters. This test also contains 63 complex requests sent both globally and addressed. The fifth and final test confirms that the DUT can properly handle request parameters in any order. This test consists of 53 complex requests.

C.6.8.1 Method of measurement (ISO Requests):

To conduct this test, two different ISO Requests, PGN 059904, requesting the ISO Address Claim PGN 060928, are sent to the DUT. The first ISO Request is sent globally while the second is sent with the DUT's specific address. Both requests are sent on both buses. Save the NAME fields of the returned ISO Address Claim PGN for use in the remaining methods of measurement in section 6.8.

C.6.8.1.1 Required results:

The required results of this test are the receipt of the Address Claim PGN 060928 for each of two ISO Requests. For each request, the corresponding Address Claim PGN response must be received on both busses with the same adjusted DLC value in the range of 9 to 15. In addition, the response to the second request must have an adjusted DLC which is the next in the DLC sequence relative to the DLC in the previous response. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.8.2 Method of measurement (Complex Requests - no parameters):

To conduct this test, two different NMEA 2000 Complex Requests, PGN 126208, containing no request parameters, are sent to the DUT. Each requests the DUT to return the ISO Address Claim PGN 060928. The first Complex request is sent globally while the second is sent with the DUT's specific address. Both requests are sent on both buses.

C.6.8.2.1 Required results:

The required results of this test are the receipt of the Address Claim PGN 060928 for each of two NMEA 2000 Complex Requests. For each request, the corresponding Address Claim PGN response must be received on both busses with the same adjusted DLC value in the range of 9 to 15. In addition, the response to the second request must have an adjusted DLC which is the next in the DLC sequence relative to the DLC in the previous response. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

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C.6.8.3 Method of measurement (Requests by ordered NAME fields):

This test is conducted with PGN 126208 complex request, by requesting the ISO Address Claim PGN 060928 with specific sets of request parameters. The test utilizes 63 valid combinations of request parameters. They will first be tested with the request being sent globally, and then with the request being sent to the DUT's specific address. The request group function shall contain the following information as specified below and will be sent globally addressed on both buses. After all sixty-three combinations have been tested globally; they will be repeated with the specific address of the DUT on both buses. The first four fields of every request shall contain the following values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 060928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})

The remaining fields shall be set according to the following table. Each row in the table identifies a specific combination of request parameters. Each row in the table as such, represents a single Request PGN. The first column contains the number of request parameters in a given request PGN. The "#" column contains the field number of the request parameter as ordered in PGN 060928. The "VALUE" column contains the value of the request parameter, as recorded from the DUT in section 6.8.1 above. Note that some padding is necessary after most parameters, as they do not end on byte boundaries. The "#" and "VALUE" columns repeat for each request parameter. Empty cells in the table indicate that that field is not used in the request PGN.

Shorthand notation is used to save space, rather than write out the name of each parameter in each cell of the table. The following single characters in the table are representative of the following fields:

A = Manufacturer's Code - (field #2 of 060928) - Requires a 5 bit pad field
B = Device Instance (lower) - (field #3 of 060928) - Requires a 5 bit pad field
C = Device Instance (upper) - (field #4 of 060928) - Requires a 3 bit pad field
D = Device Function - (field #5 of 060928) – Requires no pad field
E = Device Class - (field #7 of 060928) – Requires a 1 bit pad field
F = System Instance of Bus 1- (field #8 of 060928) – Requires a 4 bit pad field
G = Industry Group - (field #9 of 060928) – Requires a 5 bit pad field

Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
1	1	2	A												
2	2	2	A	3	B										
3	3	2	A	3	B	4	C								
4	4	2	A	3	B	4	C	5	D						
5	5	2	A	3	B	4	C	5	D	7	E				
6	6	2	A	3	B	4	C	5	D	7	E	8	F		
7	7	2	A	3	B	4	C	5	D	7	E	8	F	9	G
8	2	2	A	4	C										
9	3	2	A	4	C	5	D								
10	4	2	A	4	C	5	D	7	E						

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Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
11	5	2	A	4	C	5	D	7	E	8	F				
12	6	2	A	4	C	5	D	7	E	8	F	9	G		
13	2	2	A	5	D										
14	3	2	A	5	D	7	E								
15	4	2	A	5	D	7	E	8	F						
16	5	2	A	5	D	7	E	8	F	9	G				
17	2	2	A	7	E										
18	3	2	A	7	E	8	F								
19	4	2	A	7	E	8	F	9	G						
20	2	2	A	8	F										
21	3	2	A	8	F	9	G								
22	2	2	A	9	G										
23	1	3	B												
24	2	3	B	4	C										
25	3	3	B	4	C	5	D								
26	4	3	B	4	C	5	D	7	E						
27	5	3	B	4	C	5	D	7	E	8	F				
28	6	3	B	4	C	5	D	7	E	8	F	9	G		
29	2	3	B	5	D										
30	3	3	B	5	D	7	E								
31	4	3	B	5	D	7	E	8	F						
32	5	3	B	5	D	7	E	8	F	9	G				
33	2	3	B	7	E										
34	3	3	B	7	E	8	F								
35	4	3	B	7	E	8	F	9	G						
36	2	3	B	8	F										
37	3	3	B	8	F	9	G								
38	2	3	B	9	G										
39	1	4	C												
40	2	4	C	5	D										
41	3	4	C	5	D	7	E								
42	4	4	C	5	D	7	E	8	F						
43	5	4	C	5	D	7	E	8	F	9	G				
44	2	4	C	7	E										
45	3	4	C	7	E	8	F								
46	4	4	C	7	E	8	F	9	G						
47	2	4	C	8	F										
48	3	4	C	8	F	9	G								
49	2	4	C	9	G										
50	1	5	D												

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Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
51	2	5	D	7	E												
52	3	5	D	7	E	8	F										
53	4	5	D	7	E	8	F	9	G								
54	2	5	D	8	F												
55	3	5	D	8	F	9	G										
56	2	5	D	9	G												
57	1	7	E														
58	2	7	E	8	F												
59	3	7	E	8	F	9	G										
60	2	7	E	9	G												
61	1	8	F														
62	2	8	F	9	G												
63	1	9	G														

C.6.8.3.1 Required results:

The required results of this test are the receipt of 126 Address Claim PGN 060928 for the 63 global requests and for the 63 address specific requests. Each request response must be received on both buses and have an adjusted DLC in the range of 9 to 15. The adjusted DLC in each response should be a progression in the DLC series from the last response wrapping to 9 when 15 is reached. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.8.4 Method of measurement (Requests with invalid values):

This test is conducted by requesting the ISO Address Claim PGN 060928 with specific sets of request parameters. The test utilizes 63 valid combinations of request parameters. They will first be tested with the request being sent globally, and then with the request being sent to the DUT's specific address. The request group function shall contain the following information as specified below and will be sent globally addressed on both buses. After all sixty-three combinations have been tested globally; they will be repeated with the specific address of the DUT on both buses. The first four fields of every request shall contain the following values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 060928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})

The remaining fields shall be set according to the following table. Each row in the table identifies a specific combination of request parameters. Each row in the table as such, represents a single Request PGN. The first column contains the number of request parameters in a given request PGN. The "#" column contains the field number of the request parameter as ordered in PGN 060928. The "VALUE" column contains the value of the request parameter, as recorded from the DUT in section 3.9.1 above, with a slight modification as shown below. Note that some padding is necessary after most parameters, as they do not end on byte boundaries.

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The "#" and "VALUE" columns repeat for each request parameter. Empty cells in the table indicate that that field is not used in the request PGN.

Shorthand notation is used to save space, rather than write out the name of each parameter in each cell of the table. The following single characters in the table are be representative of the following fields:

A = Manufacturer's Code - (field #2 of 060928) - Requires a 5 bit pad field
B = Device Instance (lower) - (field #3 of 060928) - Requires a 5 bit pad field
C = Device Instance (upper) - (field #4 of 060928) - Requires a 3 bit pad field
D = Device Function - (field #5 of 060928) – Requires no pad field
E = Device Class - (field #7 of 060928) – Requires a 1 bit pad field
F = System Instance of Bus 1- (field #8 of 060928) – Requires a 4 bit pad field
G = Industry Group - (field #9 of 060928) – Requires a 5 bit pad field

Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
1	1	2	A+1												
2	2	2	A+1	3	B										
3	3	2	A	3	B+1	4	C								
4	4	2	A	3	B	4	C+1	5	D						
5	5	2	A	3	B	4	C	5	D+1	7	E				
6	6	2	A	3	B	4	C	5	D	7	E+1	8	F		
7	7	2	A	3	B	4	C	5	D	7	E	8	F+2	9	G
8	2	2	A+1	4	C										
9	3	2	A+1	4	C	5	D								
10	4	2	A	4	C	5	D+1	7	E						
11	5	2	A	4	C	5	D	7	E+1	8	F				
12	6	2	A	4	C	5	D	7	E	8	F+2	9	G		
13	2	2	A+1	5	D										
14	3	2	A	5	D	7	E+1								
15	4	2	A	5	D	7	E	8	F+2						
16	5	2	A	5	D	7	E	8	F	9	G+1				
17	2	2	A+1	7	E										
18	3	2	A	7	E	8	F+2								
19	4	2	A	7	E	8	F	9	G+1						
20	2	2	A+1	8	F										
21	3	2	A	8	F	9	G+1								
22	2	2	A+1	9	G										
23	1	3	B+1												
24	2	3	B+1	4	C										
25	3	3	B	4	C+1	5	D								
26	4	3	B	4	C	5	D+1	7	E						
27	5	3	B	4	C	5	D	7	E+1	8	F				

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Test #	Number of request parameters	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE	#	VALUE
28	6	3	B	4	C	5	D	7	E	8	F+2	9	G		
29	2	3	B+1	5	D										
30	3	3	B	5	D+1	7	E								
31	4	3	B	5	D	7	E+1	8	F						
32	5	3	B	5	D	7	E	8	F+2	9	G				
33	2	3	B+1	7	E										
34	3	3	B	7	E+1	8	F								
35	4	3	B	7	E	8	F+2	9	G						
36	2	3	B+1	8	F										
37	3	3	B	8	F+2	9	G								
38	2	3	B+1	9	G										
39	1	4	C+1												
40	2	4	C+1	5	D										
41	3	4	C	5	D+1	7	E								
42	4	4	C	5	D	7	E+1	8	F						
43	5	4	C	5	D	7	E	8	F+2	9	G				
44	2	4	C+1	7	E										
45	3	4	C	7	E	8	F+2								
46	4	4	C	7	E	8	F	9	G+1						
47	2	4	C+1	8	F										
48	3	4	C	8	F+2	9	G								
49	2	4	C+1	9	G										
50	1	5	D+1												
51	2	5	D	7	E+1										
52	3	5	D	7	E	8	F+2								
53	4	5	D	7	E	8	F	9	G+1						
54	2	5	D+1	8	F										
55	3	5	D	8	F+2	9	G								
56	2	5	D	9	G+1										
57	1	7	E+1												
58	2	7	E+1	8	F										
59	3	7	E	8	F+2	9	G								
60	2	7	E+1	9	G										
61	1	8	F+2												
62	2	8	F	9	G+1										
63	1	9	G+1												

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C.6.8.4.1 Required results:

The required results of this test are the receipt of 63 Acknowledge Group Functions, PGN 126208, in response to the destination-addressed requests. There should be no response to the globally addressed requests.

Each of the PGN 126208 responses must be received on both buses with an adjusted DLC value in the range of 9 to 15. The adjusted DLC in each response should be a progression in the DLC series from the last response wrapping to 9 when 15 is reached.

C.6.8.5 Method of measurement (Requests with random ordered fields):

This test is conducted by requesting the ISO Address Claim PGN 060928 with specific sets of request parameters in random order. The test utilizes 56 randomized combinations of request parameters. They will be tested with the request being sent to the DUT's specific address on both buses. The request group function shall contain the following information as specified below. The first four fields of every request shall contain the following values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 060928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})

The remaining fields shall consist of those identified in the table of section 6.8.3 that contain two or more request fields. Each row from the table shall have the order of the fields randomized. For instance, test 3 in the table may be ordered as B, A, C or C, B, A, or A, C, B. Any random order is valid as long as the result is properly measured against the random order used in the request. The first column contains the number of request parameters in a given request PGN. The "#" column contains the field number of the request parameter as ordered in PGN 060928. The "VALUE" column contains the value of the request parameter, as recorded from the DUT in section 6.8.1 above. Note that some padding is necessary after most parameters, as they do not end on byte boundaries. The "#" and "VALUE" columns repeat for each request parameter. Empty cells in the table indicate that that field is not used in the request PGN.

C.6.8.5.1 Required results:

The required results to this test are the reception of 56 Address Claim PGN 060928. Each request response must be received on both buses and have an adjusted DLC value in the range of 9 to 15. The adjusted DLC in each response should be a progression in the DLC series from the last response wrapping to 9 when 15 is reached. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.8.6 Reporting Requirements:

Provide detailed information as to what part (Methods of measurement) of test 6.8 passed or failed. Provide the Certification Test major section number (such as 6.8). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. The adjusted DLC value of each response will be recorded. Also include information that determines pass or failure. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.8.6.1 Example text:

The following text shows two different results. The first example shows the result when the test is passed. The second example depicts the case where the test failed. The failed example shows that the DUT improperly

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rejected complex requests containing request parameters that were not in the same order as defined in the ISO Address Claim PGN 060928. Though it is not shown here, the specific details of each complex request that failed shall be provided in the text file for use by the manufacturer and also in the Certification Results file for NMEA verification. Other results may be possible and should be similarly documented.

Test 6.8 Request for Address Claim PGN 60928 [3.2.1.7] - Mandatory - PASSED
Method 6.8.1: PGN 059904 sent bus 1 requesting PGN 060928 Globally
PGN 059904 sent bus 2 requesting PGN 060928 Globally
PGN 060928 received bus 2 DLC 13
PGN 060928 received bus 1 DLC 13
PGN 059904 sent bus 2 requesting PGN 060928 Addressed
PGN 059904 sent bus 1 requesting PGN 060928 Addressed
PGN 060928 received bus 1 DLC 14
PGN 060928 received bus 2 DLC 14
Method 6.8.2: PGN 126208 sent bus 1 requesting PGN 060928 Globally
PGN 126208 sent bus 2 requesting PGN 060928 Globally
PGN 060928 received bus 1 DLC 15
PGN 060928 received bus 2 DLC 15
PGN 126208 sent bus 1 requesting PGN 060928 Addressed
PGN 126208 sent bus 2 requesting PGN 060928 Addressed
PGN 060928 received bus 1 DLC 9
PGN 060928 received bus 2 DLC 9
Method 6.8.3: PGN 126208 sent bus 1 globally requesting PGN 060928 variation 1
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 1
PGN 060928 received bus 1 DLC 10
PGN 060928 received bus 2 DLC 10
PGN 126208 sent bus 1 globally requesting PGN 060928 variation 2
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 2
PGN 060928 received bus 1 DLC 11
PGN 060928 received bus 2 DLC 11
…
PGN 126208 sent bus 1 globally requesting PGN 060928 variation 63
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 63
PGN 060928 received bus 1 DLC 9
PGN 060928 received bus 1 DLC 9

PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 1
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 1
PGN 060928 received bus 1 DLC 10
PGN 060928 received bus 2 DLC 10
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 2
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 2
PGN 060928 received bus 1 DLC 11
PGN 060928 received bus 2 DLC 11
…
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 63

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PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 63
PGN 060928 received bus 1 DLC 9
PGN 060928 received bus 1 DLC 9

Method 6.8.4: PGN 126208 sent bus 1 globally requesting PGN 060928 variation 1
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 1
No response on bus 1 within time period.
No response on bus 2 within time period.
PGN 126208 sent bus 1 globally requesting PGN 060928 variation 2
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 2
No response on bus 1 within time period.
No response on bus 2 within time period.
…
PGN 126208 sent bus 1 globally requesting PGN 060928 variation 63
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 63
No response on bus 1 within time period.
No response on bus 2 within time period.

PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 1
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 1
PGN 126208 Acknowledge Group function received bus 1 DLC 15
PGN 126208 Acknowledge Group function received bus 2 DLC 15
Parameter error code = 0x3
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 2
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 2
PGN 126208 Acknowledge Group function received bus 1 DLC 9
PGN 126208 Acknowledge Group function received bus 2 DLC 9
Parameter error code = 0x3
…
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 63
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 63
PGN 126208 Acknowledge Group function received bus 1 DLC 14
PGN 126208 Acknowledge Group function received bus 2 DLC 14
Parameter error code = 0x3

Method 6.8.5: PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 1
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 1
PGN 060928 received bus 1 DLC 10
PGN 060928 received bus 2 DLC 10
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 2
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 2
PGN 060928 received bus 1 DLC 11
PGN 060928 received bus 2 DLC 11
…
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 56
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 56
PGN 060928 received bus 1 DLC 9
PGN 060928 received bus 1 DLC 9

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Or

Test 6.8 Request for Address Claim PGN 60928 [3.2.1.7] - Mandatory - FAILED*
Method 6.8.1: PGN 059904 sent bus 1 requesting PGN 060928 Globally
PGN 059904 sent bus 2 requesting PGN 060928 Globally
PGN 060928 received bus 2 DLC 13
PGN 060928 received bus 1 DLC 13
PGN 059904 sent bus 2 requesting PGN 060928 Addressed
PGN 059904 sent bus 1 requesting PGN 060928 Addressed
PGN 060928 received bus 1 DLC 14
PGN 060928 received bus 2 DLC 14
Method 6.8.2: PGN 126208 sent bus 1 requesting PGN 060928 Globally
PGN 126208 sent bus 2 requesting PGN 060928 Globally
PGN 060928 received bus 1 DLC 15
PGN 060928 received bus 2 DLC 15
PGN 126208 sent bus 1 requesting PGN 060928 Addressed
PGN 126208 sent bus 2 requesting PGN 060928 Addressed
PGN 060928 received bus 1 DLC 9
PGN 060928 received bus 2 DLC 9
Method 6.8.3: PGN 126208 sent bus 1 globally requesting PGN 060928 variation 1
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 1
PGN 060928 received bus 1 DLC 10
PGN 060928 received bus 2 DLC 10
PGN 126208 sent bus 1 globally requesting PGN 060928 variation 2
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 2
PGN 060928 received bus 1 DLC 11
PGN 060928 received bus 2 DLC 11
…
PGN 126208 sent bus 1 globally requesting PGN 060928 variation 63
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 63
PGN 060928 received bus 1 DLC 9
PGN 060928 received bus 1 DLC 9

PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 1
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 1
PGN 060928 received bus 1 DLC 10
PGN 060928 received bus 2 DLC 10
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 2
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 2
PGN 060928 received bus 1 DLC 11
PGN 060928 received bus 2 DLC 11
…
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 63
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 63
PGN 060928 received bus 1 DLC 9

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PGN 060928 received bus 1 DLC 9

Method 6.8.4: PGN 126208 sent bus 1 globally requesting PGN 060928 variation 1
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 1
No response on bus 1 within time period.
No response on bus 2 within time period.
PGN 126208 sent bus 1 globally requesting PGN 060928 variation 2
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 2
No response on bus 1 within time period.
No response on bus 2 within time period.
…
PGN 126208 sent bus 1 globally requesting PGN 060928 variation 63
PGN 126208 sent bus 2 globally requesting PGN 060928 variation 63
No response on bus 1 within time period.
No response on bus 2 within time period.

PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 1
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 1
PGN 126208 Acknowledge Group function received bus 1 DLC 15
PGN 126208 Acknowledge Group function received bus 2 DLC 15
Parameter error code = 0x3
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 2
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 2
PGN 126208 Acknowledge Group function received bus 1 DLC 9
PGN 126208 Acknowledge Group function received bus 2 DLC 9
Parameter error code = 0x3
…
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 63
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 63
PGN 126208 Acknowledge Group function received bus 1 DLC 14
PGN 126208 Acknowledge Group function received bus 2 DLC 14
Parameter error code = 0x3

Method 6.8.5: PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 1
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 1
PGN 126208 Acknowledge Group function received bus 1 DLC 15*
PGN 126208 Acknowledge Group function received bus 1 DLC 15*
Parameter error code = 0x1*
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 2
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 2
PGN 126208 Acknowledge Group function received bus 1 DLC 15*
PGN 126208 Acknowledge Group function received bus 1 DLC 15*
Parameter error code = 0x1*
…
PGN 126208 sent bus 1 addressed requesting PGN 060928 variation 56
PGN 126208 sent bus 2 addressed requesting PGN 060928 variation 56
PGN 126208 Acknowledge Group function received bus 1 DLC 15*
PGN 126208 Acknowledge Group function received bus 1 DLC 15*

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Parameter error code = 0x1*

C.6.9 Expanded Acknowledgment Group Function Command Response [3.2.2.4] - Mandatory

This test is designed to ensure the requirement that "*The Expanded Acknowledgment Group Function message shall be transmitted in response to each Command Message indicating acknowledgment or containing the appropriate error code.*" is met by the DUT. The purpose of this test is to verify that the DUT provides the correct response to receiving a Command Group Function PGN 126208. This is accomplished with five specific tests. The purpose of the first test (6.9.1) is to verify proper response when a mandatory PGN is commanded. The second test (6.9.2) is designed to verify that a command for a mandatory PGN that cannot be met is properly acknowledged. The third (6.9.3) test is conducted by commanding a mandatory PGN with an invalid value. The fourth test commands all supported non-mandatory PGNs to change their priority. The fifth test (6.9.5) observes the result when a command is made for a PGN not supported by the DUT. It is important to understand that the DUT does not have to accept these commands, but does have to respond with the acknowledgement PGN containing appropriate responses to the commands given.

C.6.9.1 Method of measurement (Mandatory PGN Commanded with no changes):

To conduct this test, a Command Group Function PGN 126208, commanding no fields of the NAME contained in PGN 060928, should be generated and sent as an addressed PGN to the DUT on both buses. Generate a Command Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 060928
Field 3 – Priority Setting = 8 (do not change priority)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 0

C.6.9.1.1 Required results:

The DUT should respond to this test with one transmission of the Acknowledgment Group Function PGN 126208 on both buses. Valid PGN error codes are 0, 3, and 4. Valid Priority error codes are 0 and 3. Any other returned values to this test, or no response, would constitute a failure. The adjusted DLC in the response transmissions must also be the same and have a value in the range of 9 to 15.

C.6.9.2 Method of measurement (Mandatory PGN Commanded with value):

To conduct this test, a Command Group Function PGN 126208, commanding the device instance upper field of the NAME contained in PGN 060928, should be generated and sent on both buses as an addressed PGN to the DUT. The Command Group Function PGN 126208 sent will have the data fields set as follows.

Field 1 - Complex Request Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 060928
Field 3 – Priority Setting = 8 (do not change priority)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 1
Field 6 - Field number of first parameter = 4 (field 4 from 060928)
Field 7 - Value of first command parameter = 1

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The response from the DUT on both buses will either be an Acknowledgment Group Function PGN 126208 indicating the command was not accepted or an Address Claim PGN 060928 where the unit has started the address claim process with the modified NAME value.

C.6.9.2.1 Required results:

If the DUT accepts the command, the DUT must respond by starting the address claim process. This means the response message transmitted on both buses is an Address Claim PGN 60928 containing a NAME with the new device instance value. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus). If the DUT does not accept the command, the DUT should respond with one transmission on both buses of the Acknowledgment Group Function PGN 126208. Valid PGN error codes are 0, 3, and 4. Valid Parameter error codes are 4, and 5. Any other returned values to this test, or no response, would constitute a failure. Also, the adjusted DLC on the response message received on both buses must be the same and in the range of 9 to 15.

C.6.9.3 Method of measurement (Mandatory PGN Commanded with bad value):

To conduct this test, a Command Group Function PGN 126208, commanding the reserved (ISO self-configuring) bit field of the NAME contained in PGN 060928, should be generated and sent as an addressed PGN to the DUT on both buses. Generate a Command Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 060928
Field 3 – Priority Setting = 8 (do not change priority)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 1
Field 6 - Field number of first parameter = 10 (field 10 from 060928)
Field 7 - Value of first command parameter = 0 (This value is not allowed)

C.6.9.3.1 Required results:

The DUT should respond to this test with one transmission on both buses of the Acknowledgment Group Function PGN with field 6 (First parameter error code) set to a value of 1, 4, or 5. This indicates that the Command group function was properly processed and a proper acknowledge was generated. A returned first parameter error code of 0 would not necessarily mean failure. This would indicate that the device is intended to work on other networks, and may be appropriate. A returned value of 0 requires manual validation as to the capabilities and planned markets for this device. The Acknowledge Group function's abbreviated version using a PGN error code of 4 is also an acceptable response. In addition, the response message must have the same adjusted DLC on both buses and the value must be in the range of 9 to 15.

C.6.9.4 Method of measurement (Non-Mandatory PGN Commanded):

To conduct this test, generate on both buses a Command Group Function PGN 126208 commanding each non-mandatory PGN as shown below.

Field 1 - Complex Request Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = PGN chosen from PGN 126464 response
Field 3 – Priority Setting = 7 (Set priority to 7)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 0.

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If the value of the PGN and Priority error codes in the Acknowledgement response are both 0, a request for this non-mandatory PGN will be made on both buses and the priority bits in the CAN frame header examined to verify the priority change.

C.6.9.4.1 Required results:

The required response is the Acknowledgment PGN 126208 on both buses. The Acknowledgment should contain a PGN error code value of 0 (command for this PGN are accepted), 2 (temporarily not available), 3 (access denied), or 4 (request or command not supported) meaning no commands for this PGN are accepted. The Acknowledgment should contain a Priority error code value of 0 if the priority change is accepted, or a value of 1 or 3 if not accepted. The adjusted DLC must be the same in the response message on both buses and within the range of 9 to 15.

C.6.9.5 Method of measurement (Unknown PGN Commanded):

To conduct this test, a Command Group Function PGN 126208 requesting a non-supported PGN will be generated and sent addressed to the DUT on both buses. Perform this test by selecting a PGN value of 130537, and commanding the priority to “return to default”.

C.6.9.5.1 Required results:

The DUT should respond with an acknowledgment group function PGN 126208 on both buses, with a PGN error code value of 1 indicating that the PGN requested is not supported. The adjusted DLC must be the same on both buses and in the range of 9 to 15.

C.6.9.6 Method of measurement (Global command to Unknown PGN Commanded):

To conduct this test, a Command Group Function PGN 126208 requesting a non-supported PGN should be generated and sent globally to the DUT on both buses. Perform this test by selecting a PGN value of 130537, and commanding the priority to “return to default”.

C.6.9.6.1 Required results:

The DUT should not respond on either bus since Command Group Functions are only to be sent addressed. The DUT should ignore this command.

C.6.9.7 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.9 passed or failed. Provide the Certification Test major section number (such as 6.9). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as the various error code values returned in the acknowledgment PGN 126208. The adjusted DLC values in the response messages will be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.9.7.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of six methods, is passed. The second example depicts the case where 6.9.5 is not successful. In method 6.9.5 no Acknowledgment is received within the 1-second time out period. Other results may be possible and should be similarly documented.

Test 6.9 Expanded Acknowledgment Group Function Command Response [3.2.2.4] Mandatory - PASSED

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Method 6.9.1: PGN 126208 Commanding PGN 060928 sent bus 2
PGN 126208 Commanding PGN 060928 sent bus 1
PGN 126208 Acknowledgment received bus 1 DLC 12
PGN 126208 Acknowledgment received bus 2 DLC 12
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 3 (Value from field 4 of response)

Method 6.9.2: PGN 126208 Commanding PGN 060928 sent bus 1 with Device Instance Upper value
PGN 126208 Commanding PGN 060928 sent bus 2 with Device Instance Upper value
PGN 126208 acknowledgment received bus 1 DLC 13
PGN 126208 acknowledgment received bus 2 DLC 13
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Parameter error code = 4 (Value from field 6 of response)

Method 6.9.3: PGN 126208 Commanding PGN 060928 sent bus 1 with bad self-configuring value
PGN 126208 Commanding PGN 060928 sent bus 2 with bad self-configuring value
PGN 126208 Acknowledgment received bus 2 DLC 14
PGN 126208 Acknowledgment received bus 1 DLC 14
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Parameter error code = 5 (Value from field 6 of response)

Method 6.9.4: PGN 126208 commanding priority of PGN ###### sent bus 1
PGN 126208 commanding priority of PGN ###### sent bus 2
PGN 126208 acknowledgment received bus 1 DLC 15
PGN 126208 acknowledgment received bus 2 DLC 15
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = (Value of field 3 of response)
Priority error code = 3 (Value from field 4 of response)

Method 6.9.5: PGN 126208 commanding priority of PGN 130537 sent bus 1
PGN 126208 commanding priority of PGN 130537 sent bus 2
PGN 126208 acknowledgment received bus 1 DLC 9
PGN 126208 acknowledgment received bus 2 DLC 9
PGN error code = 1, PGN not supported
PGN of message being acknowledged = 130537
Priority error code = 1 (Value from field 4 of response)

Method 6.9.6: PGN 126208 commanding priority of PGN 130537 sent bus 1
PGN 126208 commanding priority of PGN 130537 sent bus 2
No PGN 126208 acknowledgment received bus 1
No PGN 126208 acknowledgment received bus 2

Or

Test 6.9 Expanded Acknowledgment Group Function Command Response [3.2.2.4] Mandatory – FAILED*
Method 6.9.1: PGN 126208 Commanding PGN 060928 sent bus 2

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PGN 126208 Commanding PGN 060928 sent bus 1
PGN 126208 Acknowledgment received bus 1 DLC 12
PGN 126208 Acknowledgment received bus 2 DLC 12
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 3 (Value from field 4 of response)

Method 6.9.2: PGN 126208 Commanding PGN 060928 sent bus 1 with Device Instance Upper value
PGN 126208 Commanding PGN 060928 sent bus 2 with Device Instance Upper value
PGN 060928 Address Claim received bus 1 DLC 15
PGN 060928 Address Claim received bus 2 DLC 15
NAME Device Instance Upper Field 4 = 1

Method 6.9.3: PGN 126208 Commanding PGN 060928 sent bus 1 with bad self-configuring value
PGN 126208 Commanding PGN 060928 sent bus 2 with bad self-configuring value
PGN 126208 Acknowledgment received bus 2 DLC 13
PGN 126208 Acknowledgment received bus 1 DLC 13
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Parameter error code = 5 (Value from field 6 of response)

Method 6.9.4: PGN 126208 commanding priority of PGN ###### sent bus 1
PGN 126208 commanding priority of PGN ###### sent bus 2
PGN 126208 acknowledgment received bus 1 DLC 14
PGN 126208 acknowledgment received bus 2 DLC 14
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = (Value of field 3 of response)
Priority error code = 3 (Value from field 4 of response)

Method 6.9.5: PGN 126208 commanding priority of PGN 130537 sent bus 1
PGN 126208 commanding priority of PGN 130537 sent bus 2
PGN 126208 acknowledgment NOT received bus 1*
PGN 126208 acknowledgment NOT received bus 2*

Method 6.9.6: PGN 126208 commanding priority of PGN 130537 sent bus 1
PGN 126208 commanding priority of PGN 130537 sent bus 2
No PGN 126208 acknowledgment received bus 1
No PGN 126208 acknowledgment received bus 2

C.6.10 Expanded Acknowledgment Group Function [3.2.3.1] - Mandatory

This validates the requirement that "*All NMEA 2000 devices shall support the Expanded Acknowledgment Group Function message*."

C.6.10.1 Method of measurement (Previous Tests):

This is accomplished by examining the results of two previous tests, test 6.6 and test 6.9.

C.6.10.1.1 Required results:

If both tests 6.6 and 6.9 are passed then this requirement is satisfied. If either tests 6.6 or 6.9 fail, then this test is also failed.

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C.6.10.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.10 passed or failed. Provide the Certification Test major section number (such as 6.10). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.10.2.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of one method based upon previous tests, is passed. The second example depicts the case where one of the previous tests is not successful. Other results may be possible and should be similarly documented.

Test 6.10 Expanded Acknowledgment Group Function [3.2.3.1] – Mandatory – PASSED
Method 6.10.1 Test 6.6 Passed
Test 6.9 Passed

Or

Test 6.10 Expanded Acknowledgment Group Function [3.2.3.1] – Mandatory – FAILED*
Method 6.10.1 Test 6.6 Passed
Test 6.9 Failed*

C.6.11 Global Use of Expanded Acknowledgment Message [3.2.3.3] - Mandatory

This section certifies the requirement that " *The Expanded Acknowledgment message shall not be used in response to Global requests.*" The purpose of this test is to verify that the DUT does not respond to global requests with the Acknowledgement Group Function PGN 126208. This is accomplished with three specific tests. The purpose of the first test (6.11.1) is to check behavior when a global request is made for a mandatory PGN that cannot be accommodated. The second test (6.11.2) examines the same requirement with all supported non-mandatory PGNs. The third test (6.11.3) requests a PGN not supported by the DUT.

C.6.11.1 Method of measurement (Mandatory PGN with invalid field request):

This method tests the ability of the DUT to properly discriminate between an addressed request that would require a response instead of the data requested, and global request of the same. A globally addressed Request Group Function PGN 126208 is generated and sent to the DUT on both buses. The fields of the PGN are as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 3 (Out of range value)

C.6.11.1.1 Required results:

The DUT should not be able to accommodate this request. The value of the parameter supplied does not match the range of field 1 in PGN 126464. Instead of sending the PGN requested, the DUT should ignore the request

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and send nothing in response. If no response is received from the DUT for a period of 2 seconds (1 second is the maximum timeout period) on both buses, then the DUT has passed the test.

C.6.11.2 Method of measurement (Non-mandatory PGN with invalid transmission interval):

This method tests the ability of the DUT to properly discriminate between an addressed request that would require a response instead of the data requested, and global request of the same. To conduct this test, generate a globally addressed Request Group Function, PGN 126208, requesting each non-mandatory PGN as shown below on both buses.

The fields of the PGN are as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = selected non-mandatory PG Number
Field 3 - Transmission Interval = 4294967294 (According to the uint32 Type Definition, this value indicates "Out of range" and should not be used.)
Field 4 - Transmission interval offset = 0 (0000_{16} or no offset so transmit immediately)
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.6.11.2.1 Required results:

The DUT should not be able to accommodate this request. The value of the transmission interval is so large that it should not be accepted. Instead of sending the PGN requested, the DUT should ignore the request and send nothing in response. If no response is received from the DUT for a period of 2 seconds (1 second is the maximum timeout period) on both buses, then the DUT has passed the test.

C.6.11.3 Method of measurement (Non-supported PGN):

This method tests the ability of the DUT to properly discriminate between an addressed request that would require a response instead of the data requested, and global request of the same. To conduct this test, generate a globally addressed Request Group Function, PGN 126208, requesting the non-supported PGN 130537 as shown below on both buses.

The fields of the PGN are as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 130537
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.6.11.3.1 Required results:

The DUT should not be able to accommodate this request. The request for a non-supported PGN should not be accepted. Instead of sending the PGN requested, or acknowledging the request, the DUT should ignore the request and send nothing in response. If no response is received from the DUT for a period of 2 seconds (1 second is the maximum timeout period) on both buses, then the DUT has passed the test.

C.6.11.4 Reporting Requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.11 passed or failed. Provide the Certification Test major section number (such as 6.11). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of an acknowledge Group function or some or the PGN requested. The adjusted DLC of any

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received response messages will also be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.11.4.1 Example Text:

The following text shows two different results. The first example shows the result when the test, consisting of three methods is passed. The second example depicts the case where the second test (6.11.2) results in the DUT responding with an Acknowledgement Group Function containing an error code for requested transmission interval. Other results may be possible and should be similarly documented.

Test 6.11 Global Use Of Expanded Acknowledgment Message [3.2.3.3] - Mandatory – PASSED

Method 6.11.1: PGN 126208 Request Group Function requesting PGN 126464 with out of range request parameter sent globally addressed bus 1.
PGN 126208 Request Group Function requesting PGN 126464 with out of range request parameter sent globally addressed bus 2.
No response from DUT for a 2 second period bus 1.
No response from DUT for a 2 second period bus 2.

Method 6.11.2: PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN with extreme transmission interval sent globally addressed bus 1.
PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN with extreme transmission interval sent globally addressed bus 2.
No response from DUT for a 2 second period bus 1.
No response from DUT for a 2 second period bus 2.

Method 6.11.3: PGN 126208 Request Group Function requesting PGN 130537 sent globally addressed bus 1.
PGN 126208 Request Group Function requesting PGN 130537 sent globally addressed bus 2.
No response from DUT for a 2 second period bus 2.
No response from DUT for a 2 second period bus 1

Or

Test 6.11 Global Use Of Expanded Acknowledgment Message [3.2.3.3] - Mandatory – FAILED*

Method 6.11.1: PGN 126208 Request Group Function requesting PGN 126464 with out of range request parameter sent globally addressed bus 1.
PGN 126208 Request Group Function requesting PGN 126464 with out of range request parameter sent globally addressed bus 2.
No response from DUT for a 2 second period bus 1.
No response from DUT for a 2 second period bus 2.

Method 6.11.2: PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN with extreme transmission interval sent globally addressed bus 1.
PGN 126208 Request Group Function requesting PGN ###### non-mandatory PGN with extreme transmission interval sent globally addressed bus 2.
PGN 126208 Acknowledgement received bus 1 DLC 9*
PGN 126208 Acknowledgement received bus 2 DLC 9*
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = ######
Transmission Interval code = 1 (Value from field 4 of response)

Method 6.11.3: PGN 126208 Request Group Function requesting PGN 130537 sent globally addressed bus 1.

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PGN 126208 Request Group Function requesting PGN 130537 sent globally addressed bus 2.
No response from DUT for a 2 second period bus 2.
No response from DUT for a 2 second period bus 1

C.6.12 Expanded Acknowledgment Message Timing [3.2.3.4] - Mandatory

This section certifies the requirement that "*An Expanded Acknowledgment message, when required, shall be transmitted within 1 second of receiving the request or command.* " The purpose of this test is to measure the response time of the DUT when provided with destination addressed requests and commands, requiring an Acknowledgement Group Function PGN 126208 response. This is accomplished with four specific tests. The purpose of the first test (6.12.1) is to check behavior when a destination specific addressed request is made for a mandatory PGN that cannot be accommodated. The second test (6.12.2) examines the same requirement with every non-mandatory PGN supported by the DUT. The third test (6.12.3) requests a PGN not supported by the DUT. The fourth test (6.12.4) commands a mandatory PGN 060928 with an out of range parameter.

C.6.12.1 Method of measurement (Mandatory PGN with invalid field request):

This method tests the ability of the DUT to respond within the maximum allowed time of 1 second to a request that requires a response instead of the data requested. A destination specific addressed Request Group Function PGN 126208 is generated and sent to the DUT on both buses. The fields of the PGN are as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 3 (Out of range value)

C.6.12.1.1 Required results:

The DUT should not be able to accommodate this request. The value of the parameter supplied does not match the range of field 1 in PGN 126464. Instead of sending the PGN requested, the DUT should send an acknowledgement Group Function on both buses with a PGN Error code of 0 (No error/Acknowledge), Transmission Interval/Priority error code of 0 (No error/Acknowledge), and First Parameter error code of 3 (Request or Command parameter out-of-range). If a response is received on both buses from the DUT within a period of 1 second (1 second is the maximum timeout period) with the appropriate error codes, then the DUT has passed the test. An elapsed time from request to response greater than 1 second on either bus constitutes failure.

C.6.12.2 Method of measurement (Non-mandatory PGN with invalid transmission interval):

This method tests the ability of the DUT to respond within the maximum allowed time of 1 second to a request that requires a response instead of the data requested. To conduct this test, a destination specific addressed Request Group Function, PGN 126208 is generated and transmitted on both buses for each supported non-mandatory PGN as shown below.
The fields of the PGN are as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = selected non-mandatory PG Number

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Field 3 - Transmission Interval = 4294967294 ($FFFFFFFE_{16}$) (According to the uint32 Type Definition, this value indicates “Out of range” and should not be used.)
Field 4 - Transmission interval offset = 0 (0000_{16} or no offset so transmit immediately)
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.6.12.2.1 Required results:

The DUT should not be able to accommodate these requests. The value of the transmission interval is so large that each request should not be accepted. Instead of sending the PGNs requested, the DUT should send an acknowledgement Group Function with the appropriate error codes on both buses. Acceptable error code combinations are PGN error code of 0 (No error), 2 (Temporarily not available) or 3 (Access denied) with a Transmission Interval/Priority error code of 1 (Transmit interval not supported), 2 (Transmit interval less than measurement/calculation interval) or 3 (Access denied). A PGN error code of 4 (Request not supported) with a Transmission Interval/Priority error code of 4 (Request not supported) is also acceptable. The Number of requested parameters in the response must be set to 0xFF (indicating no fields to follow). If a response is received from the DUT within a period of 1 second (1 second is the maximum timeout period) on both buses, then the DUT has passed the test. An elapsed time from request to response greater than 1 second on either bus constitutes failure. Also, the adjusted DLC on the redundant responses must be the same value in the range of 9 to 15. The adjusted DLC value should walk through the DLC series wrapping from 15 to 9 appropriately.

C.6.12.7 Method of measurement (Non-supported PGN):

This method tests the ability of the DUT to respond within the maximum allowed time of 1 second to a request that requires a response instead of the data requested. To conduct this test, generate a destination specific addressed Request Group Function, PGN 126208, requesting the non-supported PGN 130537 on both buses as shown below.
The fields of the PGN are as follows:
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 130537
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 0

C.6.12.7.1 Required results:

The DUT should not be able to accommodate this request. The request for a non-supported PGN should not be accepted. Instead of sending the PGN requested, the DUT should send an acknowledgement Group Function on both buses with the PGN Error code of 1 (PGN not supported); Transmission Interval/Priority error code of 0 (No error), 1 (Transmit Interval not supported) or 3 (Access denied); and the Number of requested parameters set to 0xFF (indicating no fields to follow). If a response is received from the DUT within a period of 1 second (1 second is the maximum timeout period) on both buses, then the DUT has passed the test. An elapsed time from request to response greater than 1 second on either bus constitutes failure. Also, the adjusted DLC on the redundant responses must be the same value in the range of 9 to 15. The adjusted DLC value should walk through the DLC series wrapping from 15 to 9 appropriately.

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C.6.12.8 Method of measurement (Command Mandatory PGN with out of range parameter):

This method tests the ability of the DUT to respond with in the maximum allowed time of 1 second to a command that requires a response instead of the data requested. To conduct this test, generate and transmit on both buses a destination specific addressed Command Group Function, PGN 126208, commanding the Reserved (Self-configuring bit) bit field in the NAME to 0 as shown below.
The fields of the PGN are as follows:

Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 060928
Field 3 – Priority Setting = 8 (8_{16} meaning do not change)
Field 4 - Reserved = 15 (F_{16})
Field 5 - Number of Pairs of Commanded Parameters to follow = 1
Field 6 – Field Number of First Parameter = 10 (Reserved – ISO Self-Configurable)
Field 7 – Value of First Parameter = 0 (out of range value).

C.6.12.8.1 Required results:

The DUT should not be able to accommodate this command. The command to change the reserved bit field of the NAME entity should not be accepted. The DUT should send an acknowledgement Group Function with the appropriate error codes on both buses. Acceptable values are a PGN Error code of 0 (No error) or 3 (Access denied), the Priority Error code set to 0 (No error) or 3 (Access denied), the Number of parameters set to 1, and the First Parameter Error code set to 1 (Invalid commanded parameter field), 4 (Access denied) or 5 (Command Group Function not supported). A PGN Error of 4 (Command not supported), a Priority Error of 4 (Command not supported) and the Number of Parameters set to 0xFF is also an acceptable response. If a response is received from the DUT within a period of 1 second (1 second is the maximum timeout period) on both buses, then the DUT has passed the test. An elapsed time from request to response greater than 1 second on either bus constitutes failure. Also, the adjusted DLC on the redundant responses must be the same value in the range of 9 to 15. The adjusted DLC value should walk through the DLC series wrapping from 15 to 9 appropriately.

C.6.12.9 Reporting Requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.12 passed or failed. Provide the Certification Test major section number (such as 6.12). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of an acknowledge Group function or some or the PGN requested. The adjusted DLC in response messages must be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.12.9.1 Example Text:

The following text shows two different results. The first example shows the result when the test, consisting of four methods is passed. The second example depicts a failure to receive a response on bus 2 in test 6.12.1, a failure to receive a response on either bus for test 6.12.2 and 6.12.3, and a bad parameter error code in the response for test 6.12.4. Other results may be possible and should be similarly documented.

Test 6.12 Expanded Acknowledgment Message Timing [3.2.3.4] - Mandatory – PASSED
Method 6.12.1: PGN 126208 Request Group Function sent addressed bus 1 requesting PGN 126464 with out of range request parameter

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PGN 126208 Request Group Function sent addressed bus 2 requesting PGN 126464 with out of range request parameter
PGN 126208 Acknowledgement received within 1 second bus 1 DLC 12
PGN 126208 Acknowledgement received within 1 second bus 2 DLC 12
PGN of message being acknowledged = 126464
PGN error code = 0 (Value from field 3 of response)
Transmission Interval/Priority error code = 0 (Value from field 4 of response)
Number of requested parameters = 1 (Value from field 5 of response)
Parameter error code = 3 (Value from field 6 of response)

Method 6.12.2: PGN 126208 Request Group Function sent addressed bus 1 requesting PGN ###### non-mandatory PGN with extreme transmission interval
PGN 126208 Request Group Function sent addressed bus 2 requesting PGN ###### non-mandatory PGN with extreme transmission interval
PGN 126208 Acknowledgement received within 1 second bus 1 DLC 13
PGN 126208 Acknowledgement received within 1 second bus 2 DLC 13
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = ######
Transmission Interval code = 1 (Value from field 4 of response)
Number of requested parameters = 0xFF (Value from field 5 of response)

Method 6.12.3: PGN 126208 Request Group Function sent bus 1 requesting PGN 130537
PGN 126208 Request Group Function sent bus 2 requesting PGN 130537
PGN 126208 Acknowledgement received within 1 second bus 1 DLC 14
PGN 126208 Acknowledgement received within 1 second bus 2 DLC 14
PGN error code = 1 (Value from field 2 of response)
PGN of message being acknowledged = ######
Transmission Interval code = 1 (Value from field 4 of response)
Number of requested parameters = 0xFF (Value from field 5 of response)

Method 6.12.4: PGN 126208 Command Group Function commanding PGN 060928 sent bus 1
PGN 126208 Command Group Function commanding PGN 060928 sent bus 2
PGN 126208 acknowledgment received bus 1 DLC 15
PGN 126208 acknowledgment received bus 2 DLC 15
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Number of commanded parameters = 1 (Value from field 5 of response)
Parameter error code = 1 (Value from field 6 of response)

Or

Test 6.12 Expanded Acknowledgment Message Timing [3.2.3.4] - Mandatory – FAILED*

Method 6.12.1: PGN 126208 Request Group Function sent addressed bus 1 requesting PGN 126464 with out of range request parameter
PGN 126208 Request Group Function sent addressed bus 2 requesting PGN 126464 with out of range request parameter
PGN 126208 Acknowledgement received with 1 second bus 1 DLC 12
No response from DUT for a 2 second period bus 2*

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PGN of message being acknowledged = 126464
PGN error code = 0 (Value from field 3 of response)
Transmission Interval/Priority error code = 0 (Value from field 4 of response)
Number of requested parameters = 1 (Value from field 5 of response)
Parameter error code = 3 (Value from field 6 of response)

Method 6.12.2: PGN 126208 Request Group Function sent addressed bus 1 requesting PGN ###### non-mandatory PGN with extreme transmission interval
PGN 126208 Request Group Function sent addressed bus 2 requesting PGN ###### non-mandatory PGN with extreme transmission interval
No response from DUT for a 2 second period bus 1*
No response from DUT for a 2 second period bus 2*

Method 6.12.3: PGN 126208 Request Group Function sent bus 1 requesting PGN 130537
PGN 126208 Request Group Function sent bus 2 requesting PGN 130537
No response from DUT for a 2 second period bus 1*
No response from DUT for a 2 second period bus 2*

Method 6.12.4: PGN 126208 Command Group Function commanding PGN 060928 sent bus 1
PGN 126208 Command Group Function commanding PGN 060928 sent bus 2
PGN 126208 acknowledgment received bus 1 DLC 15
PGN 126208 acknowledgment received bus 2 DLC 15
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 060928
Priority error code = 0 (Value from field 4 of response)
Number of commanded parameters = 1 (Value from field 5 of response)
Parameter error code = 3 (Value from field 6 of response)*

C.6.13 ISO 11783-3 3.2.2 Reserved Bit ® [Table 3-1] – Mandatory

This test is intended to confirm that the DUT implements the reserved bit in bit-position five of the can frame. The requirement is this "*ISO 11783-3 Section 3.2.2 Reserved Bit, specifies that all messages shall set the reserved bit (bit number 5) in the CAN ID field to zero on transmit.*" This will be accomplished with two tests. The first test (6.13.1) will check the 5th bit of the CAN frame from the transmitted PGN List function, and then from some non-mandatory PGN.

C.6.13.1 Method of measurement (Reserved bit in Transmitted List PGN):

The purpose of this method is to verify that the reserved bit is properly set in CAN frames of mandatory messages. To conduct this test a Request PGN 126208 will be will be transmitted on both buses requesting PGN 126464, the Transmit list group function. The reserved bit, bit number 5 in the CAN frame will be examined, and the list of PGNs contained in the response will be used in method 6.13.2. The fields of the request PGN should be filled out as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 0 (Transmitted list))

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C.6.13.1.1 Required results:

The DUT should respond to the request with PGN 126464 Transmit list group function on both buses. Bit 5 of the CAN frame must be set to zero. If bit 5 is not zero in the response message received on either bus, the DUT fails this test. The adjusted DLC of the response message received on each bus must be the same and in the range of 9 to 15.

C.6.13.2 Method of measurement (Reserved bit in non-mandatory PGN):

The purpose of this method is to verify that the reserved bit is properly set in CAN frames of all non-mandatory PGNs supported by the DUT. To conduct this test, a Request PGN 126208 asking for the immediate transmission of each supported non-mandatory PGN is generated and transmitted on both buses. The reserved bit, bit number 5 in the CAN frame will be examined. The fields of the request PGN should be filled out as follows:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = ###### (random non-mandatory PGN)
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})

C.6.13.2.1 Required results:

The DUT should respond on both buses to each request with either the PGN requested or a NACK. Any other response and the DUT fails the test. The DUT response message will be checked to ensure bit 5 of the CAN frame is set to zero. If bit 5 is not zero, the DUT fails this test. Also, the adjusted DLC of each redundant response message is checked to ensure they have the same value in the range of 9 to 15. If multiple request cause the DUT to return a NACK, the adjusted DLC in the NACK response must progress through the DLC series wrapping appropriately from 15 to 9.

C.6.13.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.13 passed or failed. Provide the Certification Test major section number (such as 6.13). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs, the value of the reserved bit from bit position 5 in the CAN frame, and the adjusted DLC value in all responses. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.13.3.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of two methods is passed. The second example depicts the case where the second test (6.13.2) fails because the value of the reserved bit is a one instead of a zero. Other results may be possible and should be similarly documented.

Test 6.13 ISO 11783-3 3.2.2 Reserved Bit ® [Table 3-1] - Mandatory – PASSED
Method 6.13.1: PGN 126208 Request Group Function requesting PGN 126464 sent bus 1
PGN 126208 Request Group Function requesting PGN 126464 sent bus 2
PGN 126464 Transmit PGN List received bus 1 DLC 14
PGN 126464 Transmit PGN List received bus 2 DLC 14
Reserved bit from bit position 5 of CAN frame = 0

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Method 6.13.2: PGN 126208 Request Group Function requesting PGN ###### sent bus 2
PGN 126208 Request Group Function requesting PGN ###### sent bus 1
PGN ###### received bus 1 DLC 11
PGN ###### received bus 2 DLC 11
Reserved bit from bit position 5 of CAN frame = 0

Or

Test 6.13 ISO 11783-3 3.2.2 Reserved Bit ® [Table 3-1] - Mandatory – FAILED*
Method 6.13.1: PGN 126208 Request Group Function requesting PGN 126464 sent bus 1
PGN 126208 Request Group Function requesting PGN 126464 sent bus 2
PGN 126464 Transmit PGN List received bus 1 DLC 14
PGN 126464 Transmit PGN List received bus 2 DLC 14
Reserved bit from bit position 5 of CAN frame = 0
Method 6.13.2: PGN 126208 Request Group Function requesting PGN ###### sent bus 2
PGN 126208 Request Group Function requesting PGN ###### sent bus 1
PGN ###### received bus 1 DLC 11
PGN ###### received bus 2 DLC 11
Reserved bit from bit position 5 of CAN frame = 1*

C.6.14 ISO 11783-3 Data field reserved bytes [Table 3-1] - Mandatory

This test is to certify that the "fill-bits" placed in a multi-packet protocol message are all binary ones. The requirement is as follows: "*ISO 11783-3 Section 3.10.4 Transport Protocol - Data Transfer Messages (PGN 60160) requires that unused Data field bytes in that last packet of the multi-packet message shall be filled with* $FF_{16.}$" This test is conducted by sending an ISO multi-packet formatted Request PGN 126208 to the DUT requesting PGN 126464.

C.6.14.1 Method of measurement (ISO Multi-Packet request):

This method requires that PGN 126208 is sent on both buses using the ISO 11783 multi-packet Broadcast Message protocol, requiring the use of the Broadcast Announce Message (BAM). This method will send an ISO 060160 PGN containing the Request Group Function, PGN 126208, requesting the Transmitted PGN List Group Function, PGN 126464. Generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126464
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = 0 (Transmit PGN List Message)

C.6.14.1.1 Required results

The DUT should respond on both buses with PGN 126464 contained within ISO 060160 in accordance with the multi-packet Broadcast transfer protocol. The last packet on both buses needs to be examined, specifically, any non-data "fill-bits". The fill bits must be all binary ones. If the ISO 060160 PGN has no fill bits in the last

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packet, then this method should be repeated with a request for the Receive PGN List Group Function. Also, the adjusted DLC values of the response messages must be the same on both buses and in the range of 9 to 15.

C.6.14.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.14 passed or failed. Provide the Certification Test major section number (such as 6.14). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs, the value of the "fill-bits" from the last packet of the ISO 060160 Multi-packet PGN, and the adjusted DLC value in the DUT response messages. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.14.2.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of one method is passed. The second example depicts the case where the test fails because the values of the "fill-bits" are binary zeros instead of binary ones. Other results may be possible and should be similarly documented.

Test 6.14 ISO 11783-3 Data field reserved bytes [Table 3-1] - Mandatory – PASSED
Method 6.14.1: PGN 126208 Request Group Function requesting PGN 126464 sent with BAM protocol bus 1
PGN 126208 Request Group Function requesting PGN 126464 sent with BAM protocol bus 2
PGN 126464 Transmit PGN List received in BAM protocol bus 1 DLC 13
PGN 126464 Transmit PGN List received in BAM protocol bus 2 DLC 13 Fill-bits in last multi-packet frame = all binary ones

Or

Test 6.14 ISO 11783-3 Data field reserved bytes [Table 3-1] - Mandatory – FAILED*
Method 6.14.1: PGN 126208 Request Group Function requesting PGN 126464 sent with BAM protocol bus 1
PGN 126208 Request Group Function requesting PGN 126464 sent with BAM protocol bus 2
PGN 126464 Transmit PGN List received in BAM protocol bus 1 DLC 13
PGN 126464 Transmit PGN List received in BAM protocol bus 2 DLC 13 Fill-bits in last multi-packet frame = all binary zeros*

C.6.15 ISO 11783-3 Annex C, NACK [Table 3-1] – Mandatory

This test certifies that the ISO NACK is used as required. The requirement being tested is as follows: "*ISO 11783-3 Annex C, NACK - If the Parameter Group Number in a Command or Request is not recognized by the destination, it shall reply with the NACK message*." This is achieved with one test for proper NACK response to an ISO request.

C.6.15.1 Method of measurement (NACK - ISO Request):

This method is accomplished by sending a destination specific addressed ISO PGN 059904 Request message to the DUT on both buses. The ISO request will contain an invalid NMEA 2000 PGN number of 130537.

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C.6.15.1.1 Required results:

The DUT, upon reception of the ISO Request PGN 059904 should send back an ISO 059392 Acknowledgement indicating a negative acknowledgement on both buses. Anything else or no response constitutes failure of this test. The adjusted DLC used in the response must be the same value on both buses and be in the range of 9 to 15.

C.6.15.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.15 passed or failed. Provide the Certification Test major section number (such as 6.15). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs, the field values from the ISO Acknowledgment PGN, and the adjusted DLC value in the response messages. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.15.2.1 Example text:

The following text shows to different results. The first example shows what the text might look like if the test is passed. The second example illustrates one possibility of a failed test, where the acknowledgement contains the PGN of the ISO Request message instead of the requested PGN. Other results may be possible and should be similarly documented.

Test 6.15 ISO 11783-3 Annex C, NACK [Table 3-1] - Mandatory – PASSED
Method 6.15.1: PGN 059904 ISO Request requesting PGN 130537 sent bus 2
PGN 059904 ISO Request requesting PGN 130537 sent bus 1
PGN 059392 received bus 1 DLC 12
PGN 059392 received bus 2 DLC 12
Control Byte = 1
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537

Or

Test 6.15 ISO 11783-3 Annex C, NACK [Table 3-1] - Mandatory – FAILED*
Method 6.15.1: PGN 059904 ISO Request requesting PGN 130537 sent bus 2
PGN 059904 ISO Request requesting PGN 130537 sent bus 1
PGN 059392 received bus 1 DLC 12
PGN 059392 received bus 2 DLC 12
Control Byte = 1
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 059904*

C.6.16 NAME Support [8.3] – Mandatory

This test is designed to verify the following requirement: "*NMEA 2000 devices shall implement the NAME field, and field dependencies of the Address Claim message as defined by PGN 60928 in Appendix B database*

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file "2000vx.x.mdb"." Each field of the NAME entity will be examined to determine if they contain appropriate values. One test is used to conduct examination of the NAME fields.

C.6.16.1 Method of measurement (Examine NAME fields):

This method requests the address claim PGN 060928 containing the NAME fields from the DUT on both buses. Fields 1, 2, 5, and 7 of PGN 060928 are not tested but simply reported. These fields shall be reviewed and verified by NMEA prior to granting NMEA 2000 certification. The remaining fields (fields 3, 4, 6, 8, 9, 10) are examined and compared to the requirements to ensure proper values.

C.6.16.1.1 Required results:

The DUT must respond on both buses with the Address Claim message PGN 060928 containing identical data values except for field 8. Upon receipt of PGN 060928, the NAME fields will be examined as follows. Field 6 should always be a zero, field 9 should be set to four, and field 10 should always be set to one. Field 8 should have a value of 0 on one bus and a value of 1 on the other bus. Other values will constitute failure. Also, the adjusted DLC value in the response message must be the same on both buses and in the range of 9 to 15.

C.6.16.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.16 passed or failed. Provide the Certification Test major section number (such as 6.16). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs, all field values from PGN 060928, and the adjusted DLC value in all response messages. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.16.2.1 Example text:

The text in this section shows two of many possible outcomes. The first example text shows what a valid set of parameters may look like. The second example shows that the DUT failed the test due to an incorrect Industry Group (field 9) value. Other results may be possible and should be similarly documented.

Test 6.16 NAME Support [8.3] - Mandatory – PASSED
Method 6.16.1: PGN 059904 ISO Request requesting PGN 060928 sent bus 1
PGN 059904 ISO Request requesting PGN 060928 sent bus 2
PGN 060928 received bus 1 DLC 15
Field 1 (Unique number) = 5570 => Requires Verification
Field 2 (Manufacturer Code) = 1850 => Requires Verification
Field 3 (Device Instance Lower) = 0
Field 4 (Device Instance Upper) = 0
Field 5 (Device Function) = 150 => Requires Verification
Field 6 (Reserved) = 0
Field 7 (Device Class) = 60 => Requires Verification
Field 8 (System Instance) = 0
Field 9 (Industry Group) = 4
Field 10 (Reserved) = 1
PGN 060928 received bus 2 DLC 15
Field 1 (Unique number) = 5570 => Requires Verification
Field 2 (Manufacturer Code) = 1850 => Requires Verification

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Field 3 (Device Instance Lower) = 0
Field 4 (Device Instance Upper) = 0
Field 5 (Device Function) = 150 => Requires Verification
Field 6 (Reserved) = 0
Field 7 (Device Class) = 60 => Requires Verification
Field 8 (System Instance) = 1
Field 9 (Industry Group) = 4
Field 10 (Reserved) = 1

Or

Test 6.16 NAME Support [8.3] - Mandatory – FAILED*
Method 6.16.1: PGN 059904 ISO Request requesting PGN 060928 sent bus 1
PGN 059904 ISO Request requesting PGN 060928 sent bus 2
PGN 060928 received bus 1 DLC 15
Field 1 (Unique number) = 63953 => Requires Verification
Field 2 (Manufacturer Code) = 1851 => Requires Verification
Field 3 (Device Instance Lower) = 0
Field 4 (Device Instance Upper) = 0
Field 5 (Device Function) = 140 => Requires Verification
Field 6 (Reserved) = 0
Field 7 (Device Class) = 50 => Requires Verification
Field 8 (System Instance) = 0
Field 9 (Industry Group) = 1*
Field 10 (Reserved) = 1
PGN 060928 received bus 2 DLC 15
Field 1 (Unique number) = 63953 => Requires Verification
Field 2 (Manufacturer Code) = 1851 => Requires Verification
Field 3 (Device Instance Lower) = 0
Field 4 (Device Instance Upper) = 0
Field 5 (Device Function) = 140 => Requires Verification
Field 6 (Reserved) = 0
Field 7 (Device Class) = 50 => Requires Verification
Field 8 (System Instance) = 1
Field 9 (Industry Group) = 1*
Field 10 (Reserved) = 1

C.6.17 Product Information [8.3.2.1] – Mandatory

This test certifies the following requirement. "*All NMEA 2000 devices (except the Listen Only Device) shall support the Product Information message containing at a minimum the "NMEA 2000 Version Supported", "Manufacturer's Product Code", "NMEA 2000 Certification Level", and "Load Equivalency Number".*"

C.6.17.1 Method of measurement (Version field request parameter):

This method validates that the DUT can distinguish between valid and invalid requests for the Product Information PGN 126996 based upon the NMEA 2000 Version Supported request parameter. To conduct this

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test, an ISO request is sent on both buses requesting PGN 126996. The contents of the NMEA 2000 Database Version Number are checked and saved. Additionally, the Manufacturers Product Code, NMEA 2000 Certification Level, and Load Equivalency fields are recorded and checked. A Request PGN 126208 requesting PGN 126996 with an invalid Version number is then sent addressed to the DUT on both buses. An acknowledgement should be received from the DUT on both buses. Then a Request PGN requesting PGN 126996 with the NMEA version supported by the DUT is sent on both buses.

C.6.17.1.1 Required results:

The DUT should, upon reception of the ISO request, provide PGN 126996 on both buses. Upon receipt of the request PGN 126208, the DUT should send back on both buses an acknowledgement indicating invalid request parameter. The DUT, upon reception of a second valid request for PGN 126996 should respond by providing PGN 126996 on both buses. Also, the adjusted DLC value must be the same on redundant responses on both buses and in the range of 9 to 15. The adjusted DLC value in the PGN 126996 responses should progress through the DLC series, wrapping from 15 to 9 appropriately.

C.6.17.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.17 passed or failed. Provide the Certification Test major section number (such as 6.17). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs, applicable field values from the PGNs 126208 and 126996, and adjusted DLC values on all response messages. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.17.2.1 Example text:

The text in this section shows two of many possible outcomes. The first example text shows the results when the DUT passes the test. The second example shows that the DUT failed the test by not properly detecting a non-matching NMEA version request parameter value. Other results may be possible and should be similarly documented.

Test 6.17 Product Information [8.3.2.1] - Mandatory – PASSED
Method 6.17.1: PGN 059904 ISO Request requesting PGN 126996 sent bus 1
PGN 059904 ISO Request requesting PGN 126996 sent bus 2
PGN 126996 received bus 1 DLC 15
PGN 126996 received bus 2 DLC 15
Field 1 (NMEA 2000 Database Version) = 1000 => Requires Verification
Field 2 (NMEA Manufacturer's Product Code = 127
Field 7 (Certification Level) = 2
Field 8 (Load Equivalency) = 1 => Requires Verification
PGN 126208 Request destination addressed containing the following parameters sent bus 2
PGN 126208 Request destination addressed containing the following parameters sent bus 1
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)

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Field 7 - Value of first parameter = #### (NMEA 2000 Version + 1 from 126996 returned above)
PGN 126208 acknowledgment received bus 1 DLC 12
PGN 126208 acknowledgment received bus 2 DLC 12
PGN error code = 0 (Value from field 2 of response)
PGN of message being acknowledged = 126996
Priority error code = 0 (Value from field 4 of response)
Parameter error code = 3 (Value from field 6 of response)
PGN 126208 Request destination addressed containing the following parameters sent bus 1
PGN 126208 Request destination addressed containing the following parameters sent bus 2
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = #### (NMEA 2000 Version from 126996 returned above)
PGN 126996 received bus 2 DLC 9
PGN 126996 received bus 1 DLC 9

Or

Test 6.17 Product Information [8.3.2.1] - Mandatory – FAILED*
Method 6.17.1: PGN 059904 ISO Request requesting PGN 126996 sent bus 1
PGN 059904 ISO Request requesting PGN 126996 sent bus 2
PGN 126996 received bus 1 DLC 15
PGN 126996 received bus 2 DLC 15
Field 1 (NMEA 2000 Database Version) = 1000 => Requires Verification
Field 2 (NMEA Manufacturer's Product Code = 127
Field 7 (Certification Level) = 2
Field 8 (Load Equivalency) = 1 => Requires Verification
PGN 126208 Request destination addressed containing the following parameters sent bus 2
PGN 126208 Request destination addressed containing the following parameters sent bus 1
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = #### (NMEA 2000 Version + 1 from 126996 returned above)
PGN 126996 received bus 1 DLC 9*
PGN 126996 received bus 2 DLC 9*
PGN 126208 Request destination addressed containing the following parameters sent bus 1
PGN 126208 Request destination addressed containing the following parameters sent bus 2
Field 1 - Complex Request Group Function Code = 0 (Request group function)

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Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 65535 ($FFFF_{16}$)
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 1 (field 1 from 126464)
Field 7 - Value of first parameter = #### (NMEA 2000 Version from 126996 returned above)
PGN 126996 received bus 2 DLC 10
PGN 126996 received bus 1 DLC 10

C.6.18 Transmitted PGN List [8.4.1] - Mandatory

This section certifies the following requirement. *"All NMEA 2000 devices shall support the Transmitted PGN List message containing all transmitted messages supported by the device, including Data Link and Network Management messages. The Transmitted PGN List message shall be implemented as described in Appendix B."*

C.6.18.1 Method of measurement (Transmitted PGN List):

This method requires that the DUT provide the Transmitted PGN list. The List is then examined to ensure that all the mandatory PGNs are included. Non-mandatory PGNs require manual verification. To conduct this test, a destination specific addressed request group function PGN 126208 is sent to the DUT on both buses asking for PGN 126464 Transmit PGN List.

C.6.18.1.1 Required results:

The DUT must respond to the request with the transmitted PGN list on both buses containing all the required PGNs. This includes those that are mandatory and all other PGNs that are produced by the DUT. The entire list of PGNs in Table 9-1 of the Standard, with the exception of PGN 059904, is required to be in this list. Also, the adjusted DLC value in the response must be the same on both buses and within the range of 9 to 15.

C.6.18.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.18 passed or failed. Provide the Certification Test major section number (such as 6.18). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs, all the field values (PGNS) from PGN 126464, and the adjusted DLC in the response messages. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.18.2.1 Example text:

The example text shows two possible outcomes. The first example shows results that pass the test. The second example shows a deficiency in the mandatory PGNs listed. Note that PGN 126464 is missing. Other results may be possible and should be similarly documented.

Test 6.18 Transmitted PGN List [8.3.1] – Mandatory – PASSED
Method 6.18.1: PGN 126208 Request for Transmitted PGN List PGN 126464 sent bus 1
PGN 126208 Request for Transmitted PGN List PGN 126464 sent bus 2
PGN 126464 Transmitted PGN List received bus 1 DLC 13
PGN 126464 Transmitted PGN List received bus 2 DLC 13

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Mandatory PGNS Listed: 126208, 059392, 060160, 060416, 060928, 129996, 126464
Non-Mandatory PGNs Listed: 126998, 129025, 129026, 129029, 129044, 129045, 129284 => Require Verification

Or

Test 6.18 Transmitted PGN List [8.3.1] – Mandatory – FAILED*
Method 6.18.1: PGN 126208 Request for Transmitted PGN List PGN 126464 sent bus 1
PGN 126208 Request for Transmitted PGN List PGN 126464 sent bus 2
PGN 126464 Transmitted PGN List received bus 1 DLC 13
PGN 126464 Transmitted PGN List received bus 2 DLC 13
Mandatory PGNS Listed: 126208, 059392, 060160, 060416, 060928, 129996*
Non-Mandatory PGNs Listed: 126998, 129025, 129026, 129029, 129044, 129045, 129284 => Require Verification

C.6.19 Received PGN List [8.4.2] - Mandatory

This section certifies the following requirement, "*All NMEA 2000 devices shall support the Received PGN List message containing all received messages supported by the device, including Data Link and Network Management messages. The Received PGN List message shall be implemented as described in Appendix B.*"

C.6.19.1 Method of measurement (Transmitted PGN List):

This method requires that the DUT provide the Received PGN list. The List is then examined to ensure that all the mandatory PGNs are included. Non-mandatory PGNs require manual verification. To conduct this test, a destination specific addressed request group function PGN 126208 is sent to the DUT on both buses asking for PGN 126464 Received PGN List.

C.6.19.1.1 Required results:

The DUT must respond to the request on both buses with the received PGN list containing all the required PGNs. This includes those that are mandatory and all other PGNs that are accepted and processed by the DUT. The entire list of PGNs in Table 9-1 of the Standard is required to be in this list, with the exception of PGNs 126464 and 126996. Also, the adjusted DLC in the response message must be the same on both buses and in the range of 9 to 15.

C.6.19.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.19 passed or failed. Provide the Certification Test major section number (such as 6.19). Provide the Method of measurement section number, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs, all field values (PGNS) from PGN 126464, and the adjusted DLC values in the response messages. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.19.2.1 Example text:

The example text shows two possible outcomes. The first example shows results that pass the test. The second example shows a deficiency in the mandatory PGNs listed. Note that PGN 060160 and 060416 are missing. Other results may be possible and should be similarly documented.

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Test 6.19 Received PGN List [8.3.2] - Mandatory – PASSED
Method 6.19.1: PGN 126208 Request for Received PGN List PGN 126464 sent bus 2
PGN 126208 Request for Received PGN List PGN 126464 sent bus 1
PGN 126464 Received PGN List received bus 1
PGN 126464 Received PGN List received bus 2
Mandatory PGNS Listed: 126208, 059392, 059904, 060160, 060416, 060928
Non-Mandatory PGNs Listed: 129538 => Requires Verification

Or

Test 6.19 Received PGN List [8.3.2] - Mandatory – FAILED*
Method 6.19.1: PGN 126208 Request for Received PGN List PGN 126464 sent bus 2
PGN 126208 Request for Received PGN List PGN 126464 sent bus 1
PGN 126464 Received PGN List received bus 1
PGN 126464 Received PGN List received bus 2
Mandatory PGNS Listed: 126208, 059392, 059904, 060928*
Non-Mandatory PGNs Listed: 129538 => Requires Verification

C.6.20 Address Claim Test – Mandatory

This section certifies one of the most basic and important characteristics of the NMEA 2000 standard, Automatic Self-configuring Address Claiming, at power up. There are several very important aspects to dynamic address claiming and assignment that must be adhered to in order to ensure interoperability with other NMEA 2000 devices. This requires multiple tests. Each of the following tests has one specific goal or criteria to be satisfied.

C.6.20.1 Method of measurement (Transmit Address Claim on Startup):

The specific item of interest is that the DUT, upon power up and normal (self test or initialization) delays, sends out an address claim message, PGN 060928, on both buses. In order to conduct this test, the DUT will be powered up after the test equipment is ready to listen. The address claim PGN will be recorded, and the requested address will be documented.

C.6.20.1.1 Required results:

The DUT must provide the address claim PGN 060928 on both buses shortly after power up. The Address must be a valid NMEA 2000 address in the range of 0 to 251. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 constitutes failure of this test. The adjusted DLC value in the address claim PGN must be the same on both buses and in the range of 9 to 15. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.20.2 Method of measurement (Response to lower priority address contention):

This test verifies that the DUT properly handles address contention with devices that have a lower priority (higher value NAME means lower priority). After the DUT is powered up, and issues an address claim message on both buses, an address claim message is sent (broadcast by the test equipment) on both buses with a higher value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be

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done within the 250ms period after the DUT sends its address claim message on both buses; so that contention is achieved before the DUT actually claims an address.

C.6.20.2.1 Required results:

The DUT should receive and process the lower priority address claim message. The DUT should respond to the contention by retransmitting the same address claim message on both buses again and then beginning the 250ms waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT changes to some other address as a result of the contention or does not respond to the contention, then it has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 constitutes failure of this test. The adjusted DUT in the address claim response must be the same on both buses, in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.20.3 Method of measurement (Response to higher priority Address contention):

This test verifies that the DUT properly handles address contention with devices that have a higher priority (lower value NAME means higher priority). After the DUT is powered up, and issues an address claim message on both buses, an address claim message is sent (broadcast by the test equipment) on both buses with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done within the 250ms period after the DUT sends its address claim message on both buses; so that contention is achieved before the DUT actually claims an address.

C.6.20.3.1 Required results:

The DUT should receive and process the higher priority address claim message. The DUT should respond to the contention by choosing a different address and transmitting a new address claim message on both buses, and then begin a 250ms waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT keeps the original contested address in the second group of address claim messages, or does not respond to the contention, then the DUT has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 also constitutes failure of this test. The adjusted DLC in the address claim response must be the same on both buses, in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.20.4 Method of measurement (Response to no addresses available):

This test verifies two items. The first item is that the DUT attempts to claim a valid address across the full range of valid address space, from 0 to 251. Second, that the DUT indicates when it is unable to claim an address. The sequences of events to conduct this test are as follows. After the DUT is powered up and issues an address claim message on both buses, an address claim message is sent (broadcast by the test equipment) on both buses with a lower value NAME than the DUT's NAME and for the same address desired by the DUT. This must be done within the 250ms period after the DUT sends its address claim messages; so that contention is achieved before the DUT actually claims an address. This should cause the DUT to try for another address on both buses. The test equipment should respond with another address claim on both buses with the DUT's address, and with a lower value NAME than the DUT. This should cause the DUT to try for another address.

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This cycle should be repeated until the DUT sends a can not claim address message on both buses, or begins repeating address claim attempts with a previously attempted address.

C.6.20.4.1 Required results:

The DUT should, over the course of this test, send 252 address claim messages followed by a can not claim address message, which is an address claim message with the source address set to 254. These messages should be transmitted on both buses with the same address and adjusted DLC value. The DLC value must be in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The DUT will fail this test if it does not try all 252 addresses (order does not matter), begins repeating claims for the same address, or does not send a can not claim address message on both buses. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.20.5 Method of measurement (Address Claim Timeout):

This test is to determine if the DUT waits for 250ms after sending an address claim message for a new address before assuming that it owns the address. The DUT, upon sending its address claim message on both buses, should not respond to anything other than another address claim for the same address during the 250ms period. The test equipment should, upon receipt of the DUTs initial address claim, wait for 100ms and then send a request for the address claim PGN globally on both buses. The test equipment should not see any response from the DUT until at least an additional 150ms has elapsed.

C.6.20.5.1 Required results:

The DUT should not transmit any PGNs for a period of 250ms after transmitting the address claim PGN on both buses. Any transmission from the DUT during the 250ms constitutes failure.

C.6.20.6 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.20 passed or failed. Provide the Certification Test major section number (such as 6.20). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as Source Address, DLC values, and NAME values. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.20.6.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes all five methods. The second example shows a variety of problems. The first problem is that the address used in 6.20.1 is not valid. The problem in 6.20.2 is that the DUT changed addresses in response to a device with a lower priority, when it should not have. The problem in 6.20.3 is that the DUT did not change its address in response to a device with a higher priority. In 6.20.4 the DUT did not attempt to claim all possible addresses before sending the can not claim message. The last problem, shown in 6.20.5, is that the DUT responded to a "request for address claim message" before the 250ms period was complete. Other results may be possible and should be similarly documented.

Test 6.20 Address Claim Test – Mandatory – PASSED
Method 6.20.1: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)

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PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)

Method 6.20.2: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
PGN 060928 sent bus 2 with DLC = 14
Address = 12
NAME = 1391168962055494861 2 ($C1104000AAAAA004_{16}$)
PGN 060928 sent bus 1 with DLC = 14
Address = 12
NAME = 1383963202651702067 6 ($C0104000AAAAA004_{16}$)
PGN 060928 received bus 1 with DLC = 10
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 12
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)

Method 6.20.3: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
PGN 060928 sent bus 1 with DLC = 15
Address = 12
NAME = 1383963202651702067 4 ($C0104000AAAAA002_{16}$)
PGN 060928 sent bus 2 with DLC = 15
Address = 12
NAME = 1391168962055494861 0 ($C1104000AAAAA002_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 13
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
PGN 060928 received bus 1 with DLC = 10
Address = 13
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)

Method 6.20.4: PGN 060928 received bus 1 with DLC = 9
PGN 060928 received bus 2 with DLC = 9
252 Addresses tried bus 1
252 Addresses tried bus 2
PGN 060928 received bus 1 with Address = 254 and DLC = 13
PGN 060928 received bus 2 with Address = 254 and DLC = 13

Method 6.20.5: PGN 060928 received bus 1 with DLC = 9

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PGN 060928 received bus 2 with DLC = 9
100 ms elapsed
PGN 059904 Requesting 060928 sent Globally bus 1
PGN 059904 Requesting 060928 sent Globally bus 2
150 ms elapsed

Or

Test 6.20 Address Claim Test – Mandatory – FAILED*
Method 6.20.1: PGN 060928 received bus 1 with DLC = 9
Address = 252*
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 252*
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
Method 6.20.2: PGN 060928 received bus 2 with DLC = 11
Address = 12
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
PGN 060928 received bus 1 with DLC = 11
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 sent bus 1 with DLC = 13
Address = 12
NAME = 1383963202651702067 6 ($C0104000AAAAA004_{16}$)
PGN 060928 sent bus 2 with DLC = 13
Address = 12
NAME = 1391168962055494861 2 ($C1104000AAAAA004_{16}$)
PGN 060928 received bus 1 with DLC = 12
Address = 13*
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 12
Address = 13*
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
Method 6.20.3: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
PGN 060928 sent bus 1 with DLC = 14
Address = 12
NAME = 1383963202651702067 4 ($C0104000AAAAA002_{16}$)
PGN 060928 sent bus 2 with DLC = 14
Address = 12
NAME = 1391168962055494861 0 ($C1104000AAAAA002_{16}$)
PGN 060928 received bus 1 with DLC = 10

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Address = 12*
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 12*
NAME = 13911689620554948611 (C1104000AAAAA003$_{16}$)

Method 6.20.4: PGN 060928 received on bus 1 with DLC = 9
PGN 060928 received on bus 2 with DLC = 9
Addresses missed on bus 1: 0,1,2,3,4,5,6,7,8,9,10,11*
Addresses missed on bus 2: 0,1,2,3,4,5,6,7,8,9,10,11*
PGN 060928 received on bus 1 with Address = 254
PGN 060928 received on bus 2 with Address = 254

Method 6.20.5: PGN 060928 received bus 1 with DLC = 9
PGN 060928 received bus 2 with DLC = 9
100 ms elapsed
PGN 059904 Requesting 060928 sent Globally bus 1
PGN 059904 Requesting 060928 sent Globally bus 2
PGN 060928 received on bus 1 with DLC = 10 with Address = 254*
PGN 060928 received on bus 2 with DLC = 10 with Address = 254*

C.6.21 Minimum Message Implementation [Table 9-1] – Mandatory

This section is intended to certify, "*The messages listed in Table 9-1 are all required to be implemented*." Many previous test sections utilize these messages, both to and from the DUT, to conduct the certification. The verification of this requirement, when applicable, is based upon those other tests as described in the methods below.

C.6.21.1 Method of measurement (PGN 126208):

This method is dependant upon tests 6.5 and 6.6. If test 6.5 is passed, the DUT has implemented the Request Group Function PGN 126208. If test 6.6 is passed the DUT has implemented the Acknowledge Group Function PGN 126208. Implementation of the Command Group Function PGN 126208 is optional.

C.6.21.1.1 Required Results:

Check that tests 6.5 and 6.6 have been passed. Failure of either test constitutes failure of this method.

C.6.21.2 Method of measurement (PGN 059392):

This method is dependant upon test 6.15. If test 6.15 is passed, the DUT has implemented the ISO Acknowledgement PGN 059392.

C.6.21.2.1 Required Results:

Check that test 6.15 has passed. Failure of test 6.15 constitutes failure of this method.

C.6.21.3 Method of measurement (PGN 059904):

There are no previous tests that rely solely upon this PGN, so an additional test is needed. The test equipment will send a PGN 059904 on both buses to the DUT, both destination and globally addressed, requesting the Address Claim PGN 060928. A proper response to this test will verify that the DUT has implemented the ISO Request.

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C.6.21.3.1 Required Results:

The DUT should respond with two PGN 060928 messages, one for the destination specific, and one for the global ISO request 059904. Each response message should be received on both buses with matching address values and adjusted DLC values. The DLC value must be in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.21.4 Method of measurement (BAM - PGNs 060416 and 060160):

This test verifies both reception and transmission of the ISO transport protocol using the broadcast announcement message and multi-packet data transfer. If successfully passed, it shows that the DUT can receive, process, and transmit data using PGNs 060416 and 060160 globally addressed. The test is conducted in the following manner. First, send a PGN 060416 BAM with the Request PGN 126208 to the DUT on both buses. Then send PGN 060160 on both buses containing the contents of PGN 126208, requesting the PGN 126464 Transmit PGN List. All communications shall be in accordance with the timing requirements of the transport protocol.

C.6.21.4.1 Required Results:

The DUT should receive and decode the 060416. It should then expect to receive the Request PGN 126208 via the Transport Protocol Data Transfer PGN 060160. The DUT should then respond with a Transport Protocol Broadcast Announcement Message, PGN 060416, on both buses indicating that PGN 126464 is to follow. The DUT should then send the content of PGN 126464 on both buses, within the Transport Protocol Data Transfer PGN 060160. Failure to provide the data using the transport protocol or exceeding any of the timing requirements constitutes failure for this method. The adjusted DLC value must be the same in all PGN 060416 and 060160 frames and the value must be in the range of 9 to 15.

C.6.21.5 Method of measurement (RTS/CTS - PGNs 060416 and 060160):

This test verifies both reception and transmission of the ISO transport protocol using the destination addressed RTS/CTS connection management and multi-packet data transfer capabilities. If successfully passed, it shows that the DUT can receive, process, and transmit data using PGNs 060416 and 060160 via RTS/CTS connection management with destination specific addresses. The test is conducted by sending a PGN 060416 RTS with the Request PGN 126208 to the DUT on both buses. When the DUT responds with a CTS message, the contents of PGN 126208 requesting PGN 126464 Received PGN List will be sent on the bus the CTS was received on. The data transfer session should take place in accordance with ISO 11783-3 Annex B, Transport Protocol Transport Sequences. The DUT should process and respond to the request on both buses by providing PGN 126464 Received PGN List via the transport protocol using the RTS/CTS connection management. All communications shall be in accordance with the timing requirements of the transport protocol.

C.6.21.5.1 Required Results:

The DUT should accept and process the PGN 126208 request for PGN 126464 Received PGN List via the ISO Transport Protocol using RTS/CTS connection management on both buses. The DUT should then Initiate a RTS/CTS managed session with the test equipment on both buses, and transfer the contents of PGN 126464 in the established transport protocol session. Failure to provide the data using the transport protocol or exceeding any of the timing requirements constitutes failure for this method. The DLC value on all CTS frames and EOF

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frames must have a DLC value of 8. The DLC value for each RTS and data frames transmitted by DUT in response to the request must have the same adjusted DLC in the range of 9 to 15.

C.6.21.6 Method of measurement (PGN 060928):

This method is dependant upon tests 6.8 and 6.16. If tests 6.8 and 6.16 are passed, the DUT has implemented the ISO Address Claim PGN 060928 correctly.

C.6.21.6.1 Required Results:

Check that tests 6.8 and 6.16 have been passed. Failure of test 6.8 or 6.16 constitutes failure of this method.

C.6.21.7 Method of measurement (PGN 126996):

This method is dependant upon test 6.17. If test 6.17 is passed, the DUT has implemented the Product Information PGN 126996 with the NMEA 2000 Version request parameter correctly.

C.6.21.7.1 Required Results:

Check that test 6.17 has been passed. Failure of test 6.17 constitutes failure of this method.

C.6.21.8 Method of measurement (PGN 126464):

This method is dependant upon tests 6.18 and 6.19. If tests 6.18 and 6.19 are passed, the DUT has implemented the Transmit/Receive PGN List function, PGN 126464, correctly.

C.6.21.8.1 Required Results:

Check that tests 6.18 and 6.19 have been passed. Failure of test 6.18 or 6.19 constitutes failure of this method.

C.6.21.9 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.21 passed or failed. Provide the Certification Test major section number (such as 6.21). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. Contents of the connection management PGNs 060416 should be reported, along with all PGNs involved in the transport protocol sessions and associated DLC values. The report is intended to provide a complete record of the data transfer sequences that take place during this test. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.21.9.1 Example text:

The text in this section is merely an outline of two of many possible outcomes. The first example text outlines the results when the DUT passes the test. The second example illustrates that the DUT failed the test by not properly responding to the Transport Protocol Broadcast Announce Message in test 6.21.4. Other results may be possible and should be similarly documented.

Test 6.21 Minimum Message Implementation [Table 9-1] – Mandatory – PASSED
Method 6.21.1: Method 6.5 Passed
Method 6.6 Passed
Method 6.21.2: Method 6.15 Passed
Method 6.21.3: PGN 059904 ISO Request requesting PGN 060928 sent destination specific on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent destination specific on bus 2.
PGN 060928 received on bus 2.

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PGN 060928 received on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent globally on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent globally on bus 2.
PGN 060928 received on bus 1.
PGN 060928 received on bus 2.

Method 6.21.4: PGN 060416 BAM sent on bus 1
List contents of 060416….
PGN 060416 BAM sent on bus 2
List contents of 060416….
PGNs 060160 sent….containing PGN 126208 request for PGN 126464 on bus 1
PGNs 060160 sent….containing PGN 126208 request for PGN 126464 on bus 2
PGN 060416 received from DUT on bus 1
List and validate contents including DLC….
PGN 060416 received from DUT on bus 2
List and validate contents including DLC….
PGNs 060160 received…..containing PGN 126464 Transmit PGN List on bus 1.
PGNs 060160 received…..containing PGN 126464 Transmit PGN List on bus 2.

Method 6.21.5: PGN 060416 RTS sent on bus 1… list contents
PGN 060416 RTS sent on bus 2… list contents
List PGNs sent and received on bus 1.
List PGNs sent and received on bus 2.
List contents including DLC of PGNs 060416 sent on bus 2
List contents including DLC of PGNs 060416 sent on bus 1
List PGNs sent and received on bus 1
List PGNs sent and received on bus 2
List contents including DLC of PGNs 060416 received on bus 1
List contents including DLC of PGNs 060416 received on bus 2
PGNs 060160 received …. Containing PGN 126464 Received PGN List on bus 1
PGNs 060160 received …. Containing PGN 126464 Received PGN List on bus 2

Method 6.21.6: Method 6.8 Passed
Method 6.16 Passed

Method 6.21.7: Method 6.17 Passed

Method 6.21.8: Method 6.18 Passed
Method 6.19 Passed

Or

Test 6.21 Minimum Message Implementation [Table 9-1] – Mandatory – FAILED*

Method 6.21.1: Method 6.5 Passed
Method 6.6 Passed

Method 6.21.2: Method 6.15 Passed

Method 6.21.3: PGN 059904 ISO Request requesting PGN 060928 sent destination specific on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent destination specific on bus 2.
PGN 060928 received on bus 1.
PGN 060928 received on bus 2.
PGN 059904 ISO Request requesting PGN 060928 sent globally on bus 1.

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PGN 059904 ISO Request requesting PGN 060928 sent globally on bus 2.
PGN 060928 received on bus 1.
PGN 060928 received on bus 2.

Method 6.21.4: PGN 060416 BAM sent on bus 1
List contents of 060416….
PGN 060416 BAM sent on bus 2
List contents of 060416….
PGNs 060160 sent….containing PGN 126208 request for PGN 126464 on bus 1
PGNs 060160 sent….containing PGN 126208 request for PGN 126464 on bus 2
No response from DUT on bus 1*
No response from DUT on bus 2*

Method 6.21.5: PGN 060416 RTS sent on bus 1… list contents
PGN 060416 RTS sent on bus 2… list contents
List PGNs sent and received on bus 1.
List PGNs sent and received on bus 2.
List contents including DLC of PGNs 060416 sent on bus 1
List contents including DLC of PGNs 060416 sent on bus 2
List PGNs sent and received on bus 1
List PGNs sent and received on bus 2
List contents including DLC of PGNs 060416 received on bus 1
List contents including DLC of PGNs 060416 received on bus 2
PGNs 060160 received …. Containing PGN 126464 Received PGN List on bus 1
PGNs 060160 received …. Containing PGN 126464 Received PGN List on bus 2

Method 6.21.6: Method 6.8 Passed
Method 6.16 Passed

Method 3.21.7: Method 6.17 Passed

Method 3.21.8: Method 6.18 Passed
Method 6.19 Passed

C.6.22 Address Bumping – Mandatory

This test is intended to verify that once a DUT has claimed an address and begun normal operation, that the DUT properly handles having its address bumped (taken) by a higher priority device entering the network. Tests with lower priority claims will be made as well as testing the case when no addresses are available. Though these tests are similar to those for power-up address claim, these tests help verify that the DUT properly executes the address claim functions after having been operating in a normal state.

C.6.22.1 Method of measurement (Response to lower priority address contention):

This test verifies that the DUT properly handles address contention with devices that have a lower priority (higher value NAME means lower priority). This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) on both buses with a higher value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done after it has been confirmed that the DUT is in normal operation with a claimed address.

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C.6.22.1.1 Required results:

The DUT should receive and process the lower priority address claim message. The DUT should respond to the contention by retransmitting the same address claim message again on both buses. All PGNs are documented. If the DUT changes to some other address as a result of the contention, or does not respond to the contention on both buses, then the unit has failed this test. The adjusted DUT in the address claim response must be the same on both buses, in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.22.2 Method of measurement (Response to higher priority Address contention):

This test verifies that the DUT properly handles address contention with devices that have a higher priority (lower value NAME means higher priority). This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) on both buses with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done after it has been confirmed that the DUT is in normal operation with a claimed address.

C.6.22.2.1 Required results:

The DUT should receive and process the higher priority address claim message. The DUT should respond to the contention by choosing a different address and transmitting a new address claim message on both buses, and then begin the 250ms-waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT keeps the original contested address in its second address claim message, or does not respond to the contention, then it has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 also constitutes failure of this test. . The adjusted DLC in the address claim response must be the same on both buses, in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.22.3 Method of measurement (Response to no addresses available):

This test verifies two items. The first item is that the DUT attempts to claim a valid address across the full range of valid address space, from 0 to 251. Second, that the DUT indicates that it is unable to claim an address. The sequences of events to conduct this test are as follows. This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is then sent (broadcast by the test equipment) on both buses with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This should cause the DUT to try for another address on both buses. The test equipment should respond on both buses with another address claim with the same address as desired by the DUT, and with a lower value NAME than the DUT. This should cause the DUT to try for another address on both buses. This cycle should be repeated until the DUT sends a can not claim address message on both buses, or begins repeating address claim attempts with a previously attempted address.

C.6.22.3.1 Required results:

The DUT should, over the course of this test, send 252 address claim messages followed by a can not claim address message, which is an address claim message with the source address set to 254. These messages should be transmitted on both buses with the same address and adjusted DLC value. The DLC value must be

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in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The DUT will fail this test if it does not try all 252 addresses (order does not matter), begins repeating claims for the same address, or does not to send a can not claim address message on both buses. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.22.4 Method of measurement (reduced communication during address bumping):

This test verifies that when the DUT is bumped by a higher priority device, the DUT ceases communication until the unit's new address has been successfully claimed. This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) on both buses with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. Once the DUT has issued a new address claim for a new address on both buses, two requests will be sent before the 250ms time period expires. An ISO request, PGN 059904 is sent on both buses to the DUT's previous address, requesting the Product Information PGN 126996, and an ISO request, PGN 059904 is sent on both buses to the DUT's new attempted but not yet claimed address, requesting the Product Information PGN 126996.

C.6.22.4.1 Required results:

The DUT should receive and process the higher priority address claim message. The DUT should respond to the contention by choosing a different address and transmitting a new address claim message on both buses, and then begin the 250ms-waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT keeps the original contested address in its second address claim message, or does not respond to the contention on both buses, then it has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 also constitutes failure of this test. The address claim messages transmitted in response must have the same DLC value, be in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.22.5 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.22 passed or failed. Provide the Certification Test major section number (such as 6.22). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as Source Address, NAME, and DLC values. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.22.5.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes all four methods. The second example shows a variety of problems. The problem in 6.22.1 is that the DUT changed addresses in response to a device with a lower priority, when it should not have. The problem in 6.22.2 is that the DUT did not change its address in response to a device with a higher priority. In 6.22.3 the DUT did not attempt to claim all possible addresses before sending the can not claim message. The last problem, shown in 6.22.4, is that the DUT responded to requests for PGN 126996 while it was in transition between addresses and before the 250ms period was complete. Other results may be possible and should be similarly documented.

Test 6.22 Address Claim Test – Mandatory – PASSED

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Method 6.22.1: PGN 060928 received on bus 1 with DLC = 9
Address = 12
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 9
Address = 12
NAME = 13911689620554948611 (C1104000AAAAA003$_{16}$)
PGN 060928 sent on bus 1 with DLC = 14
Address = 12
NAME = 13839632026517020676 (C0104000AAAAA004$_{16}$)
PGN 060928 sent on bus 2 with DLC = 14
Address = 12
NAME = 13911689620554948612 (C1104000AAAAA004$_{16}$)
PGN 060928 received on bus 2 with DLC = 10
Address = 12
NAME = 13911689620554948611 (C1104000AAAAA003$_{16}$)
PGN 060928 received on bus 1 with DLC = 10
Address = 12
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
Method 6.22.2: PGN 060928 received on bus 1 with DLC = 11
Address = 12
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 11
Address = 12
NAME = 13911689620554948611 (C1104000AAAAA003$_{16}$)
PGN 060928 sent on bus 1 with DLC = 15
Address = 12
NAME = 13839632026517020674 (C0104000AAAAA002$_{16}$)
PGN 060928 sent on bus 2 with DLC = 15
Address = 12
NAME = 13911689620554948610 (C1104000AAAAA002$_{16}$)
PGN 060928 received on bus 1 with DLC = 12
Address = 13
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 12
Address = 13
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
Method 6.22.3: PGN 060928 received on bus 1 with DLC = 13
Address = 12
NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 13
Address = 12
NAME = 13911689620554948611 (C1104000AAAAA003$_{16}$)
PGN 060928 sent bus 1 with DLC = 9
PGN 060928 sent bus 2 with DLC = 9
252 Addresses tried bus 1
252 Addresses tried bus 2

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PGN 060928 received bus 1 with Address = 254 and DLC = 11
PGN 060928 received bus 2 with Address = 254 and DLC = 11
Method 6.22.4: PGN 060928 received on bus 1 with DLC = 12
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 12
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent on bus 1 with DLC = 14
Address = 12
NAME = 1383963202651702067 4 (C0104000AAAAA002$_{16}$)
PGN 060928 sent on bus 2 with DLC = 14
Address = 12
NAME = 1391168962055494861 0 (C1104000AAAAA002$_{16}$)
PGN 060928 received on bus 1 with DLC = 13
Address = 13
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 13
Address = 13
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
ISO Request PGN 050904 sent on bus 1 to Address 12 requesting PGN 126996
ISO Request PGN 050904 sent on bus 2 to Address 12 requesting PGN 126996
ISO Request PGN 050904 sent on bus 1 to Address 13 requesting PGN 126996
ISO Request PGN 050904 sent on bus 2 to Address 13 requesting PGN 126996
No response from DUT on bus 1 during the 250ms timeout period.
No response from DUT on bus 2 during the 250ms timeout period.

Or

Test 6.22 Address Claim Test – Mandatory – FAILED*
Method 6.22.1: PGN 060928 received on bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent on bus 1 with DLC = 14
Address = 12
NAME = 1383963202651702067 6 (C0104000AAAAA004$_{16}$)
PGN 060928 sent on bus 2 with DLC = 14
Address = 12
NAME = 1391168962055494861 2 (C1104000AAAAA004$_{16}$)
PGN 060928 received on bus 2 with DLC = 10
Address = 105*
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 received on bus 1 with DLC = 10

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Address = 105*
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
Method 6.22.2: PGN 060928 received on bus 1 with DLC = 11
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 11
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent on bus 1 with DLC = 15
Address = 12
NAME = 1383963202651702067 4 (C0104000AAAAA002$_{16}$)
PGN 060928 sent on bus 2 with DLC = 15
Address = 12
NAME = 1391168962055494861 0 (C1104000AAAAA002$_{16}$)
PGN 060928 received on bus 1 with DLC = 11
Address = 12*
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 11
Address = 12*
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
Method 6.22.3: PGN 060928 received on bus 1 with DLC = 13
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 13
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent bus 1 with DLC = 9
PGN 060928 sent bus 2 with DLC = 9
Addresses missed on bus 1: 10,13,21,34,44,51,62,71,80,90,101,113*
Addresses missed on bus 2: 10,13,21,34,44,51,62,71,80,90,101,113*
PGN 060928 received bus 1 with Address = 254 and DLC = 11
PGN 060928 received bus 1 with Address = 254 and DLC = 11
Method 6.22.4: PGN 060928 received on bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent on bus 1 with DLC = 14
Address = 12
NAME = 1383963202651702067 4 (C0104000AAAAA002$_{16}$)
PGN 060928 sent on bus 2 with DLC = 14
Address = 12
NAME = 1391168962055494861 0 (C1104000AAAAA002$_{16}$)
PGN 060928 received on bus 1 with DLC 10
Address = 13

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NAME = 13839632026517020675 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC 10
Address = 13
NAME = 13911689620554948611 (C1104000AAAAA003$_{16}$)
ISO Request PGN 050904 sent on bus 1 to Address 12 requesting PGN 126996
ISO Request PGN 050904 sent on bus 2 to Address 12 requesting PGN 126996
PGN 126996 received on bus 1 from DUT before 250 ms timeout from old address*
PGN 126996 received on bus 2 from DUT before 250 ms timeout from old address*
ISO Request PGN 050904 sent on bus 1 to Address 13 requesting PGN 126996
ISO Request PGN 050904 sent on bus 2 to Address 13 requesting PGN 126996
PGN 126996 received on bus 1 from DUT before 250 ms timeout from new address*
PGN 126996 received on bus 2 from DUT before 250 ms timeout from new address*

C.6.23 Default Reporting Intervals – Mandatory

The purpose of this is to verify that the DUT, operating with factory default settings, adheres to the default reporting intervals for the NMEA 2000 published PGNs. Reporting intervals of Proprietary PGNs will be observed if possible.

C.6.23.1 Method of measurement (Default reporting intervals):

This test requires the DUT to have claimed an address and be operating in normal mode. The test consists of listening to the normal PGN outputs of the DUT and recording the time between like PGN transmissions. A Complex Request PGN 126208 is sent on both buses to the DUT requesting PGN 126464 Transmit PGN List. This forms the basis of what PGN outputs are expected from the DUT. Any anomalies, i.e. PGNs in the list but not transmitted by the DUT, PGNs not in the list but transmitted by the DUT, or PGNs transmitted by the DUT on only one bus should be noted.

C.6.23.1.1 Required results:

The PGNs provided by the DUT under normal operation should be transmitted on both buses at the default rates listed in the NMEA 2000 PGN descriptions. Reporting intervals or update rates that are smaller, i.e. less time between PGN reports that the default constitutes failure. Receiving a message on only one bus also constitutes failure. The results will require manual verification by NMEA.

C.6.23.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.23 passed or failed. Provide the Certification Test major section number (such as 6.23). Provide the Method of measurement section numbers, with the PGN number(s) that were received. Also include information that determines pass or failure, such as the time between like PGN transmissions. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.23.2.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes. The second example shows a PGN being transmitted at much faster intervals on the network than the required defaults. Other results may be possible and should be similarly documented.

Test 6.23 Default Reporting Intervals – Mandatory – PASSED

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Method 6.23.1: PGN 129025 with expected period of 100 milliseconds was received 50 times on both buses
Shortest Period: 99 msec Longest Period: 101 msec Average Period: 100 msec
PGN 129029 with expected period of 1000 milliseconds was received 5 times on both buses
Shortest Period: 998 msec Longest Period: 1002 msec Average Period: 1000 msec

Or

Test 6.23 Default Reporting Intervals – Mandatory – FAILED*
Method 6.23.1: PGN 129025 with expected period of 100 milliseconds was received 50 times on both buses
Shortest Period: 49 msec Longest Period: 51 msec Average Period: 50 msec*
=> Requires Validation
PGN 129029 with expected period of 1000 milliseconds was received 5 times on both buses
Shortest Period: 950 msec Longest Period: 1020 msec Average Period: 990 msec *
=> Requires Validation

C.6.24 Default Priority – Mandatory

The purpose of this is to verify that the DUT, operating with factory default settings, adheres to the default priority settings for the NMEA 2000 published PGNs. Priority values of Proprietary PGNs will be observed if possible.

C.6.24.3 Method of measurement (Default reporting intervals):

This test requires the DUT to have claimed an address and be operating in normal mode. The test consists of listening to the normal PGN outputs of the DUT and recording the priority value of each PGN sent by the DUT. A Complex Request PGN 126208 is sent on both buses to the DUT requesting PGN 126464 Transmit PGN List. This forms the basis of what PGN outputs are expected from the DUT. Any anomalies, i.e. PGNs in the list but not transmitted by the DUT, PGNs not in the list but transmitted by the DUT, or PGNs transmitted on only one bus should be noted.

C.6.24.3.1 Required results:

The PGNs provided by the DUT under normal operation should be transmitted on both buses with the default priority listed in the NMEA 2000 PGN descriptions. Higher Priority (lower values) than the default constitutes failure. The results will require manual verification by NMEA

C.6.24.4 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.24 passed or failed. Provide the Certification Test major section number (such as 6.24). Provide the Method of measurement section numbers, with the PGN number(s) that were received. Also include information that determines pass or failure, such as the priority value of PGN transmissions. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.24.4.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes. The second example shows a PGN being transmitted with a higher priority (lower value) than the required defaults. Other results may be possible and should be similarly documented.

Test 6.24 Default Reporting Intervals – Mandatory – PASSED

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Method 6.24.1: PGN 129025 received on bus 1 with Priority 2
PGN 129025 received on bus 2 with Priority 2
PGN 129029 received on bus 2 with Priority 3
PGN 129029 received on bus 1 with Priority 3
PGN 129539 received on bus 1 with Priority 7
PGN 129539 received on bus 2 with Priority 7

Or

Test 6.24 Default Reporting Intervals – Mandatory – FAILED*
Method 6.24.1: PGN 129025 received on bus 1 with Priority 2
PGN 129025 received on bus 2 with Priority 2
PGN 129029 received on bus 2 with Priority 2* => Requires Validation
PGN 129029 received on bus 1 with Priority 2* => Requires Validation
PGN 129539 received on bus 1 with Priority 6
PGN 129539 received on bus 2 with Priority 6

C.6.25 Unique NAME Entities [3.6.2] - Mandatory

This test validates the NAME value transmitted on both buses is identical except for the System Instance field. The System Instance field must have a value of 0 on one bus and a value of 1 on the other bus.

C.6.25.1 Method of measurement (Previous Tests):

This is accomplished by examining the results test 6.16.

C.6.25.1.1 Required results:

If test 6.16 is passed then this requirement is satisfied. If test 6.16 is failed, then this test is also failed.

C.6.25.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.25 passed or failed. Provide the Certification Test major section number (such as 6.25). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.25.2.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of one method based upon the previous test, is passed. The second example depicts the case where the previous test is not successful. Other results may be possible and should be similarly documented.

Test 6.25 Unique NAME Entities [3.6.2] – Mandatory – PASSED
Method 6.25.1 Test 6.16 Passed

Or

Test 6.25 Unique NAME Entities [3.6.2] – Mandatory – FAILED*
Method 6.25.1 Test 6.16 Failed*

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C.6.26 Address Claim Class 1 Contention [3.6.3] - Mandatory

The purpose of this test is to validate the unit's response to address claim contention from a Class 1 device. Sections C.6.20 and C.6.22 already define tests that cover address claim contention from a Class 2 device as defined by 3.6.3. The tests defined here will be a repeat of the tests in both of these previous sections, but with all address contention appearing as Class 1 messages. Specifically, during these tests all Address Claim messages (PGN 60928) generated by the test program shall be in Class 1 format (unmodified DLC values). The first three tests cover contention that occurs when the DUT is claiming an address for the first time after being powered on. The final three tests cover contention that occurs after the DUT has successfully claimed an address and started normal operation.

C.6.26.1 Method of measurement (Response to lower priority address contention):

This test verifies that the DUT properly handles address contention with Class 1 devices that have a lower priority (higher value NAME means lower priority). After the DUT is powered up, and issues an address claim message on both buses, an address claim message is sent (broadcast by the test equipment) on bus 1 with a higher value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done within the 250ms period after the DUT sends its address claim message on both buses; so that contention is achieved before the DUT actually claims an address. This should cause the DUT to reclaim the same address. The test equipment will respond by transmitting an address claim message on bus 2 with the DUT's address and a higher value NAME than the DUT.

C.6.26.1.1 Required results:

The DUT should receive and process each lower priority address claim message. The DUT should respond to each contention by retransmitting the same address claim message on both buses again and then beginning the 250ms waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT changes to some other address as a result of the contention or does not respond to the contention, then it has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 constitutes failure of this test. The adjusted DUT in the address claim response must be the same on both buses, in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.26.2 Method of measurement (Response to higher priority Address contention):

This test verifies that the DUT properly handles address contention with Class 1 devices that have a higher priority (lower value NAME means higher priority). After the DUT is powered up, and issues an address claim message on both buses, an address claim message is sent (broadcast by the test equipment) on bus 1 with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done within the 250ms period after the DUT sends its address claim message on both buses; so that contention is achieved before the DUT actually claims an address. This should cause the DUT to attempt to claim a new address. The test equipment will respond by transmitting an address claim message on bus 2 with the DUT's address and a lower value NAME than the DUT.

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C.6.26.2.1 Required results:

The DUT should receive and process each higher priority address claim message. The DUT should respond to each contention by choosing a different address and transmitting a new address claim message on both buses, and then begin a 250ms waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT keeps the original contested address in the second or third group of address claim messages, or does not respond to the contention, then the DUT has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 also constitutes failure of this test. The adjusted DLC in the address claim response must be the same on both buses, in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.26.3 Method of measurement (Response to no addresses available):

This test verifies two items. The first item is that the DUT attempts to claim a valid address across the full range of valid address space, from 0 to 251. Second, that the DUT indicates when it is unable to claim an address. The sequences of events to conduct this test are as follows. After the DUT is powered up and issues an address claim message on both buses, an address claim message is sent (broadcast by the test equipment) on bus 1 with a lower value NAME than the DUT's NAME and for the same address desired by the DUT. This must be done within the 250ms period after the DUT sends its address claim messages; so that contention is achieved before the DUT actually claims an address. This should cause the DUT to try for another address on both buses. The test equipment should respond with an address claim on bus 2 with the DUT's address, and with a lower value NAME than the DUT. This should cause the DUT to try for another address. This cycle should be repeated with the test equipment alternating transmissions between bus 1 and bus 2 until the DUT sends a can not claim address message on both buses, or begins repeating address claim attempts with a previously attempted address.

C.6.26.3.1 Required results:

The DUT should, over the course of this test, send 252 address claim messages followed by a can not claim address message, which is an address claim message with the source address set to 254. These messages should be transmitted on both buses with the same address and adjusted DLC value. The DLC value must be in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The DUT will fail this test if it does not try all 252 addresses (order does not matter), begins repeating claims for the same address, or does not send a can not claim address message on both buses. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.26.4 Method of measurement (Normal operation lower priority address contention):

This test verifies that the DUT properly handles address contention with Class 1 devices that have a lower priority (higher value NAME means lower priority) after the DUT has entered normal operation mode. This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) on bus 1 with a higher value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done after it has been confirmed that the DUT is in normal operation with a claimed address. This should cause the DUT to reclaim the same address. The test equipment will respond by transmitting an address claim message on bus 2 with the DUT's address and a higher value NAME than the DUT.

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C.6.26.4.1 Required results:

The DUT should receive and process each lower priority address claim message. The DUT should respond to each contention by retransmitting the same address claim message again on both buses. All PGNs are documented. If the DUT changes to some other address as a result of the contention, or does not respond to the contention on both buses, then the unit has failed this test. The adjusted DUT in the address claim response must be the same on both buses, in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.26.5 Method of measurement (Normal operation higher priority address contention):

This test verifies that the DUT properly handles address contention with Class 1 devices that have a higher priority (lower value NAME means higher priority) after the DUT has entered normal operation mode. This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) on bus 1 with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This must be done after it has been confirmed that the DUT is in normal operation with a claimed address. This should cause the DUT to attempt to claim a new address. The test equipment will respond by transmitting an address claim message on bus 2 with the DUT's address and a lower value NAME than the DUT.

C.6.26.5.1 Required results:

The DUT should receive and process each higher priority address claim message. The DUT should respond to each contention by choosing a different address and transmitting a new address claim message on both buses, and then begin the 250ms-waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT keeps the original contested address, or does not respond to the contention, then the unit has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 also constitutes failure of this test. The adjusted DLC in the address claim response must be the same on both buses, in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.26.6 Method of measurement (Response to no addresses available):

This test verifies that the DUT in normal operation mode will, upon losing its address, attempt to claim a valid address across the full range of valid address space, from 0 to 251. This test also verifies that the DUT indicates when it is unable to claim an address. The sequences of events to conduct this test are as follows. This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is then sent (broadcast by the test equipment) on bus 1 with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. This should cause the DUT to try for another address on both buses. The test equipment should respond on bus 2 with another address claim with the same address as desired by the DUT, and with a lower value NAME than the DUT. This should cause the DUT to try for another address on both buses. This cycle should be repeated with the test equipment alternating transmissions between bus 1 and bus 2 until the DUT sends a can not claim address message on both buses, or begins repeating address claim attempts with a previously attempted address.

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C.6.26.6.1 Required results:

The DUT should, over the course of this test, send 252 address claim messages followed by a can not claim address message, which is an address claim message with the source address set to 254. These messages should be transmitted on both buses with the same address and adjusted DLC value. The DLC value must be in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The DUT will fail this test if it does not try all 252 addresses (order does not matter), begins repeating claims for the same address, or does not to send a can not claim address message on both buses. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.26.7 Method of measurement (reduced communication during address bumping):

This test verifies that when the DUT is bumped by a higher priority Class 1 device, the DUT ceases communication until the unit's new address has been successfully claimed. This test should not begin until the DUT has been powered up, successfully claimed an address, and begun normal operation. An address claim message is sent (broadcast by the test equipment) on bus 2 with a lower value NAME than the DUT's NAME, and for the same address desired by the DUT. Once the DUT has issued a new address claim for a new address on both buses, two requests will be sent before the 250ms time period expires. An ISO request, PGN 059904 is sent on both buses to the DUT's previous address, requesting the Product Information PGN 126996, and an ISO request, PGN 059904 is sent on both buses to the DUT's new attempted but not yet claimed address, requesting the Product Information PGN 126996.

C.6.26.7.1 Required results:

The DUT should receive and process the higher priority address claim message. The DUT should respond to the contention by choosing a different address and transmitting a new address claim message on both buses, and then begin the 250ms-waiting period before concluding that the address is successfully claimed. All PGNs are documented. If the DUT keeps the original contested address in its second address claim message, or does not respond to the contention on both buses, then it has failed this test. Any other message from the DUT during this test, other than an ISO Request for the Address Claim message with a source address of 254 also constitutes failure of this test. The address claim messages transmitted in response must have the same DLC value, be in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.6.26.8 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.26 passed or failed. Provide the Certification Test major section number (such as 6.26). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as Source Address, DLC values, and NAME values. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.26.8.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes all six methods. The second example shows a variety of problems. The problem in 6.26.1 is that the DUT changed addresses in response to a device with a lower priority, when it should not have. The problem in 6.26.2 is that the DUT did not change its address in response to a device with a higher priority. In 6.26.3 the DUT did not attempt to claim all possible addresses before sending the can not claim message. The problem in 6.26.4 is that the DUT

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changed addresses in response to a device with a lower priority, when it should not have. The problem in 6.26.5 is that the DUT did not change its address in response to a device with a higher priority. In 6.26.6 the DUT did not attempt to claim all possible addresses before sending the can not claim message. The last problem, shown in 6.26.7, is that the DUT responded to requests for PGN 126996 while it was in transition between addresses and before the 250ms period was complete. Other results may be possible and should be similarly documented.

Test 6.26 Address Claim Class 1 Contention Test – Mandatory – PASSED
Method 6.26.1: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 13839632026517020675 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 13911689620554948611 ($C1104000AAAAA003_{16}$)
PGN 060928 sent bus 1 with DLC = 8
Address = 12
NAME = 13839632026517020676 ($C0104000AAAAA004_{16}$)
PGN 060928 received bus 1 with DLC = 10
Address = 12
NAME = 13839632026517020675 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 12
NAME = 13911689620554948611 ($C1104000AAAAA003_{16}$)
PGN 060928 sent bus 2 with DLC = 8
Address = 12
NAME = 13911689620554948610 ($C1104000AAAAA004_{16}$)
PGN 060928 received bus 1 with DLC = 11
Address = 12
NAME = 13839632026517020675 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 11
Address = 12
NAME = 13911689620554948611 ($C1104000AAAAA003_{16}$)
Method 6.26.2: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 13839632026517020675 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 13911689620554948611 ($C1104000AAAAA003_{16}$)
PGN 060928 sent bus 2 with DLC = 8
Address = 12
NAME = 13911689620554948610 ($C1104000AAAAA002_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 13
NAME = 13911689620554948611 ($C1104000AAAAA003_{16}$)
PGN 060928 received bus 1 with DLC = 10
Address = 13

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NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 sent bus 1 with DLC = 8
Address = 13
NAME = 1383963202651702067 4 (C0104000AAAAA002$_{16}$)
PGN 060928 received bus 2 with DLC = 11
Address = 14
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 received bus 1 with DLC = 11
Address = 14
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)

Method 6.26.3: PGN 060928 received bus 1 with DLC = 9
PGN 060928 received bus 2 with DLC = 9
252 Addresses tried bus 1
252 Addresses tried bus 2
PGN 060928 received bus 1 with Address = 254 and DLC = 13
PGN 060928 received bus 2 with Address = 254 and DLC = 13

Method 6.26.4: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent bus 1 with DLC = 8
Address = 12
NAME = 1383963202651702067 6 (C0104000AAAAA004$_{16}$)
PGN 060928 received bus 1 with DLC = 10
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent bus 2 with DLC = 8
Address = 12
NAME = 1391168962055494861 2 (C1104000AAAAA004$_{16}$)
PGN 060928 received bus 1 with DLC = 11
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 11
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)

Method 6.26.5: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)

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PGN 060928 sent bus 2 with DLC = 8
Address = 12
NAME = 1391168962055494861O (C1104000AAAAA002$_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 13
NAME = 1391168962055494861I (C1104000AAAAA003$_{16}$)
PGN 060928 received bus 1 with DLC = 10
Address = 13
NAME = 1383963202651702067S (C0104000AAAAA003$_{16}$)
PGN 060928 sent bus 1 with DLC = 8
Address = 13
NAME = 1383963202651702067O (C0104000AAAAA002$_{16}$)
PGN 060928 received bus 2 with DLC = 11
Address = 14
NAME = 1391168962055494861I (C1104000AAAAA003$_{16}$)
PGN 060928 received bus 1 with DLC = 11
Address = 14
NAME = 1383963202651702067S (C0104000AAAAA003$_{16}$)

Method 6.26.6: PGN 060928 received bus 1 with DLC = 9
PGN 060928 received bus 2 with DLC = 9
252 Addresses tried bus 1
252 Addresses tried bus 2
PGN 060928 received bus 1 with Address = 254 and DLC = 13
PGN 060928 received bus 2 with Address = 254 and DLC = 13

Method 6.26.7: PGN 060928 received on bus 1 with DLC = 12
Address = 12
NAME = 1383963202651702067S (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 12
Address = 12
NAME = 1391168962055494861I (C1104000AAAAA003$_{16}$)
PGN 060928 sent on bus 1 with DLC = 8
Address = 12
NAME = 1391168962055494861O (C1104000AAAAA002$_{16}$)
PGN 060928 received on bus 1 with DLC = 13
Address = 13
NAME = 1383963202651702067S (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 13
Address = 13
NAME = 1391168962055494861I (C1104000AAAAA003$_{16}$)
ISO Request PGN 050904 sent on bus 1 to Address 12 requesting PGN 126996
ISO Request PGN 050904 sent on bus 2 to Address 12 requesting PGN 126996
ISO Request PGN 050904 sent on bus 1 to Address 13 requesting PGN 126996
ISO Request PGN 050904 sent on bus 2 to Address 13 requesting PGN 126996
No response from DUT on bus 1 during the 250ms timeout period.
No response from DUT on bus 2 during the 250ms timeout period.

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Or

Test 6.26 Address Claim Class 1 Contention Test – Mandatory – FAILED*
Method 6.26.1: PGN 060928 received bus 2 with DLC = 11
Address = 12
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
PGN 060928 received bus 1 with DLC = 11
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 sent bus 1 with DLC = 8
Address = 12
NAME = 1383963202651702067 6 ($C0104000AAAAA004_{16}$)
PGN 060928 received bus 1 with DLC = 12
Address = 13*
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 12
Address = 13*
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
PGN 060928 sent bus 2 with DLC = 8
Address = 13
NAME = 1391168962055494861 2 ($C1104000AAAAA004_{16}$)
PGN 060928 received bus 1 with DLC = 13
Address = 14*
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 13
Address = 14*
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
Method 6.26.2: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
PGN 060928 sent bus 1 with DLC = 8
Address = 12
NAME = 1383963202651702067 4 ($C0104000AAAAA002_{16}$)
PGN 060928 received bus 1 with DLC = 10
Address = 12*
NAME = 1383963202651702067 5 ($C0104000AAAAA003_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 12*
NAME = 1391168962055494861 1 ($C1104000AAAAA003_{16}$)
PGN 060928 sent bus 2 with DLC = 8
Address = 12
NAME = 1391168962055494861 0 ($C1104000AAAAA002_{16}$)
PGN 060928 received bus 1 with DLC = 10

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Address = 12*
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 12*
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)

Method 6.26.3: PGN 060928 received on bus 1 with DLC = 9
PGN 060928 received on bus 2 with DLC = 9
Addresses missed on bus 1: 0,1,2,3,4,5,6,7,8,9,10,11*
Addresses missed on bus 2: 0,1,2,3,4,5,6,7,8,9,10,11*
PGN 060928 received on bus 1 with Address = 254
PGN 060928 received on bus 2 with Address = 254

Method 6.26.4: PGN 060928 received bus 2 with DLC = 11
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 received bus 1 with DLC = 11
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 sent bus 1 with DLC = 8
Address = 12
NAME = 1383963202651702067 6 (C0104000AAAAA004$_{16}$)
PGN 060928 received bus 1 with DLC = 12
Address = 13*
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 12
Address = 13*
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent bus 2 with DLC = 8
Address = 13
NAME = 1391168962055494861 2 (C1104000AAAAA004$_{16}$)
PGN 060928 received bus 1 with DLC = 13
Address = 14*
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 13
Address = 14*
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)

Method 6.26.5: PGN 060928 received bus 1 with DLC = 9
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 9
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent bus 1 with DLC = 8
Address = 12
NAME = 1383963202651702067 4 (C0104000AAAAA002$_{16}$)
PGN 060928 received bus 1 with DLC = 10
Address = 12*

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NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 12*
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent bus 2 with DLC = 8
Address = 12
NAME = 1391168962055494861 0 (C1104000AAAAA002$_{16}$)
PGN 060928 received bus 1 with DLC = 10
Address = 12*
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received bus 2 with DLC = 10
Address = 12*
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)

Method 6.26.6: PGN 060928 received on bus 1 with DLC = 9
PGN 060928 received on bus 2 with DLC = 9
Addresses missed on bus 1: 0,1,2,3,4,5,6,7,8,9,10,11*
Addresses missed on bus 2: 0,1,2,3,4,5,6,7,8,9,10,11*
PGN 060928 received on bus 1 with Address = 254
PGN 060928 received on bus 2 with Address = 254

Method 6.26.7: PGN 060928 received on bus 1 with DLC = 12
Address = 12
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC = 12
Address = 12
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
PGN 060928 sent on bus 1 with DLC = 8
Address = 12
NAME = 1383963202651702067 4 (C0104000AAAAA002$_{16}$)
PGN 060928 received on bus 1 with DLC 13
Address = 13
NAME = 1383963202651702067 5 (C0104000AAAAA003$_{16}$)
PGN 060928 received on bus 2 with DLC 13
Address = 13
NAME = 1391168962055494861 1 (C1104000AAAAA003$_{16}$)
ISO Request PGN 050904 sent on bus 1 to Address 12 requesting PGN 126996
ISO Request PGN 050904 sent on bus 2 to Address 12 requesting PGN 126996
PGN 126996 received on bus 1 from DUT before 250 ms timeout from old address*
PGN 126996 received on bus 2 from DUT before 250 ms timeout from old address*
ISO Request PGN 050904 sent on bus 1 to Address 13 requesting PGN 126996
ISO Request PGN 050904 sent on bus 2 to Address 13 requesting PGN 126996
PGN 126996 received on bus 1 from DUT before 250 ms timeout from new address*
PGN 126996 received on bus 2 from DUT before 250 ms timeout from new address*

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C.6.27 Address Claim Single Bus Failure and Recovery [3.6.3.1, 3.6.3.2, 3.6.3.3] - Mandatory

This test verifies that the DUT completes the address claim process and begins normal operation when one bus is broken. This test also verifies that the DUT recovers correctly when the broken bus is restored. To achieve this test, the measurement methods are performed as an unbroken sequence of events. To simulate a broken bus, we will turn the network power on one bus off.

C.6.27.1 Method of measurement (Normal operation minus one bus)

The DUT must be powered off at the start of this. Likewise, the network power supply for bus 1 must be powered off. The test begins by powering on the DUT. When the DUT attempts to claim a network address, an Address Claim PGN 60928 for the DUT's address with a higher priority NAME will be transmitted in response. This contention will be repeated 4 times moving the DUT 5 addresses from its starting point.

C.6.27.1.1 Required results:

The DUT should transmit six Address Claim PGNs on bus 2. The first transmission is a normal startup attempt to claim an address. The remaining five transmissions are a response to the address contention and should increment to the next available network address on each transmission. The adjusted DLC of each PGN 60928 transmission must be in the range of 9 to 15 and progress through the series on each successive transmission. After the sixth transmission of PGN 60928, the DUT should wait 250 milliseconds and then start transmitting PGNs like it would during normal operation. The only other message that can be transmitted during the time between the DUT being powered on and the completion of the 250 millisecond delay is an ISO Request for PGN 60928 from the source address 254.

C.6.27.2 Method of measurement (Bus recover with contention)

The DUT is allowed to continue in the state from the previous section for a minimum of four minutes. This is to allow some time to pass before recovering the bus. After the delay, the network power supply for bus 1 is to be powered on. The DUT should transmit an address claim message on bus 1 for the source address that was successfully claimed on bus 2. In response, a Class 1 address claim message will be transmitted on bus 1 for the DUT's address with a higher priority.

C.6.27.2.1 Required results:

When network power is restored to bus 1, the DUT should transmit the Address Claim PGN claiming the the source address successfully claimed on bus 2. This transmission should have the adjusted DLC value next in the series from the last received on bus 2. There should not be a matching transmission on bus 2. When the DUT receives the Class 1 address claim contention on bus 1, the normal operation PGN transmissions on bus 2 should stop. The DUT should transmit and Address Claim PGN on both buses for the next available network address and with matching adjusted DLC values (the next in the series). The DUT should then wait for 250 milliseconds to determine there is no contention and then start transmitting normal operation PGNs again. The only other message that can be transmitted during the time between the DUT stopping normal operation and the completion of the 250 millisecond delay is an ISO Request for PGN 60928 from the source address 254. All Address Claim PGN adjusted DLC values must be in the range of 9 to 15 and progress through the series with each successive transmission. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

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C.6.27.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.27 passed or failed. Provide the Certification Test major section number (such as 6.27). Provide the Method of measurement section numbers, with the PGN number(s) that were received. Also include information that determines pass or failure, such as what source address was claimed and what DLC value was used. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.27.3.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes. The second example shows 6.27.2 failing because the DUT started transmitting PGNs without claiming an address on the restored bus. Other results may be possible and should be similarly documented.

Test 6.27 Address Claim Single Bus Failure and Recovery – Mandatory – PASSED
Method 6.27.1: Waiting for DUT to be powered on….
PGN 060928 received on bus 2 for source address = 12 with DLC = 9
PGN 060928 sent on bus 2 for source address = 12 with DLC = 14
PGN 060928 received on bus 2 for source address = 13 with DLC = 10
PGN 060928 sent on bus 2 for source address = 13 with DLC = 15
PGN 060928 received on bus 2 for source address = 14 with DLC = 11
PGN 060928 sent on bus 2 for source address = 14 with DLC = 9
PGN 060928 received on bus 2 for source address = 15 with DLC = 12
PGN 060928 sent on bus 2 for source address = 15 with DLC = 10
PGN 060928 received on bus 2 for source address = 16 with DLC = 13
PGN 060928 sent on bus 2 for source address = 16 with DLC = 11
PGN 060928 received on bus 2 for source address = 17 with DLC = 14
No messages received for 250 milliseconds
PGNs now being received on bus 2
Method 6.27.2: Waiting 4 minutes.
Turn bus 1 network power supply on.
PGN 060928 received on bus 1 for source address = 17 with DLC = 15
PGN 060928 sent on bus 1 for source address = 17 with DLC = 8
PGN 060928 received on bus 1 for source address = 18 with DLC = 9
PGN 060928 received on bus 2 for source address = 18 with DLC = 9
No messages received for 250 milliseconds on bus 1
No messages received for 250 milliseconds on bus 2

Or

Test 6.27 Address Claim Single Bus Failure and Recovery – Mandatory – FAILED*
Method 6.27.1: Waiting for DUT to be powered on….
PGN 060928 received on bus 2 for source address = 12 with DLC = 9
PGN 060928 sent on bus 2 for source address = 12 with DLC = 14
PGN 060928 received on bus 2 for source address = 13 with DLC = 10
PGN 060928 sent on bus 2 for source address = 13 with DLC = 15
PGN 060928 received on bus 2 for source address = 14 with DLC = 11

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PGN 060928 sent on bus 2 for source address = 14 with DLC = 9
PGN 060928 received on bus 2 for source address = 15 with DLC = 12
PGN 060928 sent on bus 2 for source address = 15 with DLC = 10
PGN 060928 received on bus 2 for source address = 16 with DLC = 13
PGN 060928 sent on bus 2 for source address = 16 with DLC = 11
PGN 060928 received on bus 2 for source address = 17 with DLC = 14
No messages received for 250 milliseconds
PGNs now being received on bus 2

Method 6.27.2: Waiting 4 minutes.
Turn bus 1 network power supply on.
PGN other than 060928 received on bus 1*

C.6.28 Default Redundant Transmissions [3.6.4.1] - Mandatory

The purpose of this test is to verify that the DUT transmits all PGNs on both buses. Though covered by C.6.23 and C6.24, the test is repeated to isolate the failure to non-redundant transmissions.

C.6.28.1 Method of measurement (Default reporting intervals):

This test requires the DUT to have claimed an address and be operating in normal mode. The test consists of listening to the normal PGN outputs of the DUT, matching the redundant transmissions by PGN and DLC, and recording which PGNs are sent by the DUT.

C.6.28.1.1 Required results:

The PGNs provided by the DUT under normal operation should be transmitted on both buses.

C.6.28.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.28 passed or failed. Provide the Certification Test major section number (such as 6.28). Provide the Method of measurement section numbers, with the PGN number(s) that were received. Also include information that determines pass or failure, such as which bus the PGN was transmitted on. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.28.2.1 Example text:

The example text shows two results. The first example depicts the results when a DUT passes. The second example shows a PGN being transmitted only on bus 1. Other results may be possible and should be similarly documented.

Test 6.28 Default Redundant Transmissions – Mandatory – PASSED
Method 6.28.1: PGN 129025 received on bus 1 and bus 2
PGN 129029 received on bus 1 and bus 2
PGN 129539 received on bus 1 and bus 2

Or

Test 6.28 Default Redundant Transmissions – Mandatory – FAILED*
Method 6.28.1: PGN 129025 received on bus 1 and bus 2
PGN 129029 received on bus 1*

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PGN 129539 received on bus 1 and bus 2

C.6.29 Adjusted DLC Sequencing [3.6.4.2, 3.6.4.3, 3.6.4.4] - Mandatory

This section certifies that each PGN transmitted by a Class 2 device is transmitted with an adjusted DLC in the range of 9 to 15. This test certifies that on successive transmissions of the PGN the adjusted DLC increments through the series, restarting at 9 when the value 15 has been used. This test also certifies that fast packet transmissions use the same adjusted DLC value for every frame of the transmission. And finally, this test certifies that a separate DLC series is used for each PGN transmitted by the DUT.

C.6.29.1 Method of measurement (Multiple PGNs):

This test is performed with every PGN transmitted by the DUT as identified during the automated PGN determination process. A PGN is selected from the list of PGNs transmitted by the DUT. An ISO Request for the selected PGN is broadcast on both buses. When the DUT responds with the requested message, ISO Requests messages for PGN 60928 are broadcast on both buses until adjusted DLC value is a minimum of 2 units different from the selected PGN's last adjusted DLC value. Alternating ISO Request messages are transmitted on both buses, one for the selected PGN and one for PGN 60928. These alternating requests are transmitted eight times, walking both PGNs through the entire adjusted DLC sequence. This process is repeated until every PGN transmitted by the DUT has been selected.

C.6.29.1.1 Required results:

The adjusted DLC behavior should be independent of PGNs, that is, the series of numbers from one PGN should have no relationship to those of another PGN. The adjusted DLC on each response message should be in the range of 9 to 15, should increment up to 15, should wrap from 15 to 9, and should be the same on both buses for a given redundant transmission. If the response is a fast packet, every frame of the fast packet must contain the same adjusted DLC value as the first frame.

C.6.29.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.29 passed or failed. Provide the Certification Test major section number (such as 6.29), and each Method of measurement section number, with the PGNs that were utilized in each Method of measurement, and the values of the adjusted DLCs recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.29.2.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the results showing that the Multiple PGN measurements have failed because the adjusted DLC values did not wrap properly. The third example shows that the adjusted DLC sequences were not unique across multiple PGNs. The fourth example shows a failure from a fast packet using the wrong adjusted DLC values on successive frames. Other results may be possible and should be similarly documented.

Test 6.29 Adjusted DLC Sequencing - PASSED
Method 6.29.1: PGN 126996 bus 1 DLC 11
PGN 126996 bus 2 DLC 11
PGN 060928 bus 1 DLC 14
PGN 060928 bus 1 DLC 14

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PGN 126996 bus 1 DLC 12
PGN 126996 bus 2 DLC 12
PGN 060928 bus 1 DLC 15
PGN 060928 bus 1 DLC 15
PGN 126996 bus 1 DLC 13
PGN 126996 bus 2 DLC 13
PGN 060928 bus 1 DLC 9
PGN 060928 bus 1 DLC 9
PGN 126996 bus 1 DLC 14
PGN 126996 bus 2 DLC 14
PGN 060928 bus 1 DLC 10
PGN 060928 bus 1 DLC 10
PGN 126996 bus 1 DLC 15
PGN 126996 bus 2 DLC 15
PGN 060928 bus 1 DLC 11
PGN 060928 bus 1 DLC 11
PGN 126996 bus 1 DLC 9
PGN 126996 bus 2 DLC 9
PGN 060928 bus 1 DLC 12
PGN 060928 bus 1 DLC 12
PGN 126996 bus 1 DLC 10
PGN 126996 bus 2 DLC 10
PGN 060928 bus 1 DLC 13
PGN 060928 bus 1 DLC 13
PGN 126996 bus 1 DLC 11
PGN 126996 bus 2 DLC 11
PGN 060928 bus 1 DLC 14
PGN 060928 bus 1 DLC 14
PGN 126996 bus 1 DLC 12
PGN 126996 bus 2 DLC 12
PGN 060928 bus 1 DLC 15
PGN 060928 bus 1 DLC 15

Or

Test 6.29 Adjusted DLC Sequencing - FAILED*
Method 6.29.1: PGN 126996 bus 1 DLC 11
PGN 126996 bus 2 DLC 11
PGN 060928 bus 1 DLC 14
PGN 060928 bus 1 DLC 14
PGN 126996 bus 1 DLC 12
PGN 126996 bus 2 DLC 12
PGN 060928 bus 1 DLC 15
PGN 060928 bus 1 DLC 15
PGN 126996 bus 1 DLC 13
PGN 126996 bus 2 DLC 13

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PGN 060928 bus 1 DLC 8*
PGN 060928 bus 1 DLC 8*
PGN 126996 bus 1 DLC 14
PGN 126996 bus 2 DLC 14
PGN 060928 bus 1 DLC 9
PGN 060928 bus 1 DLC 9
PGN 126996 bus 1 DLC 15
PGN 126996 bus 2 DLC 15
PGN 060928 bus 1 DLC 10
PGN 060928 bus 1 DLC 10
PGN 126996 bus 1 DLC 8*
PGN 126996 bus 2 DLC 8*
PGN 060928 bus 1 DLC 11
PGN 060928 bus 1 DLC 11
PGN 126996 bus 1 DLC 9
PGN 126996 bus 2 DLC 9
PGN 060928 bus 1 DLC 12
PGN 060928 bus 1 DLC 12
PGN 126996 bus 1 DLC 10
PGN 126996 bus 2 DLC 10
PGN 060928 bus 1 DLC 13
PGN 060928 bus 1 DLC 13
PGN 126996 bus 1 DLC 11
PGN 126996 bus 2 DLC 11
PGN 060928 bus 1 DLC 14
PGN 060928 bus 1 DLC 14

Or

Test 6.29 Adjusted DLC Sequencing - FAILED*
Method 6.29.1: PGN 126996 bus 1 DLC 11
PGN 126996 bus 2 DLC 11
PGN 060928 bus 1 DLC 12
PGN 060928 bus 1 DLC 12
PGN 126996 bus 1 DLC 13 *
PGN 126996 bus 2 DLC 13 *
PGN 060928 bus 1 DLC 14 *
PGN 060928 bus 1 DLC 14 *
PGN 126996 bus 1 DLC 15 *
PGN 126996 bus 2 DLC 15 *
PGN 060928 bus 1 DLC 9 *
PGN 060928 bus 1 DLC 9 *
PGN 126996 bus 1 DLC 10 *
PGN 126996 bus 2 DLC 10 *
PGN 060928 bus 1 DLC 11 *
PGN 060928 bus 1 DLC 11 *

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PGN 126996 bus 1 DLC 12 *
PGN 126996 bus 2 DLC 12 *
PGN 060928 bus 1 DLC 13 *
PGN 060928 bus 1 DLC 13 *
PGN 126996 bus 1 DLC 14 *
PGN 126996 bus 2 DLC 14 *
PGN 060928 bus 1 DLC 15 *
PGN 060928 bus 1 DLC 15 *
PGN 126996 bus 1 DLC 9 *
PGN 126996 bus 2 DLC 9 *
PGN 060928 bus 1 DLC 10 *
PGN 060928 bus 1 DLC 10 *
PGN 126996 bus 1 DLC 11 *
PGN 126996 bus 2 DLC 11 *
PGN 060928 bus 1 DLC 12 *
PGN 060928 bus 1 DLC 12 *
PGN 126996 bus 1 DLC 13 *
PGN 126996 bus 2 DLC 13 *
PGN 060928 bus 1 DLC 14 *
PGN 060928 bus 1 DLC 14 *

Or

Test 6.29 Adjusted DLC Sequencing - FAILED*
Method 6.29.1: PGN 126996 bus 1 frame 1 DLC 11
PGN 126996 bus 1 frame 2 DLC 12 *
PGN 126996 bus 1 frame 3 DLC 13 *
PGN 126996 bus 2 frame 1 DLC 11
PGN 126996 bus 2 frame 2 DLC 12 *
PGN 126996 bus 2 frame 3 DLC 13 *
Test aborted.

C.6.30 Transport Protocol Connection Managed Adjusted DLC Usage [3.6.4.6] - Mandatory

This test verifies that the DUT uses the adjusted DLC sequence associated with the contained PGN instead of establishing a separate sequence for the Transport Protocol control and data frames. The proper application of the adjusted DLC to the RTS frame and data frames only is also verified.

C.6.30.1 Method of measurement (RTS/CTS - PGNs 060416 and 060160):

An ISO Request for the Product Information PGN 126996 is broadcast on both buses. The DUT should respond with the Product Information PGN. The adjusted DLC and Field 2 (NMEA Manufacturer's Product Code) of the response are recorded. A Request Group Function PGN 126208 is constructed with the following field values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)

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Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code)

This request is then transmitted on both buses using the Transport Protocol Connection Management method. The DUT should respond with the Product Information message contained in a Transport Protocol Connection Management package. The request will then be transmitted on both buses using the fast packet protocol. The DUT should respond with the Product Information message in a fast packet package. The request will then be sent again using the Transport Protocol Connection Management package.

C.6.30.1.1 Required Results:

In response to the initial ISO Request, the DUT should transmit the PGN 126996 Product Information message using the fast packet protocol. The DUT should then accept and process the first PGN 126208 request for PGN 126996 via the ISO Transport Protocol using RTS/CTS connection management on both buses. The DUT should then Initiate a RTS/CTS managed session with the test equipment on both buses, and transfer the contents of PGN 126996 in the established transport protocol session. The DUT should respond to the second PGN 126208 request for PGN 126996 by transmitting PGN 126996 on both buses using the fast packet protocol. The DUT should then accept and process the third PGN 126208 request for PGN 126996 via the ISO Transport Protocol using RTS/CTS connection management on both buses. The DUT should respond to this request by initiating a RTS/CTS managed session with the test equipment on both buses, and transfer the contents of PGN 126996 in the established transport protocol session. Failure to provide the data using the proper protocol or exceeding any of the timing requirements constitutes failure for this method. The DLC value on all CTS frames and EOM frames must have a DLC value of 8. The DLC value for the RTS and data frames of a given response transmitted by DUT must have the same adjusted DLC in the range of 9 to 15. Each successive response by the DUT containing a Product Information PGN should increment through the DLC series wrapping from 15 to 9.

C.6.30.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.30 passed or failed. Provide the Certification Test major section number (such as 6.30). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. Contents of the connection management PGNs 060416 should be reported, along with all PGNs involved in the transport protocol sessions and associated DLC values. The report is intended to provide a complete record of the data transfer sequences that take place during this test. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.30.2.1 Example text:

The text in this section is merely an outline of two of many possible outcomes. The first example text outlines the results when the DUT passes the test. The second example illustrates that the DUT failed the test by not properly using the adjusted DLC sequence. Other results may be possible and should be similarly documented.

Test 6.30 Transport Protocol Connection Managed Adjusted DLC – Mandatory – PASSED
Method 6.30.1: PGN 059904 sent on bus 1 requesting PGN 126996
PGN 059904 sent on bus 2 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 13

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PGN 126996 received on bus 2 DLC = 13
PGN 060416 RTS sent on bus 1… list contents
PGN 060416 RTS sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 1 DLC = 14 … list contents
PGN 060416 received on bus 2 DLC = 14 … list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 126208 sent on bus 1 requesting PGN 126996
PGN 126208 sent on bus 2 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 15
PGN 126996 received on bus 2 DLC = 15
PGN 060416 RTS sent on bus 1… list contents
PGN 060416 RTS sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 1 DLC = 9 … list contents
PGN 060416 received on bus 2 DLC = 9 … list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.

Or

Test 6.30 Transport Protocol Connection Managed Adjusted DLC – Mandatory – FAILED*
Method 6.30.1: PGN 059904 sent on bus 1 requesting PGN 126996
PGN 059904 sent on bus 2 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 13
PGN 126996 received on bus 2 DLC = 13
PGN 060416 RTS sent on bus 1… list contents
PGN 060416 RTS sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 1 DLC = 12 … list contents *
PGN 060416 received on bus 2 DLC = 12 … list contents *
List Transport Protocol PGNs sent and received on bus 1. *
List Transport Protocol PGNs sent and received on bus 2. *
PGN 126208 sent on bus 1 requesting PGN 126996
PGN 126208 sent on bus 2 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 14 *
PGN 126996 received on bus 2 DLC = 14 *
PGN 060416 RTS sent on bus 1… list contents
PGN 060416 RTS sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 1 DLC = 13 … list contents *

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PGN 060416 received on bus 2 DLC = 13 … list contents *
List Transport Protocol PGNs sent and received on bus 1. *
List Transport Protocol PGNs sent and received on bus 2. *

C.6.31 Transport Protocol Broadcast Method Adjusted DLC Usage [3.6.4.7] - Mandatory

This test verifies that the DUT uses the adjusted DLC sequence associated with the contained PGN instead of establishing a separate sequence for the Transport Protocol control and data frames. The proper application of the adjusted DLC to the BAM frame and data frames only is also verified.

C.6.31.1 Method of measurement (BAM - PGNs 060416 and 060160):

An ISO Request for the Product Information PGN 126996 is broadcast on both buses. The DUT should respond with the Product Information PGN. The adjusted DLC and Field 2 (NMEA Manufacturer's Product Code) of the response are recorded. A Request Group Function PGN 126208 is constructed with the following field values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code)

This request is then transmitted on both buses using the Transport Protocol Broadcast method. The DUT should respond with the Product Information message contained in a Transport Protocol Broadcast package. The request will then be transmitted on both buses using the fast packet protocol. The DUT should respond with the Product Information message in a fast packet package. The request will then be sent again using the Transport Protocol Broadcast package.

C.6.31.1.1 Required Results:

In response to the initial ISO Request, the DUT should transmit the PGN 126996 Product Information message using the fast packet protocol. The DUT should then accept and process the first PGN 126208 request for PGN 126996 via the ISO Transport Protocol Broadcast method on both buses. The DUT should respond with a Transport Protocol Broadcast Announcement Message (PGN 060416) on both buses indicating that PGN 126996 is to follow. The DUT should then transmit the contents of PGN 126996 within the Transport Protocol Data Transfer PGN 060160 on both buses. The DUT should respond to the second PGN 126208 request for PGN 126996 by transmitting PGN 126996 on both buses using the fast packet protocol. The DUT should then accept and process the third PGN 126208 request for PGN 126996 via the ISO Transport Protocol Broadcast method on both buses. The DUT should respond with a Transport Protocol Broadcast Announcement Message for PGN 126996 on both buses followed by the contents of PGN 126996 within the Transport Protocol Data Transfer PGN. Failure to provide the data using the proper protocol or exceeding any of the timing requirements constitutes failure for this method. The DLC value for the BAM and data frames of a given response transmitted by DUT must have the same adjusted DLC in the range of 9 to 15. Each successive response by the DUT containing a Product Information PGN should increment through the DLC series wrapping from 15 to 9.

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C.6.31.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.31 passed or failed. Provide the Certification Test major section number (such as 6.31). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. The contents of PGN 060416 should be reported, along with all PGNs involved in the transport protocol sessions and associated DLC values. The report is intended to provide a complete record of the data transfer sequences that take place during this test. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.31.2.1 Example text:

The text in this section is merely an outline of two of many possible outcomes. The first example text outlines the results when the DUT passes the test. The second example illustrates that the DUT failed the test by not properly using the adjusted DLC sequence. Other results may be possible and should be similarly documented.

Test 6.31 Transport Protocol Broadcast Adjusted DLC – Mandatory – PASSED
Method 6.31.1: PGN 059904 sent on bus 1 requesting PGN 126996
PGN 059904 sent on bus 2 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 13
PGN 126996 received on bus 2 DLC = 13
PGN 060416 BAM sent on bus 1… list contents
PGN 060416 BAM sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 1 DLC = 14 … list contents
PGN 060416 received on bus 2 DLC = 14 … list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 126208 sent on bus 1 requesting PGN 126996
PGN 126208 sent on bus 2 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 15
PGN 126996 received on bus 2 DLC = 15
PGN 060416 BAM sent on bus 1… list contents
PGN 060416 BAM sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 1 DLC = 9 … list contents
PGN 060416 received on bus 2 DLC = 9 … list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.

Or

Test 6.31 Transport Protocol Broadcast Adjusted DLC – Mandatory – FAILED*
Method 6.31.1: PGN 059904 sent on bus 1 requesting PGN 126996
PGN 059904 sent on bus 2 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 13

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PGN 126996 received on bus 2 DLC = 13
PGN 060416 BAM sent on bus 1… list contents
PGN 060416 BAM sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 1 DLC = 12 … list contents *
PGN 060416 received on bus 2 DLC = 12 … list contents *
List Transport Protocol PGNs sent and received on bus 1. *
List Transport Protocol PGNs sent and received on bus 2. *
PGN 126208 sent on bus 1 requesting PGN 126996
PGN 126208 sent on bus 2 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 14 *
PGN 126996 received on bus 2 DLC = 14 *
PGN 060416 BAM sent on bus 1… list contents
PGN 060416 BAM sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 1 DLC = 13 … list contents *
PGN 060416 received on bus 2 DLC = 13 … list contents *
List Transport Protocol PGNs sent and received on bus 1. *
List Transport Protocol PGNs sent and received on bus 2. *

C.6.32 NMEA Request/Command Group Function from Class 2 [3.6.5.1] - Mandatory

This validates that the DUT responds to Class 2 NMEA Request and Command Group Function PGNs on both buses. Sections C.6.6 and C.6.9 already cover the cases where the request or command is transmitted from Class 2 devices on both buses. This test covers the condition where the DUT receives only one copy of the message due to bus failure. This is simulated by transmitting the trigger messages with a Class 2 format (adjusted DLC values) on one bus. The testing machine may alternate between buses but will not send redundant copies. Each request or command transmitted by the testing machine will cause the DLC series associated with PGN 126208 to progress. The first test (C.6.32.1) verifies proper response to a request for a mandatory PGN. The second test (C.6.32.2) verifies the proper response to an improper request for a valid PGN. The third test (C.6.32.3) verifies the proper response to an improper request for an invalid PGN. The fourth test (C.6.32.4) verifies the proper response to an improper command for an invalid PGN.

C.6.32.1 Method of measurement (PGN Request):

To conduct this test, a Request Group Function PGN 126208, requesting the NAME contained in PGN 060928, is used to trigger responses from the DUT. First, an ISO Request PGN 059904 requesting the ISO Address Claim PGN 060928 is transmitted on both buses. The value of field two (Manufacturer code) returned in PGN 060928 is recorded. A Request Group Function PGN 126208 is generated with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 60928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1

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Field 6 - Field number of first parameter = 2 (field 2 from 060928)
Field 7 - Value of first parameter = value of field 2, Manufacturer code

This request is transmitted destination specific on bus 1 to the DUT's address with an adjusted DLC. The DUT should transmit the Address Claim PGN in response on both buses. The Request Group Function adjusted DLC is modified to the next in the series. The request is then transmitted destination specific on bus 2 to the DUT's address with the new adjusted DLC. Again, the DUT should respond with the Address Claim PGN on both buses. The Request Group Function adjusted DLC is modified to the next in the series and the request message is transmitted broadcast on bus 1. The DUT should respond with the Address Claim PGN on both buses. The Request Group Function adjusted DLC is modified to the next in the series and the request message is transmitted broadcast on bus 2.

C.6.32.1.1 Required results:

The DUT should respond to this test with five transmissions of the ISO Address Claim PGN 060928 on both buses. The first response is to an addressed ISO request. The other four responses are to the Request group function containing a specific Manufacturer code parameter. All of the responses should be received on both buses with the redundant copies having the same adjusted DLC in the range of 9 to 15. The response's adjusted DLC should be the next in the DLC series relative to the DLC value returned in the previous response.

C.6.32.2 Method of measurement (PGN Acknowledgment):

To conduct this test, a Request Group Function PGN 126208, requesting the NAME contained in PGN 060928, is used to trigger responses from the DUT. First, an ISO Request PGN 059904 requesting the ISO Address Claim PGN 060928 is transmitted on both buses. The value of field two (manufacturer code) returned in PGN 060928 is recorded. A Request Group Function PGN 126208 is generated with the data fields set as follows (Note that the value of the manufacturer code returned in PGN 060928 is different than what is being requested by PGN 126208 below):

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 60928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field 2 from 060928)
Field 7 - Value of first parameter = value of field 2 (Manufacturer code) minus 1

This request is transmitted destination specific on bus 1 to the DUT's address with an adjusted DLC. The DUT should transmit the Acknowledgement Group Function PGN 126208 in response on both buses. The Request Group Function adjusted DLC is modified to the next in the series. The request is then transmitted destination specific on bus 2 to the DUT's address with the new adjusted DLC. Again, the DUT should respond with the Acknowledgement Group Function PGN on both buses. The Request Group Function adjusted DLC is modified to the next in the series and the request message is transmitted broadcast on bus 1. The DUT should not respond so the test tool will monitor the bus for two seconds before proceeding. The Request Group Function adjusted DLC is modified to the next in the series and the request message is transmitted broadcast on bus 2.

C.6.32.2.1 Required results:

The DUT should respond to this test with the ISO Address Claim PGN 060928 and two Acknowledgment Group Function PGNs (126208) on both buses. The ISO Address Claim PGN is in response to the ISO Request. The Acknowledgment Group Function PGNs are in response to the destination specific requests and

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should have field 6 (First parameter error code) set to a value of 3 meaning "Request or command parameter field out-of-range". This indicates that the request group function was properly processed. The adjusted DLC values in the redundant responses must be the same and in the range of 9 to 15 and progress through the DLC series with successive transmissions.

C.6.32.3 Method of measurement (Invalid PGN Requested):

To conduct this test, a Request Group Function PGN 126208 requesting PGN 130537 is generated with the destination field set to the network address of the DUT. The request is then transmitted on bus 2. The DUT should respond with the Acknowledgement Group Function PGN. The Request Group Function PGN adjusted DLC will progress to the next in the series and the request will be transmitted on bus 1.

C.6.32.3.1 Required results:

The DUT should respond with an acknowledgment group function PGN 126208 on both buses to each request, with a PGN error code value of 1 indicating that the PGN requested is not supported. The adjusted DLC in the response messages must be the same on both buses, in the range of 9 to 15, and progress through the series with each successive transmission.

C.6.32.4 Method of measurement (Unknown PGN Commanded):

To conduct this test, a Command Group Function PGN 126208 for PGN 130537 is generated commanding the priority to “return to default”. This command message is transmitted destination specific to the DUT on bus 1. The DUT should respond with the Acknowledgement Group Function PGN. The Command Group Function PGN adjusted DLC will progress to the next in the series and the command message will be transmitted on bus 2.

C.6.32.4.1 Required results:

The DUT should respond with an acknowledgment group function PGN 126208 on both buses to each command, with a PGN error code value of 1 indicating that the PGN requested is not supported. The adjusted DLC in the response messages must be the same on both buses, in the range of 9 to 15, and progress through the series with each successive transmission.

C.6.32.5 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.32 passed or failed. Provide the Certification Test major section number (such as 6.32). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as the various error code values returned in the acknowledgment PGN 126208. The adjusted DLC value in the response messages will also be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.32.5.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of five methods, is passed. The second example shows errors with each test. Method 6.32.1 shows that the DUT did not recognize or utilize the parameter provided with the request. Method 6.32.2 should have resulted in an acknowledgment with an error code, but instead the DUT provided the PGN. In method 6.32.3, the DUT provides an incorrect error code. In method 6.32.4, the DUT only responds on the bus the command was received on. Other results may be possible and should be similarly documented.

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Test 6.32 NMEA Request/Command Group Function from Class 2 - Mandatory - PASSED

Method 6.32.1: PGN 059904 requesting PGN 060928 sent bus 1
PGN 059904 requesting PGN 060928 sent bus 2
PGN 060928 received bus 1 DLC 12
PGN 060928 received bus 2 DLC 12
PGN 126208 request sent destination specific bus 1 with Manufacturer code ### DLC = 9
PGN 060928 received bus 1 DLC 13
PGN 060928 received bus 2 DLC 13
PGN 126208 request sent destination specific bus 2 with Manufacturer code ### DLC = 10
PGN 060928 received bus 1 DLC 14
PGN 060928 received bus 2 DLC 14
PGN 126208 request sent broadcast bus 1 with Manufacturer code ### DLC = 11
PGN 060928 received bus 1 DLC 15
PGN 060928 received bus 2 DLC 15
PGN 126208 request sent broadcast bus 2 with Manufacturer code ### DLC = 12
PGN 060928 received bus 1 DLC 9
PGN 060928 received bus 2 DLC 9

Method 6.32.2: PGN 059904 requesting PGN 060928 sent bus 1
PGN 059904 requesting PGN 060928 sent bus 2
PGN 060928 received bus 1 DLC 14
PGN 060928 received bus 2 DLC 14
PGN 126208 request sent destination specific bus 1 with Manufacturer code ### DLC = 15
PGN 126208 acknowledgment received bus 1 DLC 9
PGN 126208 acknowledgment received bus 2 DLC 9
First parameter error code = 3, Request Parameter out of range
PGN 126208 request sent destination specific bus 2 with Manufacturer code ### DLC = 9
PGN 126208 acknowledgment received bus 1 DLC 10
PGN 126208 acknowledgment received bus 2 DLC 10
First parameter error code = 3, Request Parameter out of range
PGN 126208 request sent broadcast bus 1 with Manufacturer code ### DLC = 10
No response received in 2 seconds on bus 1
No response received in 2 seconds on bus 2
PGN 126208 request sent broadcast bus 2 with Manufacturer code ### DLC = 11
No response received in 2 seconds on bus 1
No response received in 2 seconds on bus 2

Method 6.32.3: PGN 126208 requesting PGN 130537 sent bus 2 DLC = 12
PGN 126208 acknowledgment received bus 2 DLC 11
PGN 126208 acknowledgment received bus 1 DLC 11
PGN error code = 1, PGN not supported
PGN 126208 requesting PGN 130537 sent bus 1 DLC = 13
PGN 126208 acknowledgment received bus 2 DLC 12
PGN 126208 acknowledgment received bus 1 DLC 12
PGN error code = 1, PGN not supported

Method 6.32.4: PGN 126208 commanding priority PGN 130537 sent bus 1 DLC = 14
PGN 126208 acknowledgment received bus 2 DLC 13
PGN 126208 acknowledgment received bus 1 DLC 13

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PGN error code = 1, PGN not supported
PGN 126208 commanding priority PGN 130537 sent bus 2 DLC = 15
PGN 126208 acknowledgment received bus 2 DLC 14
PGN 126208 acknowledgment received bus 1 DLC 14
PGN error code = 1, PGN not supported

Or

Test 6.32 NMEA Request/Command Group Function from Class 2 - Mandatory - FAILED*
Method 6.32.1: PGN 059904 requesting PGN 060928 sent bus 1
PGN 059904 requesting PGN 060928 sent bus 2
PGN 060928 received bus 1 DLC 12
PGN 060928 received bus 2 DLC 12
PGN 126208 request sent destination specific bus 1 with Manufacturer code ### DLC = 9
PGN 126208 acknowledgment received bus 1 DLC 13*
PGN 126208 acknowledgment received bus 2 DLC 13*
First parameter error code = 1, Invalid request parameter field*
PGN 126208 request sent destination specific bus 2 with Manufacturer code ### DLC = 10
PGN 126208 acknowledgment received bus 1 DLC 14*
PGN 126208 acknowledgment received bus 2 DLC 14*
First parameter error code = 1, Invalid request parameter field*
PGN 126208 request sent broadcast bus 1 with Manufacturer code ### DLC = 11
No response received in 2 seconds on bus 1*
No response received in 2 seconds on bus 2*
PGN 126208 request sent broadcast bus 2 with Manufacturer code ### DLC = 12
No response received in 2 seconds on bus 1*
No response received in 2 seconds on bus 2*
Method 6.32.2: PGN 059904 requesting PGN 060928 sent bus 1
PGN 059904 requesting PGN 060928 sent bus 2
PGN 060928 received bus 1 DLC 14
PGN 060928 received bus 2 DLC 14
PGN 126208 request sent destination specific bus 1 with Manufacturer code ### DLC = 15
PGN 060928 received bus 1 DLC 15*
PGN 060928 received bus 2 DLC 15*
PGN 126208 request sent destination specific bus 2 with Manufacturer code ### DLC = 9
PGN 060928 received bus 1 DLC 9*
PGN 060928 received bus 2 DLC 9*
PGN 126208 request sent broadcast bus 1 with Manufacturer code ### DLC = 10
PGN 060928 received bus 1 DLC 10*
PGN 060928 received bus 2 DLC 10*
PGN 126208 request sent broadcast bus 2 with Manufacturer code ### DLC = 11
PGN 060928 received bus 1 DLC 11*
PGN 060928 received bus 2 DLC 11*
Method 6.32.3: PGN 126208 requesting PGN 130537 sent bus 2 DLC = 12
PGN 126208 acknowledgment received bus 2 DLC 11
PGN 126208 acknowledgment received bus 1 DLC 11

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PGN error code = 2, PGN temporarily not available*
PGN 126208 requesting PGN 130537 sent bus 1 DLC = 13
PGN 126208 acknowledgment received bus 2 DLC 12
PGN 126208 acknowledgment received bus 1 DLC 12
PGN error code = 2, PGN temporarily not available*

Method 6.32.4: PGN 126208 commanding priority PGN 130537 sent bus 1 DLC = 14
PGN 126208 acknowledgment received bus 1 DLC 13
PGN 126208 acknowledgment NOT received bus 2*
PGN error code = 1, PGN not supported
PGN 126208 commanding priority PGN 130537 sent bus 2 DLC = 15
PGN 126208 acknowledgment received bus 2 DLC 14
PGN 126208 acknowledgment NOT received bus 1*
PGN error code = 1, PGN not supported

C.6.33 ISO Request from Class 2 [3.6.5.2] - Mandatory

This section validates that the DUT responds on both buses to Class 2 transmissions of ISO Request PGN 059904. Sections C.6.15 and C.6.21 already cover the cases where the request is transmitted from Class 2 devices on both buses. This test covers the condition where the DUT receives only one copy of the request message due to bus failure. This is simulated by transmitting the trigger messages on only one bus. The testing machine may alternate between buses but will not send redundant copies. The first test (C.6.33.1) verifies proper response to a request for valid PGN. The second test (C.6.33.2) verifies the proper response to a request for an invalid PGN.

C.6.33.1 Method of measurement (Valid PGN Request):

An ISO Request PGN 059904 requesting the ISO Address Claim PGN 060928 is broadcast on bus 1. The DUT should respond with the Address Claim PGN 060928 on both buses. An ISO Request PGN requesting the ISO Address Claim PGN is transmitted destination specific to the DUT on bus 1. The DUT should respond with the Address Claim PGN on both buses. These steps are repeated using bus 2.

C.6.33.1.1 Required results:

The DUT should respond with four transmissions of the ISO Address Claim PGN 060928 on both buses. All of the responses should be received on both buses with the redundant copies having the same adjusted DLC in the range of 9 to 15. The response's adjusted DLC should be the next in the DLC series relative to the DLC value returned in the previous response.

C.6.33.2 Method of measurement (Invalid PGN Requested):

An ISO Request PGN 059904 requesting PGN 130537 is broadcast on bus 1. After waiting 2 minutes for a response, an ISO Request PGN requesting PGN 130537 is transmitted destination specific to the DUT on bus 1. The DUT should respond with an ISO Acknowledgement PGN 059392 on both buses. These steps are repeated using bus 2.

C.6.33.2.1 Required results:

If the DUT responds with a NACK on either bus to the first request, the test is failed. The DUT should respond to the second request with an ISO Acknowledgement PGN 059392 on both buses indicating a negative acknowledgment. If the DUT responds with a NACK on either bus to the third request, the test is failed. The DUT should respond to the fourth request with PGN 059392 on both buses indicating a negative

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acknowledgment. The adjusted DLC in the response messages must be the same on both buses, in the range of 9 to 15, and progress through the series with each successive transmission.

C.6.33.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.33 passed or failed. Provide the Certification Test major section number (such as 6.33). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as the various error code values returned in the Acknowledgment PGN 059392. The adjusted DLC value in the response messages will also be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.33.3.1 Example text:

The following text shows two different results. The first example shows the result when the test is passed. The second example shows errors with each test. Method 6.33.1 shows that the DUT only responded on the bus the request was received on. In Method 6.33.2, the DUT provides an improper field value in the Acknowledgement PGN. Other results may be possible and should be similarly documented.

Test 6.33 ISO Request from Class 2 - Mandatory - PASSED
Method 6.33.1: PGN 059904 requesting PGN 060928 sent broadcast bus 1
PGN 060928 received bus 1 DLC 12
PGN 060928 received bus 2 DLC 12
PGN 059904 requesting PGN 060928 sent destination specific bus 1
PGN 060928 received bus 1 DLC 13
PGN 060928 received bus 2 DLC 13
PGN 059904 requesting PGN 060928 sent broadcast bus 2
PGN 060928 received bus 1 DLC 14
PGN 060928 received bus 2 DLC 14
PGN 059904 requesting PGN 060928 sent destination specific bus 2
PGN 060928 received bus 1 DLC 15
PGN 060928 received bus 2 DLC 15
Method 6.33.2: PGN 059904 requesting PGN 130537 sent broadcast bus 1
No response for 2 minutes
PGN 059904 requesting PGN 130537 sent destination specific bus 1
PGN 059392 received bus 1 DLC 12
PGN 059392 received bus 2 DLC 12
Control Byte = 1
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537
PGN 059904 requesting PGN 130537 sent broadcast bus 2
No response for 2 minutes
PGN 059904 requesting PGN 130537 sent destination specific bus 2
PGN 059392 received bus 1 DLC 13
PGN 059392 received bus 2 DLC 13
Control Byte = 1
Group Function Value = 255 (FF_{16})

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Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537

Or

Test 6.33 ISO Request from Class 2 - Mandatory - FAILED*
Method 6.33.1: PGN 059904 requesting PGN 060928 sent broadcast bus 1
PGN 060928 received bus 1 DLC 12
No response received on bus 2 *
PGN 059904 requesting PGN 060928 sent destination specific bus 1
PGN 060928 received bus 1 DLC 13
No response received on bus 2 *
PGN 059904 requesting PGN 060928 sent broadcast bus 2
PGN 060928 received bus 2 DLC 14
No response received on bus 1 *
PGN 059904 requesting PGN 060928 sent destination specific bus 2
PGN 060928 received bus 2 DLC 15
No response received on bus 1 *
Method 6.33.2: PGN 059904 requesting PGN 130537 sent broadcast bus 1
No response for 2 minutes
PGN 059904 requesting PGN 130537 sent destination specific bus 1
PGN 059392 received bus 1 DLC 12
PGN 059392 received bus 2 DLC 12
Control Byte = 0 *
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537
PGN 059904 requesting PGN 130537 sent broadcast bus 2
No response for 2 minutes
PGN 059904 requesting PGN 130537 sent destination specific bus 2
PGN 059392 received bus 1 DLC 13
PGN 059392 received bus 2 DLC 13
Control Byte = 0 *
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537

C.6.34 Redundant ISO Request Timeout [3.6.5.3] - Mandatory

The purpose of this test is to verify the DUT correctly identifies redundant ISO Request received from a Class 2 device. This test consists of two methods in which ISO Requests for the Product Information PGN 126996 are used to trigger responses from the DUT. Delays are inserted between the same requests on different buses to check whether the required timeout is observed.

C.6.34.1 Method of measurement (Redundant)

An ISO Request PGN 059904 requesting PGN 126996 is transmitted destination specific to the DUT on bus 1. After delaying 45 milliseconds, the redundant copy of the request is sent on bus 2. The test tool then waits for

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2 seconds for the response from the DUT. An ISO Request PGN 059904 requesting PGN 126996 is transmitted destination specific to the DUT on bus 2. After delaying 45 milliseconds, the redundant copy of the request is sent on bus 1. The test tool then waits for 2 seconds for the response from the DUT. An ISO Request PGN 059904 requesting PGN 126996 is transmitted broadcast to the DUT on bus 1. After delaying 45 milliseconds, the redundant copy of the request is sent on bus 2. The test tool then waits for 2 seconds for the response from the DUT. An ISO Request PGN 059904 requesting PGN 126996 is transmitted broadcast to the DUT on bus 2. After delaying 45 milliseconds, the redundant copy of the request is sent on bus 1. The test tool then waits for 2 seconds for the response from the DUT.

C.6.34.1.1 Required results:

Each pair of requests separated by a 45 millisecond delay should be interpreted by the DUT as a single redundant request and produce a single transmission of PGN 126996 on both buses. Thus, the test tool should receive four copies of PGN 126996 on each bus. If the DUT responds to each individual request of the redundant pair, the DUT fails the test. The adjusted DLC in the response messages must be the same on both buses, in the range of 9 to 15, and progress through the series with each successive transmission.

C.6.34.2 Method of measurement (Non-redundant)

An ISO Request PGN 059904 requesting PGN 126996 is transmitted destination specific to the DUT on bus 1. After delaying 55 milliseconds, the redundant copy of the request is sent on bus 2. The test tool then waits for 2 seconds for the response from the DUT. An ISO Request PGN 059904 requesting PGN 126996 is transmitted destination specific to the DUT on bus 2. After delaying 55 milliseconds, the redundant copy of the request is sent on bus 1. The test tool then waits for 2 seconds for the response from the DUT. An ISO Request PGN 059904 requesting PGN 126996 is transmitted broadcast to the DUT on bus 1. After delaying 55 milliseconds, the redundant copy of the request is sent on bus 2. The test tool then waits for 2 seconds for the response from the DUT. An ISO Request PGN 059904 requesting PGN 126996 is transmitted broadcast to the DUT on bus 2. After delaying 55 milliseconds, the redundant copy of the request is sent on bus 1. The test tool then waits for 2 seconds for the response from the DUT.

C.6.34.2.1 Required results:

Each request should be interpreted by the DUT as a single redundant request and produce a single transmission of PGN 126996 on both buses. Thus, the test tool should receive eight copies of PGN 126996 on each bus. If the DUT responds to each pair of requests separated by 55 milliseconds as a single request, the DUT fails the test. The adjusted DLC in the response messages must be the same on both buses, in the range of 9 to 15, and progress through the series with each successive transmission.

C.6.34.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.34 passed or failed. Provide the Certification Test major section number (such as 6.34). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. The DLC value in the response messages will also be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.34.3.1 Example text:

The following text shows two different results. The first example shows the result when the test is passed. The second example shows errors with each test. Method 6.34.1 shows that the DUT treating the destination specific requests as individual requests from a Class 1 device instead of redundant requests from a Class 2

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device. Method 6.34.2 shows the DUT failing to respond to every request. Other results may be possible and should be similarly documented.

Test 6.34 Redundant ISO Request Timeout - Mandatory - PASSED

Method 6.34.1: PGN 059904 requesting PGN 126996 sent destination specific bus 1
Delaying 45 milliseconds
PGN 059904 requesting PGN 126996 sent destination specific bus 2
PGN 126996 received on bus 1 with DLC = 9
PGN 126996 received on bus 2 with DLC = 9
2 second delay completed without receiving additional 126996 messages
PGN 059904 requesting PGN 126996 sent destination specific bus 2
Delaying 45 milliseconds
PGN 126996 received on bus 1 with DLC = 10
PGN 126996 received on bus 2 with DLC = 10
PGN 059904 requesting PGN 126996 sent destination specific bus 1
2 second delay completed without receiving additional 126996 messages
PGN 059904 requesting PGN 126996 sent broadcast bus 1
Delaying 45 milliseconds
PGN 059904 requesting PGN 126996 sent broadcast bus 2
PGN 126996 received on bus 1 with DLC = 11
PGN 126996 received on bus 2 with DLC = 11
2 second delay completed without receiving additional 126996 messages
PGN 059904 requesting PGN 126996 sent broadcast bus 2
Delaying 45 milliseconds
PGN 059904 requesting PGN 126996 sent broadcast bus 1
PGN 126996 received on bus 1 with DLC = 12
PGN 126996 received on bus 2 with DLC = 12
2 second delay completed without receiving additional 126996 messages

Method 6.34.2: PGN 059904 requesting PGN 126996 sent destination specific bus 1
Delaying 55 milliseconds
PGN 126996 received on bus 1 with DLC = 13
PGN 126996 received on bus 2 with DLC = 13
PGN 059904 requesting PGN 126996 sent destination specific bus 2
PGN 126996 received on bus 1 with DLC = 14
PGN 126996 received on bus 2 with DLC = 14
2 second delay completed
PGN 059904 requesting PGN 126996 sent destination specific bus 2
Delaying 55 milliseconds
PGN 059904 requesting PGN 126996 sent destination specific bus 1
PGN 126996 received on bus 1 with DLC = 15
PGN 126996 received on bus 2 with DLC = 15
PGN 126996 received on bus 1 with DLC = 9
PGN 126996 received on bus 2 with DLC = 9
2 second delay completed
PGN 059904 requesting PGN 126996 sent broadcast bus 1
Delaying 55 milliseconds

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PGN 126996 received on bus 1 with DLC = 10
PGN 126996 received on bus 2 with DLC = 10
PGN 059904 requesting PGN 126996 sent broadcast bus 2
PGN 126996 received on bus 1 with DLC = 11
PGN 126996 received on bus 2 with DLC = 11
2 second delay completed
PGN 059904 requesting PGN 126996 sent broadcast bus 2
Delaying 55 milliseconds
PGN 059904 requesting PGN 126996 sent broadcast bus 1
PGN 126996 received on bus 1 with DLC = 12
PGN 126996 received on bus 2 with DLC = 12
PGN 126996 received on bus 1 with DLC = 13
PGN 126996 received on bus 2 with DLC = 13
2 second delay completed

Or

Test 6.34 Redundant ISO Request Timeout - Mandatory - FAILED*
Method 6.34.1: PGN 059904 requesting PGN 126996 sent destination specific bus 1
Delaying 45 milliseconds
PGN 126996 received on bus 1 with DLC = 9*
PGN 059904 requesting PGN 126996 sent destination specific bus 2
PGN 126996 received on bus 2 with DLC = 10*
2 second delay completed
PGN 059904 requesting PGN 126996 sent destination specific bus 2
Delaying 45 milliseconds
PGN 126996 received on bus 2 with DLC = 11*
PGN 059904 requesting PGN 126996 sent destination specific bus 1
PGN 126996 received on bus 1 with DLC = 12*
2 second delay completed
PGN 059904 requesting PGN 126996 sent broadcast bus 1
Delaying 45 milliseconds
PGN 126996 received on bus 1 with DLC = 13
PGN 126996 received on bus 2 with DLC = 13
PGN 059904 requesting PGN 126996 sent broadcast bus 2
PGN 126996 received on bus 1 with DLC = 14*
PGN 126996 received on bus 2 with DLC = 14*
2 second delay completed
PGN 059904 requesting PGN 126996 sent broadcast bus 2
Delaying 45 milliseconds
PGN 059904 requesting PGN 126996 sent broadcast bus 1
PGN 126996 received on bus 1 with DLC = 15
PGN 126996 received on bus 2 with DLC = 15
2 second delay completed without receiving additional 126996 messages
Method 6.34.2: PGN 059904 requesting PGN 126996 sent destination specific bus 1
Delaying 55 milliseconds

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PGN 126996 received on bus 1 with DLC = 9
PGN 126996 received on bus 2 with DLC = 9
PGN 059904 requesting PGN 126996 sent destination specific bus 2
2 second delay completed without receiving additional 126996 messages*
PGN 059904 requesting PGN 126996 sent destination specific bus 2
Delaying 55 milliseconds
PGN 059904 requesting PGN 126996 sent destination specific bus 1
PGN 126996 received on bus 1 with DLC = 10
PGN 126996 received on bus 2 with DLC = 10
PGN 126996 received on bus 1 with DLC = 11
PGN 126996 received on bus 2 with DLC = 11
2 second delay completed
PGN 059904 requesting PGN 126996 sent broadcast bus 1
Delaying 55 milliseconds
PGN 126996 received on bus 1 with DLC = 10
PGN 126996 received on bus 2 with DLC = 10
PGN 059904 requesting PGN 126996 sent broadcast bus 2
PGN 126996 received on bus 1 with DLC = 11
PGN 126996 received on bus 2 with DLC = 11
2 second delay completed
PGN 059904 requesting PGN 126996 sent broadcast bus 2
Delaying 55 milliseconds
PGN 059904 requesting PGN 126996 sent broadcast bus 1
PGN 126996 received on bus 1 with DLC = 12
PGN 126996 received on bus 2 with DLC = 12
2 second delay completed without receiving additional 126996 messages*

C.6.35 Responding to Class 1 Destination Specific Requests/Commands [3.6.5.4] - Mandatory

The purpose of this test is to verify the DUT's response to Class 1 requests and command where the response would be transmitted destination specific. The test confirms that the response is transmitted with a Class 1 DLC. The test also confirms that the response is only transmitted on the bus the request or command is received on. The first method (C.6.35.1) covers the response to an ISO request for an invalid PGN. The second method (C.6.35.2) covers the response to a Complex Request for an invalid PGN. The third method (C.6.35.3) covers the response to a Complex Command for an invalid PGN. The fourth method (C.6.35.4) covers the Transport Protocol Connection Management transmission.

C.6.35.1 Method of measurement (ISO Request)

An ISO Request PGN 059904 requesting PGN 130537 is transmitted destination specific to the DUT on bus 1. The DUT should respond with an ISO Acknowledgement PGN 059392. The test tool will then transmit an ISO Request PGN 059904 requesting PGN 130537 destination specific to the DUT on bus 2.

C.6.35.1.1 Required results:

The DUT should respond to the first request with an ISO Acknowledgement PGN 059392 on bus 1 indicating a negative acknowledgment. If the DUT responds to the first request on bus 2, the DUT fails the test. The DUT should respond to the second request with PGN 059392 on bus 2 indicating a negative acknowledgement.

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If the DUT responds to the second request on bus 1, the DUT fails the test. If any of the responses have a DLC greater than 8, the DUT fails the test.

C.6.35.2 Method of measurement (Complex Request)

To conduct this test, a Request Group Function PGN 126208 requesting PGN 130537 is generated with the destination field set to the network address of the DUT. The request is then transmitted on bus 1. After waiting 2 minutes, the Request Group Function PGN will be transmitted on bus 2.

C.6.35.2.1 Required results:

The DUT should respond to the first request with an acknowledgment group function PGN 126208 on bus 1, with a PGN error code value of 1 indicating that the PGN requested is not supported. If the DUT responds to the first request on bus 2, the DUT fails the test. The DUT should respond to the second request with an acknowledgment group function PGN 126208 on bus 2, with a PGN error code value of 1 indicating that the PGN requested is not supported. If the DUT responds to the second request on bus 1, the DUT fails the test. If either of the responses has a DLC greater than 8, the DUT fails the test.

C.6.35.3 Method of measurement (Complex Command)

To conduct this test, a Command Group Function PGN 126208 for PGN 130537 is generated commanding the priority to “return to default”. This command message is transmitted destination specific to the DUT on bus 1. The DUT should respond with the Acknowledgement Group Function PGN. The Command Group Function PGN will then be transmitted on bus 2.

C.6.35.3.1 Required results:

The DUT should respond to the first command with an acknowledgment group function PGN 126208 on bus 1, with a PGN error code value of 1 indicating that the PGN requested is not supported. If the DUT responds on bus 2 to the first command, the DUT fails the test. The DUT should respond to the second command with an acknowledgment group function PGN 126208 on bus 2, with a PGN error code value of 1 indicating that the PGN requested is not supported. If the DUT responds on bus 1 to this command, the DUT failed the test. If either response has a DLC greater than 8, the DUT failed the test.

C.6.35.4 Method of measurement (Transport Protocol CM)

An ISO Request for the Product Information PGN 126996 is broadcast on bus 1. The DUT should respond with the Product Information PGN. Field 2 (NMEA Manufacturer’s Product Code) of the response is recorded. A Request Group Function PGN 126208 is constructed with the following field values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT’s Manufacturers Product code)

This request is then transmitted on bus 1 using the Transport Protocol Connection Management method. The DUT should respond with the Product Information message contained in a Transport Protocol Connection Management package. After waiting two additional minutes, the request will be transmitted on bus 2 using the Transport Protocol Connection Management method.

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C.6.35.4.1 Required Results:

In response to the initial ISO Request, the DUT should transmit the PGN 126996 Product Information message using the fast packet protocol on both buses. The DUT should then accept and process the first PGN 126208 request for PGN 126996 via the ISO Transport Protocol using RTS/CTS connection management on bus 1. If the DUT transmits any of the Connection Management flow control frames on bus 2, the DUT fails the test. The DUT should then Initiate a RTS/CTS managed session with the test equipment on bus 1, and transfer the contents of PGN 126996 in the established transport protocol session. If the DUT attempts an RTS/CTS managed session on bus 2 containing PGN 126996, the DUT fails the test. The DUT should then accept and process the second PGN 126208 request for PGN 126996 via the ISO Transport Protocol using RTS/CTS connection management on bus 2. If the DUT transmits any of the Connection Management flow control frames on bus 1, the DUT fails the test. The DUT should respond to this request by initiating a RTS/CTS managed session with the test equipment on bus 2, and transfer the contents of PGN 126996 in the established transport protocol session. If the DUT attempt an RTS/CTS managed session on bus 1 containing PGN 126996, the DUT fails the test. Failure to provide the data using the proper protocol or exceeding any of the timing requirements constitutes failure for this method. The DLC value on all RTS, CTS, EOM and data frames must have a DLC value of 8.

C.6.35.5 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.35 passed or failed. Provide the Certification Test major section number (such as 6.35). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as the various error code values returned in the Acknowledgment PGN 059392. The DLC value in the response messages will also be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.35.5.1 Example text:

The following text shows two different results. The first example shows the result when the test is passed. The second example shows errors with each test. Method 6.35.1 shows that the DUT responded on both buses with a Class 2 DLC. Method 6.35.2 shows the DUT responding on both buses. Method 6.35.3 shows the DUT failing to use a Class 1 DLC. Method 6.35.4 shows the DUT failing to respond to the initial broadcast request properly. Other results may be possible and should be similarly documented.

Test 6.35 Responding to Class 1 Destination Specific Requests/Commands - Mandatory - PASSED
Method 6.35.1: PGN 059904 requesting PGN 130537 sent destination specific bus 1
PGN 059392 received bus 1 DLC 8
Control Byte = 1
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537
No response on bus 2 for 2 minutes
PGN 059904 requesting PGN 130537 sent destination specific bus 2
PGN 059392 received bus 2 DLC 8
Control Byte = 1
Group Function Value = 255 (FF_{16})

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Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537
No response on bus 1 for 2 minutes
Method 6.35.2: PGN 126208 requesting PGN 130537 sent bus 1 DLC = 8
PGN 126208 acknowledgment received bus 1 DLC 8
PGN error code = 1, PGN not supported
No response on bus 2 for 2 minutes
PGN 126208 requesting PGN 130537 sent bus 2 DLC = 8
PGN 126208 acknowledgment received bus 2 DLC 8
PGN error code = 1, PGN not supported
No response on bus 1 for 2 minutes
Method 6.35.3: PGN 126208 commanding priority PGN 130537 sent bus 1 DLC = 8
PGN 126208 acknowledgment received bus 1 DLC 8
PGN error code = 1, PGN not supported
No response on bus 2 for 2 minutes
PGN 126208 commanding priority PGN 130537 sent bus 2 DLC = 8
PGN 126208 acknowledgment received bus 2 DLC 8
PGN error code = 1, PGN not supported
No response on bus 1 for 2 minutes
Method 6.35.4: PGN 059904 sent on bus 1 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 13
PGN 126996 received on bus 2 DLC = 13
PGN 060416 RTS sent on bus 1… list contents
List Transport Protocol PGNs sent and received on bus 1.
PGN 060416 received on bus 1 DLC = 8 … list contents
List Transport Protocol PGNs sent and received on bus 1.
No response received on bus 2 for 2 minutes.
PGN 060416 RTS sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 2 DLC = 8 … list contents
List Transport Protocol PGNs sent and received on bus 2.
No response received on bus 1 for 2 minutes.

Or

Test 6.35 Responding to Class 1 Destination Specific Requests/Commands - Mandatory - FAILED*
Method 6.35.1: PGN 059904 requesting PGN 130537 sent destination specific bus 1
PGN 059392 received bus 1 DLC 12*
PGN 059392 received bus 2 DLC 12*
Control Byte = 1
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537
PGN 059904 requesting PGN 130537 sent destination specific bus 2
PGN 059392 received bus 1 DLC 13*
PGN 059392 received bus 2 DLC 13*

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Control Byte = 1
Group Function Value = 255 (FF_{16})
Reserved bits = 16777215 ($FFFFFF_{16}$)
PGN of requested information = 130537

Method 6.35.2: PGN 126208 requesting PGN 130537 sent bus 1 DLC = 8
PGN 126208 acknowledgment received bus 1 DLC 8
PGN 126208 acknowledgment received bus 2 DLC 8*
PGN error code = 1, PGN not supported
No response on bus 2 for 2 minutes
PGN 126208 requesting PGN 130537 sent bus 2 DLC = 8
PGN 126208 acknowledgment received bus 2 DLC 8
PGN 126208 acknowledgment received bus 1 DLC 8*
PGN error code = 1, PGN not supported

Method 6.35.3: PGN 126208 commanding priority PGN 130537 sent bus 1 DLC = 8
PGN 126208 acknowledgment received bus 1 DLC 10*
PGN error code = 1, PGN not supported
No response on bus 2 for 2 minutes
PGN 126208 commanding priority PGN 130537 sent bus 2 DLC = 8
PGN 126208 acknowledgment received bus 2 DLC 11*
PGN error code = 1, PGN not supported
No response on bus 1 for 2 minutes

Method 6.35.4: PGN 059904 sent on bus 1 requesting PGN 126996
PGN 126996 received on bus 1 DLC = 8*
PGN 126996 Not received on bus 2*
PGN 060416 RTS sent on bus 1… list contents
List Transport Protocol PGNs sent and received on bus 1.
PGN 060416 received on bus 1 DLC = 8 … list contents
List Transport Protocol PGNs sent and received on bus 1.
No response received on bus 2 for 2 minutes.
PGN 060416 RTS sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 2 DLC = 8 … list contents
List Transport Protocol PGNs sent and received on bus 2.
No response received on bus 1 for 2 minutes.

C.6.36 Responding to Class 1 Broadcast Requests [3.6.5.5] - Mandatory

The purpose of this test is to verify the DUT's response to Class 1 requests where the response would be transmitted broadcast. The test confirms that the response is transmitted with a Class 2 adjusted DLC. The test also confirms that the response transmitted on both buses. The first method (C.6.36.1) covers the response to an ISO request for a valid PGN. The second method (C.6.36.2) covers the response to a Complex Request for a valid PGN. The third method (C.6.36.3) covers the Transport Protocol Broadcast transmission.

C.6.36.1 Method of measurement (ISO Request)

An ISO Request PGN 059904 requesting the Product Information PGN 126996 is broadcast on bus 1. The DUT should respond with the Product Information PGN on both buses. An ISO Request PGN requesting the Product Information PGN is broadcast on bus 2.

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C.6.36.1.1 Required results:

The DUT should respond with two transmissions of the Product Information PGN 126996 on both buses. Each redundant transmission of the Product Information PGN should have the same adjusted DLC in the range of 9 to 15. The response's adjusted DLC should be the next in the DLC series relative to the DLC value returned in the previous response.

C.6.36.2 Method of measurement (Complex Request)

A Request Group Function PGN 126208 is generated with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 0

This request is broadcast on bus 1. The test tool waits 2 minutes for a response from the DUT on both buses. Then, the same request is broadcast on bus 2.

C.6.36.2.1 Required results:

The DUT should respond to both requests by transmitting the Product Information PGN 126996 on both buses. The redundant responses should have the same adjusted DLC in the range of 9 to 15. The response's adjusted DLC should be the next in the DLC series relative to the DLC value returned in the previous response.

C.6.36.3 Method of measurement (Transport Protocol BAM)

A Request Group Function PGN 126208 is constructed with the following field values:

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 0

This request is transmitted on bus 1 using the Transport Protocol Broadcast method. The test tool will wait 2 minutes for the DUT to respond. The request will then be transmitted on bus 2 using the Transport Protocol Broadcast method.

C.6.36.3.1 Required Results:

The DUT should accept and process the first PGN 126208 request for PGN 126996 via the ISO Transport Protocol Broadcast method on bus 1. The DUT should respond with a Transport Protocol Broadcast Announcement Message (PGN 060416) on both buses indicating that PGN 126996 is to follow. The DUT should then transmit the contents of PGN 126996 within the Transport Protocol Data Transfer PGN 060160 on both buses. The DUT should then accept and process the second PGN 126208 request for PGN 126996 via the ISO Transport Protocol Broadcast method on bus 2. The DUT should respond with a Transport Protocol Broadcast Announcement Message for PGN 126996 on both buses followed by the contents of PGN 126996 within the Transport Protocol Data Transfer PGN. Failure to provide the data using the proper protocol or exceeding any of the timing requirements constitutes failure for this method. The DLC value for the BAM and data frames of a given response transmitted by DUT must have the same adjusted DLC in the range of 9 to 15.

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Each successive response by the DUT containing a Product Information PGN should increment through the DLC series wrapping from 15 to 9.

C.6.36.4 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.36 passed or failed. Provide the Certification Test major section number (such as 6.36). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. The contents of PGN 060416 should be reported, along with all PGNs involved in the transport protocol sessions. The adjusted DLC value in the response messages will also be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.36.4.1 Example text:

The following text shows two different results. The first example shows the result when the test is passed. The second example shows errors with each test. Method 6.36.1 shows that the DUT only responded on the bus the request was received on. Method 6.36.2 shows a DUT responding on both buses with non-matching adjusted DLC values. Method 6.36.3 shows a DUT responding like a Class 1 device. Other results may be possible and should be similarly documented.

Test 6.36 Responding to Class 1 Broadcast Requests - Mandatory - PASSED
Method 6.36.1: PGN 059904 requesting PGN 126996 sent broadcast bus 1
PGN 126996 received bus 1 DLC 12
PGN 126996 received bus 2 DLC 12
PGN 059904 requesting PGN 126996 sent broadcast bus 2
PGN 126996 received bus 1 DLC 13
PGN 126996 received bus 2 DLC 13
Method 6.36.2: PGN 126208 request sent broadcast requesting PGN 126996 on bus 1
PGN 126996 received bus 1 DLC 14
PGN 126996 received bus 2 DLC 14
PGN 126208 request sent broadcast requesting PGN 126996 on bus 2
PGN 126996 received bus 1 DLC 15
PGN 126996 received bus 2 DLC 15
Method 6.36.3: PGN 060416 BAM sent on bus 1… list contents
List Transport Protocol PGNs sent and received on bus 1.
PGN 060416 received on bus 1 DLC = 9 … list contents
PGN 060416 received on bus 2 DLC = 9 … list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 BAM sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 1 DLC = 10 … list contents
PGN 060416 received on bus 2 DLC = 10 … list contents
List Transport Protocol PGNs sent and received on bus 1.
List Transport Protocol PGNs sent and received on bus 2.

Or

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Test 6.36 Responding to Class 1 Broadcast Requests - Mandatory - FAILED*

Method 6.36.1: PGN 059904 requesting PGN 126996 sent broadcast bus 1
PGN 126996 received bus 1 DLC 12
No response received on bus 2 *
PGN 059904 requesting PGN 126996 sent broadcast bus 2
PGN 060928 received bus 2 DLC 13
No response received on bus 1 *

Method 6.36.2: PGN 126208 request sent broadcast requesting PGN 126996 on bus 1
PGN 126996 received bus 1 DLC 14
PGN 126996 received bus 2 DLC 15*
PGN 126208 request sent broadcast requesting PGN 126996 on bus 2
PGN 126996 received bus 1 DLC 9
PGN 126996 received bus 2 DLC 10*

Method 6.36.3: PGN 060416 BAM sent on bus 1… list contents
List Transport Protocol PGNs sent and received on bus 1.
PGN 060416 received on bus 1 DLC = 8 … list contents*
List Transport Protocol PGNs sent and received on bus 1.
No response on bus 2 for 2 minutes. *
PGN 060416 BAM sent on bus 2… list contents
List Transport Protocol PGNs sent and received on bus 2.
PGN 060416 received on bus 2 DLC = 8 … list contents*
List Transport Protocol PGNs sent and received on bus 2.
No response received on bus 1 for 2 minutes*

C.6.37 Address Claim Request Node Independence [3.6.5.6] - Mandatory

The purpose of this test is to verify the DUT responds to a request for the address claim PGN 60928 when the request applies to either node, not just the node the request was received on. This test is accomplished with a single method that utilizes the NMEA Request Group Function PGN to trigger Address Claim PGN 60928 responses based on parameter content.

C.6.37.1 Method of measurement (Complex Request)

A Request Group Function PGN 126208 is generated with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 060928
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 – Field Number of First Requested Parameter = 8
Field 7 – Value of First Requested Parameter = System Instance of DUT’s bus 2 node

This request is transmitted to the DUT destination specific on bus 1 with appropriate adjusted DLC. The test tool waits 4 seconds for a response from the DUT on both buses. Then, Field 7 is changed to the System Instance of the DUT’s bus 1 node and the adjusted DLC is advanced to the next in the series. The request is transmitted to the DUT destination specific on bus 2. The test tool waits 4 seconds for a response from the DUT on both buses.

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C.6.37.1.1 Required results:

The DUT should respond to both requests by transmitting the Address Claim PGN 060928 on both buses. The redundant responses should have the same adjusted DLC in the range of 9 to 15. The response's adjusted DLC should be the next in the DLC series relative to the DLC value returned in the previous response. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus). If the DUT responds with a NACK or only responds on one bus, the DUT fails the test.

C.6.37.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 6.37 passed or failed. Provide the Certification Test major section number (such as 6.37). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. The adjusted DLC value in the response messages will also be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.6.37.2.1 Example text:

The following text shows two different results. The first example shows the result when the test is passed. The second example shows errors with each test. Method 6.37.1 shows the DUT responding with a NMEA Acknowledge Group Function on the bus the request was received on. Other results may be possible and should be similarly documented.

Test 6.37 Address Claim Request Node Independence - Mandatory - PASSED
Method 6.37.1: PGN 126208 requesting PGN 060928 with System Instance 1 sent on bus 1
PGN 060928 received bus 1 DLC 12 with System Instance 0
PGN 060928 received bus 2 DLC 12 with System Instance 1
PGN 126208 requesting PGN 060928 with System Instance 0 sent on bus 2
PGN 060928 received bus 1 DLC 13 with System Instance 0
PGN 060928 received bus 2 DLC 13 with System Instance 1

Or

Test 6.37 Address Claim Request Node Independence - Mandatory - FAILED *
Method 6.37.1: PGN 126208 requesting PGN 060928 with System Instance 1 sent on bus 1
PGN 126208 acknowledgment received bus 1 DLC 8 *
PGN error code = 0
Transmission Interval error code = 0
Number of Requested Parameters = 1
First Parameter error code = 1*
PGN 126208 requesting PGN 060928 with System Instance 0 sent on bus 2
PGN 126208 acknowledgment received bus 2 DLC 8 *
PGN error code = 0
Transmission Interval error code = 0
Number of Requested Parameters = 1
First Parameter error code = 1*

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C.7 Class 2 Level B - Automated Test Methods

This section contains the requirements and methods of measurement that can be performed automatically via a suite of software and hardware designed to certify compliance with the Level B minimum requirements of the NMEA 2000 standard for Class 2 devices. If the DUT contains capabilities beyond the Level B minimum requirements, such as the Request/Command/Acknowledge Group Function PGN 126208, then the corresponding Level A certification test will be conducted to verify that implementation.

To perform these tests, you must assemble the network as shown in section 6.

The first part of performing a Level B certification test is to determine what PGNs are provided by the Classs 2 Level B DUT. This determination includes both NMEA 2000 defined PGNs and Manufacturers Proprietary PGNs. Since the TX/RX List PGN 126464 is not required, other methods of finding out which data PGNs are provided are implemented in the Certification Tool for Level B. The Tool will do the following:

1. It should prompt the user to identify ALL (including proprietary) the PGNs (enter the PGNs) that are generated by the DUT.
2. It can query the DUT on both buses for each and every NMEA 2000 defined PGN and Proprietary PGN (including all PGN numbers slated for use by NMEA 2000 but not yet assigned) with the ISO request.

The responses to this extensive interrogation provide several important things. This adds some extensibility to the Certification Tool as the PGN data base changes or grows, in between updates to the Tool, the Tool can now create a list of all PGNS supported by the device, the default priority of provided PGNs can be checked, and the consistency between user's notification of what PGNS are supported versus actual responses to ISO Request can be made. Further, any fast packet PGNS that are identified as being provided by the DUT, other than Product Information, can be used to conduct additional tests. The results of this PGN determination shall be provided in a table or tabular form with the certification test results.

C.7.1 Fast Packet Sequence Counter [3.1.3.3] – Mandatory

This section certifies the requirement that "*The modulo 8 sequence counter shall increment each time the same PGN is transmitted from the same source address*." This is accomplished with two specific tests.

The purpose of the first test (7.1.1) is to verify proper sequence number behavior with a mandatory PGN.

The second test (7.1.2) examines the same requirement with a non-mandatory Fast Packet PGN. This second test can only be performed if the DUT transmits a non-mandatory Fast Packet PGN. Determination of the actual PGN to be tested is dependent upon either a user entered Fast Packet PGN number(s) or a Fast Packet PGN as determined by the certification tool. If the DUT does not transmit any non-mandatory PGN, then the second test is not necessary, no failure will be indicated for test 7.1.2.

C.7.1.1 Method of measurement (Mandatory PGN):

Using the ISO Request PGN 059904, a request for Product Information, PGN 126996 is transmitted on both buses. Record the sequence number and adjusted DLC values. Send eight more requests for PGN 126996 on both buses, and verify that the sequence number and adjusted DLC are incremented by one for each successive PGN. The sequence number must properly wrap from 7 to 0. The adjusted DLC must wrap from 15 to 9.

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C.7.1.1.1 Required results:

The sequence counter should be seen to start at some value between 0 and 7, and should increment up to 7 and then should start back at 0. . The adjusted DLC must start at some value between 9 and 15, should increment up to 15, and should start back at 9. The adjust DLC should be the same on both buses for a given redundant transmission.

C.7.1.2 Method of measurement (Non-Mandatory PGN):

This test is only performed if the DUT transmits one or more non-mandatory Fast Packet PGNs. This is determined by user entry or automatic tool determination efforts. Perform the same test procedure described in 7.1.1 above for every non-mandatory PGN. This will verify that all supported PGNs, not just the mandatory PGNs, are implemented properly, with respect to the sequence number. If the DUT does not transmit any Fast Packet PGNs, then this test is not performed.

C.7.1.2.1 Required results:

If the DUT does not transmit any non-mandatory Fast Packet PGNs, there are no required results. If the DUT does transmit any non-mandatory Fast Packet PGNs, then see 7.1.1.1

C.7.1.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 7.1 passed or failed. Provide the Certification Test major section number (such as 7.1), and each Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement, and the values of the sequence numbers and adjusted DLCs recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.7.1.3.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of two methods, is passed. The second example depicts the results showing that the non-mandatory Fast Packet PGN measurements have failed. Notice that the sequence numbers were not incremented, but fixed at zero. The third example illustrates the results with a Level B device that does not transmit any non-mandatory Fast Packet PGNs. Other results may be possible and should be similarly documented.

Test 7.1 Fast Packet Sequence Counter [3.1.3.3] - PASSED
Method 7.1.1: PGN 126996 bus 1 DLC 10 sequence 2
PGN 126996 bus 2 DLC 10 sequence 2
PGN 126996 bus 1 DLC 11 sequence 3
PGN 126996 bus 2 DLC 11 sequence 3
PGN 126996 bus 2 DLC 12 sequence 4
PGN 126996 bus 1 DLC 12 sequence 4
PGN 126996 bus 1 DLC 13 sequence 5
PGN 126996 bus 2 DLC 13 sequence 5
PGN 126996 bus 1 DLC 14 sequence 6
PGN 126996 bus 2 DLC 14 sequence 6
PGN 126996 bus 2 DLC 15 sequence 7
PGN 126996 bus 1 DLC 15 sequence 7
PGN 126996 bus 1 DLC 9 sequence 0
PGN 126996 bus 2 DLC 9 sequence 0

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PGN 126996 bus 1 DLC 10 sequence 1
PGN 126996 bus 2 DLC 10 sequence 1
PGN 126996 bus 1 DLC 11 sequence 2
PGN 126996 bus 2 DLC 11 sequence 2
Method 7.1.2: PGN 126998 bus 2 DLC 15 sequence 0
PGN 126998 bus 1 DLC 15 sequence 0
PGN 126998 bus 2 DLC 9 sequence 1
PGN 126998 bus 1 DLC 9 sequence 1
PGN 126998 bus 2 DLC 10 sequence 2
PGN 126998 bus 1 DLC 10 sequence 2
PGN 126998 bus 2 DLC 11 sequence 3
PGN 126998 bus 1 DLC 11 sequence 3
PGN 126998 bus 2 DLC 12 sequence 4
PGN 126998 bus 1 DLC 12 sequence 4
PGN 126998 bus 2 DLC 13 sequence 5
PGN 126998 bus 1 DLC 13 sequence 5
PGN 126998 bus 2 DLC 14 sequence 6
PGN 126998 bus 1 DLC 14 sequence 6
PGN 126998 bus 2 DLC 15 sequence 7
PGN 126998 bus 1 DLC 15 sequence 7
PGN 126998 bus 2 DLC 9 sequence 0
PGN 126998 bus 1 DLC 9 sequence 0

Or

Test 7.1 Fast Packet Sequence Counter [3.1.3.3] - FAILED*
Method 7.1.1: PGN 126996 bus 1 DLC 10 sequence 2
PGN 126996 bus 2 DLC 10 sequence 2
PGN 126996 bus 1 DLC 11 sequence 3
PGN 126996 bus 2 DLC 11 sequence 3
PGN 126996 bus 2 DLC 12 sequence 4
PGN 126996 bus 1 DLC 12 sequence 4
PGN 126996 bus 1 DLC 13 sequence 5
PGN 126996 bus 2 DLC 13 sequence 5
PGN 126996 bus 1 DLC 14 sequence 6
PGN 126996 bus 2 DLC 14 sequence 6
PGN 126996 bus 2 DLC 15 sequence 7
PGN 126996 bus 1 DLC 15 sequence 7
PGN 126996 bus 1 DLC 9 sequence 0
PGN 126996 bus 2 DLC 9 sequence 0
PGN 126996 bus 1 DLC 10 sequence 1
PGN 126996 bus 2 DLC 10 sequence 1
PGN 126996 bus 1 DLC 11 sequence 2
PGN 126996 bus 2 DLC 11 sequence 2
Method 7.1.2: PGN 126998 bus 2 DLC 15 sequence 0
PGN 126998 bus 1 DLC 15 sequence 0

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PGN 126998 bus 2 DLC 9 sequence 0*
PGN 126998 bus 1 DLC 9 sequence 0*
PGN 126998 bus 2 DLC 10 sequence 0*
PGN 126998 bus 1 DLC 10 sequence 0*
PGN 126998 bus 2 DLC 11 sequence 0*
PGN 126998 bus 1 DLC 11 sequence 0*
PGN 126998 bus 2 DLC 12 sequence 0*
PGN 126998 bus 1 DLC 12 sequence 0*
PGN 126998 bus 2 DLC 13 sequence 0*
PGN 126998 bus 1 DLC 13 sequence 0*
PGN 126998 bus 2 DLC 14 sequence 0*
PGN 126998 bus 1 DLC 14 sequence 0*
PGN 126998 bus 2 DLC 15 sequence 0*
PGN 126998 bus 1 DLC 15 sequence 0*
PGN 126998 bus 2 DLC 9 sequence 0*
PGN 126998 bus 1 DLC 9 sequence 0*

Or

Test 7.1 Fast Packet Sequence Counter [3.1.3.3] - PASSED
Method 7.1.1: PGN 126996 bus 1 DLC 10 sequence 2
PGN 126996 bus 2 DLC 10 sequence 2
PGN 126996 bus 1 DLC 11 sequence 3
PGN 126996 bus 2 DLC 11 sequence 3
PGN 126996 bus 2 DLC 12 sequence 4
PGN 126996 bus 1 DLC 12 sequence 4
PGN 126996 bus 1 DLC 13 sequence 5
PGN 126996 bus 2 DLC 13 sequence 5
PGN 126996 bus 1 DLC 14 sequence 6
PGN 126996 bus 2 DLC 14 sequence 6
PGN 126996 bus 2 DLC 15 sequence 7
PGN 126996 bus 1 DLC 15 sequence 7
PGN 126996 bus 1 DLC 9 sequence 0
PGN 126996 bus 2 DLC 9 sequence 0
PGN 126996 bus 1 DLC 10 sequence 1
PGN 126996 bus 2 DLC 10 sequence 1
PGN 126996 bus 1 DLC 11 sequence 2
PGN 126996 bus 2 DLC 11 sequence 2
Method 7.1.2: Not Applicable

C.7.2 Unique Fast Packet Sequence Counter [3.1.3.4] – Mandatory

This section certifies the requirement that "*A separate sequence counter shall be utilized for each different PGN transmitted from the same source address.*" The purpose of this test is to ensure there is no linkage of the

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sequence number between Fast Packet PGNs. This test is only performed on devices that transmit non-mandatory Fast Packet PGNs in addition to the mandatory Product Information PGN 126996. Determination of the actual non-mandatory Fast Packet PGN to be tested is dependent upon either user entered Fast Packet PGN number(s) or a Fast Packet PGN as determined by the certification tool. If the DUT does not transmit any non-mandatory PGN, then this test cannot be conducted, and no failure will be indicated for test 7.2.1.

C.7.2.1 Method of measurement (Multiple PGNs):

Using the ISO Request PGN 059904, send one or more requests on both buses for the Product Information PGN 126996, and one or more requests on both buses for a supported non-mandatory Fast Packet PGN, until both have different sequence numbers. Then, send alternate requests on both buses, one for PGN 126996 and one for the selected non-mandatory Fast Packet PGN, eight times. This will verify that the behavior is proper given interleaved requests. Repeat this procedure for every supported non-mandatory Fast Packet PGN.

C.7.2.1.1 Required results:

The sequence number behavior should be independent of PGNs, that is, the series of sequence numbers from one PGN should have no relationship to those of another PGN. The adjusted DLC on each response message should be in the range of 9 to 15, should increment up to 15, should wrap from 15 to 9, and should be the same on both buses for a given redundant transmission. If the DUT does not transmit any non-mandatory Fast Packet PGNs, there are no required results.

C.7.2.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 7.2 passed or failed. Provide the Certification Test major section number (such as 7.2), and each Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement, and the values of the sequence numbers and adjusted DLCs recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.7.2.2.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the results showing that the Multiple PGN measurements have failed. Notice that the sequence numbers were not incremented, but fixed at zero. The third example shows that the sequence numbers were not unique across multiple PGNs. The fourth example illustrates the documented result when the test can not be run due to no non-mandatory Fast Packet PGNs supported by the DUT. Other results may be possible and should be similarly documented.

Test 7.2 Unique Fast Packet Sequence Counter [3.1.3.4] - PASSED
Method 7.2.1: PGN 126996 on bus 1 with DLC = 11 and sequence = 3
PGN 126996 on bus 2 with DLC = 11 and sequence = 3
PGN 126998 on bus 1 with DLC = 9 and sequence = 1
PGN 126998 on bus 2 with DLC = 9 and sequence = 1
PGN 126996 on bus 1 with DLC = 12 and sequence = 4
PGN 126996 on bus 2 with DLC = 12 and sequence = 4
PGN 126998 on bus 1 with DLC = 10 and sequence = 2
PGN 126998 on bus 2 with DLC = 10 and sequence = 2
PGN 126996 on bus 1 with DLC = 13 and sequence = 5
PGN 126996 on bus 2 with DLC = 13 and sequence = 5

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PGN 126998 on bus 1 with DLC = 11 and sequence = 3
PGN 126998 on bus 2 with DLC = 11 and sequence = 3
PGN 126996 on bus 1 with DLC = 14 and sequence = 6
PGN 126996 on bus 2 with DLC = 14 and sequence = 6
PGN 126998 on bus 1 with DLC = 12 and sequence = 4
PGN 126998 on bus 2 with DLC = 12 and sequence = 4
PGN 126996 on bus 1 with DLC = 15 and sequence = 7
PGN 126996 on bus 2 with DLC = 15 and sequence = 7
PGN 126998 on bus 1 with DLC = 13 and sequence = 5
PGN 126998 on bus 2 with DLC = 13 and sequence = 5
PGN 126996 on bus 1 with DLC = 9 and sequence = 0
PGN 126996 on bus 2 with DLC = 9 and sequence = 0
PGN 126998 on bus 1 with DLC = 14 and sequence = 6
PGN 126998 on bus 2 with DLC = 14 and sequence = 6
PGN 126996 on bus 1 with DLC = 10 and sequence = 1
PGN 126996 on bus 2 with DLC = 10 and sequence = 1
PGN 126998 on bus 1 with DLC = 15 and sequence = 7
PGN 126998 on bus 2 with DLC = 15 and sequence = 7
PGN 126996 on bus 1 with DLC = 11 and sequence = 2
PGN 126996 on bus 2 with DLC = 11 and sequence = 2
PGN 126998 on bus 1 with DLC = 9 and sequence = 0
PGN 126998 on bus 2 with DLC = 9 and sequence = 0
PGN 126996 on bus 1 with DLC = 12 and sequence = 3
PGN 126996 on bus 2 with DLC = 12 and sequence = 3
PGN 126998 on bus 1 with DLC = 10 and sequence = 1
PGN 126998 on bus 2 with DLC = 10 and sequence = 1

Or

Test 7.2 Unique Fast Packet Sequence Counter [3.1.3.4] - FAILED*
Method 7.2.1: PGN 126996 on bus 1 with DLC = 11 and sequence = 3
PGN 126996 on bus 2 with DLC = 11 and sequence = 3
PGN 126998 on bus 1 with DLC = 9 and sequence = 0
PGN 126998 on bus 2 with DLC = 9 and sequence = 0
PGN 126996 on bus 1 with DLC = 12 and sequence = 4
PGN 126996 on bus 2 with DLC = 12 and sequence = 4
PGN 126998 on bus 1 with DLC = 10 and sequence = 0*
PGN 126998 on bus 2 with DLC = 10 and sequence = 0*
PGN 126996 on bus 1 with DLC = 13 and sequence = 5
PGN 126996 on bus 2 with DLC = 13 and sequence = 5
PGN 126998 on bus 1 with DLC = 11 and sequence = 0*
PGN 126998 on bus 2 with DLC = 11 and sequence = 0*
PGN 126996 on bus 1 with DLC = 14 and sequence = 6
PGN 126996 on bus 2 with DLC = 14 and sequence = 6
PGN 126998 on bus 1 with DLC = 12 and sequence = 0*
PGN 126998 on bus 2 with DLC = 12 and sequence = 0*

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PGN 126996 on bus 1 with DLC = 15 and sequence = 7
PGN 126996 on bus 2 with DLC = 15 and sequence = 7
PGN 126998 on bus 1 with DLC = 13 and sequence = 0*
PGN 126998 on bus 2 with DLC = 13 and sequence = 0*
PGN 126996 on bus 1 with DLC = 9 and sequence = 0
PGN 126996 on bus 2 with DLC = 9 and sequence = 0
PGN 126998 on bus 1 with DLC = 14 and sequence = 0*
PGN 126998 on bus 2 with DLC = 14 and sequence = 0*
PGN 126996 on bus 1 with DLC = 10 and sequence = 1
PGN 126996 on bus 2 with DLC = 10 and sequence = 1
PGN 126998 on bus 1 with DLC = 15 and sequence = 0*
PGN 126998 on bus 2 with DLC = 15 and sequence = 0*
PGN 126996 on bus 1 with DLC = 11 and sequence = 2
PGN 126996 on bus 2 with DLC = 11 and sequence = 2
PGN 126998 on bus 1 with DLC = 9 and sequence = 0*
PGN 126998 on bus 2 with DLC = 9 and sequence = 0*
PGN 126996 on bus 1 with DLC = 12 and sequence = 3
PGN 126996 on bus 2 with DLC = 12 and sequence = 3
PGN 126998 on bus 1 with DLC = 10 and sequence = 0*
PGN 126998 on bus 2 with DLC = 10 and sequence = 0*

Or

Test 7.2 Unique Fast Packet Sequence Counter [3.1.3.4] - FAILED*
Method 7.2.1: PGN 126996 on bus 1 with DLC = 11 and sequence = 4
PGN 126996 on bus 2 with DLC = 11 and sequence = 4
PGN 126998 on bus 1 with DLC = 9 and sequence = 5*
PGN 126998 on bus 2 with DLC = 9 and sequence = 5*
PGN 126996 on bus 1 with DLC = 12 and sequence = 6*
PGN 126996 on bus 2 with DLC = 12 and sequence = 6*
PGN 126998 on bus 1 with DLC = 10 and sequence = 7*
PGN 126998 on bus 2 with DLC = 10 and sequence = 7*
PGN 126996 on bus 1 with DLC = 13 and sequence = 0*
PGN 126996 on bus 2 with DLC = 13 and sequence = 0*
PGN 126998 on bus 1 with DLC = 11 and sequence = 1*
PGN 126998 on bus 2 with DLC = 11 and sequence = 1*
PGN 126996 on bus 1 with DLC = 14 and sequence = 2*
PGN 126996 on bus 2 with DLC = 14 and sequence = 2*
PGN 126998 on bus 1 with DLC = 12 and sequence = 3*
PGN 126998 on bus 2 with DLC = 12 and sequence = 3*
PGN 126996 on bus 1 with DLC = 15 and sequence = 4*
PGN 126996 on bus 2 with DLC = 15 and sequence = 4*
PGN 126998 on bus 1 with DLC = 13 and sequence = 5*
PGN 126998 on bus 2 with DLC = 13 and sequence = 5*
PGN 126996 on bus 1 with DLC = 9 and sequence = 6*
PGN 126996 on bus 2 with DLC = 9 and sequence = 6*

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PGN 126998 on bus 1 with DLC = 14 and sequence = 7*
PGN 126998 on bus 2 with DLC = 14 and sequence = 7*
PGN 126996 on bus 1 with DLC = 10 and sequence = 0*
PGN 126996 on bus 2 with DLC = 10 and sequence = 0*
PGN 126998 on bus 1 with DLC = 15 and sequence = 1*
PGN 126998 on bus 2 with DLC = 15 and sequence = 1*
PGN 126996 on bus 1 with DLC = 11 and sequence = 2*
PGN 126996 on bus 2 with DLC = 11 and sequence = 2*
PGN 126998 on bus 1 with DLC = 9 and sequence = 3*
PGN 126998 on bus 2 with DLC = 9 and sequence = 3*
PGN 126996 on bus 1 with DLC = 12 and sequence = 4*
PGN 126996 on bus 2 with DLC = 12 and sequence = 4*
PGN 126998 on bus 1 with DLC = 10 and sequence = 5*
PGN 126998 on bus 2 with DLC = 10 and sequence = 5*

Or

Test 7.2 Unique Fast Packet Sequence Counter [3.1.3.4] – Not Applicable

C.7.3 Fast Packet Message Time Out [3.1.4.8] – Mandatory

This section certifies the requirement that "*A fast-packet message shall not be considered as incomplete until a time-out of 750 ms from the last received frame occurs*." The purpose of this test is to verify that the DUT will function properly on a heavily loaded network, i.e. that it does not drop messages prematurely, when dealing with significant yet acceptable delays between a given message's frames. Though this test is conducted with only a single fast packet message, this situation will occur in cases where devices may need to handle multiple concurrent fast packet transmissions on the network. This test is intended to verify that the DUT adheres to the timeout period on reception of fast packet PGNs, however, Level B devices are not required to receive any Fast Packet PGNs, but if they do they must do so in accordance with this standard. This test and verification that the DUT adheres to the specification can only be accomplished if the Fast Packet Request Group Function PGN 126208 has been implemented.

C.7.3.1 Method of measurement (Fast Packet Message Time Out):

This test requires the ability of the test hardware and software to be able to delay or not complete sending all the frames of a message for a specified period of time. To conduct this test, a Request Group Function PGN 126208, requesting the Product Information PGN 126996, should be generated on both buses. Some portion of the request PGN should be sent on both buses, and then a delay of 745ms should occur, followed by the remaining frames of the request PGN on both buses.

C.7.3.1.1 Required results:

If the DUT supports the Request Group Function PGN, then upon receiving the completed request, it should provide the Product Information PGN 126996 on both buses. The adjusted DLC on the response message should be the same on both buses and in the range of 9 to 15. If the DUT does not support the Request Group Function PGN, than adherence to this requirement cannot be validated, this test is not run, and no failure is indicated.

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C.7.3.2 Reporting requirements:

Provide detailed information as to what part (Method of measurement) of test 7.3 passed or failed. Provide the Certification Test major section number (such as 7.3), and the Method of measurement section number, with the PGN number(s) that were utilized in each Method of measurement. The DLC of the response messages must be recorded. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.7.3.2.1 Example text:

The following text shows three different results. The first example shows the result when the test, consisting of one method, is passed. The second example depicts the results of a failure to respond with the Product Information PGN. The third example illustrates the situation when the DUT does not support the PGNs necessary to conduct this test. Other results may be possible and should be similarly documented.

Test 7.3 Fast Packet Message Time Out [3.1.4.8] – Mandatory - PASSED
Method 7.3.1: PGN 126208 requesting PGN 126996 sent on bus 1
PGN 126208 requesting PGN 126996 sent on bus 2
PGN 126996 Product Information PGN received on bus 1 with DLC = 13
PGN 126996 Product Information PGN received on bus 2 with DLC = 13

Or

Test 7.3 Fast Packet Message Time Out [3.1.4.8] - Mandatory - FAILED*
Method 7.3.1: PGN 126208 requesting PGN 126996 sent on bus 1
PGN 126208 requesting PGN 126996 sent on bus 2
PGN 126996 Product Information PGN NOT received on bus 1*
PGN 126996 Product Information PGN NOT received on bus 2*

Or

Test 7.3 Fast Packet Message Time Out [3.1.4.8] – Not Applicable

C.7.4 ISO 11783-3 3.2.2 Reserved Bit ® [Table 3-1] – Mandatory

This test is intended to confirm that the DUT implements the reserved bit in bit-position five of the can frame. The requirement is this "*ISO 11783-3 Section 3.2.2 Reserved Bit, specifies that all messages shall set the reserved bit (bit number 5) in the CAN ID field to zero on transmit*." This will be accomplished with two tests. The first test (7.4.1) will check the 5th bit of the CAN frame from the Product Information PGN 126996, and then from some non-mandatory PGN.

C.7.4.1 Method of measurement (Reserved bit in Product Information PGN):

The purpose of this method is to verify that the reserved bit is properly set in CAN frames of mandatory messages. To conduct this test an addressed ISO Request PGN 059904 is transmitted on both buses requesting the Product Information PGN 126996. The reserved bit, bit number 5, in the CAN frame will be examined.

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C.7.4.1.1 Required results:

The DUT should respond to the ISO Request with PGN 126996 Product Information on both buses. Bit 5 of the CAN frame must be set to zero. If bit 5 is not zero in the response message on either bus, the DUT fails this test. The adjusted DLC of the response message received on each bus must be the same and in the range of 9 to 15.

C.7.4.2 Method of measurement (Reserved bit in non-mandatory PGN):

The purpose of this method is to verify that the reserved bit is properly set in CAN frames of all supported non-mandatory messages. To conduct this test, generate and send on both buses an ISO Request PGN 059904 asking for the immediate transmission of each supported non-mandatory PGN. The reserved bit, bit number 5, in the CAN frame will be examined.

C.7.4.2.1 Required results:

The DUT should respond to the request with the requested PGN or an ISO NACK on both buses. Any other response, the DUT fails the test. Bit 5 of the CAN frame must be set to zero. If bit 5 is not zero, the DUT fails this test. No response also constitutes failure. Also, the adjusted DLC of each redundant response message is checked to ensure they have the same value in the range of 9 to 15. If multiple requests cause the DUT to return a NACK, the adjusted DLC in the NACK response must progress through the DLC series wrapping appropriately from 15 to 9.

C.7.4.3 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 7.4 passed or failed. Provide the Certification Test major section number (such as 7.4). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Also include information that determines pass or failure, such as receipt of PGNs, the value of the reserved bit from bit position 5 in the CAN frame, and the adjusted DLC value in all responses. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.7.4.3.1 Example text:

The following text shows two different results. The first example shows the result when the test, consisting of two methods is passed. The second example depicts the case where the second test (7.4.2) fails because the value of the reserved bit is a one instead of a zero. Other results may be possible and should be similarly documented.

Test 7.4 ISO 11783-3 3.2.2 Reserved Bit [Table 3-1] - Mandatory – PASSED
Method 7.4.1: PGN 059904 ISO Request requesting PGN 126996 sent on bus 1
PGN 059904 ISO Request requesting PGN 126996 sent on bus 2
PGN 126996 Product Information PGN received on bus 1 with DLC = 14
Reserved bit from bit position 5 of CAN frame = 0
PGN 126996 Product Information PGN received on bus 2 with DLC = 14
Reserved bit from bit position 5 of CAN frame = 0
Method 7.4.2: PGN 059904 ISO Request requesting PGN ###### non-mandatory PGN sent bus 1
PGN 059904 ISO Request requesting PGN ###### non-mandatory PGN sent bus 2
PGN ###### received on bus 1 with DLC = 11
Reserved bit from bit position 5 of CAN frame = 0
PGN ###### received on bus 2 with DLC = 11

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Reserved bit from bit position 5 of CAN frame = 0

Or

Test 7.4 ISO 11783-3 3.2.2 Reserved Bit [Table 3-1] - Mandatory – FAILED*
Method 7.4.1: PGN 059904 ISO Request requesting PGN 126996 sent bus 1
PGN 059904 ISO Request requesting PGN 126996 sent bus 2
PGN 126996 Product Information PGN received on bus 1 with DLC = 14
Reserved bit from bit position 5 of CAN frame = 0
PGN 126996 Product Information PGN received on bus 2 with DLC = 14
Reserved bit from bit position 5 of CAN frame = 0
Method 7.4.2: PGN 059904 ISO Request requesting PGN ###### non-mandatory PGN sent bus 1
PGN 059904 ISO Request requesting PGN ###### non-mandatory PGN sent bus 2
PGN ###### received on bus 1 with DLC = 11
Reserved bit from bit position 5 of CAN frame = 1*
PGN ###### received on bus 2 with DLC = 11
Reserved bit from bit position 5 of CAN frame = 1*

C.7.5 ISO 11783-3 Annex C, NACK [Table 3-1] – Mandatory – Refer to C.6.15

This test certifies that the ISO NACK is used as required. The requirement being tested is as follows: "*ISO 11783-3 Annex C, NACK - If the Parameter Group Number in a Command or Request is not recognized by the destination, it shall reply with the NACK message*." This is achieved with one test for proper NACK response to an ISO request. See section C.6.15 for specific test methods, results, and reporting requirements.

C.7.6 NAME Support [8.3] – Mandatory - Refer to C.6.16

This test is designed to verify the following requirement: "*NMEA 2000 devices shall implement the NAME field, and field dependencies of the Address Claim message as defined by PGN 60928 in Appendix B database file "2000vx.x.mdb"*." Each field of the NAME entity will be examined to determine if they contain appropriate values. One test is used to conduct examination of the NAME fields. See section C.6.16 for specific test methods, results, and reporting requirements.

C.7.7 Product Information [8.3.2] – Mandatory

This test certifies the following requirement. "All NMEA 2000 devices shall support the Product Information message containing at a minimum the "NMEA 2000 Version Supported" data field. The Product Information message shall be implemented as described in Appendix B." If the Level B DUT supports the Request Group Function PGN 126208 see section C.6.17 for specific test methods, results, and reporting requirements. If the DUT does not support the Request Group Function PGN 126208, the ISO Request portion of test C.6.17 will be utilized to satisfy then this test.

C.7.8 Address Claim Test – Mandatory – Refer to C.6.20

This section certifies one of the most basic and important characteristics of the NMEA 2000 standard, automatic self-configuring Address Claiming, at power up. There are several very important aspects to

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dynamic address claiming and assignment that must be adhered to in order to ensure interoperability with other NMEA 2000 devices. This requires multiple tests. Each of the following tests has one specific goal or criteria to be satisfied. See section C.6.20 for specific test methods, results, and reporting requirements.

C.7.9 Minimum Message Implementation [Table 9-2] – Mandatory

This section is intended to certify, "*The messages listed in Table 9-2 are all required to be implemented*." Many previous test sections utilize these messages, both to and from the DUT, to conduct the certification. The verification of this requirement, when applicable, is based upon those other tests as described in the methods below.

C.7.9.1 Method of measurement (PGN 059392):

This method is dependent upon test 7.5. If test 7.5 is passed, the DUT has implemented the ISO Acknowledgement PGN 059392.

C.7.9.1.1 Required Results:

Check that test 7.5 has passed. Failure of test 7.5 constitutes failure of this method.

C.7.9.2 Method of measurement (PGN 059904):

There are no previous tests that rely solely upon this PGN, so an additional test is needed. The test equipment will send a PGN 059904 on both buses to the DUT, both destination and globally addressed, requesting the Address Claim PGN 060928. A proper response to this test will verify that the DUT has implemented the ISO Request.

C.7.9.2.1 Required Results:

The DUT should respond with two PGN 060928 messages, one for the destination specific, and one for the global ISO request 059904. Each response message should be received on both buses with matching adjusted DLC values and matching address values. The adjusted DLC value must be in the range of 9 to 15, progress through the DLC series on successive transmissions, and wrap from 15 to 9. The NAME values returned should have the appropriate system instance value for the bus received on (i.e. the NAME value must be different on each bus).

C.7.9.3 Method of measurement (PGN 060928):

This method is dependant upon tests 7.6 and 7.8. If tests 7.6 and 7.8 are passed, the DUT has implemented the ISO Address Claim PGN 060928 correctly.

C.7.9.3.1 Required Results:

Check that tests 7.6 and 7.8 have been passed. Failure of test 7.6 or 7.8 constitutes failure of this method.

C.7.9.4 Method of measurement (PGN 126996):

This method is dependant upon test 7.1. If test 7.1 is passed, the DUT has implemented the Product Information PGN 126996.

C.7.9.4.1 Required Results:

Check that test 7.1 has been passed. Failure of test 7.1 constitutes failure of this test.

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C.7.9.5 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 7.9 passed or failed. Provide the Certification Test major section number (such as 7.9). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.7.9.5.1 Example text:

The text in this section is merely an outline of two of many possible outcomes. The first example text outlines the results when the DUT passes the test. The second example illustrates that the DUT failed test 7.5. Other results may be possible and should be similarly documented.

Test 7.9 Minimum Message Implementation [Table 9-2] – Mandatory – PASSED
Method 7.9.1: Method 7.5 Passed
Method 7.9.2: PGN 059904 ISO Request requesting PGN 060928 sent destination specific on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent destination specific on bus 2.
PGN 060928 received on bus 2.
PGN 060928 received on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent globally on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent globally on bus 2.
PGN 060928 received on bus 1.
PGN 060928 received on bus 2.
Method 7.9.3: Method 7.6 Passed
Method 7.8 Passed
Method 7.9.4: Method 7.1 Passed

Or

Test 7.9 Minimum Message Implementation [Table 9-2] – Mandatory – PASSED
Method 7.9.1: Method 7.5 Failed*
Method 7.9.2: PGN 059904 ISO Request requesting PGN 060928 sent destination specific on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent destination specific on bus 2.
PGN 060928 received on bus 2.
PGN 060928 received on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent globally on bus 1.
PGN 059904 ISO Request requesting PGN 060928 sent globally on bus 2.
PGN 060928 received on bus 1.
PGN 060928 received on bus 2.
Method 7.9.3: Method 7.6 Passed
Method 7.8 Passed
Method 7.9.4: Method 7.1 Passed

C.7.10 Address Bumping – Mandatory – Refer to C.6.22

This test is intended to verify that once a DUT has claimed an address, and begun normal operation, that it properly handles having its address bumped (taken) by a higher priority device entering the network. Tests

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with lower priority claims will be made as well as testing the case when no addresses are available. Though these tests are similar to those for power-up address claim, these tests help verify that the DUT properly executes the address claim functions after having been operating in a normal state. See section C.6.22 for specific test methods, results, and reporting requirements.

C.7.11 Default Reporting Intervals – Mandatory

The purpose of this is to verify that the DUT, operating with factory default settings, adheres to the default reporting intervals for the NMEA 2000 published PGNs. Reporting intervals of Proprietary PGNs will be observed if possible.

C.7.11.1 Method of measurement (Default reporting intervals):

This test requires the DUT to have claimed an address and be operating in normal mode. The test consists of listening to the normal PGN outputs of the DUT and recording the time between like PGN transmissions. Determination of the actual PGNs expected is dependent upon the user entered PGN number(s) or the PGNs as determined by the certification tool. This forms the basis of what PGN outputs are expected from the DUT. Any anomalies, i.e. PGNs in the list but not transmitted by the DUT, PGNs not in the list but transmitted by the DUT, or PGNs transmitted by the DUT on only one bus should be noted. See test C.6.23 for results and reporting requirements.

C.7.12 Default Priority – Mandatory

The purpose of this is to verify that the DUT, operating with factory default settings, adheres to the default priority settings for the NMEA 2000 published PGNs. Priority values of Proprietary PGNs will be observed if possible.

C.7.12.1 Method of measurement (Default reporting intervals):

This test requires the DUT to have claimed an address and be operating in normal mode. The test consists of listening to the normal PGN outputs of the DUT and recording the priority value of each PGN sent by the DUT. Determination of the actual PGNs expected is dependent upon the user entered PGN number(s) or the PGNs as determined by the certification tool. This forms the basis of what PGN outputs are expected from the DUT. Any anomalies, i.e. PGNs in the list but not transmitted by the DUT or PGNs not in the list but transmitted by the DUT should be noted. See test C.6.24 for results and reporting requirements.

C.7.13 Unique NAME Entities [3.6.2] – Mandatory - Refer to C.6.25

This test validates the NAME value transmitted on both buses is identical except for the System Instance field. The System Instance field must have a value of 0 on one bus and a value of 1 on the other bus. See section C.6.25 for specific test methods, results, and reporting requirements.

C.7.14 Address Claim Class 1 Contention [3.6.3] – Mandatory - Refer to C.6.26

The purpose of this test is to validate the unit's response to address claim contention from a Class 1 device. See section C.6.26 for specific test methods, results, and reporting requirements.

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C.7.15 Address Claim Single Bus Failure and Recovery [3.6.3.1, 3.6.3.2, 3.6.3.3] – Mandatory - Refer to C.6.27

This test verifies that the DUT completes the address claim process and begins normal operation when one bus is broken. This test also verifies that the DUT recovers correctly when the broken bus is restored. See section C.6.27 for specific test methods, results, and reporting requirements.

C.7.16 Default Redundant Transmissions [3.6.4.1] – Mandatory - Refer to C.6.28

The purpose of this test is to verify that the DUT transmits all PGNs on both buses. See section C.6.28 for specific test methods, results, and reporting requirements.

C.7.17 Adjusted DLC Sequencing [3.6.4.2, 3.6.4.3, 3.6.4.4] – Mandatory - Refer to C.6.29

This section certifies that each PGN transmitted by a Class 2 device is transmitted with an adjusted DLC in the range of 9 to 15. This test certifies that on successive transmissions of the PGN the adjusted DLC increments through the series, restarting at 9 when the value 15 has been used. This test also certifies that fast packet transmissions use the same adjusted DLC value for every frame of the transmission. And finally, this test certifies that a separate DLC series is used for each PGN transmitted by the DUT. See section C.6.29 for specific test methods, results, and reporting requirements.

C.7.18 ISO Request from Class 2 [3.6.5.2] – Mandatory - Refer to C.6.33

This section validates that the DUT responds on both buses to Class 2 transmissions of ISO Request PGN 059904. See section C.6.33 for specific test methods, results, and reporting requirements.

C.7.19 Redundant ISO Request Timeout [3.6.5.3] – Mandatory - Refer to C.6.34

The purpose of this test is to verify the DUT correctly identifies redundant ISO Request received from a Class 2 device. See section C.6.34 for specific test methods, results, and reporting requirements.

C.7.20 Verification of unsupported Level A capabilities – Mandatory

The purpose of this test is verify that level B devices respond consistently and properly to any requests, commands, or attempts to access capabilities, that may be supported by the device.

If the device implements and supports any capability of the NMEA 2000 Request and Command Group Function PGN 126208, then it must respond properly to any requests or commands received. If the device does not support any form of PGN 126208, which is optional for Level B, then the device will be checked.

If the device implements and supports the Transmit/Received PGN List, PGN 126464, then the DUT must respond properly to requests received for PGN 126464. If the device does not support PGN 126464, which is optional for Level B, then the device will be checked. If the device supports PGN 126464, but does not support PGN 126208, a test will be conducted to confirm this. If the device supports PGN 126464 and PGN 126208, then the Level A tests of C.6.18 and C.6.19 shall be conducted.

The following test methods are designed to confirm proper behavior by the device under these circumstances.

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C.7.20.1 Method of measurement (Acknowledgement of Request Group Function):

The test is conducted by sending a request for a Level B mandatory PGN using the Request Group Function, PGN 126208. The Request Group Function PGN 126208 requesting the Product Information PGN 126996 will be sent to the DUT on both buses. Though the Product Information PGN 126996 is a mandatory PGN for all NMEA 2000 devices, Level B devices are only required to provide this PGN from requests made using the ISO Request PGN 059904. Generate a Request Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)

C.7.20.1.1 Required Results:

There are three acceptable responses, depending upon the support for PGN 126208.

No PGN 126208 Support:

If the DUT does not support any PGN 126208 capability, then the required response is no response. Any Acknowledgment on either bus under this circumstance is considered a failure.

Any PGN 126208 Support:

If the DUT supports any PGN 126208 Request or Command capabilities, then it must fully support PGN 126208 Acknowledgements to ALL Requests and Commands. There are two types or forms of acknowledgement responses. The first form is the full response with individual error codes for each item requested or commanded. The second form is an abbreviated acknowledgment with no individual field error codes for each item requested or commanded. See the Example Text. Regardless of form, the response must be received on both buses. The adjusted DLC in the response messages must be the same on both buses and be in the range of 9 to 15.

C.7.20.2 Method of measurement (Acknowledgement of Command Group Function):

The test is conducted by sending a command to change the priority and manufacturers product code value of a Level B mandatory PGN using the Command Group Function, PGN 126208. The Command Group Function PGN 126208 attempting to modify the priority and manufacturers product code of the Product Information PGN 126996 will be sent to the DUT. Generate a Command Group Function PGN 126208, with the data fields set as follows.

Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)

C.7.20.2.1 Required Results:

There are three acceptable responses, depending upon the support for PGN 126208.

No PGN 126208 Support:

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If the DUT does not support any PGN 126208 capability, then the required response is no response. Any Acknowledgment on either bus under this circumstance is considered a failure.

Any PGN 126208 Support:

If the DUT supports any PGN 126208 Request or Command capabilities, then it must fully support PGN 126208 Acknowledgements to ALL Requests and Commands. There are two types or forms of acknowledgement responses. The first form is the full response with individual error codes for each item requested or commanded. The second form is an abbreviated acknowledgment with no individual field error codes for each item requested or commanded. See the Example Text. Regardless of form, the response must be received on both buses. The adjusted DLC in the response messages must be the same on both buses and be in the range of 9 to 15.

C.7.20.3 Method of measurement (PGN 126464 Transmit/Received PGN List):

This method validates the DUT's implementation of the Transmit/Received PGN list. The contents of the PGN if supported are examined to ensure that all the mandatory PGNs are included. Non-mandatory PGNs require manual verification. To conduct this test, a destination specific addressed ISO request PGN 059904 is sent to the DUT on both buses asking for PGN 126464 Transmit/Received PGN List.

C.7.20.3.1 Required Results:

There are three acceptable responses, depending upon the support for PGN 126464 and PGN 126208.

No PGN 126464 Support:

The required response is an ISO Acknowledgement PGN 059392 indicating a negative acknowledgement on both buses.

PGN 126464 Supported but no support for PGN 126208:

The required response is the transmission on both buses of PGN 126464 Transmit PGN List and PGN 126464 Received PGN list. The adjusted DLC in the response messages must be the same on both buses and be in the range of 9 to 15, increment through the DLC series with successive transmissions, and wrap from 15 to 9.

Both PGN 126464 and PGN 126208 supported:

In this case, tests C.6.18 and C.6.19 shall be conducted and the results of those tests shall determine pass or failure of this test.

C.7.20.4 Reporting Requirements:

Provide detailed information as to what parts (Methods of measurement) of test 7.20 passed or failed. Provide the Certification Test major section number (such as 7.20). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.7.20.4.1 Example Text:

The text in this section shows four of many possible outcomes. The first example text shows the results when the DUT does not support PGN 126208 or PGN 126464 and passes the tests. The second example shows the result when the DUT does not support PGN 126208, but does support PGN 126464. The third example shows the result when the DUT supports both PGN 126208 and PGN 126464, and passes all the tests. The fourth example shows that though the DUT supports PGN 126208 and PGN 126464, it failed the test by not responding properly to the Command Group Function PGN 126208 Command. Other results may be possible and should be similarly documented.

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Example 1 – No support for PGN 126208 or PGN 126464
Test 7.20 Verification of unsupported Level A capabilities - Mandatory – PASSED
Method 7.20.1: PGN 126208 Request destination addressed containing the following parameters sent bus 1:
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Request destination addressed containing the following parameters sent bus 2:
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT on bus 1
No Response from DUT on bus 2
Method 7.20.2: PGN 126208 Command destination addressed containing the following parameters sent bus 1:
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Command destination addressed containing the following parameters sent bus 2:
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT on bus 1
No Response from DUT on bus 2
Method 7.20.3: PGN 059904 ISO Request requesting PGN 126464 sent bus 1
PGN 059904 ISO Request requesting PGN 126464 sent bus 2
PGN 059392 received from DUT on bus 1
Field 1 - Control Byte = 1
Field 2 - Group Function Value = 255 (FF_{16})
Field 3 - Reserved bits = 16777215 ($FFFFFF_{16}$)
Field 4 - PGN of requested information = 126464

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PGN 059392 received from DUT on bus 2
Field 1 - Control Byte = 1
Field 2 - Group Function Value = 255 (FF_{16})
Field 3 - Reserved bits = 16777215 ($FFFFFF_{16}$)
Field 4 - PGN of requested information = 126464

Example 2 – No support for PGN 126208, but PGN 126464 supported.
Test 7.20 Verification of unsupported Level A capabilities - Mandatory – PASSED
Method 7.20.1: PGN 126208 Request destination addressed containing the following parameters sent bus 1:
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Request destination addressed containing the following parameters sent bus 2:
Field 1 - Complex Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT on bus 1
No Response from DUT on bus 2
Method 7.20.2: PGN 126208 Command destination addressed containing the following parameters sent bus 1
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Command destination addressed containing the following parameters sent bus 2
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT on bus 1
No Response from DUT on bus 2
Method 7.20.3: PGN 059904 ISO Request requesting PGN 126464 sent bus 1

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PGN 059904 ISO Request requesting PGN 126464 sent bus 2
PGN 126464 Transmit PGN List received on bus 1 with DLC = 9:
Mandatory PGNS Listed: 059392, 060928, 129996
Non-Mandatory PGNs Listed: 126464, 126998, 129025, 129026, 129029, 129044, 129045, 129284 => Require Verification
PGN 126464 Transmit PGN List received on bus 2 with DLC = 9:
Mandatory PGNS Listed: 059392, 060928, 129996
Non-Mandatory PGNs Listed: 126464, 126998, 129025, 129026, 129029, 129044, 129045, 129284 => Require Verification
PGN 126464 Received PGN List received on bus 1 with DLC = 10.
Mandatory PGNS Listed: 059392, 059904, 060928
Non-Mandatory PGNs Listed: 129538 => Requires Verification
PGN 126464 Received PGN List received on bus 2 with DLC = 10.
Mandatory PGNS Listed: 059392, 059904, 060928
Non-Mandatory PGNs Listed: 129538 => Requires Verification

Example 3 – Support for PGN 126208 and PGN 126464
Test 7.20 Verification of unsupported Level A capabilities - Mandatory – PASSED
Method 7.20.1: PGN 126208 Request destination addressed containing the following parameters sent bus 1:
Field 1 - Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Request destination addressed containing the following parameters sent bus 2:
Field 1 - Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Acknowledgment received on bus 1 with DLC = 10:
Acknowledgment Group Function Code = 2
PGN of message being acknowledged = 126996
PGN error code = 0
Transmission Interval code = 0
Number of Requested or Commanded Parameters = 1
Parameter error code = 3
PGN 126208 Acknowledgment received on bus 2 with DLC = 10:
Acknowledgment Group Function Code = 2
PGN of message being acknowledged = 126996
PGN error code = 0

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Transmission Interval code = 0
Number of Requested or Commanded Parameters = 1
Parameter error code = 3

Method 7.20.2: PGN 126208 Command destination addressed containing the following parameters sent bus 1:
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Command destination addressed containing the following parameters sent bus 2:
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Commanded PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Acknowledgment received on bus 1:
Acknowledgment Group Function Code = 2
PGN of message being acknowledged = 126996
PGN error code = 4
Priority error code = 4
Number of Requested or Commanded Parameters = FF
PGN 126208 Acknowledgment received on bus 2:
Acknowledgment Group Function Code = 2
PGN of message being acknowledged = 126996
PGN error code = 4
Priority error code = 4
Number of Requested or Commanded Parameters = FF

Method 7.20.3: Method 6.18 Passed (Full results of 6.18 to be listed)
Method 6.19 Passed (Full results of 6.19 to be listed)

Example 4 – Support for 126208
Test 7.20 Verification of unsupported Level A capabilities - Mandatory – FAILED*
Method 7.20.1: PGN 126208 Request destination addressed containing the following parameters sent bus 1:
Field 1 - Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Request destination addressed containing the following parameters sent bus 2:

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Field 1 - Request Group Function Code = 0 (Request group function)
Field 2 - Requested PGN = 126996
Field 3 - Transmission Interval = 4294967295 ($FFFFFFFF_{16}$)
Field 4 - Transmission interval offset = 0 (0000_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Acknowledgment received on bus 1:
Acknowledgment Group Function Code = 2
PGN of message being acknowledged = 126996
PGN error code = 0
Transmission Interval code = 0
Number of Requested or Commanded Parameters = 1
Parameter error code = 3
PGN 126208 Acknowledgment received on bus 2:
Acknowledgment Group Function Code = 2
PGN of message being acknowledged = 126996
PGN error code = 0
Transmission Interval code = 0
Number of Requested or Commanded Parameters = 1
Parameter error code = 3

Method 7.20.2: PGN 126208 Command destination addressed containing the following parameters sent bus 1:
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Requested PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
PGN 126208 Command destination addressed containing the following parameters sent bus 2:
Field 1 - Complex Command Group Function Code = 1 (Command group function)
Field 2 - Requested PGN = 126996
Field 3 – Priority Setting = 5 (5_{16})
Field 4 – Reserved Bits = 15 (F_{16})
Field 5 - Number of Pairs of Request Parameters to follow = 1
Field 6 - Field number of first parameter = 2 (field number 2 from 126996)
Field 7 - Value of first parameter = (the value of the DUT's Manufacturers Product code + 1)
No Response from DUT on bus 1*
No Response from DUT on bus 2*

Method 7.20.3: Method 6.18 Passed (Full results of 3.18 to be listed)
Method 6.19 Passed (Full results of 3.19 to be listed)

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C.7.21 NMEA Request/Command Group Function from Class 2 [3.6.5.1] – Mandatory - Refer to C.6.32

This test is only performed if the DUT additionally supports the NMEA Request/Command Group Function PGN. This test validates that the DUT responds to Class 2 NMEA Request and Command Group Function PGNs on both buses. See section C.6.32 for specific test methods, results, and reporting requirements.

C.7.22 Responding to Class 1 Destination Specific Requests/Commands [3.6.5.4] – Mandatory - Refer to C.6.35

The purpose of this test is to verify the DUT's response to Class 1 requests and commands where the response would be transmitted destination specific. See section C.6.35 for specific test methods, results, and reporting requirements. If the DUT does not support the NMEA Request Group Function PGN, only test method 6.35.1 is performed. If the DUT does support the NMEA Request Group Function PGN, test methods 6.35.1, 6.35.2, and 6.35.3 are performed.

C.7.23 Responding to Class 1 Broadcast Requests/Commands [3.6.5.5] – Mandatory - Refer to C.6.36

The purpose of this test is to verify the DUT's response to Class 1 requests and commands where the response would be transmitted broadcast. See section C.6.36 for specific test methods, results, and reporting requirements. If the DUT does not support the NMEA Request Group Function PGN, only test method 6.36.1 is performed. If the DUT does support the NMEA Request Group Function PGN, test methods 6.36.1 and 6.36.2 are performed.

C.7.24 Address Claim Request Node Independence [3.6.5.6] – Mandatory - Refer to C.6.37

This test is only performed if the DUT additionally supports the NMEA Request/Command Group Function PGN. The purpose of this test is to verify the DUT responds to a request for the address claim PGN 60928 when the request applies to either node, not just the node the request was received on. See section C.6.37 for specific test methods, results, and reporting requirements.

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C.8 Class 2 Listen Only - Automated Test Methods

This section contains the requirements and methods of measurement that can be performed automatically via a suite of software and hardware designed to certify compliance with the minimum requirements of the NMEA 2000 standard for Listen Only Devices.

To perform these tests, you must assemble the network as shown in section C.6.

C.8.1 No Response Verifiaction – Mandatory

The purpose of this test is to verify that the Listen Only device does not transmit any acknowledge or error frames on the CAN bus.

C.8.1.1 Method of measurement (Repsonse to Address Claim):

This method validates that the DUT does not respond at all when an address claim PGN is placed on the bus. To conduct this test there should be only one other device on the bus, the certification tool. The tool will generate an address claim PGN on both buses, and look for any response from the DUT.

C.8.1.1.1 Required Results:

The only acceptable response from the DUT is no response at all and the address claim PGN transmission from the tool should fail on both buses. The DUT will fail this test if the address claim PGN transmission succeeds, because this can only happen if the DUT provides an acknowledgment frame on the bus.

C.8.1.2 Reporting requirements:

Provide detailed information as to what parts (Methods of measurement) of test 8.1 passed or failed. Provide the Certification Test major section number (such as 8.1). Provide the Method of measurement section numbers, with the PGN number(s) that were utilized. Include information that determines pass or failure. This information shall be placed in a text file for use by the manufacturer and also in the Certification Results file for NMEA verification.

C.8.1.2.1 Example Text:

The text is this section shows two possible results. The first result illustrates the DUT passing the test while the second result shows the DUT failing the test.

Example 1 – DUT does not respond to address claim message
Test 8.1 No Response Verification – Mandatory - PASSED
Method 8.1.1: PGN 060928 Address Claim PGN transmitted on bus 1
PGN 060928 Address Claim PGN transmitted on bus 2
Transmission of PGN 060928 failed due to no response from DUT on bus 1
Transmission of PGN 060928 failed due to no response from DUT on bus 2

Example 2 – DUT responds to address claim message
Test 8.1 No Response Verification – Mandatory – FAILED*
Method 8.1.1: PGN 060928 Address Claim PGN transmitted on bus 1
PGN 060928 Address Claim PGN transmitted on bus 2

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Transmission of PGN 060928 successful due to acknowledge frame response from DUT on bus 1*
Transmission of PGN 060928 successful due to acknowledge frame response from DUT on bus 2*

C.9 Revision History

Revision	Date	Description
1.000	Sept 2001	Initial Release
1.001	Oct 2001	This Revision History Log added
	Oct 2001	Sections C.3.4 & C.3.5 reference to main document 3.1.3.7 & 3.1.3.8 corrected to 3.1.4.7 & 3.1.4.8 respectively.
		Spelling corrections: C.3.4 & C.3.9 – "parameters" C.3.9 – "complex" C.3.10.4.1 – "accepted"
1.002	Nov 2002	Added Title to Cover Page
1.003	May 2003	Version number changed
1.200	October 2004	Version number changed. Expanded C.1.1 Scope to include Level B and Listen Only devices. Corrected the following sections: C.2.9, C.2.10, C.2.23, C.3.1.1, C.3.1.1.1, C.3.7.3.1, C.3.10.1.1, C.3.10.2.1, C.3.10.3.1, C.3.10.4.1, C.3.13.1.1, C.3.13.2.1, C.3.13.3.1, C.3.13.4.1, C.3.13.5.1, C.3.14.2.1, C.3.18.1, C.3.18.2.1, and C.3.21.4 Expanded the following sections to test all supported non-mandatory PGNs instead of testing a single randomly chosen non-mandatory PGN: C.3.2.2, C. 3.2.2.1, C.3.2.3.1, C.3.3.1, C.3.7.3, C.3.7.3.1, C.3.7.6.1, C.3.10, C.3.10.4, C.3.10.4.1, C.3.10.7.1, C.3.12, C.3.12.2, C.3.12.4.1, C.3.13, C.3.13.2, C.3.13.5.1, and C.3.14.2 Added new section 4 Level B Automated Test Methods. Added new section 5 Listen Only Automated Test Methods. Added new test 2.42 and moved existing 2.42 test to 2.43.

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1.201	March 2016	Editorial heading number adjustments: Section 3.2 Main Document reference was changed to 3.1.6.3 Section 3.2 reference modulo 7 changed to modulo 8 Section 3.2.3.1 Main Document reference was changed to 3.1.6.3 Section 3.3 Main Document reference was changed to 3.1.6.4 Section 3.3.2.1 reference was changed to 3.1.6.4 Section 3.4 Main Document reference was changed to 3.2.6.7 Section 3.5 Main Document reference was changed to 3.2.6.8 Section 3.6 was followed by 3.6.3 Changed to 3.6.1 Section 3.6.3.1 was changed to 3.6.6.1 Section 3.6.4 to was changed to 3.6.2 Section 3.6.4.1 was changed to 3.6.2.1 Section C 3.9 was followed by 3.9. 3. Changed to 3.9.1 Section 3.9 Main Document reference was changed to 3.2.1.7 Section 3.14 Main Document reference was changed to Table 4 Section 3.15 Main Document reference was changed to Table 4 Section 3.21 Second Test number was changed to 3.23 in example test 3.23.5.1 Section C 4.1 reference modulo 7 changed to modulo 8 Section C 6.2 reference modulo 7 changed to modulo 8 Section C 6.6 was followed by 6.6.3. Changed to 6.6.1 Section C 6.12 was followed by 6.12.5. Changed to 6.12.1 Section C 6.29 was followed by 6.29.3. Changed to 6.29.1 Section C 7.1 reference modulo 7 changed to modulo 8

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